dune-rover report

Predicting slip, sinkage and traversability of wheeled robots on dry sand and dunes: a literature review, a gap analysis for Arabian dunes, and a minimal reproducible experiment plan

Author: Huda Mueen. Repository: `hudasol/dune-rover`. Retrieval dates: 2026-10-11 for the original sources; 2026-10-13 for five papers the author supplied as PDFs (per row in `research/evidence.csv`). Related work, not repeated here: `hudasol/dhill` (heat and dust operating envelope). This project is about terrain mobility only.

> **Read this first.** This is a research document. It is not engineering approval, a safety case or a performance claim for any real vehicle. Three kinds of content appear and are always labelled: **PUBLISHED**: a number read in a retrieved source (every one is a row in the evidence table, Appendix A, with URL, passage and retrieval date). **SIMULATED BY ME**: output of code in this repository (section 8). It depends on stated model assumptions and is **never field evidence**. **PLANNED**: a protocol that has not been run (section 9).

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0. Executive summary

Question. How well can slip, sinkage and traversability of wheeled (secondarily legged) robots on dry sand and dunes be predicted from terrain data and models; how well does that transfer from lab to field; and what would a minimal open data planner for Arabian dunes need that the literature does not provide?

Method. A written protocol (`docs/PROTOCOL.md`) came first. Five row extraction passes (terramechanics, simulation, learning, field and analog, legged and wheel design) collected 144 evidence rows from retrieved sources (26 more were read on 2026-10-13 from five papers the author supplied as PDFs, giving 170; one of them was split into a lab row and a field row on 2026-10-18, giving 187; 323 more were added on 2026-10-20, section 13, giving 510),

and a sixth pass audited open data, dune morphology and UAE facts. Separate agents then re opened the sources for **every one of the 510 rows** (30 randomly drawn plus 7 flagged in phase 1, the other 107 in phase 2, the 26 supplied PDF rows in phase 3, and the two rows split from one of them in phase 4 on 2026-10-18, and the 323 rows added in phase 5 on 2026-10-20). A simulation only slope and slip traversability study was run on real Copernicus GLO-30 elevation data from the Liwa and Rub' al Khali areas, with one at a time and global (Latin hypercube) sensitivity analysis. Phase 2 added a sub grid slip face detector, a three class map (TRAVERSABLE, BLOCKED, UNKNOWN) with Monte Carlo over the four assumed parameters, a second morphology (linear dunes), and a pre registered field kit.

Headline findings (published evidence).

1. **Classical and granular models work only after per soil fitting, and mostly on level ground.** Bekker-Wong style drawbar pull errors are about 24 to 27 % (median) on two lunar simulant wheel datasets and sinkage errors 28 to 52 %, but reach 135 % and 385 % on a firm soil with a small light wheel (SIM-025 to SIM-027). Granular RFT gives drawbar mean absolute errors of roughly 5 to 7 N on medium sand and 15 N on a fine simulant (TM-001 to TM-004). The Janosi shear modulus K, which controls slip, differs about 75 fold between the papers first reviewed (0.0006 to 0.045 m, the top of the range from poppy seeds) and up to about 117 fold if the Toyoura sand model input of Ishigami et al. 2009 (kx 0.07 m, TM-IS09-2) is included, so it is tuned rather than transferred.
2. **Speed and slope break the quasi static assumptions.** On olivine sand the drawbar coefficient fell 8 % between 0.01 and 0.2 m/s and 63.7 % between 0.2 and 1 m/s (TM-018, TM-019). A rover scale Bekker-Wong slope model had untuned side slip errors of 14.8 % at 10 degrees and 25.9 % at 20 degrees on Toyoura sand (TM-022, TM-023, a thesis); the 0.49 % at 20 degrees came after tuning on that same case (TM-024, in sample).
3. **Field slip on sand is large, patchy and not predictable from slope or appearance alone.** On natural Earth dunes, the only slope curve found is from Dumont Dunes, California: a 340 kg test rover went from below 50 % slip at 12.5 degrees to about 75 % at 16 degrees and about 90 % at 20 degrees, with no soil parameters reported (FLD-HV13-1, FLD-HV13-2). Spirit measured 15 to 20 % slip at 19 degrees and 95 % slip at 16 degrees about 30 m later on terrain that looked alike (FLD-011, FLD-012). Opportunity's visual odometry slip stayed at 98.9 to 99.5 % at Purgatory ripple (FLD-002). Perseverance exceeded 94 % at about 20 degrees (FLD-025). Visual odometry, the field slip sensor, itself degrades on bland sand (FLD-041).

4. **Learned predictors are validated mostly in distribution.** The strongest held out result is a Perseverance model (AUROC 0.874 on held out sols, LRN-010) on Mars. No retrieved paper holds out terrain type (train without sand, test on sand), and none reports a learned slip or sinkage error on Earth dunes.
5. **Arabian evidence is almost empty.** The only Arabian wheeled robot field evidence found is qualitative: a rover in Dhofar, Oman, met looser sand than expected and got stuck (FLD-042). No Bekker or RFT parameters for Arabian dune sand, no grain size map of the Empty Quarter and no open LiDAR or drone elevation model of its dunes were found. The UAE authored wheel work found is single wheel lunar simulant testing (not sand).
6. **Open elevation data resolve dunes but not slip faces.** Copernicus GLO-30 (about 28 by 31 m posts at 23.5 N, free licence with attribution) resolves dune crests and interdune corridors for dunes tens to hundreds of metres high. Slip faces of the large Rub' al Khali dunes are only 1 to 4 posts wide, so slope read from the grid is biased low; 90 m products remove 80 to 100 % of the cells steeper than 25 degrees in the three windows used here.

Headline findings (SIMULATED BY ME, model assumption dependent, not field evidence). With a stated friction circle slip model and nominal assumptions, the 30 m grid blocks only 0.2 to 1 % of travel directions in three real dune windows, because few cells exceed the model's nominal climbing limit of 25.5 degrees. That is mostly a statement about grid resolution and assumptions, not evidence that the dunes are easy. Across the plausible assumption ranges, under the original ranges (i0 0.05 to 0.40, f_r 0.05 to 0.25), the model's climbing limit varies from 7.6 to 34.7 degrees (5 to 95 %: about 15 to 32 degrees). The assumptions that change the map most are the friction angle, the form of the slip law, the DEM posting, the slip scale i0 and the motion resistance coefficient. DEM noise and the planning slip limit matter least (section 8).

Headline findings, phase 2 (SIMULATED BY ME, model assumption dependent; detector validated on synthetic truth only in phase 2, tested on real lidar on 2026-10-20, section 13). A 30 m DEM cannot see slip faces, so each cell now carries a modelled probability that an unresolved active face lies inside it and the map has three classes. At 30 m the **UNKNOWN share is 8.6 % in A south Liwa, 10.9 % in B (megadunes) and 2.5 % in C (Saudi Rub' al Khali linear dunes)**; BLOCKED is under 0.1 % everywhere, because the DEM hardly shows slopes the model calls impassable. If a face within one post of the cell counts (a post hoc variant) the UNKNOWN share is 34.4 %, 32.2 % and 22.0 %. 99.1 % (A), 97.0 % (B) and 100.0 % (C) of cells have the same slope based verdict in at least 95 % of the 256 draws of the four assumed slip parameters, so, under the original ranges (i0 0.05 to

0.40, f_r 0.05 to 0.25), the three class map is far more sensitive to the sub grid morphology assumptions and the resolution than to those four parameters (section 8.7); with the widened ranges of 2026-10-18 this no longer holds (below). Of 153 pre registered predictions (stage 1: 1 hit, 2 mixed, 1 missed; stage 2: 3 hit, 1 mixed, 4 missed; stage 3: 2 hit, 1 mixed, 3 missed; stage 4: 1 hit, 0 mixed, 3 missed; stage 5: 9 hit, 1 mixed, 0 missed; stage 6: 5 hit, 0 mixed, 1 missed; stage 7: 4 hit, 0 mixed, 2 missed; stage 8: 3 hit, 0 mixed, 0 missed; stage 9: 6 hit, 0 mixed, 2 missed; stage 10: 5 hit, 0 mixed, 0 missed; stage 11: 3 hit, 0 mixed, 0 missed; stage 12: 3 hit, 0 mixed, 3 missed; stage 13: 4 hit, 0 mixed, 3 missed; stage 14: 5 hit, 0 mixed, 2 missed; stage 16: 2 hit, 0 mixed, 2 missed; stage 17: 1 hit, 0 mixed, 4 missed; stage 18: 3 hit, 0 mixed, 2 missed; stage 15: 4 hit, 0 mixed, 3 missed; stage 19: 6 hit, 0 mixed, 2 missed; stage 21: 2 hit, 0 mixed, 3 missed; stage 22: 3 hit, 0 mixed, 1 missed; stage 20: 5 hit, 0 mixed, 2 missed; stage 23: 4 hit, 0 mixed, 1 missed; stage 24: 2 hit, 0 mixed, 1 missed; stage 25: 7 hit, 0 mixed, 2 missed; stage 26: 5 hit, 0 mixed, 3 missed), the hits and misses are in `docs/PREDICTIONS.md`; most of the quantitative ones about the phase 2 and 3 simulation missed, and the stage 5 hits were partly informed by hand arithmetic on the model equation.

Headline finding, published anchors (SIMULATED BY ME against PUBLISHED values; comparison of unlike vehicles and soils; section 8.8). Run on the same slopes as the Dumont Dunes curve, the nominal model crosses 50 % slip at 25.5 degrees against 12.6 degrees for the published curve, and the curve lies above the 95th percentile of the model's assumption range between 10.7 to 17.9 degrees. With the Ishigami et al. 2009 Table I friction angle the model gives 0.01 to 0.20 slip on a 10 degree side slope (0.22 to 0.40 with the Table I shear modulus), where a slip ratio of about 0.5 was measured. **The model predicts too little slip on sand slopes; a post hoc refit at the edge of its ranges still misses the shape of the Dumont curve. Widened slip scale range (2026-10-18, sections 8.4, 8.7, 8.8):** with i0 widened to 0.05 to 1.20 (nominal 0.81, the lower bound implied by Ishigami 2009 Table I; shorter contact lengths imply larger values) the nominal model crosses 50 % slip at 10.2 degrees, now before the Dumont curve, and UNKNOWN at 30 m rises to 33.9 % (A), 39.2 % (B) and 11.2 % (C); with f_r also widened to 0.35 it reaches 92.6 %, 90.1 % and 97.4 %. In the second set 9 to 12 % of draws cannot move on level ground at all (`results/range_set_diagnostics/immobile_on_level.csv`), a combination the original resistance range excluded, and that explains most of its UNKNOWN share. Widening the ranges brackets the Dumont curve and the measured Ishigami slip, though not Ishigami's own model value at 5 degrees, and leaves the map much less certain.

Headline findings, 2026-10-20 (section 13; SIMULATED BY ME against PUBLISHED data; every main test pre registered in stages 12 to 18, post hoc analyses labelled). The evidence table grew to 510 verified rows from 176 sources. Calibrated on other studies, 4 slip model families predicted a held out study with median errors of 0.20 to 0.24 slip, against a pre registered bar of 0.15 (exponential fails (held out median 0.24); Mohr Coulomb ratio fails (held out median 0.23); logistic fails (held out median 0.24); Bekker Wong fails (held out median 0.20)). Project Chrono with published soils gives far less slip than the Dumont curve and disagrees with a direct implementation of the same Bekker and Janosi laws. Against one metre lidar of three North American dune fields, the slip face detector and three class map work much better at 10 m than at 30 m, where 25.6 % to 28.6 % of TRAVERSABLE cells contain steep faces and 36 % to 73 % of face cells are called TRAVERSABLE; route planning on 30 m maps was no safer than naive planning. With slip ranges calibrated on published curves, UNKNOWN at 30 m becomes 62.8 % (A), 44.3 % (B) and 46.0 % (C), or 25.1 %, 30.9 % and 4.2 % if draws that cannot move on level ground are rejected (post hoc).

Verification. All 510 rows of the current table have an independent re check against the source (section 11). In the first check of the 170 rows as extracted, material errors were **9 of 170 (5.3 %)**: 6 in the 107 rows checked in phase 2 (5.6 %), 3 in the 26 rows from supplied PDFs checked in phase 3 (11.5 %, all on validation type) and none in the 37 rows of phase 1; any discrepancy including minor label errors: 51 of 170 (30.0 %). All are corrected in the table. For the current table, taking each row's latest verdict (the two rows split from FLD-HV13-5 were checked again on 2026-10-18), material errors are **38 of 510 (7.5 %, Wilson 95 % interval 5.5 to 10.1 %)** and any discrepancy 136 of 510 (26.7 %). The phase 1 random sample (0 of 30 material) had understated the rate. The unit of the slope term in SIM-013 is not stated in its source, so that row is flagged and must not be used numerically outside the tested envelope.

Biggest gap and first experiment. There is no published slip versus slope dataset for a wheeled robot on dune sand with full soil characterisation; the nearest, from Dumont Dunes, reports no soil parameters (FLD-HV13-1), and none is from Arabian sand. The first field experiment to run is the **slope ladder** (section 9): one small rover with fixed wheels, one dune stoss slope, runs at about 0, 5, 10, 12.5, 15, 17.5 and 20 degrees (bins revised on 2026-10-15; 25 degrees only if allowed, and an open decision remains, section 9) with RTK ground truth and full soil description, calibrate on the gentle slopes and predict the steep slopes blind.

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1. Scope, protocol and method

Protocol. Written before searching, as the file itself states (`docs/PROTOCOL.md`; no independent timestamp): search strategy by query family, inclusion criteria, comparability rules, extraction fields, a verification plan and a stopping rule. One deviation is documented: the verification pass was run on the draft evidence table, before this report's prose was written (an ordering from my recollection of the workflow, not logged), so the report quotes corrected values. A second, independent pass on this report's own claims is described in section 11.

Evidence classes are kept visibly separate (see the banner). Published results also carry a validation type: `lab` (bench, soil bin, tilt bed), `field` (real vehicle on real terrain, including Mars), `simulation-only`, and `sim-vs-lab` or `sim-vs-field` where a physical comparison is reported. Of 510 rows: 166 lab, 133 field, 40 simulation-only, 35 sim-vs-lab, 23 sim-vs-field, 113 none (an unreferenced capability statement kept for the record). A physical test on a prepared bed, tilt platform or engineered analog site (for example the JPL Mars Yard) is classed `lab`; a real vehicle on natural terrain, including Mars, is `field`.

Inclusion. A result became a row only if the number was found in retrieved text (table, caption or body), the terrain is dry granular or sand like (or flagged as a comparison), at least one mobility metric is reported, and vehicle and soil are described or marked "not reported". Paywalled sources not retrieved are listed in the per topic notes under `research/raw/` with the exact URL to fetch; no number is taken from them.

Comparability. Rows are never averaged or ranked across setups. Each row has a `comparability_warning` where slip definition (wheel odometry, visual odometry, velocity based, distance based), slope definition (terrain versus pitch, or emulated by horizontal pull), gravity, wheel geometry, soil, speed or validation type differ.

Parameters. No soil parameter is invented. Each is either cited with its location or labelled ASSUMPTION with a sensitivity range (section 8, Table 8.1).

Quotation and licences. Sources are paraphrased. The licence of every dataset and library used is in `docs/DATA_AND_LICENSES.md` (summarised in section 5).

Limits of the search. The search tool was US only and returned a limited number of hits per query. Arabic language and regional literature, paywalled journals (for example the Journal of Terramechanics, Science) and several standard references (Bekker 1969, Wong's textbook, Wong and Reece 1967)

were not retrieved (five journal papers supplied as PDFs by the author on 2026-10-13 were read and added); absence of a result is therefore a statement about this search, not about the world.

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2. Background in brief (for newcomers; see `docs/GLOSSARY.md`)

A wheel on loose sand sinks, pushes sand aside (motion resistance) and shears the sand beneath it to push the vehicle forward (traction). **Slip** measures how much the wheel turns without advancing. **Sinkage** is the depth the wheel penetrates. **Drawbar pull** is the traction left over after resistance is paid. On a slope, part of the vehicle's weight must be paid for with drawbar pull, so slip rises with slope until the vehicle can no longer climb (entrapment). Dry sand has almost no cohesion and a friction angle of roughly 28 to 38 degrees in the lab papers retrieved; wind blown dune slip faces sit near the angle of repose, 32 to 35 degrees (M05, Kroy et al.; `research/raw/morphology_rows.csv`).

The older family of models (Bekker; Wong and Reece; Janosi and Hanamoto) describes the soil with a pressure sinkage law (constants kc, kphi, n) and a shear law (cohesion, friction angle, shear modulus K), then integrates stress under the wheel. A newer family (granular resistive force theory, RFT) sums depth- and orientation dependent forces on small wheel elements. Discrete element methods (DEM) simulate individual grains, and Project Chrono offers fast deformable terrain (SCM) and continuum (CRM) alternatives. Learning methods predict slip or hazard from images, proprioception or both.

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3. Evidence synthesis

Row identifiers refer to Appendix A. Every number below is **PUBLISHED** unless marked.

3.1 Classical Bekker-Wong style models

- **Parameters are soil-, geometry- and tuning dependent.** On dry Tottori sand the fitted kphi roughly doubles (about 430 to 800-880) when plate aspect ratio drops from 6 to 2-1.4, and kc rises about four fold (from about 2.5 to 3, up to 9 to 10); sinkage goodness of fit ranged 65 to 95 % (TM-029 and its table). Shear moduli K found in the retrieved papers range from 0.0006 m to 0.045 m (75-fold; the top is poppy seeds), so K behaves as a tuning knob. The Wong textbook table of dry sand constants was not retrieved; one secondary table found carries an apparent transcription error and is not used.

- **Error on benchmark wheels.** RoverDevKit's own evaluation of a Bekker-Wong wheel kernel against published single wheel data reports median drawbar errors of 24 % (Rashid-1 style wheel on FJS-1 simulant) and 27 % (Ding wheel data), sinkage errors of 28 and 52 %, and 135 % (drawbar) and 385 % (sinkage) for a small light wheel on dense simulant (SIM-025 to SIM-027). These are the authors' evaluations of other authors' data, sim-vs-lab, not field.
- **Slope.** Three sources report slope results. At Dumont Dunes a 340 kg test rover went from below 50 % slip at 12.5 degrees to about 75 % at 16 and about 90 % at 20 degrees, with no soil parameters reported (FLD-HV13-1, FLD-HV13-2), and two rovers embedded at about 17 degrees on prepared Marsyard sand (FLD-HV13-3). A 13.5 kg rover traversing a 10 degree side slope of Toyoura sand in a tilting bed showed a slip ratio of about 0.5 (TM-IS09-3). In a thesis, a Bekker-Wong-Ishigami model for a 7 kg rover on Toyoura sand gave side slip errors of 14.76 % at 10 degrees and 25.91 % at 20 degrees with parameters borrowed from another lab, and 0.49 % at 20 degrees after tuning on that case (TM-022 to TM-024). The error metric is a side slip distance ratio, not slip ratio, and the sandbox is 0.08 m deep.
- **Rate dependence.** Measured drawbar coefficient on olivine sand fell 8 % from 0.01 to 0.2 m/s and 63.7 % from 0.2 to 1 m/s at 20 % slip (TM-018, TM-019), against the quasi static assumption. Slip rate effects were also reported for grousered wheels (TM-021; the 85 % "accuracy" there is an in sample goodness of fit, per the verification).
- **Steering on loose sand (supplied paper, read 2026-10-13).** A Bekker type wheel model with lateral terms matched single wheel drawbar pull and side force well for slip angles of 5 to 20 degrees and less well at 25 degrees and above (TM-IS07-3, qualitative), and a rover model built on it reproduced four lab steering runs with position rms errors of 0.015 to 0.039 m (TM-IS07-4). Slip ratios of 0.1 to 0.3 were observed in those runs (TM-IS07-5). All are lab results on a prepared bed of lunar regolith simulant.

3.2 Granular RFT and scaling

- RFT predicts rigid wheel drawbar pull and sinkage from a small parameter set. On Quikrete medium sand the mean absolute drawbar error was 5.55 to 7.11 N across loads (TM-001, TM-002); on Mars Mojave Simulant (fines) 15.11 N and 18.41 mm sinkage at 190 N (TM-003, TM-004). Error grows with load; RFT under predicts drawbar at low positive slip and over predicts at |slip| above 0.4 (per the source's discussion). The response surfaces for both the medium sand and the fines were scaled from poppy seed data rather than measured directly, so the medium sand results carry that caveat too.

- On poppy seeds (not sand), RFT sinkage error was 2.73 mm against 26.07 mm for the Bekker type model but drawbar error was larger for RFT (5.05 N versus 3.41 N) (TM-005, TM-006; the plate in the Bekker fit was far smaller than the wheel patch).
- Granular scaling laws held within about 3 % on level Quikrete sand for pairs of wheels (TM-007). Their extension to slopes and lunar gravity has been checked only in DEM (slip RMSE 0.015, TM-020, simulation only). No physical sloped or low gravity scaling validation was found.
- Legged and plate RFT follow ups agree with controlled box experiments within roughly 3 to 15 % (up to 30 % for reversed legs; TM-008, TM-009, TM-011, TM-012), two of the four on poppy seeds rather than sand and 10.5 and 11.5 % mean speed and height error against a beach (TM-010, sim-vs-field, legged). A humanoid trained with an RFT based simulator walked on real dry sand at up to 2.5 m/s commanded (LEG-015).
- **3D RFT for rover wheels (supplied paper, read 2026-10-13).** Combined with a sand deformation model, 3D RFT gave mean absolute errors of 2.88 (smooth wheel) and 6.25 (grousered wheel) in the forward force over 45 slip and slip angle conditions, against 4.47 for plain 3D RFT on the smooth wheel; the table gives no unit (the force axes elsewhere in the paper are in newtons, not verified here), and the predicted trench profile was within 2.2 mm on average of lidar scans (TM-YU1, TM-YU2, TM-YU5). One prepared bed of Soil Direct #90 sand, small wheels on a gantry at 1 cm/s; no slope.

3.3 DEM, multibody and Project Chrono

- **Accuracy where a number exists.** A 2D DEM of a grouser wheel on Toyoura sand gave 12.1 % slip against 14.7 +/- 3.0 % in the lab (SIM-018). A 36-million-particle DEM of a rigid tyre gave pull 15 % of weight against 17 % lab and gross traction 20 % against 31 % at 20 % slip, under predicting sinkage and resistance (SIM-019). A soft foot in silica sand: mean pressure sinkage difference 13.5 % (SIM-021).
- **Many validations are plots only.** Chrono CRM and DEM wheel and rover comparisons against lab data (NASA flat and tilted beds, SIM-001, SIM-002, SIM-009; the Rashid-1 wheel rig, SIM-015) are described as close agreement but give no scalar error. Slope climbing limit error is not reported anywhere in the retrieved set.
- **Calibration is not independent evidence.** SCM parameters fitted to a DEM gave a negative Kc and a friction angle of about 21 degrees (a numerical fit, SIM-005); DEM used scaled particles and tuned friction (SIM-015, SIM-018).
- **Cost.** Wall clock to simulated time ranges from about 9 million (fine particle CPU DEM, SIM-020, derived) to hundreds or thousands (GPU DEM, SIM-016, SIM-024) to about 11 to 23 (Chrono CRM) and 18 to 51

(Chrono SCM) on one benchmark (SIM-011), and about 0.5 ms per closed form Bekker-Wong wheel solve (SIM-028; no simulated time ratio reported). The SCM versus CRM ordering depends on the benchmark.

- **Gravity.** Simulating a 440 kg rover at lunar gravity predicted far higher slip than a light 1 g comparison (about 85 % versus about 42 % at 30 degrees, SIM-003, where the source of the 42 % is not stated; about 75 % versus 18 % at 10 degrees, SIM-004), with no lunar validation (simulation only).
- Project Chrono is BSD-3 licensed (verified from its LICENSE file). It was not used in phases 1 and 2; from 2026-10-20 a single wheel Soil Contact Model sweep is run with it (section 13.4). No retrieved source compares Chrono, DEM or any multibody simulator with a full rover on real dune sand.
- **High speed grouser wheels (supplied paper, read 2026-10-13).** In single wheel tests on a prepared bed of Tohoku silica sand at up to 1.0 m/s, a slick wheel reached a traction coefficient of only about 0.02 at slip 0.5 and 1.0 m/s, and its traction efficiency fell from about 0.2 at 0.2 m/s to about 0.03 at 1.0 m/s (TM-TK1, TM-TK2). The DEM simulation predicted negative traction for the slick wheel at every slip ratio and both speeds, where the experiment did not, and the paper gives no numeric error between DEM and experiment (TM-TK4). Speed matters for sand traction, and DEM did not capture it for the slick wheel.

3.4 Learning based slip and traversability

- **Data needs.** Visual slip prediction needed 1000 to 2000 stereo frames per terrain with slip labels from visual odometry (LRN-003, LRN-004). The Perseverance model used about 43,000 self labelled images from about 500 sols and 45 km (LRN-010). A proprioceptive slip classifier used about 13,000 samples (9235 train, 3958 test) from one single wheel testbed (LRN-009).
- **Best held out evidence.** AUROC 0.874, F1 0.758 on held out Perseverance sols, including a sand dune patch at about 0.3 slip (LRN-010, LRN-011). Held out by sol, not by terrain type; Mars, not Earth sand. Removing proprioceptive alignment dropped AUROC to 0.723 (LRN-011): appearance alone does not fix mechanics.
- **Documented failures.** Test error on gravel rose to 27.5 % against 7.16 % in training because the test roll/pitch combinations were absent from training (LRN-003). Misclassifying soil or gravel as sand inflated whole pipeline error from 11.2 to 21.8 % (LRN-004). The moderate slip class was the weakest in proprioceptive classification, about 75 % against about 85 % for low and high slip (LRN-009), and that test was a random hold out on one testbed.

- **Transfer.** A policy trained in simulation transferred zero shot to a rover on simulant at 20 degrees: travel loss fell from 81.9 to 43.0 % on dry sand and from 100 % (entrapment) to 30 % on wet sand (LRN-019, sim-vs-lab, trial count not stated).
- **Datasets.** No public dataset with Earth dune or desert sand and measured wheel or foot slip labels was found. TAIL (sandy sites, SLAM benchmark, GPLv3, no slip labels) and the Perseverance driving set (Martian, release pending, no licence seen) are the closest.

3.5 Field and analog evidence

- **Mars rovers.** Spirit: 15 to 20 % slip on 19 degrees, 95 % on 16 degrees (FLD-011, FLD-012), up to 50 % at 10 to 13 degrees (FLD-010), over 100 % in places (FLD-009, FLD-013). Opportunity: about 50 m of wheel travel for 2 m of track at Purgatory ripple (FLD-001), slip 98.9 to 99.5 % (FLD-002), ripple 30 cm high, trapped several weeks (FLD-040). Curiosity: 3D slip above 70 % on shallow megaripples (FLD-018). Perseverance: above 94 % at about 20 degrees (FLD-025), slip limits above 50 % needed on a 15 to 20 degree hill (FLD-024).
- **Not a clean function of slope.** Segment averaged Spirit slippage on sand sized terrain rose from 4.0 % (0 to 5 degrees) to 9.9 % (5 to 10 degrees) (FLD-015, FLD-016), while fine grain terrain approached 100 % at 10 to 15 degrees (FLD-014). The authors warn that steep segments were avoided or visual odometry corrected, so higher slopes sometimes show less slippage (warning on FLD-015). Do not fit a clean slip versus slope line to these aggregates.
- **Lab to field.** Engineering slip tables built from a dry sand tilt bed worked on pure sand but failed on mixed or hidden weak deposits (Maimone et al.); separately, Lamarre and Kelly report that vision based slip prediction fails on sand (both summarised in the field notes). The tilt bed figures that are quantitative (90 % slip at 20 degrees, 25 degrees climbing limit, FLD-006, FLD-007) come from a press release, so FLD-006 is excluded from the anchor comparison in section 8.
- **Tilt beds on Earth.** A smooth wheel 35 kg rover (El-Dorado-II) showed slip close to 0.9 at 4 degrees without grousers and about 0.2 with grousers (FLD-030, FLD-031); a 23 kg rover with 296 mm wheels made less than 10 mm progress per revolution at 30 degrees, the angle of repose, whatever the grouser length (FLD-035); a 400 kg Scarab exceeded 90 % at 20 degrees and had no uphill progress at 25 degrees (FLD-027, FLD-028).
- **Arabian and Gulf.** One qualitative report (FLD-042): the AMADEE-18 Husky rover in Dhofar, Oman, met much looser and softer sand than expected. Nothing with slip, sinkage or slope numbers was found for Rub' al Khali, Saudi, Qatari, Kuwaiti or Emirati dunes. The UAE authored single wheel rig results (slip 0 to 0.75, sinkage about 1 to 2 cm) use lunar

simulant and are not evidence for desert sand. See also `research/raw/field_notes.md` for non Arabian desert trials, which were qualitative.

3.6 Legged and tracked comparisons (secondary)

- **Legs.** Gait dominates: the same SandBot ran about 2 cm/s with a hard ground gait and 8 cm/s with a soft ground gait (LEG-001, LEG-002), and about 20 cm/s versus 2 cm/s for a 5 % change in packing (LEG-003, LEG-004). On slopes a 350 g hexapod lost about two thirds of its speed between 0 and 20 to 24 degrees (LEG-007 to LEG-009; the same row set appears as FLD-037, FLD-038). On a 22 kg quadruped (SpaceBok) planar feet slipped near the soil friction angle while point feet sank up to 120 mm (LEG-010, LEG-012, LEG-013; testbed limit 25 degrees). All lab beds, none dune sand.
- **Wheel geometry.** Larger and wider wheels cut slip strongly on Toyoura sand: slip 0.741 at D = 116 mm versus 0.056 at D = 327 mm at an equivalent 12 degrees (LEG-017, LEG-018), and 0.471 at narrow versus 0.247 at wide at 14 degrees (LEG-019, LEG-020), with slope emulated by horizontal pull and from slides, not a paper. Grouser benefit saturates (LEG-021). A flexible metallic wheel gave more than twice the drawbar pull of a rigid one at 1 m/s on olivine sand (LEG-022).
- **Tracks.** The one direct wheel versus track comparison found could not be attributed unambiguously and is not entered as a row. No extractable numeric tracked vehicle sand data were retrieved. No same vehicle wheel versus leg versus track comparison on dry sand was found.

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4. Taxonomy of prediction approaches and when each is appropriate

The table is a judgement made from the evidence above, not a ranking from head to head tests. No retrieved study compares these families on the same vehicle and the same dune sand.

Family	Needs	Strengths (published)	Weaknesses (published)	Appropriat when
Slope threshold or friction circle rules (the model in section 8)	Slope, a friction angle, a few tuned constants	Costs almost nothing, runs on any elevation model, transparent	Constants for slip and resistance are not transferable (K varies about 75-fold, section 3.1); no sinkage; blind to sand state	Coarse screening c large areas, ranking assumption planning where entrapment the worst case

Family	Needs	Strengths (published)	Weaknesses (published)	Appropriat when
Classical Bekker-Wong, Wong-Reece	Bevameter or plate tests, shear tests, wheel geometry and load	Interpretable, closed form, fast (SIM-028)	24 to 52 % median error on benchmark wheels, worse on firm soil (SIM-025 to SIM-027); parameters depend on plate geometry (TM-029); quasi static (TM-018, TM-019)	Single whe design trad studies on a characteris soil, with tuning on th soil
Granular RFT	Depth and orientation force law from a plate or intruder test	Few parameters; wheel drawbar mean absolute error 5.6 to 15 N and sinkage error 18 mm at 190 N on sand and fines (TM-001 to TM-004); relative errors of about 2.6 to 15 %, up to 30 % for reversed legs, in legged and plate tests (TM-008 to TM-013)	Error grows with load; response surfaces scaled from poppy seeds; no sloped wheel field validation found	Slow quasi static locomotion on dry grai in a controlled bed, legge robots, fast what if studies
Scaling laws	Matched wheel pairs	Collapse level ground data within about 3 % (TM-007)	Slopes and low gravity unvalidated physically (TM-020 is DEM only)	Transferrin a lab result between wheel sizes on level sar

Family	Needs	Strengths (published)	Weaknesses (published)	Appropriat when
DEM	Particle contact calibration (friction, rolling resistance, cohesion), often scaled particles	Captures flow, avalanching and high slip behaviour; about 13 to 18 % where a scalar error exists (SIM-018, SIM-021), with larger gaps for gross traction (20 against 31 % of weight, SIM-019)	Wall clock hundreds to thousands of times simulated time with scaled particles on GPU (SIM-016, SIM-024) and about 9 million for fine particles on CPU (SIM-020), all derived; calibration is a tuning step, not independent evidence	Mechanism studies, generating training dat checking simpler models
Chrono SCM / CRM, multibody	Bekker-Wong constants (SCM) or density and friction angle (CRM)	Two to several orders of magnitude cheaper than DEM (SIM-007, SIM-008, SIM-011); BSD-3	Validation mostly plots, no scalar errors (SIM-001, SIM-002, SIM-009); fitted SCM constants can be unphysical (SIM-005)	Rover scale simulation, controller testing, generating labelled dat
Learning from images or terrain class	Slip labels per terrain from visual odometry or encoders	Held out AUROC 0.874 on Perseverance (LRN-010)	Needs labelled runs on each terrain; class confusion inflates error (LRN-004); fails outside training roll/pitch (LRN-003)	Onboard hazard ranking where the vehicle has driven simil sand
Learning from proprioception	Wheel speeds, currents, IMU	Detects slip as it happens (LRN-009, LRN-015, LRN-024)	Moderate slip is weakest (LRN-009); mostly binary events; random hold outs	Online slip detection a adaptation

Family	Needs	Strengths (published)	Weaknesses (published)	Appropriat when
Hybrid or physics informed learning	Physical model plus onboard adaptation	Online soil adaptation lifted physical grass to sand completion in a small trial (NeSAM, notes; the physical dataset size was not located)	Few physical tests, small trial counts	Promising f adapting a planner to unseen san
Empirical field tables (Mars slip tables)	Prior drives and operational constraints	Used in practice on Mars (FLD-020, FLD-021, FLD-024)	Failed on mixed terrain and hidden weak deposits (field notes)	Conservati operations with a huma in the loop

Selection rule used in this project. Use a cheap slope and slip rule to map where the answer is sensitive to assumptions (section 8), use a calibrated single wheel model or RFT to choose wheel geometry on a characterised sand, and use proprioceptive or image based learning on board only after the vehicle has logged enough runs on that sand. Use DEM only to understand mechanisms.

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5. Open data for an Arabian dune planner

All retrievals 2026-10-11. Licences are as read from provider pages; legal reading is the user's responsibility. Full table: `research/raw/data_audit.csv`; licences: `docs/DATA_AND_LICENSES.md`.

Dataset	Resolution	Coverage	Licence (as read)
Copernicus DEM GLO-30 (DSM, TanDEM-X, 2010 to 2015)	1 arcsec; about 28.3 m E-W by 30.8 to 30.9 m N-S at 23.5 N (my arithmetic from tile geometry; the code uses 30.76 m)	Global land	Free with attribution (c) DLR and Airbus, EU/ESA; controlling licenc (COP-DEM-GLO-30-F) read in f phase 2: no commercial restrictio attribution and no liability notices required (`docs/DATA_AND_LICENSES`

Dataset	Resolution	Coverage	Licence (as read)
Copernicus GLO-90	3 arcsec (about 90 m)	Global land	As GLO-30
SRTM derived "skadi" tiles	1 arcsec	Global (mixed SRTM and other sources per the tiles page)	Joerd project attribution, SRTM courtesy USGS
NASADEM, ASTER GDEM v3	1 arcsec	NASADEM 60 N to 56 S; ASTER 83 N to 83 S	Openly shared; download needs a Earthdata login
ALOS AW3D30 v4.1	1 arcsec	Global land within about 82 degrees latitude	Free after registration, JAXA term not read
FABDEM v1-2	1 arcsec	Global	CC BY-NC-SA 4.0 (non commer
TanDEM-X 90 m (DLR)	3 arcsec	Global	Scientific use, registration
ICESat-2 ATL08, GEDI L2A	100 m segments; 25 m footprints	Tracks only; GEDI 54 N to 54 S	Free, Earthdata login
Sentinel-2 L2A	10, 20, 60 m	Global, frequent	Free, full and open
Landsat 8/9	30 m	Global	No restrictions stated

Dataset	Resolution	Coverage	Licence (as read)
Maxar Open Data	sub metre class (figure not stated on the page read)	Event based; no Gulf event found	CC BY-NC 4.0
Planet NICFI, Pleiades Neo	n/a	n/a	Planet terms not verified; Pleiade Neo commercial
Abu Dhabi SDI open data (habitat and geology polygons)	vector	Abu Dhabi emirate	Licence field empty in service metadata

Not found. Open airborne LiDAR or drone elevation models of Liwa or the Rub' al Khali dunes; any gridded sand colour or grain size map of the Empty Quarter; any open sub metre stereo for the region.

Can slip faces and crests be resolved? Evidence for the answer, all of it my arithmetic or computation unless marked published:

- Published dune sizes: Liwa dunes 50 to 80 m high (Lorenz et al., abstract); northern Rub' al Khali complex linear dunes 50 to 100 m (Lancaster et al., abstract); published mean wavelengths 0.63 to 3.17 km (USGS Professional Paper 1052 Landsat morphometry; `morphology_rows.csv`). At 30 m posting that is 20 to 110 posts per wavelength: well resolved. Small simple linear dunes of 10 to 25 m height and 200 to 300 m spacing (a secondary summary, to be re verified) are 7 to 10 posts at 30 m and 2 to 3 at 90 m.
- A slip face at 32 degrees has a horizontal run of about 1.6 times its height, so a 50 m dune face spans about 80 m, 2.7 posts at 30 m. In the Rub' al Khali window, runs of consecutive posts steeper than 25 degrees had a median length of 2 posts (90th percentile 4; 45 % a single post).
- A synthetic test of an ideal 32 degree face sampled at 30 m with random phase (SIMULATED BY ME, optimistic because it samples points rather than averaging over cells) recovers full steepness only if the face is 30 m or higher by forward difference (50 m or higher by central difference). A 10 m face reads as 9 to 18 degrees and a 20 m face as 18 to 26 degrees. At 90 m, even 50 m faces read 16 to 23 degrees.
- **Verdict.** Dune crests, ridges and interdune corridors of the large Liwa and Rub' al Khali dunes: resolved at 30 m. Slip faces of dunes above about 40 to 50 m: marginal, located as thin bands of high slope but biased low. Slip faces of dunes below about 20 m, brink sharpness, ripples (centimetre wavelength) and small barchans of 1 to 7 m: not resolved, and the barchans are at or below the published vertical error. A planner reading 30 m slope alone will under flag lee faces.
- **Caveats.** GLO-30 is a radar surface model; the size of X-band bias over dry sand was not retrieved. The epoch (2010 to 2015) is stale for fast small dunes (over 20 m/yr in Kuwait). No lidar truth of Arabian dunes was

available (North American lidar is used from 2026-10-20, section 13.5), so slip face bias is estimated by synthetic and cross product comparison only.

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6. UAE context (factual background; every item checked against retrieved sources, see `research/raw/uae_facts.csv`)

- **Rashid 1** (MBRSC, about 10 kg, four wheels, designed for slopes up to 20 degrees and rocks 10 cm high) never operated: the ispace Hakuto-R Mission 1 lander failed to land on 2023-04-26 (JST), which ispace attributed to altitude sensor handling in software (UAE01, UAE02, UAE10, UAE11).
- **Rashid 2** is built and was shipped to the US for Firefly's Blue Ghost Mission 2 (lunar far side). Launch dates conflict between sources (2026 in late-2025 coverage, not earlier than 2027 in an October 2026 report), so no flight data exist and the date is unresolved (UAE14, UAE15, UAE17).
- **Wheel work.** Khalifa University and MBRSC published single wheel rig results for the Rashid wheel on a lunar regolith sample (slip 0 to 0.75, sinkage about 1 to 2 cm, drawbar 5.2 to 24.1 N, UAE04, UAE06) and wheel and grouser studies (UAE07, UAE08). These concern lunar simulant, not Arabian sand.
- **Desert ground vehicles.** A TII desert unmanned ground vehicle (page date ambiguous, 2023 or 2025) and, at UMEX in January 2026, SteerAI's xRift (debuted 20 January 2026) and a Space42 vehicle (25 degree slope capability stated) were shown; the retrieved pages give no slip, sinkage or dune traversability data (UAE22 to UAE25).
- **Programmes that are not mobility evidence.** Hope is an orbiter (UAE19, UAE20); Mars Science City was announced as a simulated settlement with 3D-printed desert sand domes and its article does not mention rover testing (UAE21); the Emirates Robotics Competition task is indoor (UAE26).
- **Not verified or not found:** Rashid 2 wheel design, current Rashid 2 launch date, any rover testing on UAE dunes by UAE universities, current Mars Science City status. The 1963 archive item on wheeled vehicle sand trials could not be read.
- Thermal and dust effects on these vehicles are out of scope; see `hudasol/dhill`.

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7. Gap analysis

7.1 Where lab to field transfer fails or is untested

1. **Every quantitative sand slope result is lab, a prepared Earth sand bed (SIM-012, SIM-013; the unit of the slope term is not stated in its source) or Mars, except one Earth dune curve from Dumont Dunes, California, for a 340 kg rover with no soil parameters reported (FLD-HV13-1, FLD-HV13-2); none is Arabian dune sand.** Lab slope beds use simulants or Toyoura sand, mostly small wheels (exception: the 400 kg Scarab, FLD-027, FLD-028) and slow speeds (FLD-027 to FLD-035, LEG-017 to LEG-020, TM-022 to TM-024). Mars field data are low gravity and on megaripples, crusts and sulfates (FLD-001 to FLD-025, apart from the Earth tilt bed rows FLD-006 to FLD-008).
2. **Engineering tables transferred from a lab bed work on uniform sand and fail on mixed or hidden weak ground** (field notes, Maimone et al.). The Spirit pair (15 to 20 % versus 95 % slip on terrain with no visible difference, FLD-011, FLD-012) shows the pattern; the source does not state the cause.
3. **Model validation uses in sample or borrowed parameters.** The rover scale slope error fell from 25.9 to 0.49 % only after tuning on the same case (TM-023, TM-024); DEM and SCM fits are calibrations (SIM-005, SIM-015, SIM-018).
4. **The quasi static models are outside their validated envelope when driving is fast or transient** (TM-018, TM-019, TM-021).
5. **Learning evidence is in distribution.** Hold outs are random, temporal or by sol, never by terrain type (LRN-003, LRN-007, LRN-009, LRN-015). The one held out soil result is a simulation (LRN-016).
6. **The field slip sensor is unreliable on exactly the terrain of interest.** Visual odometry degrades on featureless sand (FLD-041), and different papers define slip differently (wheel odometry, visual odometry, velocity based, distance based), so numbers are not interchangeable.

7.2 What is missing for dune terrain specifically

1. A **slip versus slope dataset with a wheeled vehicle on dune sand with complete soil and wheel description** and independent ground truth position. None found.
2. **Behaviour on and near the slip face**: avalanching and shear collapse near the angle of repose. Slope induced shear collapse is shown only for a foot on silica sand (sinkage 13 mm at 0 degrees to 36 mm at 35 degrees, SIM-022, SIM-023) and in a few legged tests. Simulators that omit it will under predict.
3. **Arabian sand properties**: grain size, rounding, packing, moisture and cementation below the surface, and Bekker or RFT constants. Nearest values are Kuwaiti barchans (mean grain 0.23 mm) and a Brazilian dune

(d50 about 0.2 mm), not Gulf dunes (`morphology_rows.csv`).
4. **Dune scale terrain data at the resolution that matters** (open, metre scale elevation). None open for the region; GLO-30 under resolves slip faces (section 5).
5. **A common slip definition and a shared benchmark** so that rows can be compared.
6. **Dynamic effects**: crosswind and lateral slip on crests, slope at speed, repeated passes on the same track (not covered by any retrieved row).
7. **Tracked and legged vehicles on real dunes**, and any same vehicle comparison of the three.

7.3 What a minimal open data planner needs that the literature does not provide

- A slip law whose constants come from **measurements on the target sand and the target vehicle**, with stated uncertainty. This project supplies the structure and an analysis plan (sections 8 and 9) but not the constants.
- A **slip face and crest detector** that does not rely on a 30 m slope value: aspect, curvature, dune height and optical imagery are the open candidates, but none has been validated here.
- A **way to carry the assumption uncertainty into the plan**, for example by treating a route as acceptable only if it is acceptable across the plausible parameter range, which is what the sensitivity analysis in section 8 suggests is necessary.
- An **onboard log** (section 9.5) that lets the vehicle correct the model as it drives.

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8. Simulation study (SIMULATED BY ME): slope and slip traversability from an open elevation model

> **SIMULATION ONLY, model assumption dependent. Nothing in this section is field evidence.** The model is a deliberately minimal planning screen, not a vehicle model. It omits sinkage, wheel geometry, vehicle mass, speed, avalanching and cohesion. The published evidence in section 3 shows that the slip it tries to represent varies from near 0 to near 1 at similar slopes, so the outputs below describe the **model**, not any real rover on any real dune.

8.1 What was run

- **Data.** Copernicus DEM GLO-30 tiles N23E053 and N22E054 (licence in section 5; tile hashes in `results/summary.json`). Three windows: **A-south Liwa** (23.02 to 23.17 N, 53.55 to 53.95 E, about 17 by 41 km, relief 139 m), **A north Liwa** (23.25 to 23.45 N, same longitudes; smaller crescentic dunes, relief 85 m) and **B Rub' al Khali megadunes** (22.20 to 22.50 N, 54.20 to 54.60 E, relief 211 m). The windows were chosen by me for visible dune texture, not at random.
- **Slope.** Horn 3 by 3 gradient with per latitude metric spacing (`src/dunerover/terrain.py`). A central difference check on the same A-south window gave 1.70 % of cells above 20 degrees in the earlier data audit against 1.45 % here (Horn): slope tails depend on the kernel by roughly 15 % relative.
- **Slip model** (`src/dunerover/slipmodel.py`). For an edge between neighbouring cells with along heading slope angle theta_a and cross slope theta_c: demand D = sqrt((tan theta_a + f_r)^2 + tan^2 theta_c) (friction circle), capacity mu_max = tan(phi), slip i = -i0 ln(1 - D / mu_max), entrapment if D >= mu_max. An edge is passable if i <= i_max; travel cost = distance / (1 - i). The exponential comes from the Janosi-Hanamoto shear law applied at the contact patch level.
- **Planner.** 8-neighbour directed graph, Dijkstra least cost routes (`scipy.sparse.csgraph`), ten deterministic origin destination pairs per window picked on gentle low ground (seed 20261011). Slope in each direction is taken from the elevation difference along the edge, cross slope from the Horn gradient at the edge midpoint.
- **Seeds and reproducibility.** Origin destination 20261011; Latin hypercube 20261012 (A core) and 20261013 (B core); 256 draws each, on core windows of 0.07 by 0.16 degrees (A core 23.06 to 23.13 N, 53.62 to 53.78 E; B core 22.28 to 22.35 N, 54.30 to 54.46 E); the full study was run twice and the area and sensitivity entries of the two `results/summary.json` files were identical (a one off comparison that is not stored in the repository; re run to confirm). Runtime about 330 s.

Table 8.1: parameters of the slip model. Only the friction angle has a published basis; the other three are assumptions with ranges. DEM noise is an additional assumption used in the sensitivity analysis only (spatially correlated, correlation length 3 posts, standard deviation 0 to 2 m).

Parameter	Nominal	Range	Status	Basis
`phi_deg`	33.0	28.0 to 38.0	published range of laboratory sands (not measured for Arabian sand)	Internal friction angles printed in retrieved papers: 28 (Irani dry sand), 34 (Quikrete sand), 35 (Mojave simulant), about 35 (olivine sand), 38 (Toyoura, Tottori); angle of repose 32-35 deg (Kroy et al. 2002). See research/raw/terramechanics_notes.md and morphology_rows.csv (M05).

Parameter	Nominal	Range	Status	Basis
`f_r`	0.12	0.05 to 0.25	ASSUMPTION	Motion-resistance coefficient (resistance/weight) lumped for a small wheeled UGV in loose sand. No retrieved source gives a transferable value; range chosen to span light-to-heavy sinkage while keeping the vehicle able to move on flat sand. Replace with a measured value for the real vehicle.
`i0`	0.2	0.05 to 0.4	ASSUMPTION	Slip scale K/L. Published Janosi shear modulus K spans 0.0006-0.045 m between papers (terramechanics_notes.md), contact length L is vehicle specific, so i0 is a tuning parameter.
`i_max`	0.5	0.4 to 0.7	ASSUMPTION (planning threshold)	Slip above which an edge is treated as impassable. A Mars rover autonav slip table needed limits above 50 % on 15-20 deg terrain (evidence FLD-024); that is Mars-specific, used only as a pointer.
`noise_sigma_m` (sensitivity only)	0	0 to 2 m	ASSUMPTION	Spatially correlated DEM error, correlation length 3 posts; no measured value for dune DEM error was retrieved.

8.2 Elevation statistics (computed from the data, not from the slip model)

Window	Posting, m	Block	Relief, m	Max slope, deg	% cells >10	>15	>20	>25	>30
A_south_Liwa	29.6	1	139	37.0	14.6	3.89	1.45	0.57	0.0549
A_south_Liwa	59.3	2	136	29.5	6.86	2.04	0.632	0.0931	0
A_south_Liwa	88.9	3	133	25.5	5.23	1.25	0.163	0.00463	0
A_north_Liwa	29.6	1	85	35.1	15.3	4.34	1.39	0.304	0.0242
A_north_Liwa	59.2	2	81	26.1	6.52	0.917	0.061	0.000386	0
A_north_Liwa	88.8	3	77	19.4	1.72	0.0365	0	0	0
B_RubAlKhali_megadunes	29.7	1	211	42.7	33.1	12.6	4.97	2.33	0.45
B_RubAlKhali_megadunes	59.4	2	205	33.8	25.4	7.94	3.09	1.15	0.0283
B_RubAlKhali_megadunes	89.1	3	200	29.9	20.8	5.86	1.91	0.437	0

Coarsening to 60 and 90 m (block mean) removes most steep cells: the share of A-south cells above 25 degrees falls from 0.57 % at about 30 m to 0.093 % at about 60 m and 0.0046 % at about 90 m. This is the resolution effect discussed in section 5, reproduced on real tiles.

8.3 Nominal results

Under nominal parameters the model's critical direct ascent slope (slip equal to i_max) is **25.5 degrees**; the slip curve reaches entrapment at about 28 degrees. Model slip at 0, 5, 10, 15, 20 and 25 degrees of direct ascent is about 0.04, 0.08, 0.12, 0.18, 0.27 and 0.47.

Window	Grid (rows x cols)	Blocked edges	Reachable share	Median cost ratio	Median length ratio	Steepest uphill on any route, deg	Origin destination pairs
A_south_Liwa	541 x 1441	0.25 %	1.0000	1.116	1.063	20.4	10 (0 unreachable)
A_north_Liwa	721 x 1441	0.20 %	1.0000	1.115	1.064	18.1	10 (0 unreachable)
B_RubAlKhali_megadunes	1081 x 1441	0.97 %	0.9999	1.115	1.056	23.1	10 (0 unreachable)

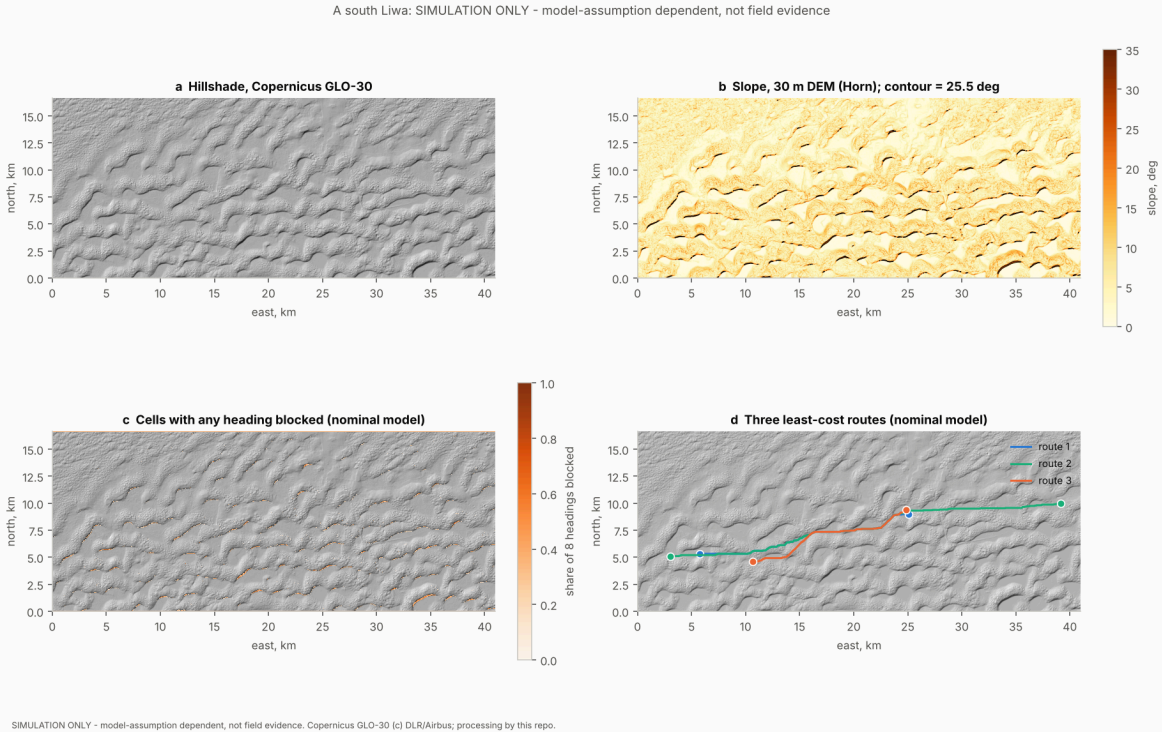


Figure 8.1. (a) hillshade of GLO-30, (b) slope with the nominal critical angle contoured, (c) cells where at least one of eight headings is blocked, (d) three least cost routes. SIMULATION ONLY, model assumption dependent, not field evidence. The Rub' al Khali window is `results/figures/fig_area_B_RubAlKhali_megadunes.png`.

How to read these numbers.

- The blocked share is tiny (0.2 to 1 % of edges). That follows from two things, not from the dunes being easy: at 30 m posting only a small share of cells exceed the nominal critical angle (section 8.2), and slip face slopes are biased low by the grid (section 5). A real rover facing the steepest faces would be blocked far more often than this map shows.
- The median route cost ratio (about 1.115 in all three windows) is **almost entirely a grid and baseline effect**: an 8-neighbour path is about 6 % longer than the straight line (column "Median length ratio" above) and nominal flat sand slip of about 0.04 adds a factor near 1.04. It should not be read as a dune effect. The routes avoid steep cells at small cost because the origin and destination are on gentle ground.
- Reachability is about 0.9999 in every window: the interdune corridors connect everything at nominal parameters.

8.4 Sensitivity analysis: which assumptions change the map most

Two analyses were run. **One at a time (OAT):** each parameter to its low and high value, plus structural alternatives (slip law form, cross slope rule, DEM posting, DEM noise). **Global:** 256 Latin hypercube draws of the five continuous assumptions on a core window of A and of B, with Spearman

rank correlations against each output (ranks are used because the outputs are monotonic but non linear; the rank regression R squared values in `results/summary.json` are 0.86 to 0.96 for blocked share, reachability and climbing limit, so the ranking explains most of the variance).

Table 8.2: one at a time sensitivity, area A-south (SIMULATED BY ME).

Case	A-south blocked share	x nominal	A-south median cost ratio	B blocked share	B median cost ratio	Critical uphill, deg
nominal	0.0025	1	1.116	0.0097	1.115	25.5
phi_deg=28.0	0.0084	3.38x	1.132	0.0271	1.139	20.2
phi_deg=38.0	0.0002	0.07x	1.106	0.0009	1.103	30.8
f_r=0.05	0.0010	0.40x	1.093	0.0051	1.093	28.6
f_r=0.25	0.0083	3.35x	1.186	0.0243	1.188	19.1
i0=0.05	0.0011	0.44x	1.076	0.0048	1.070	27.9
i0=0.4	0.0110	4.43x	1.177	0.0344	1.198	18.9
i_max=0.4	0.0038	1.51x	1.116	0.0136	1.115	23.8
i_max=0.7	0.0015	0.61x	1.116	0.0064	1.115	27.0
form=janosi_mean	0.0199	8.02x	1.185	0.0561	1.217	16.2
cross=additive	0.0097	3.92x	1.126	0.0325	1.134	25.5
dem_block=2	0.0006	0.25x	1.114	0.0049	1.113	25.5
dem_block=3	0.0001	0.05x	1.113	0.0023	1.111	25.5
dem_noise=1.0 m	0.0025	1.00x	1.116	0.0096	1.115	25.5
dem_noise=2.0 m	0.0025	1.01x	1.117	0.0096	1.115	25.5

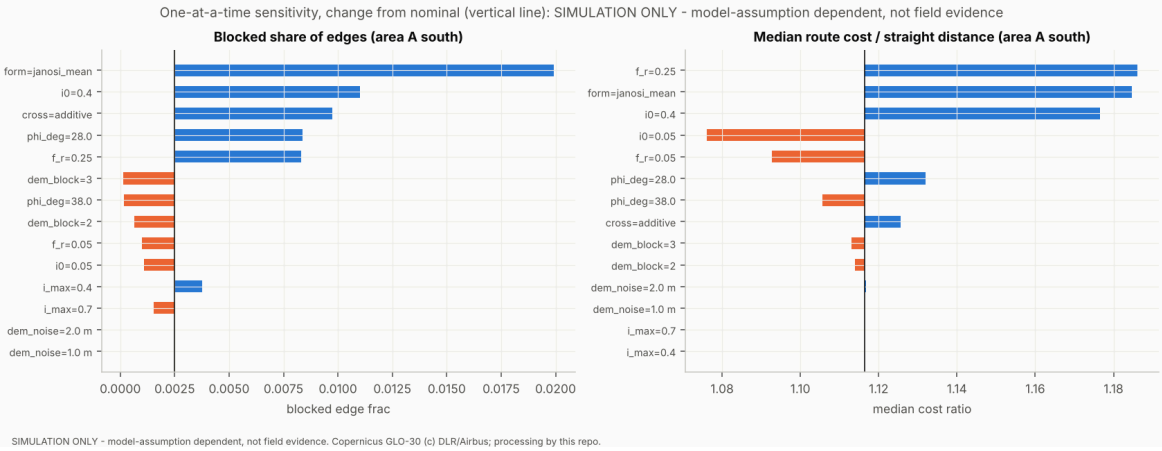


Figure 8.2. Change from nominal in blocked share (left) and median route cost ratio (right), area A-south. SIMULATION ONLY.

Table 8.3: global sensitivity, Spearman rank correlation of each assumption with each output (Latin hypercube, n = 256, SIMULATED BY ME).

Assumption	A: blocked share	A: cost ratio	A: critical uphill	B: blocked share	B: cost ratio	B: critical uphill
friction angle phi	-0.67	-0.29	+0.63	-0.69	-0.26	+0.63
resistance f_r	+0.46	+0.58	-0.53	+0.36	+0.53	-0.51
slip scale i0	+0.56	+0.75	-0.54	+0.59	+0.79	-0.54
slip limit i_max	-0.27	-0.04	+0.26	-0.12	+0.08	+0.08

Assumption	A: blocked share	A: cost ratio	A: critical uphill	B: blocked share	B: cost ratio	B: critical uphill
DEM noise	+0.06	+0.02	-0.05	-0.11	-0.08	+0.10

Table 8.4: spread of outputs across the assumption ranges (5th, 50th and 95th percentile, and extremes; SIMULATED BY ME).

Core window	Output	min	5 %	median	95 %	max
A_core	blocked share	0.0000	0.0002	0.0043	0.0240	0.0509
A_core	median cost ratio	1.079	1.087	1.133	1.272	1.369
A_core	reachable share	0.9964	0.9994	1.0000	1.0000	1.0000
A_core	critical uphill, deg	11.3	14.7	23.4	31.4	34.7
B_core	blocked share	0.0001	0.0003	0.0132	0.0550	0.1950
B_core	median cost ratio	1.090	1.099	1.152	1.295	1.581
B_core	reachable share	0.9757	0.9895	1.0000	1.0000	1.0000
B_core	critical uphill, deg	7.6	15.3	23.4	31.6	34.5

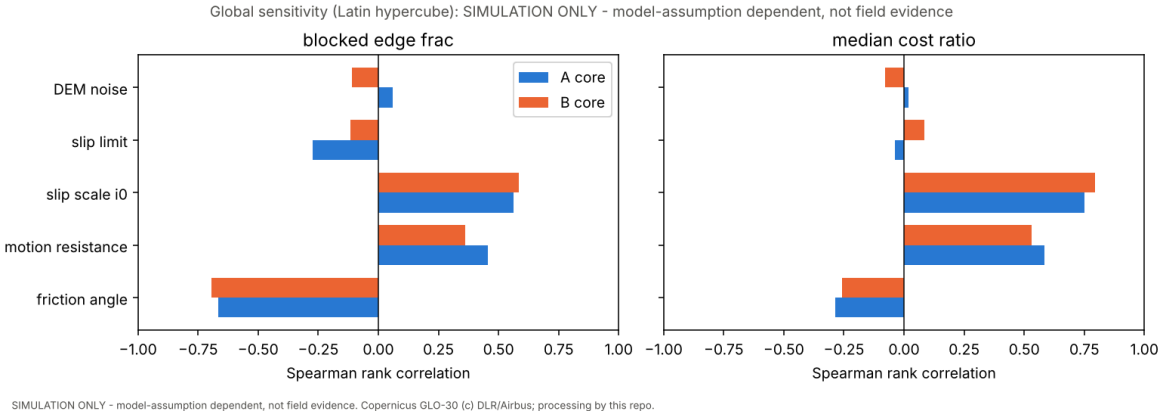


Figure 8.3. Spearman rank correlations (Latin hypercube). SIMULATION ONLY.

Reading of the sensitivity results.

1. **Structural modelling choices move the map as much as the parameter ranges.** Replacing the point wise exponential with the contact patch averaged Janosi form raises the blocked share in A-south from 0.0025 to 0.0199 (8-fold, the largest single upward change in Table 8.2) and lowers the model's critical slope to 16.2 degrees (independent of the DEM). The additive cross slope rule (instead of the friction circle) raises the blocked share about 4-fold. Neither choice has published validation for dune sand. For comparison, moving the friction angle from 28 to 38 degrees changes the blocked share about 50-fold (0.0084 to 0.0002), so the friction angle gives the largest swing of any single parameter.
2. **DEM posting matters as much as the parameters.** Block averaging to 60 m and 90 m cuts the blocked share to 0.25 and 0.05 times nominal in A-south and to about 0.5 and 0.24 times in B (Table 8.2), because slip faces disappear (section 5). A map built from a coarser product would look easier than the 30 m map, and the 30 m map is itself likely too easy.

3. **Among the continuous parameters, the slip scale i0, the friction angle and the resistance coefficient dominate.** In both cores, the friction angle has the strongest rank correlation with blocked share (about -0.67 to -0.69), then i0 (+0.56 to +0.59), then f_r (+0.36 to +0.46). For the route cost ratio, i0 and f_r come first. The model's climbing limit spans 7.6 to 34.7 degrees across the draws under the original ranges (i0 0.05 to 0.40, f_r 0.05 to 0.25) (5 to 95 %: 14.7 to 31.4 in A, 15.3 to 31.6 in B). Two of these three (i0, f_r) are assumptions with no transferable published value, and the third (phi) differs by 10 degrees between published lab sands, so this is where measurements pay off most.
4. **The slip limit i_max and spatially correlated DEM noise matter least** in this setup (rank correlation magnitudes at most 0.27 and 0.11). The noise result is conditional on the assumed noise model; real DEM error over dunes (X-band penetration, slip face bias) is not modelled.
5. **Blocked share varies by about two orders of magnitude** between the 5th and 95th percentiles of the assumption space (A core 0.02 to 2.4 %; B core 0.03 to 5.5 %), with extremes of 0 to 5 % (A) and 0.01 to 19.5 % (B), while the OAT ranking is similar in the two areas. The conclusion that matters for a planner is that **the map is controlled by assumptions that field data have not yet measured**.

Widened slip scale range (2026-10-18; SIMULATED BY ME, simulation only, model assumption dependent). At the author's decision the phase 1 study was run again with the i0 range widened to 0.05 to 1.20 and a nominal value of 0.81, the lower bound implied by Ishigami et al. 2009 Table I under the model definition (TM-IS09-2), and once more with f_r also widened to 0.35 as a test. Same seeds; the original results above are unchanged and stay the reference (`results/widened_i0/`, `results/widened_i0_fr/`; pre registered as stage 9 in `docs/PREDICTIONS.md`).

Range set	Ranges	Nominal climbing limit, deg	Climbing limit across draws, 5th / median / 95th, deg	Blocked edges, A south (nominal)	Blocked edges, B (nominal)
original	i0 0.05 to 0.4 (nominal 0.2); f_r 0.05 to 0.25	25.5	14.9 / 23.4 / 31.6	0.25 %	0.97 %
widened_i0	i0 0.05 to 1.2 (nominal 0.81); f_r 0.05 to 0.25	10.2	2.3 / 13.5 / 28.6	7.80 %	16.56 %
widened_i0_fr	i0 0.05 to 1.2 (nominal 0.81); f_r 0.05 to 0.35	10.2	0.0 / 11.4 / 27.0	7.80 %	16.56 %

With the widened range the nominal climbing limit falls from 25.5 to about 10 degrees and the median across draws from 23.4 to 13.5 degrees, and the nominal map blocks many more travel directions. With the widened range the assumed i0 range appears to dominate the map; only a measured slip

against slope curve for the vehicle can narrow it. In the widened_i0_fr set 9 to 12 % of draws cannot move on level ground (stage 11), a case the original f_r basis excluded.

8.5 Comparison with published slip (descriptive only, nothing fitted)

Thirteen published slip versus slope points were transcribed by hand (`data/anchor_points.csv`: nine from FLD-010, FLD-011, FLD-012, FLD-025 and FLD-027 to FLD-031, and four from LEG-017 to LEG-020). They differ in slip definition, gravity, wheel and soil (see the warning column) and **should not be used as a test of the model**. The comparison shows only whether the assumption envelope overlaps them.

Row	Slope, deg	Published validation	Published slip	Model nominal (SIM)	Model envelope min to max (SIM)	Envelope overlaps	Definition warning
FLD-010	11.5	field	0.00 to 0.50 (upper_bound)	0.14	0.02 to 0.67	yes	slip = 1 - achieved/commanded distance; Mars; slope range; 'up to 50 %' used as an upper bound
FLD-011	19.0	field	0.15 to 0.20 (range)	0.25	0.04 to 1.00	yes	VO-based slip; Mars; appearance-identical terrain to FLD-012
FLD-012	16.0	field	0.95	0.20	0.03 to 1.00	yes	VO-based slip; Mars
FLD-025	20.0	field	0.94 to 1.00 (lower_bound)	0.27	0.04 to 1.00	yes	rover-level VO slip; slope 'about 20'; '>94 %' used as a lower bound
FLD-027	20.0	lab	0.90 to 1.00 (lower_bound)	0.27	0.04 to 1.00	yes	velocity-based slip; lunar simulant, rigid wheels without grousers; '>90 %' used as a lower bound
FLD-028	25.0	lab	1.00	0.47	0.06 to 1.00	yes	no uphill progress at 25 deg direct ascent, treated as slip = 1
FLD-029	15.0	lab	0.20 to 0.30 (range)	0.18	0.03 to 1.00	yes	descriptive capability statement; lunar simulant
FLD-030	4.0	lab	0.85 to 0.95 (range)	0.07	0.01 to 0.33	NO	text says close to 0.9; band +/-0.05 is mine
FLD-031	4.0	lab	0.15 to 0.25 (range)	0.07	0.01 to 0.33	yes	text says around 0.2; band +/-0.05 is mine
LEG-017	12.0	lab	0.74	0.14	0.02 to 0.71	NO	slope emulated by horizontal pull; slides not a paper
LEG-018	12.0	lab	0.06	0.14	0.02 to 0.71	yes	slope emulated by horizontal pull; diameter and lug height varied together
LEG-019	14.0	lab	0.47	0.17	0.03 to 0.93	yes	slope emulated by horizontal pull
LEG-020	14.0	lab	0.25	0.17	0.03 to 0.93	yes	slope emulated by horizontal pull

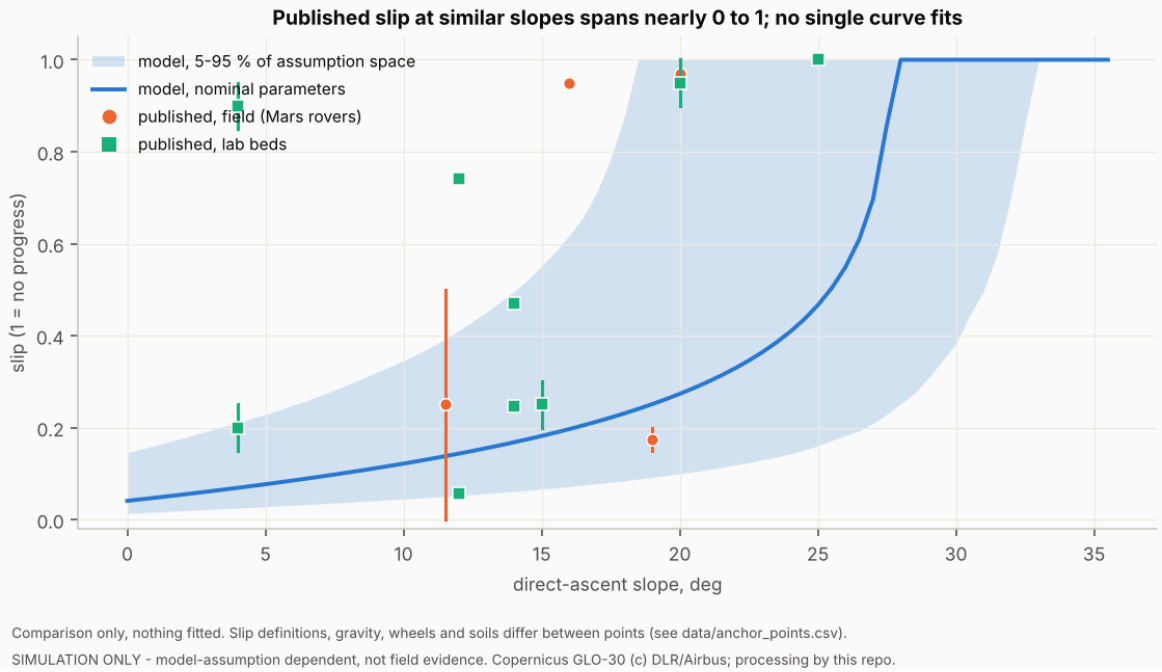


Figure 8.4. Blue line and band: model at nominal parameters and its 5 to 95 % range over the assumption space (SIMULATED BY ME). Orange circles: published field results (Mars rovers). Green squares: published lab beds. Bars show reported ranges or bounds. Published points at similar slopes span nearly 0 to 1 and no single curve fits them.

The assumption envelope overlaps 11 of the 13 published points, which says mostly that it is wide, not that the model is valid. The nominal curve is more than 0.5 slip away from six of them (FLD-012, FLD-025, FLD-027, FLD-028, FLD-030, LEG-017), and the envelope misses two (FLD-030, a smooth wheeled lab rover at 0.9 slip at 4 degrees, and LEG-017, a 116 mm wheel at 0.741 at 12 degrees). These are cases a slope only model with lumped constants cannot represent.

8.6 Limitations specific to this study

- Quasi static, point mass, no sinkage, speed or avalanching; no wheel geometry or vehicle mass; i0 and f_r absorb everything.
- Slope is taken from a 30 m radar surface model whose slip face slopes are biased low (section 5); the blocked shares of phase 1 are lower bounds in that sense. Section 8.7 adds a modelled sub grid face probability and an UNKNOWN class to address this, with its own assumptions.
- The windows are not random samples of any dune field. Origins and destinations sit on gentle low ground, so route statistics say nothing about crossing slip faces.
- Descent is treated with the same symmetric friction circle demand (|tan theta_a + f_r|); descending stability and avalanching are not modelled.
- Parameters were not fitted to any data. No claim is made that the nominal values represent Arabian sand or any vehicle.

- Code is tested (`tests/test_core.py`: 15 tests covering exact planar slope, aspect, slip monotonicity and entrapment, cross slope behaviour, graph blocking, flat ground cost ratio, determinism of origin destination pairs and sampling, coarsening, noise reproducibility and metres per degree), but passing tests show the code does what it says, not that the model is right.

8.7 Phase 2: what the elevation model cannot see (SIMULATED BY ME, model assumption dependent)

> **SIMULATION ONLY, model assumption dependent.** Everything below describes a synthetic detector and a model, validated only against synthetic truth in phase 2 (tested against North American lidar on 2026-10-20, section 13.5). No lidar, drone or field profile of a real Arabian slip face was retrieved, so the detector has never been tested against an Arabian dune. The predictions for this stage were committed before the code was run (`docs/PREDICTIONS.md`, stage 2) and the hits and misses are recorded there.

Problem. A 30 m GLO-30 post spacing cannot resolve an active slip face, whose horizontal run is only about 1.4 to 1.7 times its height (morphology row PM10): a few metres to a few tens of metres. A cell that contains a face reads as a gentle slope. The phase 1 map was therefore binary and optimistic.

Sub grid face probability. For each cell I estimate the probability that an unresolved active face (28 degrees or steeper) lies inside the cell from four features the DEM does see: slope, crest convexity along the steepest direction, slope break (largest change of the signed slope component between the cell and its eight neighbours) and local relief in a 7 by 7 window (a proxy for dune height). The mapping is a logistic regression fitted on a synthetic ensemble of straight crested asymmetric dunes (stoss, brink, active face, apron, flat interdune) that are sampled the way a DEM samples them (mean over a post, or a point sample), separately for each posting (`src/dunerover/subgrid.py`). Truth in the ensemble is known exactly. Every range used is in Table 8.7a; only the stoss slope, the active face slope and part of the face height fraction have a published basis (retrieved morphology rows), and the rest are labelled assumptions. The share of dune crests that carry an active face is **not published** (not found), so the baseline assumes all of them do (pi = 1, the worst case) and the result is shown for pi = 0.6 and 0.3.

Table 8.7a: ranges of the sub grid ensemble.

Sub grid quantity	Range used	Status	Basis (morphology row ids in research/phase2/phase2_morphology.csv)
`stoss_slope_deg`	6.0 to 20.0	published (barchan and transverse dunes, small and medium)	PM01 (barchan 8 to 20), PM02/PM06 (transverse about 6). No distribution for dunes of tens of metres was found, so extrapolated.
`face_slope_deg`	29.5 to 32.5	published (lidar, two US fields)	PM18: sharp peak at 32.5 deg (White Sands) and 29.5 deg (Great Sand Dunes). Not shown to hold in Arabia.
`face_height_fraction`	0.2 to 1.0	partly published, mostly ASSUMPTION	PM08: 0.54 to 1.0 for six transverse dunes of 7 to 10 m. The 0.2 lower end is an assumption for tall compound dunes (no source found).
`dune_height_m`	5.0 to 120.0	published range, log-uniform is an ASSUMPTION	PM29 to PM35: Rub' al Khali dunes from under 5 m to over 155 m; 50 to 130 m for the linear and star types.
`interdune_flat_ratio`	0.0 to 2.0	ASSUMPTION	Flat interdune width as a multiple of the dune footprint; no source used.
`apron_slope`	equal to stoss slope	ASSUMPTION	Slope of the slip face base below the face; no source.
`no_face_lee_slope_deg`	10.0 to 25.0	ASSUMPTION	Lee slope when the face is inactive; the morphology search found no source for it (PM20).
`p_active_face (pi)`	0.3 to 1.0	ASSUMPTION	Share of dune crests with an active face; no source found. Baseline 1.0 (worst case).

Three classes. For each of 256 Latin hypercube draws of the four assumed slip parameters (friction angle, f_r, i0, i_max; fixed seeds 20261014, 20261015 and 20261016) a cell is passable if at least 4 of its 8 headings are passable; q is the share of draws in which it is passable. **BLOCKED** if q is at most 0.05. **TRAVERSABLE** if q is at least 0.95 and the face probability is below tau = 0.20. **UNKNOWN** otherwise, with the cause recorded: assumption sensitive (q between 0.05 and 0.95), sub grid face risk (face probability at least tau), or both. The threshold tau is a free choice; the coverage versus certainty table shows what it trades. Third area **C** is a window of linear dunes in the Saudi Arabian Rub' al Khali (20.40 to 20.55 N, 52.30 to 52.70 E, tile N20E052; not in the UAE), chosen because the UAE tiles held here show star and network dunes. It has a different morphology from windows A and B and is a robustness check, not a study of the UAE.

Table 8.7b: class shares by DEM posting. Coarser postings are block means of the 30 m grid (a stand in for GLO-90; the real GLO-90 product differs).

Area	Posting, m	TRAVERSABLE	BLOCKED	UNKNOWN total	UNKNOWN: assumption sensitive only	UNKNOWN: sub-grid face risk only	UNKNOWN both
A_south_Liwa	29.6	91.4 %	0.00 %	8.6 %	0.0 %	7.8 %	0.8 %
A_south_Liwa	59.2	90.9 %	0.00 %	9.1 %	0.0 %	8.9 %	0.2 %
A_south_Liwa	88.8	92.9 %	0.00 %	7.1 %	0.0 %	7.0 %	0.0 %
B_RubAlKhali_megadunes	29.7	89.1 %	0.01 %	10.9 %	0.0 %	7.9 %	2.9 %
B_RubAlKhali_megadunes	59.4	80.1 %	0.00 %	19.9 %	0.1 %	18.3 %	1.5 %
B_RubAlKhali_megadunes	89.1	85.6 %	0.00 %	14.4 %	0.0 %	13.6 %	0.7 %
C_RubAlKhali_linear	29.9	97.5 %	0.00 %	2.5 %	0.0 %	2.5 %	0.0 %

Area	Posting, m	TRAVERSABLE	BLOCKED	UNKNOWN total	UNKNOWN: assumption sensitive only	UNKNOWN: sub-grid face risk only	UNKNOWN both
C_RubAlKhali_linear	59.7	89.8 %	0.00 %	10.2 %	0.0 %	10.2 %	0.0 %
C_RubAlKhali_linear	89.6	72.5 %	0.00 %	27.5 %	0.0 %	27.5 %	0.0 %

At 30 m the UNKNOWN share is 8.6 % in A south Liwa, 10.9 % in B (megadunes) and 2.5 % in C (linear dunes); BLOCKED is 0.00 %, 0.01 % and 0.00 %. The BLOCKED class is almost empty because the 30 m map hardly contains slopes the model calls impassable in every draw: that is a statement about what the DEM can show, not about the dunes (section 5 and the sub grid problem above). At 60 and 90 m the UNKNOWN share is 9.1 % and 7.1 % (A), 19.9 % and 14.4 % (B), 10.2 % and 27.5 % (C). It is not monotone in A, because coarser cells average out relief and slope break and the detector, trained per posting, also shifts.

How much of the map is robust to the four assumed parameters. Slope based verdicts (before the sub grid step) are unanimous across the draws for 99.1 % (A), 97.0 % (B) and 100.0 % (C) of cells at 30 m; only 0.85 %, 2.96 % and 0.00 % are sensitive to them. Under the original ranges (i0 0.05 to 0.40, f_r 0.05 to 0.25), the slip parameters therefore matter far less for the three class map than the **resolution and the face question** (with the widened ranges of 2026-10-18 this no longer holds; see the end of this section), because almost every 30 m cell is gentle enough to pass under every assumed parameter set. This does not contradict the phase 1 sensitivity result (the climbing limit itself varies from 7.6 to 34.7 degrees across the ranges): the DEM simply rarely shows slopes near that limit.

Table 8.7c: robustness to the four assumed slip parameters.

Area	Posting, m	Robust: same verdict in at least 95 % of draws	Sensitive to the four assumptions	Passable in all 256 draws	Passable in no draw
A_south_Liwa	29.6	99.15 %	0.85 %	98.33 %	0.00 %
A_south_Liwa	59.2	99.82 %	0.18 %	99.50 %	0.00 %
A_south_Liwa	88.8	99.95 %	0.05 %	99.88 %	0.00 %
B_RubAlKhali_megadunes	29.7	97.04 %	2.96 %	94.95 %	0.00 %
B_RubAlKhali_megadunes	59.4	98.39 %	1.61 %	97.00 %	0.00 %
B_RubAlKhali_megadunes	89.1	99.25 %	0.75 %	98.23 %	0.00 %
C_RubAlKhali_linear	29.9	100.00 %	0.00 %	99.95 %	0.00 %
C_RubAlKhali_linear	59.7	100.00 %	0.00 %	99.99 %	0.00 %
C_RubAlKhali_linear	89.6	100.00 %	0.00 %	99.99 %	0.00 %

Sensitivity to the sub grid assumptions. The UNKNOWN share (30 m) when the share of crests with an active face falls from 1.0 to 0.3 is 2.9 % (A), 5.9 % (B) and 0.2 % (C). The most important choice is the definition of "inside the cell". The pre registered definition counts a face only if it lies within the cell itself. A **post hoc variant** (added after the baseline result was

seen, so not pre registered) counts a face within one post of the cell, which is closer to the question a route cares about; it gives UNKNOWN shares of 34.4 % (A), 32.2 % (B) and 22.0 % (C) at 30 m. Both are shown because the plausible range is wide.

Table 8.7d: UNKNOWN share under the sub grid variants.

Area (30 m)	Sub grid variant	pi (share of crests with an active face)	UNKNOWN total	Mean face probability
A_south_Liwa	base	1.0	8.6 %	7.3 %
A_south_Liwa	base	0.6	5.0 %	4.9 %
A_south_Liwa	base	0.3	2.9 %	3.1 %
A_south_Liwa	gentle	1.0	12.3 %	8.9 %
A_south_Liwa	steep	1.0	8.6 %	7.8 %
A_south_Liwa	POST HOC: face within +/-1 post (3-post footprint)	1.0	34.4 %	21.7 %
A_south_Liwa	base, point sampling	1.0	8.2 %	7.1 %
B_RubAlKhali_megadunes	base	1.0	10.9 %	9.6 %
B_RubAlKhali_megadunes	base	0.6	8.1 %	7.2 %
B_RubAlKhali_megadunes	base	0.3	5.9 %	4.8 %
B_RubAlKhali_megadunes	gentle	1.0	13.4 %	9.5 %
B_RubAlKhali_megadunes	steep	1.0	22.4 %	11.9 %
B_RubAlKhali_megadunes	POST HOC: face within +/-1 post (3-post footprint)	1.0	32.2 %	23.1 %
B_RubAlKhali_megadunes	base, point sampling	1.0	10.9 %	9.3 %
C_RubAlKhali_linear	base	1.0	2.5 %	4.4 %
C_RubAlKhali_linear	base	0.6	0.8 %	2.7 %
C_RubAlKhali_linear	base	0.3	0.2 %	1.5 %
C_RubAlKhali_linear	gentle	1.0	3.5 %	3.4 %
C_RubAlKhali_linear	steep	1.0	3.7 %	6.4 %
C_RubAlKhali_linear	POST HOC: face within +/-1 post (3-post footprint)	1.0	22.0 %	14.4 %
C_RubAlKhali_linear	base, point sampling	1.0	2.1 %	4.3 %

Coverage versus certainty. Coverage is the share of cells that are not UNKNOWN. Certainty is the share of cells declared face free that are truly face free. On the synthetic ensemble (known truth) and on the real maps (model expected certainty, not a measured accuracy):

Posting, m	tau	Coverage (declared face free)	Certainty (truly face free among declared)	Missed face rate among declared
29.6	0.05	32.3 %	99.9 %	0.1 %
29.6	0.1	55.5 %	97.7 %	2.3 %
29.6	0.2	66.3 %	95.0 %	5.0 %
29.6	0.5	83.0 %	90.4 %	9.6 %
59.2	0.05	18.6 %	99.9 %	0.1 %
59.2	0.1	25.3 %	99.4 %	0.6 %
59.2	0.2	34.4 %	97.5 %	2.5 %
59.2	0.5	70.4 %	77.2 %	22.8 %
88.8	0.05	14.6 %	99.8 %	0.2 %

Posting, m	tau	Coverage (declared face free)	Certainty (truly face free among declared)	Missed face rate among declared
88.8	0.1	19.4 %	99.1 %	0.9 %
88.8	0.2	26.1 %	97.1 %	2.9 %
88.8	0.5	50.9 %	84.6 %	15.4 %

Area (30 m)	Face probability threshold tau	Coverage (not UNKNOWN)	Model expected certainty of TRAVERSABLE cells (1 minus mean face probability)
A_south_Liwa	0.02	42.3 %	99.1 %
A_south_Liwa	0.05	67.3 %	98.2 %
A_south_Liwa	0.1	83.6 %	97.2 %
A_south_Liwa	0.2	91.4 %	96.2 %
A_south_Liwa	0.3	94.3 %	95.6 %
A_south_Liwa	0.5	97.0 %	94.6 %
A_south_Liwa	0.8	98.7 %	93.7 %
B_RubAlKhali_megadunes	0.02	37.8 %	99.1 %
B_RubAlKhali_megadunes	0.05	55.5 %	98.4 %
B_RubAlKhali_megadunes	0.1	82.4 %	96.4 %
B_RubAlKhali_megadunes	0.2	89.1 %	95.7 %
B_RubAlKhali_megadunes	0.3	91.8 %	95.1 %
B_RubAlKhali_megadunes	0.5	94.5 %	94.1 %
B_RubAlKhali_megadunes	0.8	96.4 %	93.0 %
C_RubAlKhali_linear	0.02	36.8 %	98.8 %
C_RubAlKhali_linear	0.05	75.5 %	97.8 %
C_RubAlKhali_linear	0.1	91.6 %	97.0 %
C_RubAlKhali_linear	0.2	97.5 %	96.3 %
C_RubAlKhali_linear	0.3	98.8 %	96.0 %
C_RubAlKhali_linear	0.5	99.7 %	95.7 %
C_RubAlKhali_linear	0.8	100.0 %	95.6 %

Detector skill on held out synthetic profiles. The detector separates cells with and without a face better than chance (AUC well above 0.5) and better than always predicting the base rate (Brier). It was also tested on ensembles with a different prior than the one it was trained on (gentle and steep dunes, and a lower share of active faces), where the skill changes only moderately. **This is skill against the synthetic world it was trained on; it is not evidence about real dunes.**

Posting, m	Held out synthetic ensemble (detector trained on base, pi 1.0)	Face rate	AUC	Brier	Brier of always predicting the base rate
29.6	base pi=1.0	22.2 %	0.924	0.086	0.173
29.6	gentle pi=1.0	13.7 %	0.954	0.057	0.118
29.6	steep pi=1.0	34.2 %	0.892	0.119	0.225
29.6	base pi=0.3	6.8 %	0.886	0.084	0.063
59.2	base pi=1.0	35.0 %	0.843	0.149	0.227
59.2	gentle pi=1.0	23.0 %	0.827	0.135	0.177
59.2	steep pi=1.0	49.0 %	0.846	0.161	0.250
59.2	base pi=0.3	10.7 %	0.778	0.151	0.095
88.8	base pi=1.0	44.7 %	0.857	0.151	0.247
88.8	gentle pi=1.0	31.6 %	0.794	0.170	0.216

Posting, m	Held out synthetic ensemble (detector trained on base, pi 1.0)	Face rate	AUC	Brier	Brier of always predicting the base rate
88.8	steep pi=1.0	59.1 %	0.893	0.135	0.242
88.8	base pi=0.3	13.5 %	0.760	0.216	0.117

Effect on routes. Treating UNKNOWN cells as impassable, the share of the 10 origin destination pairs per window that stay reachable is:

Area	Posting, m	Pairs reachable (phase 1 rule, 10 pairs)	Pairs reachable with UNKNOWN impassable	Endpoints not TRAVERSABLE
A_south_Liwa	29.6	100 %	100 %	0 %
A_south_Liwa	59.2	100 %	70 %	20 %
A_south_Liwa	88.8	100 %	80 %	10 %
B_RubAlKhali_megadunes	29.7	100 %	100 %	0 %
B_RubAlKhali_megadunes	59.4	100 %	20 %	55 %
B_RubAlKhali_megadunes	89.1	100 %	40 %	45 %
C_RubAlKhali_linear	29.9	100 %	100 %	0 %
C_RubAlKhali_linear	59.7	100 %	80 %	10 %
C_RubAlKhali_linear	89.6	100 %	60 %	20 %

Widened slip scale range (2026-10-18; SIMULATED BY ME, simulation only, model assumption dependent). The three class map was recomputed with the widened i0 range (0.05 to 1.20) and with f_r also widened (0.05 to 0.35), same seeds, detector and rules (`results/subgrid_widened_i0/`, `results/subgrid_widened_i0_fr/`; stage 9). The original shares above are unchanged.

Area	Posting, m	UNKNOWN / BLOCKED, original	UNKNOWN / BLOCKED, widened_i0	UNKNOWN / BLOCKED, widened_i0_fr
A	30	8.6 % / 0.00 %	33.9 % / 0.00 %	92.6 % / 0.00 %
A	59	9.1 % / 0.00 %	29.3 % / 0.00 %	94.2 % / 0.00 %
A	89	7.1 % / 0.00 %	23.7 % / 0.00 %	96.5 % / 0.00 %
B	30	10.9 % / 0.01 %	39.2 % / 0.02 %	90.1 % / 0.02 %
B	59	19.9 % / 0.00 %	44.6 % / 0.00 %	88.8 % / 0.00 %
B	89	14.4 % / 0.00 %	35.9 % / 0.00 %	88.4 % / 0.00 %
C	30	2.5 % / 0.00 %	11.2 % / 0.00 %	97.4 % / 0.00 %
C	60	10.2 % / 0.00 %	13.8 % / 0.00 %	98.8 % / 0.00 %
C	90	27.5 % / 0.00 %	30.2 % / 0.00 %	99.1 % / 0.00 %

At 30 m the UNKNOWN share in A rises from 8.6 % to 33.9 % with the widened i0 range, and to 92.6 % when f_r is widened as well; BLOCKED stays near zero. The pre registered prediction of a small rise missed badly (W9-5, W9-7). The added UNKNOWN is assumption sensitive, not sub grid face risk: once the assumed ranges are this wide, most cells change verdict between draws. In the widened_i0_fr set 12.1 % (A), 9.0 % (B) and 11.3 % (C) of draws cannot move on level ground (`results/range_set_diagnostics/immobile_on_level.csv`), more than the 5 % a TRAVERSABLE cell may fail, so its 90 % shares mostly reflect those draws,

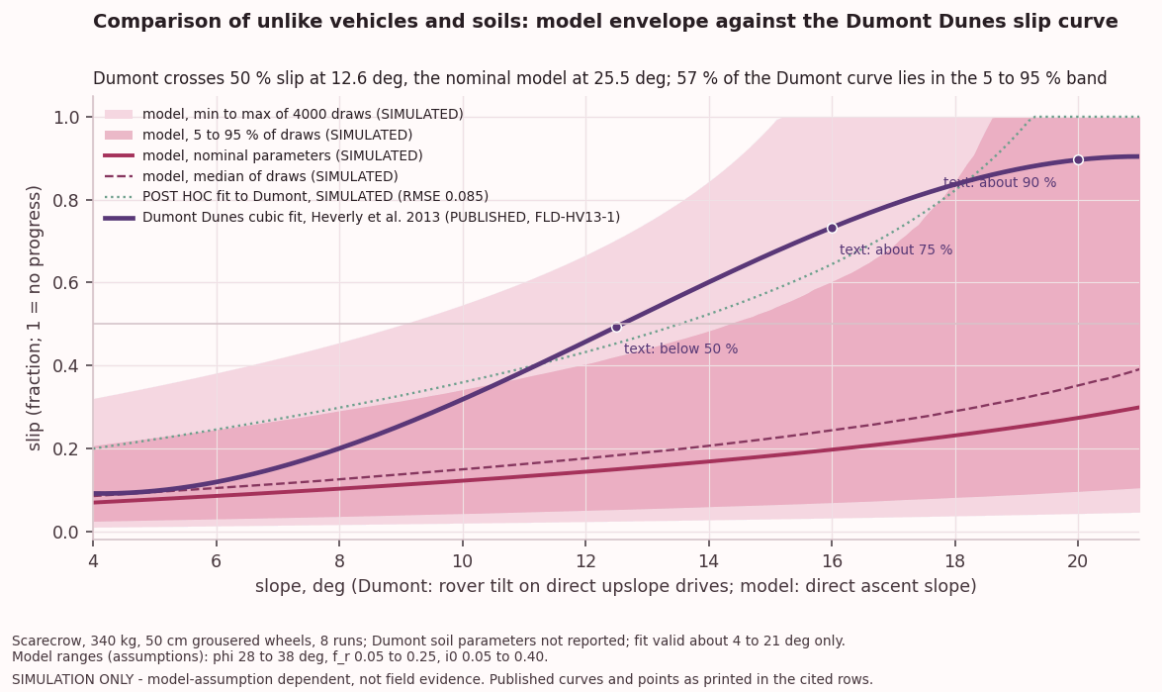
not the terrain; in the widened_i0 set the share is 1 to 2 %. The map's certainty is therefore set by how well the slip parameters of the actual vehicle are known.

What this section establishes and does not. It establishes that, under stated assumptions about dune shape, a 30 m DEM leaves a modest share of cells (about a tenth in the Liwa and megadune windows, a few percent in the linear dune window) with a face probability above 0.20, that this share rises to roughly a quarter to a third if a face within one post counts, and that it is driven by the assumed morphology and the cell definition far more than by the four slip parameters. It does **not** establish that any cell is safe: a TRAVERSABLE cell has a face probability below 0.20, not zero. No claim is made that the detector would find real faces; the way to test it is to compare it with an RTK or drone profile of a real dune (planned in `docs/FIELD_VALIDATION_PLAN.md` and as milestone M5 of `docs/PROPOSAL_EDGE.md`).

8.8 Two published anchors: Dumont Dunes and Ishigami 2009 (SIMULATED BY ME against PUBLISHED values; simulation only, model assumption dependent)

Both comparisons are of unlike vehicles and soils. They test whether the unchanged slip model (section 8.1, `src/dunerover/slipmodel.py`) and its assumed parameter ranges are consistent with two published slope results, not whether the model is right for any rover on Arabian sand. Design, seeds and tolerances were committed before the script existed (`docs/PREDICTIONS.md`, stage 5). Script `scripts/anchor_comparisons.py`; outputs `results/anchors/` (CSVs, `summary.json`, `run_log.txt`, `figures/fig_dumont_vs_model.png`, `figures/fig_ishigami2009_vs_model.png`).

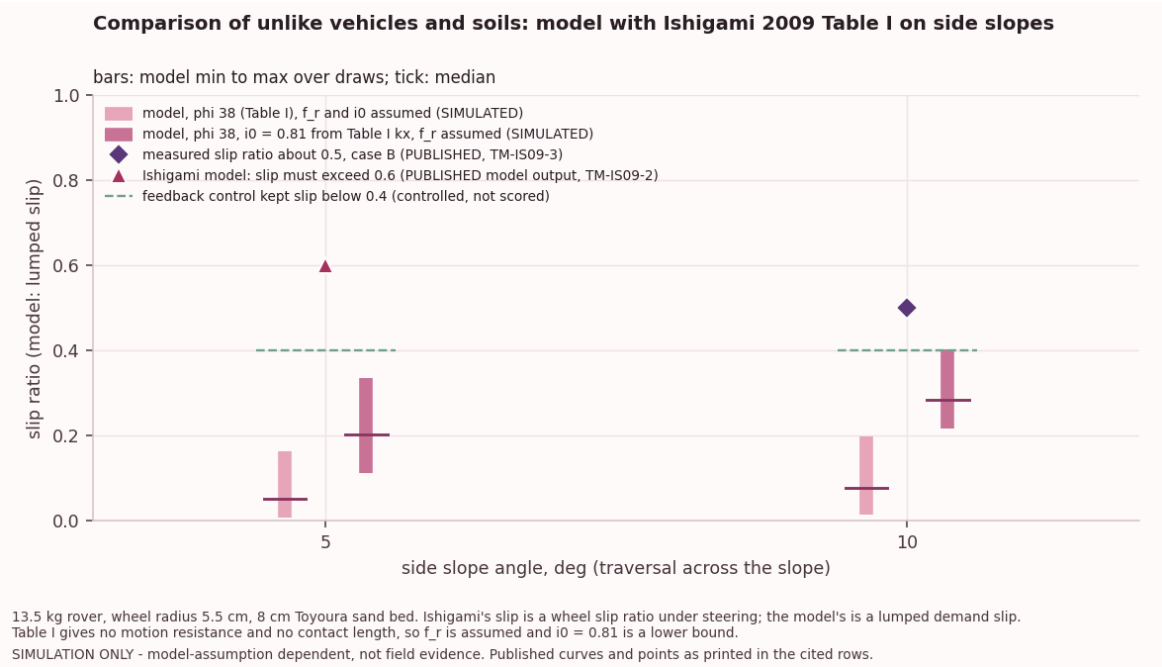
(a) Dumont Dunes (PUBLISHED: Heverly et al. 2013, Sec 6.3, rows FLD-HV13-1 and FLD-HV13-2). The published curve (FLD-HV13-1; slope as rover tilt and the S shape explanation are in that row's notes) is a cubic fit of slip (%) against slope (deg) from eight direct upslope runs of the Scarecrow rover (340 kg, six 50 cm grousered wheels) in natural dune sand, valid only from about 4 to 21 degrees. No soil parameters are reported, slope is the rover tilt, and slip is one minus traverse over commanded distance. The model was evaluated on the same slope grid at nominal parameters and over 4000 Latin hypercube draws of its assumed ranges (phi 28 to 38 deg, f_r 0.05 to 0.25, i0 0.05 to 0.40).



Slope, deg	Dumont cubic (PUBLISHED fit, FLD-HV13-1)	Model, nominal (SIMULATED)	Model, 95th percentile of draws (SIMULATED)
4.0	0.09	0.07	0.21
10.0	0.32	0.12	0.34
12.5	0.49	0.15	0.42
16.0	0.73	0.20	0.60
20.0	0.90	0.27	1.00
21.0	0.90	0.30	1.00

- The Dumont curve crosses 50 % slip at 12.6 degrees; the nominal model crosses it at 25.5 degrees and the median of the draws at 22.9 degrees. Against this vehicle and dune, the nominal model is far too permissive above about 10 degrees.
- 100.0 % of the Dumont curve lies inside the min to max envelope, which only shows how wide the assumed ranges are. 57.3 % lies inside the 5 to 95 % band. **The envelope fails** between 10.7 to 17.9 degrees, where Dumont slip is above the 95th percentile of the draws; the band rejoins the curve near 18 degrees only because many draws reach entrapment (slip 1) there.
- 0 of 4000 draws stay within 0.10 of the Dumont curve at every slope.
- **POST HOC recalibration test (not a validation).** A bounded least squares fit of the three parameters to the Dumont curve ends at the edges of the ranges (phi 28.0 deg, f_r 0.139, i0 0.40) with RMSE 0.085 and a largest error of 0.13; it crosses 50 % at 13.5 degrees. The model form rises smoothly to entrapment and cannot reproduce the S shaped Dumont curve (slow rise, rapid transition between about 10 and 17 degrees, then saturation near 90 %), which Heverly et al. attribute to sinkage and bulldozing growing with slip. No parameter of the model was changed on the basis of this fit.

(b) Ishigami et al. 2009 (PUBLISHED: Table I and Sec 3.2.2, 4.3, 6.3; rows TM-IS09-1 to TM-IS09-5). A 13.5 kg four wheel rover (wheel radius 5.5 cm) traversed 5 and 10 degree side slopes in a tilting bed of Toyoura sand, steering to hold a straight line. Table I gives a friction angle of 38 degrees and zero cohesion, which map directly onto the model; it gives no motion resistance and no contact length, so f_r stays assumed. The measured slip ratio of about 0.5 is recorded in the passage and notes of row TM-IS09-3 (Sec 4.3, p. 275), whose value field holds the slip angle of about 23 degrees from the same run; the verifier confirmed both on p. 275. Variant b1 keeps i0 over its assumed range. Variant b2 sets i0 = kx / (largest contact arc) = 0.07 m / (r pi / 2) = 0.81, a lower bound on i0 under the model's own definition (i0 = K/L), which lies outside the assumed range of 0.05 to 0.40. The model was run across the slope (no along slope component). Ishigami's slip is a wheel slip ratio under steering; the model's is a lumped demand slip.



Row	Side slope, deg	Published	What it is	Model variant	Model min to max (SIMULATED)	Compatible	Scored
TM-IS09-3	10	0.5 (point)	measured slip ratio, about 0.5, feed forward control, case B	b1	0.01 to 0.20	NO	yes
TM-IS09-3	10	0.5 (point)	measured slip ratio, about 0.5, feed forward control, case B	b2	0.22 to 0.40	NO	yes
TM-IS09-2	5	0.6 (lower bound)	Ishigami model: slip ratio must exceed 0.6 for positive thrust (model to model)	b1	0.01 to 0.16	NO	yes
TM-IS09-2	5	0.6 (lower bound)	Ishigami model: slip ratio must exceed 0.6 for positive thrust (model to model)	b2	0.11 to 0.34	NO	yes
TM-IS09-5	5	0.4 (upper bound)	feedback control kept slip ratio below 0.4 (controlled, not scored)	b1	0.01 to 0.16	yes	no

Row	Side slope, deg	Published	What it is	Model variant	Model min to max (SIMULATED)	Compatible	Scored
TM-IS09-5	5	0.4 (upper bound)	feedback control kept slip ratio below 0.4 (controlled, not scored)	b2	0.11 to 0.34	yes	no
TM-IS09-5	10	0.4 (upper bound)	feedback control kept slip ratio below 0.4 (controlled, not scored)	b1	0.01 to 0.20	yes	no
TM-IS09-5	10	0.4 (upper bound)	feedback control kept slip ratio below 0.4 (controlled, not scored)	b2	0.22 to 0.40	yes	no

- At 10 degrees the model spans 0.01 to 0.20 (b1) and 0.22 to 0.40 (b2). **Neither reaches the measured slip ratio of about 0.5** (case B, TM-IS09-3). At 5 degrees the model spans 0.01 to 0.16 (b1) and 0.11 to 0.34 (b2), far below Ishigami's own model statement that slip must exceed 0.6 for positive thrust (a model to model comparison).
- The feedback runs (slip kept below 0.4) are compatible with every variant but are a controlled slip, so they test nothing and are not scored.
- Using the Table I shear modulus pushes i0 above the assumed range: the published value kx = 0.07 m is also larger than the spread of Janosi moduli recorded in `research/raw/terramechanics_notes.md` (0.0006 to 0.045 m), so the assumed i0 range may be too narrow for small wheels in loose sand.

With the widened slip scale range (2026-10-18; stage 9). Both comparisons were repeated with the widened ranges (`results/anchors_widened_i0/`, `results/anchors_widened_i0_fr/`); still comparisons of unlike vehicles and soils.

Range set	Nominal crosses 50 % at, deg	Median of draws crosses 50 % at, deg	Dumont crosses 50 % at, deg (PUBLISHED fit)	Dumont inside 5 to 95 % band	Dumont above the band, deg	POST HOC fit RMSE	Ishigami b1 at 10 deg (measured about 0.5)	Ishigami b2 (i0 0.81) at 10 deg	Ishigami b1 at 5 deg (his model: above 0.6)
original	25.5	22.9	12.6	57.3 %	10.7 to 17.9	0.085	0.01 to 0.20	0.22 to 0.40	0.01 to 0.16
widened_i0	10.2	12.3	12.6	100.0 %	none	0.071	0.01 to 0.59	0.22 to 0.40	0.01 to 0.49
widened_i0_fr	10.2	10.0	12.6	100.0 %	none	0.071	0.01 to 0.83	0.22 to 0.56	0.01 to 0.73

With nominal i0 = 0.81 the model crosses 50 % slip at 10.2 degrees, earlier than the Dumont curve: the nominal error has changed sign rather than disappeared. The envelope now contains the Dumont curve and the measured Ishigami value (b1 at 10 degrees: 0.01 to 0.59), but only because it is very wide. The post hoc fit improves only from 0.085 to 0.071: the model form still misses the S shape.

Verdict. On both anchors the model, at nominal parameters and over most of its assumed ranges, predicts too little slip on sand slopes, by a wide margin above about 10 degrees. Its envelope contains the Dumont curve only because the ranges are wide, with the original f_r range it does not reach the slip Ishigami measured at 10 degrees even with the Table I shear modulus (with f_r widened to 0.35 the b2 variant does, see the table below), and a post hoc fit at the edge of the ranges still misses the shape of the Dumont curve. With the original ranges this is a negative result for the model as a predictor on sand slopes; with the widened ranges the model brackets both anchors but no longer predicts anything narrowly, which is a negative result of another kind and is consistent with the stage 3 finding that the calibrate then predict protocol is weak above 15 degrees. It does not change the published inputs to section 8.7 or the earlier results, which stay as they were; it strengthens the case for measuring slip against slope directly (section 9). The Dumont cubic also gives the slope ladder a published comparison curve for a different and heavier vehicle (`docs/NEXT_STEPS.md`).

8.9 Stretch study: the Dumont curve as a prior for i0 (SIMULATED BY ME; simulation only, model assumption dependent)

Pre registered as stage 7 in `docs/PREDICTIONS.md`; script `scripts/study_dumont_prior.py`; outputs `results/dumont_prior/`. In each of the 256 phase 2 draws, i0 was replaced by the value that best fits the Dumont curve given that draw's friction angle and resistance, treating a heavier vehicle on a different dune as if it described the rover the map is for (an unsupported assumption, which is what is being tested). In area A the fitted i0 has a median of 0.58, above the assumed range of 0.05 to 0.40 in 81 % of draws, and even the best i0 leaves a median RMSE of 0.10 against the curve. The 30 m map hardly changes: in area A, UNKNOWN goes from 8.60 % to 8.95 % and BLOCKED from 0.00 % to 0.32 %; in area B from 10.89 % to 12.24 % and from 0.01 % to 1.68 %; in area C from 2.54 % to 2.56 % (`results/dumont_prior/class_shares_30m.csv`). The baseline reproduced phase 2 exactly. Two pre registered predictions, a large rise in UNKNOWN and in BLOCKED, missed. The reading: at 30 m the elevation model shows few slopes in the range where a higher slip scale changes the verdict, so the map's uncertainty is dominated by what the grid cannot see (section 8.7), not by the slip law. A larger slip scale matters for a rover on a real slope; when it is fitted to the Dumont curve in each draw it matters little for this map, but drawn independently over a widened range it matters a great deal (section 8.7, stage 9).

Stage 7 against stage 9. Stage 7 replaced i0 by a value fitted to the Dumont curve for each draw's friction angle and resistance, so within a draw the parameters agreed with one curve and the map barely moved. Stage 9 draws i0 independently over the widened range, and UNKNOWN in area A rises to 33.9 %. Stage 9 changed both the centre and the width of the range,

so this is not a controlled comparison; the difference suggests that it is the spread of plausible parameter combinations, not a shift of the slip scale on its own, that makes the map uncertain.

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9. Field validation plan (PLANNED; nothing here has been run)

Full plan: `docs/FIELD_VALIDATION_PLAN.md`. Summary:

1. **Vehicle.** One small four wheel rover (5 to 25 kg, wheels at least 150 mm, mass and wheel load known and kept fixed). Wheel diameter and width alter slip by large factors in the published lab data (LEG-017 to LEG-020), so the wheel must be fully described.
2. **Measurements.** Per wheel encoders and motor current at 50 Hz or better, RTK GNSS ground truth (not rover odometry), a 6-axis IMU, rut depth by gauge at three stations per run, and tow tension if a load cell is available. Slip is computed against ground truth speed with a calibrated effective radius, and distance based slip is reported alongside because published studies use both.
3. **Soil, at every station.** Sieve grain size, bulk density, moisture at two depths, crust thickness, angle of repose, hand penetrometer profile and a plate pressure sinkage set, with local slope from the RTK profile.
4. **Run design (about 60 runs).** Slope bins at 0, 5, 10, 15, 20 and 25 degrees (plus or minus 2) in the original plan, revised on 2026-10-15 (below), randomised order within a day, fresh track each run; a 12-run pilot to measure run to run scatter, then about 30 main runs; 9 cross slope, 6 tethered descent and 3 repeated pass runs. Run counts follow n = (1.96 sigma / E)^2: for sigma = 0.10 to 0.15 and E = 0.10 that is 4 to 9 runs per bin.
5. **Pre registered analysis.** Calibrate the section 8 model on 0 to 10 degree runs only, **record dated predictions** for 15, 20 and 25 degrees (original bins; 12.5 to 20 degrees in the revision below) before running them, then compare mean and root mean square error, the calibration slope and the entrapment slope, against a constant slip baseline and a slope threshold baseline. **Pre data predictions are committed** (`docs/predictions/slope_ladder_predictions_2026-10-12.csv`, SIMULATED BY ME, from the assumption ranges only) and a script writes the post calibration predictions from the gentle slope runs before the steep runs. **Tolerance rule (assumption, fixed before the main block):** tolerance = the larger of 3 times the pilot sigma and 0.05, with pilot sigma the pooled within bin standard deviation of the 12 pilot runs; the model is called adequate if at least 80 % of held out runs fall within the tolerance and the entrapment slope is within 3 degrees; if 3 times the pilot sigma exceeds 0.30 the

experiment is flagged uninformative. **Synthetic test with known truth (SIMULATED BY ME, `scripts/field_analysis_power.py`)** shows this rule discriminates weakly:

Synthetic truth	Noise sigma	Replicates	Called adequate	Failed	Flagged uninformative (3 x pilot sigma above 0.30)	Median tolerance
model	0.03	300	46 %	54 %	0 %	0.098
model	0.06	300	39 %	61 %	0 %	0.178
model	0.1	300	34 %	66 %	42 %	0.285
janosi_mean	0.03	300	24 %	76 %	0 %	0.114
janosi_mean	0.06	300	56 %	44 %	7 %	0.183
janosi_mean	0.1	300	78 %	22 %	43 %	0.286
collapse	0.03	300	31 %	69 %	20 %	0.208
collapse	0.06	300	35 %	65 %	34 %	0.255
collapse	0.1	300	38 %	62 %	64 %	0.340

The rule passes an exactly right model in under half of the replicates and fails a model that is wrong by construction (`collapse`: slip jumps to 0.85 from 22 degrees) in less than 70 %; at high noise it is lenient and flagged. **So an adequacy pass or fail from one ladder must not be over read. Report the per bin predicted against observed slip with intervals and the entrapment slope as the primary result, and treat the pass or fail as secondary.** Table 9.1 shows what the protocol can and cannot predict before any data:

Table 9.1: pre data slope ladder predictions (prior predictive from the assumption ranges, and the error band of the calibrate then predict protocol on synthetic truths).

Slope, deg	Role	Prior predictive slip, 5 to 95 % (phi 28 to 38)	Same with phi 33 plus or minus 2	P(slip at least 0.8), phi unknown	Protocol error band (predicted minus true), sigma 0.06
0	calibration	0.01 to 0.14 (median 0.05)	0.01 to 0.14	0 %	n/a (calibration slope)
5	calibration	0.03 to 0.23 (median 0.10)	0.03 to 0.22	0 %	n/a (calibration slope)
10	calibration	0.05 to 0.34 (median 0.15)	0.05 to 0.33	0 %	n/a (calibration slope)
15	held out prediction	0.07 to 0.53 (median 0.23)	0.07 to 0.49	1 %	-0.05 to +0.07
20	held out prediction	0.10 to 1.00 (median 0.36)	0.11 to 0.87	14 %	-0.08 to +0.41
25	held out prediction	0.16 to 1.00 (median 0.76)	0.18 to 1.00	48 %	-0.40 to +0.49

Calibrating on 0 to 10 degrees constrains the 15 degree prediction well, the 20 degree prediction weakly, and the 25 degree prediction hardly at all: the slip diverges near entrapment and gentle runs do not fix where that happens. The 25 degree bin is therefore mainly a test of the entrapment slope, not of the slip value. The same check found that calibration needs bounds on f_r and i0, because the two trade off at gentle slopes.

Revision of 2026-10-15 (PLANNED; pre registered as stage 6 in `docs/PREDICTIONS.md`). The Dumont Dunes curve (section 8.8) and the Marsyard sand result (FLD-HV13-3) put the change from moderate to high slip between about 10 and 17 degrees, so the bins became 0, 5, 10, 12.5, 15, 17.5 and 20 degrees (plus or minus 1 degree from 10 to 20), with 25 degrees only if fewer than half of the 20 degree runs entrap. The original file is kept; the revised expected ranges add the Dumont value and widen the upper end of the planning range where it lies above the model band (`docs/predictions/slope_ladder_predictions_2026-10-15.csv`):

Slope, deg	Role	Tolerance, deg	Model prior 5 to 95 % (SIMULATED, unchanged)	Dumont cubic (PUBLISHED, heavier vehicle)	Planning range (PLANNED)
0	calibration	plus or minus 2	0.01 to 0.14	outside 4 to 21 deg	0.01 to 0.14
5	calibration	plus or minus 2	0.03 to 0.23	0.10	0.03 to 0.23
10	calibration	plus or minus 1	0.05 to 0.34	0.32	0.05 to 0.34
12.5	held out prediction	plus or minus 1	0.06 to 0.42	0.49	0.06 to 0.49
15	held out prediction	plus or minus 1	0.07 to 0.53	0.67	0.07 to 0.67
17.5	held out prediction	plus or minus 1	0.08 to 0.76	0.81	0.08 to 0.81
20	held out prediction	plus or minus 1	0.10 to 1.00	0.90	0.10 to 1.00
25	held out prediction (conditional: only if fewer than half the 20 deg runs entrap)	plus or minus 2	0.16 to 1.00	outside 4 to 21 deg	0.16 to 1.00

The revision has a cost that the synthetic re run exposed (`results/field_kit/power_study_2026-10-15.csv`, SIMULATED BY ME, synthetic data only):

Synthetic truth	Noise sigma	Replicates	Called adequate	Failed	Flagged uninformative (3 x pilot sigma above 0.30)	Median tolerance
model	0.03	300	1 %	99 %	0 %	0.090
model	0.06	300	0 %	100 %	0 %	0.169
model	0.1	300	1 %	99 %	39 %	0.272
janosi_mean	0.03	300	2 %	98 %	0 %	0.092
janosi_mean	0.06	300	34 %	66 %	2 %	0.162
janosi_mean	0.1	300	65 %	35 %	40 %	0.277
collapse	0.03	300	53 %	47 %	1 %	0.158
collapse	0.06	300	48 %	52 %	13 %	0.217
collapse	0.1	300	45 %	55 %	56 %	0.312
dumont	0.03	300	0 %	100 %	0 %	0.107
dumont	0.06	300	0 %	100 %	12 %	0.179
dumont	0.1	300	5 %	95 %	60 %	0.321

Without a bin beyond the synthetic truth's entrapment slope (26.3 degrees), the observed entrapment slope is usually undefined and the adequacy rule fails even a correct model. The rule and the revised bins are incompatible as written; the choice between amending the rule, keeping a tethered 25 degree bin, or returning to the original bins is left to the author (`docs/QUESTIONS_FOR_HUDA.md`). The Dumont shaped truth is failed in nearly every replicate, but so is a correct model, so that detection is uninformative. **Amended entrapment rule (2026-10-18; PLANNED rule, tested on SIMULATED BY ME synthetic data only).** At the author's decision, and after seeing the 1.3 % result above, the rule was amended and pre registered again (stage 10 in `docs/PREDICTIONS.md`): an entrapment slope that is not reached is reported as "not reached", never as a failure and never as a pass. A run set is adequate only if the slip test and the entrapment test both pass; "slip within tolerance, entrapment not reached" is its own verdict. Re run on the revised bins (`results/field_kit/power_study_2026-10-15_rule_2026-10-18.csv`):

Synthetic truth	Noise sigma	Replicates	Adequate	Slip within tolerance, entrapment not reached	Not adequate	Entrapment not reached	Flagged uninformative
model	0.03	300	1 %	74 %	24 %	98 %	0 %
model	0.06	300	0 %	79 %	21 %	98 %	0 %
model	0.1	300	1 %	80 %	20 %	97 %	39 %
janosi_mean	0.03	300	2 %	0 %	98 %	0 %	0 %
janosi_mean	0.06	300	34 %	0 %	66 %	0 %	2 %
janosi_mean	0.1	300	65 %	0 %	35 %	0 %	40 %
collapse	0.03	300	53 %	0 %	47 %	0 %	1 %
collapse	0.06	300	48 %	1 %	51 %	1 %	13 %
collapse	0.1	300	45 %	7 %	48 %	7 %	56 %
dumont	0.03	300	0 %	0 %	100 %	0 %	0 %
dumont	0.06	300	0 %	0 %	100 %	0 %	12 %
dumont	0.1	300	5 %	0 %	95 %	0 %	60 %

For a correct model at sigma 0.03, entrapment is not reached in 97.7 % of replicates; 74.3 % end as "slip within tolerance, entrapment not reached", 1.3 % as adequate and 24.3 % as not adequate. The Dumont shaped truth is called adequate in 0.0 %. **The wrong collapse model is still called adequate in 52.7 % of replicates at sigma 0.03, and the mildly wrong janosi_mean model in 65.3 % at sigma 0.10.** The rule was not tuned after this run; the next decision is the author's (`docs/QUESTIONS_FOR_HUDA.md`, question 7). Both bin designs stay documented; whether the 1 degree tolerance of the revised bins can be achieved on a real dune is unconfirmed. The calibration bounds in the field analysis and the synthetic truths of this power study still use the original ranges (i0 up to 0.40, f_r up to 0.25); a real rover whose slip scale lies near the widened range would push the calibration to its bound, and the power result does not cover such truths. 6. **Safety and scope.** Tether and recovery line, a second person at all times; slip faces near repose can avalanche. Not a safety case. Heat and dust are handled in `hudasol/dhill`. 7.

What to log for future prediction. Wheel speeds and currents, RTK and IMU synchronised, effective radius, DEM and RTK slope and aspect, the soil state list, wheel state, ambient conditions, pass number and outcome flags.

The helper script `scripts/analyze_field_runs.py` and the log template `docs/field_log_template.csv` implement the analysis and logging; the script is tested on synthetic data only (including a wrong model case) and has never seen a field measurement. A one page checklist for the field day is `docs/FIELD_DAY_CHECKLIST.md`; low cost rover and measurement kit options, with costs only where a source was retrieved, are in `docs/ROVER_OPTIONS.md`; a proposal outline for a funded validation is `docs/PROPOSAL_SEED.md` (a proposal, not an achievement).

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10. Limitations

What this work cannot say without field data.

- It cannot say how well any model predicts slip, sinkage or entrapment of any rover on Arabian dune sand. No published number does either, and the simulation here is not evidence.
- It cannot say that any route is traversable. The model map is lower bound optimistic (30 m grid) and its assumptions are unmeasured.
- It cannot say which model family (RFT, Bekker-Wong, SCM, DEM, learned) is best for dunes; no head to head test exists.
- It cannot give soil parameters for Gulf sand. The only grain sizes of natural dune sand retrieved are from Kuwait and Brazil.

Limits of the review.

- Retrieval was constrained to sources reachable from the sandbox and found by a US only search tool. Paywalled journals and standard textbooks were not read, except five papers the author supplied as PDFs on 2026-10-13; the per topic notes list the exact URLs to fetch (`research/raw/*_notes.md`, sections NOT_RETRIEVED).
- Several values are secondary: tilt bed slip numbers for a Mars rover engineering model from a JPL press release (FLD-006, excluded from the comparison), a thesis (TM-022 to TM-024), conference slides (LEG-017 to LEG-020), and RoverDevKit's own re evaluation of other authors' data (SIM-025 to SIM-027).
- Slip is defined differently across papers; several rows in the same figure measure different things. The warning column says how.
- Many simulation validations are plot only. Rows with "qualitative" values carry no error.

- The rows' extraction and the first verification were done by language model agents using the same search tools; shared blind spots (especially on figure only values) are possible.
- UAE items are as of 2026-10-11 and some, such as the Rashid 2 launch date, conflict between sources.

Not done. DEM and RFT were reviewed but not run; Chrono was run from 2026-10-20 (section 13.4). No dataset with slip labels from dunes was available, so no learning model was trained. The GLO-30 licence (COP-DEM-GLO-30-F) was read in full in phase 2 (`docs/DATA_AND_LICENSES.md`). Ishigami et al. 2007 and 2009, Heverly et al. 2013, Yu et al. 2024 and Takehana et al. 2025 were supplied by the author as PDFs on 2026-10-13, read in full and added as 26 rows (verified 2026-10-15). Still not retrieved: Wong and Reece 1967 and the Wong textbook parameter table; neither is cited for any number. Until 2026-10-20 the sub grid detector (section 8.7) had not been compared with any real lidar or drone profile; it has since been tested against one metre lidar of three North American dune fields (section 13.5), never against an Arabian slip face; the share of dune crests with an active face and the face height fraction of tall dunes are not published and are assumptions.

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11. Verification

Evidence table, the 187 rows before 2026-10-20 (the 323 rows of phase 5 are described in section 13.1). Separate agents with no access to the extraction notes re read sources. Phase 1 checked **30 rows drawn at random** (seed 20261011) from the 137 rows the extractors had not flagged, plus all 7 flagged rows; phase 2 checked the **remaining 107 rows** in four batches; phase 3 (2026-10-15) checked the **26 rows read from the PDFs the author supplied**, one verifier per source, each seeing the claim fields only; phase 4 (2026-10-18) checked the **two rows split from FLD-HV13-5**. The table gives two views: the first check of the 170 rows as extracted (rows labelled ALL 144 and ALL 170), and the current table of 171 rows with each row's latest verdict (rows labelled CURRENT). Material error means a wrong value, unit, metric meaning, validation type, vehicle, terrain or slope; minor error means a correct number with a wrong label, section reference or descriptive detail.

Group	n	correct	minor	material	material rate	any discrepancy rate
random sample, phase 1 (n=30 of 137 unflagged)	30	24	6	0	0 % (exact 95 % upper bound 9.5 %)	20.0 % (Wilson 95 % CI 9.5 to 37.3 %)
flagged sample, phase 1	7	4	3	0	0 % (exact 95 % upper bound 34.8 %)	42.9 % (Wilson 95 % CI 15.8 to 75.0 %)
remaining rows, phase 2 full re-check	107	80	21	6	5.6 % (Wilson 95 % CI 2.6 to 11.7 %)	25.2 % (Wilson 95 % CI 18.0 to 34.2 %)
ALL 144 ROWS	144	108	30	6	4.2 % (Wilson 95 % CI 1.9 to 8.8 %)	25.0 % (Wilson 95 % CI 18.6 to 32.7 %)
supplementary rows from supplied PDFs, phase 3	26	11	12	3	11.5 % (Wilson 95 % CI 4.0 to 29.0 %)	57.7 % (Wilson 95 % CI 38.9 to 74.5 %)
ALL 170 ROWS	170	119	42	9	5.3 % (Wilson 95 % CI 2.8 to 9.8 %)	30.0 % (Wilson 95 % CI 23.6 to 37.3 %)
split rows, phase 4	2	1	1	0	0 % (exact 95 % upper bound 77.6 %)	50.0 % (Wilson 95 % CI 9.5 to 90.5 %)
new rows of the 2026-10-20 evidence expansion, phase 5	323	241	52	30	9.3 % (Wilson 95 % CI 6.6 to 12.9 %)	25.4 % (Wilson 95 % CI 20.9 to 30.4 %)
new rows of 2026-10-21 (scaling law sources), phase 6	16	13	3	0	0 % (exact 95 % upper bound 17.1 %)	18.8 % (Wilson 95 % CI 6.6 to 43.0 %)
CURRENT TABLE, latest verdict per row	510	374	98	38	7.5 % (Wilson 95 % CI 5.5 to 10.1 %)	26.7 % (Wilson 95 % CI 23.0 to 30.7 %)
CURRENT SUPPLEMENTARY ROWS, latest verdict per row	27	12	13	2	7.4 % (Wilson 95 % CI 2.1 to 23.4 %)	55.6 % (Wilson 95 % CI 37.3 to 72.4 %)

All errors were corrected through `research/corrections.csv` (old values in `research/dedup_log.md`). The six material errors found in phase 2: LRN-017 (a three wheeled prototype labelled four wheel), SIM-012 and LRN-006 (validation type: a prepared sand bed and the engineered JPL Mars Yard are `lab`, not `field`), LRN-019 (physical rover results only, `lab`, not `sim-vs-lab`), TM-020 (0.015 was a best case grid entry, the worst case is 0.070) and FLD-029 (an unreferenced capability statement presented as a lab measurement; now validation type `none`). SIM-013 was set to `lab` for consistency with SIM-012 and carries the unit warning. The phase 1 sample had suggested zero material errors; the 107 rows showed that estimate was optimistic, which is why all rows were checked. Verifiers are language model agents using the same search tools, so shared blind spots (for example on figure only values) remain possible; a human spot check of a few rows is advisable. Phase 3 found 3 material errors in 26 rows (11.5 %), all on validation type and none on a value: TM-TK5 and TM-IS07-5 were labelled as model comparisons although the source compares no simulation with them (now `lab`), and FLD-HV13-5 was labelled `field` although its bedrock limit and its 75 % slip at 20 degrees come from the JPL Marsyard (set to `lab and field`, then split on 2026-10-18 at the author's request into FLD-HV13-5a, `lab`, and FLD-HV13-5b, `field`, both checked again in phase 4). Any discrepancy was 15 of 26 (57.7 %), mostly page labels. The phase 3 verifiers read the PDFs through their text layer and could not render page images, so figure only values were checked against the text. Duplicate rows between agents were removed (`research/dedup_log.md`); seven Kern et al. candidate rows read in phase 2 duplicate SIM-012 to SIM-014 and were not added.

Predictions register. Predictions were written and committed before each run (`docs/PREDICTIONS.md`). Hits and misses so far:

Stage	HIT	MIXED (part hit)	MISSED	pending
Stage 1	1	2	1	0
Stage 2	3	1	4	0
Stage 3	2	1	3	0
Stage 4	1	0	3	0
Stage 5	9	1	0	0
Stage 6	5	0	1	0
Stage 7	4	0	2	0
Stage 8	3	0	0	0
Stage 9	6	0	2	0
Stage 10	5	0	0	0
Stage 11	3	0	0	0
Stage 12	3	0	3	0
Stage 13	4	0	3	0
Stage 14	5	0	2	0
Stage 16	2	0	2	0
Stage 17	1	0	4	0
Stage 18	3	0	2	0

Stage	HIT	MIXED (part hit)	MISSED	pending
Stage 15	4	0	3	0
Stage 19	6	0	2	0
Stage 21	2	0	3	0
Stage 22	3	0	1	0
Stage 20	5	0	2	0
Stage 23	4	0	1	0
Stage 24	2	0	1	0
Stage 25	7	0	2	0
Stage 26	5	0	3	0

Report claims, second pass. A separate subagent that had not written the report checked the prose of the 2026-10-11 draft of this report (everything before Appendix A at that time) against `research/evidence.csv`, `research/raw/*` and `results/*`, after the draft was complete. It examined 225 claim groups (every number in the simulation tables was compared individually and the statistics were recomputed): 188 correct, 26 minor (right number but a wrong label, row identifier or imprecise wording), 5 wrong and 6 unverifiable (WRONG rate 2.2 %; wrong plus minor 13.8 %). The five wrong items were a kc statement that belonged to another source (TM-029), percentage units applied to drawbar errors given in newtons (TM-001 to TM-004), a data-coverage cell that belonged to another product, a UAE vehicle date, and a tilt-bed row called a Mars measurement (FLD-006). **All 31 wrong and minor items were corrected in this text**; the six unverifiable statements were softened or removed. The full finding table is `research/factcheck_report_pass.md`. This checker is also a language-model agent working from the repository files, so it shares any error that is present in those files; it cannot catch a mistake that traces back to a source misread twice. That pass did not cover text added later: sections 8.7 and 8.8, the 2026-10-15 revision in section 9, phase 3 in section 11 and the slope evidence added to section 3; those parts were checked only by the adversarial review of 2026-10-15 (`docs/REVIEW_2026-10-15.md`).

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12. Open questions

1. **Sand variability.** How do grain size and shape distribution, packing, moisture and subsurface cementation (Liwa has sabkha interdunes with a shallow water table) change slip and sinkage, at one vehicle and slope? No quantitative Gulf data were found.
2. **Sim to field gaps.** Does a model calibrated on a flat or gentle slope predict slip at 15 to 25 degrees on the same dune sand? Is slope induced shear collapse near repose important enough that a model must include avalanching?

3. **Resolution.** How far does a 30 m radar DEM understate the slope a rover feels at a slip face, measured against RTK profiles? Can aspect, curvature and dune height infer slip faces better than slope?
4. **Slip measurement.** Can a low cost rover (RTK or total station, wheel encoders, IMU) give a slip estimate that agrees across the definitions used in the literature?
5. **Learning.** Does a model trained on non sand terrain transfer to sand? Does a physical prior hybrid generalise to Arabian dunes with a handful of onboard runs?
6. **Legged and tracked.** What are the same vehicle differences on the same dune slope?
7. **Dynamics.** Crosswind, lateral slip on crests, speed and repeated passes.

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13. Validation and extension of 2026-10-20

Everything in this section was pre registered in `docs/PREDICTIONS.md` (stages 12 to 18) before its run; hits, misses and post hoc analyses are listed there. The narrative version is `manuscript/MANUSCRIPT.md`. Labels as above.

13.1 Evidence base (PUBLISHED)

The evidence table now has **510 rows from 176 distinct sources** (166 lab, 133 field, 113 none, 40 simulation only, 35 simulation against laboratory, 23 simulation against field). 323 rows were added on 2026-10-20 under a written protocol (`docs/SEARCH_PROTOCOL.md`: five topic agents, a live DOI or URL check for every source, licences recorded, a not retrieved list per topic) in seven groups: planetary and analog traverses (PLN), single wheel terramechanics (TM3), empirical mobility models such as the cone index family and NRMM (MOB), Arabian dune sand properties (SND), dune morphology, wind and DEM error (DUN), terrain mapping and planning (MAP), and heat and dust effects (ENV). Every new row was re checked against its full text by a fresh agent: **30 of 323 merged rows had a material error before correction (9.3 %, Wilson 95 % interval 6.6 to 12.9 %)**, most of them a wrong validation type; any discrepancy 25.4 %. One row was dropped as unverifiable. A blinded random sample of 20 rows for a human rater is in `research/spot_check_inter_rater.md`. Arabian items now include grain size, sorting, mineralogy and density of UAE, Qatari, Saudi and Omani dune sands and Gulf drift potentials; no friction angle or slip face angle measured on UAE dunes was found.

13.2 Published curves (PUBLISHED, digitised)

26 curves (235 points) from 7 sources were digitised reproducibly (`scripts/digitise_figures.py`, calibration files in `research/phase3/digitise/`), with uncertainty, conditions and a comparability judgement per curve

(`docs/DATA_CATALOG.md`). Nine published soil parameter sets are in `research/phase3/sand_parameters.csv`. The Wong table images supplied earlier remain unusable (no legible title and page) and were not used.

13.3 Model families (SIMULATED BY ME against PUBLISHED curves; stage 15)

Table 1. Model families on the published slip against slope curves (SIMULATED BY ME against PUBLISHED curves). Bar fixed before the run: median held out error at most 0.15 and coverage at least 0.80.

Family	Held out curves	Median error, nominal	Median error, own fit	Median error, held out	90 % interval coverage	Bar
Exponential (existing model)	17	0.31	0.06	0.24	0.89	fails
Bekker Wong rigid wheel	8	0.26	0.08	0.20	0.71	fails
Mohr Coulomb ratio	17	0.24	0.10	0.23	0.85	fails
Logistic, empirical	17	0.19	0.03	0.24	0.86	fails

No family meets the bar fixed before the run. Held out errors are about 0.2 to 0.3 slip for every family, and the empirical logistic curve transfers as well as the physics based ones. The posterior of the existing exponential model over the 12 curves judged comparable to dune sand is phi 36.93 (30.57 to 43.74) degrees, f_r 0.13 (0.03 to 0.32), i0 0.63 (0.33 to 1.21); used as map ranges (`posterior_2026_10_20`) it gives UNKNOWN at 30 m of 62.8 % (A), 44.3 % (B) and 46.0 % (C), or 25.1 %, 30.9 % and 4.2 % when the about 5 % of draws that cannot move on level ground are rejected (post hoc); the UNKNOWN share under this set depends strongly on that choice. The Bekker Wong family with the published Ishigami 2007 simulant set does not reproduce that paper's drawbar pull curve (`results/model_validation/bw_vs_ishigami2007.csv`). Question 8: without draws that cannot move on level ground, the widened resistance range gives UNKNOWN of 33.3 %, 40.6 % and 8.5 %, close to the widened i0 set, so the earlier 90 % shares came from immobile draws.

13.4 Independent simulation (SIMULATED BY ME; stage 14)

Project Chrono's Soil Contact Model with published wheel sizes and soil sets gives slip 0.21 at 12.5 degrees for a Scarecrow sized wheel against about 0.49 on the Dumont curve (root mean square difference 0.35), and differs from the repository's Bekker Wong family with the same inputs by 0.44 to 0.75. Both use the Bekker and Janosi laws, so this is model form and implementation uncertainty, not independent physics. The "Toyoura" label of soil set S1 in the stage 14 files is wrong: that set is the FJS-1 lunar simulant of Ishigami et al. 2007.

13.5 Real terrain test (SIMULATED BY ME on PUBLISHED lidar; stages 12 and 13)

Table 2. Slip face detector and three class map against one metre lidar truth (SIMULATED BY ME on PUBLISHED USGS 3DEP data; North American dune fields).

Site	Posting, m	AUC	True face rate	Mean predicted probability	TRAVERSABLE share	Share of TRAVERSABLE cells with a face	Share of UNKNOWN cells with a face	Miss rate (face cells called TRAVERSABLE)
Dumont	5	0.98	23.0 %	35.5 %	59.3 %	0.2 %	51.4 %	0 %
Dumont	10	0.97	32.0 %	29.9 %	62.0 %	3.6 %	77.1 %	7 %
Dumont	30	0.86	48.0 %	23.1 %	67.8 %	25.6 %	95.1 %	36 %
WhiteSands	5	0.98	5.0 %	10.2 %	87.3 %	0.1 %	38.4 %	3 %
WhiteSands	10	0.97	10.8 %	9.9 %	86.0 %	2.5 %	61.6 %	20 %
WhiteSands	30	0.66	35.7 %	14.6 %	75.9 %	28.6 %	58.0 %	61 %
Kelso	5	0.98	5.5 %	15.7 %	79.1 %	0.1 %	26.4 %	1 %
Kelso	10	0.95	10.7 %	13.0 %	80.8 %	1.8 %	48.4 %	13 %
Kelso	30	0.70	34.0 %	8.6 %	88.4 %	28.0 %	79.5 %	73 %

The detector and the map work much better at 5 and 10 m than at 30 m on three real dune fields (Dumont Dunes, White Sands, Kelso Dunes; USGS 3DEP one metre lidar): at 30 m about one TRAVERSABLE cell in four contains at least 10 square metres steeper than 28 degrees, 36 % to 73 % of the cells containing a face are called TRAVERSABLE (7 % to 20 % at 10 m), and the detector underestimates face prevalence by about 0.2. Slip face direction against ERA5 resultant drift direction: 4 of 6 sites within 45 degrees (`results/wind_azimuth/face_vs_wind.csv`).

13.6 Planning (SIMULATED BY ME; stage 17)

On 30 m maps no policy was reliably safer than naive least cost planning against lidar truth (Dumont: 70.0 % naive and 72.5 % conservative routes met a 3 m grade above the climbing limit). At 10 m, compared on matched pairs (only the pairs the conservative policy could route; post hoc after the midpoint review), the conservative policy failed less often than naive at White Sands (61.1 % against 66.7 % with the registered 3 m baseline; 13.9 % against 25.0 % with a 10 m baseline) and at Kelso (36.1 % against 63.9 %; 2.8 % against 11.1 %), and at Dumont it routed only 15 of 40 pairs and did not do better (26.7 % against 13.3 %). The Dumont window contains steep ground that is not sand, which affects all Dumont numbers.

13.7 Sensitivity (SIMULATED BY ME; stage 16)

Total Sobol indices of the passable share in area A at 30 m: original ranges i0 0.55, phi 0.45, i_max 0.17, f_r 0.04; widened i0 range i0 0.67, f_r 0.48, i_max 0.31, phi 0.26. First order indices sum to well below one: the parameters act through interactions.

13.8 Field rule options (synthetic data only; stage 18; the current rule stays in force)

Rule	Correct model accepted (sigma 0.03)	Collapse model accepted (sigma 0.03)	Dumont truth accepted (sigma 0.10)
Current (stage 10)	76 %	53 %	5 %
A, stricter tolerance	43 %	30 %	0 %
B, steep runs	39 %	18 %	5 %
C, calibration slope	44 %	20 %	2 %

Every option that catches the collapse model also rejects a correct model more than half the time; more runs in the steep bins would be needed. The choice is Huda's (question 7).

13.9 Wheeled, tracked and legged (PUBLISHED rows only)

Wheeled, tracked and legged locomotion on sand (PUBLISHED rows only; vehicles, soils and metrics differ, so the rows are not comparable with each other).

Row	Vehicle	Slope, deg	Metric	Value	Validation
MOB-043	Military wheeled vehicles (and tracked for comparison)	0 (level)	maximum drawbar pull on level sand as fraction of gross vehicle test weight (wh...	20 and 40; 50 and 60 percent (ranges between)	field
TM3-067	Four track platform rover with robotic arm, 120 kg, L153 x W154 x H11...	0 (level) and 18	steady state track slip on level ground and on an 18 deg sand slope	approximately 0.05 (level); approximately 0.3 (18 deg slope...	lab
TM3-071	Model wheeled vehicle with independently driven front and back wheels...	0 (level)	slip ratio of the wheeled vehicle without and with the intermediate crawler on...	1 (wheels only, did not move); 0.065 (with crawler) ratio	lab
FLD-HV13-1	Scarecrow (Mars weight MSL test rover, 340 kg, 50 cm diameter rigid g...	4 to 21 (fit validity r...	Cubic fit coefficients c0; c1; c2; c3 (slip vs slope)	26.197; -8.849; 1.281; -0.034 dimensionless polynomial coef...	field
FLD-025	Perseverance (Mars 2020)	about 20	slip	in excess of 94 %	field
LEG-007	Hexapod with C-shaped legs (350 g, 22 cm x 10 cm; leg radius 4 cm, wi...	0	step-averaged speed parallel to surface	15.7 cm/s	lab
LEG-008	Same 350 g hexapod	10	step-averaged speed parallel to surface	11.0 cm/s	lab
LEG-009	Same 350 g hexapod	24	step-averaged speed parallel to surface	5.6 cm/s	lab
LEG-013	SpaceBok quadruped (22 kg), planar and point feet, static and trot	25	maximum slope traversed (testbed limit)	25 deg	lab

The published rows suggest that tracks and wheel and crawler hybrids keep moving on sand where plain wheels stall (MOB-043, TM3-071), and that small legged robots climb sand slopes slowly (LEG-007 to LEG-009). None of these rows shares a vehicle size, sand or slip definition with another, so no ranking of locomotion types follows from them.

14. Completion work of 2026-10-21

Every test in this section was registered in `docs/PREDICTIONS.md` (stages 19 to 24) and pushed before it was run; analyses decided after a result are labelled post hoc. The plan and its success criteria are in `docs/process/PLAN_2026-10-21.md`, and every number in the manuscript, the brief and the proposals is traced in `docs/NUMBER_LEDGER.csv`.

14.1 Evidence and curves (PUBLISHED)

The evidence table now has 510 rows from 176 sources. The 16 new rows come from the granular scaling law literature (`research/phase4/scaling_laws_notes.md`); a fresh verifier found 0 material and 3 minor errors (location labels, corrected). 36 further slip against slope curves (164 points) from 7 new sources were digitised with calibration files (`research/phase4/`). A separate agent, without access to the first reading but with the same rasters and tools, digitised a random third (12 curves) again, which measures the repeatability of the reading, not an independent extraction: all 63 of the 63 original points were matched, the median of the per curve median slip differences was 0.0007 (largest 0.0041), and the second reading found 13 further points on partly hidden markers that the first had left out (`research/phase4/redigitise_agreement.csv`). The largest single point difference was 0.30 slip, on the curve hu2025goff fig4b trec singlewheel grc3, where overlapping markers of several curves in one panel were assigned differently by the two readings. Two curve notes carry caveats: the Caruso 2023 points pool several driving modes, and the load of the Hu 2025 single wheel curve is not derived.

14.2 Model families on the expanded curve set (SIMULATED BY ME against PUBLISHED curves; stages 21 and 22)

Rerun unchanged on the expanded set, no family passes the bar of 0.15: median held out error and pooled 90 % coverage are 0.22 and 0.84 (exponential), 0.20 and 0.86 (Mohr Coulomb), and 0.21 and 0.86 (logistic). Against stage 15 the median held out errors fell by 0.02 to 0.03, and the exponential family's posterior moved its friction angle to the upper bound of its prior (median 44.3 against a bound of 45 degrees; stage 15: 36.9), so the friction angle is not identified by these curves. On the 33 held out curves with wheel size, width and load, the Bekker Wong family did much worse than in the first round: median held out error 0.41 with coverage 0.46,

against 0.22 for the logistic family on the same curves. Holding out the two Scarab campaigns together (post hoc, because the Inotsume 2016 and Creager 2017 curves come from the same vehicle and tilt bed) gave 0.20 to 0.22; also grouping the Yoshida 2010, Nakashima 2010 and Sutoh 2012 Toyoura sand studies as one laboratory (post hoc, after the final review) gave 0.20 to 0.22. Two families built on the strength to load number of the granular scaling laws did worse than a curve with no covariates: 0.31 to 0.34 against 0.22 for the logistic on the same curves. A post hoc check found that in the one study that varies only wheel width, wider wheels slipped more on the lunar simulant, the opposite of what the number implies. Three caveats weaken this test of the scaling laws: the narrow Watanabe wheels may not meet the plane flow condition the laws assume, the density parser takes the lower end of a reported range, and two soils without a reported density share the 1600 kg/m3 default. The families use 33 held out curves (34 loaded curves have wheel radius, width and load; the Dumont cubic fit is a target only and is not held out), and the logistic comparison uses the stage 21 logistic, which was calibrated on more curves. The scaling laws stay what their authors call them: similarity laws for geometrically similar wheels in one material, not predictors across vehicles and sands.

14.3 Simulator calibrated on a field curve (SIMULATED BY ME against a PUBLISHED curve; stage 24)

Project Chrono's Soil Contact Model, calibrated on the Dumont Dunes curve up to 12.5 degrees over a grid of the shear modulus and the friction angle, reached a calibration error of 0.08 (best K 0.08 m, friction angle 32 degrees). Asked to predict 15 to 20 degrees, it gave 0.48, 0.44 and 0.24 against 0.67, 0.81 and 0.90 on the curve (error 0.45), with slip falling as the slope steepens. Post hoc grid and speed checks at those slopes (`results/chrono_scm_calibration/posthoc_checks.csv`) show the run is not numerically stable there: at 17.5 degrees a fourfold lower wheel speed gave 0.18 against 0.44, and at 20 degrees a finer soil grid gave 0.53 against 0.24 while the lower speed made the wheel slide backwards (slip 31.30). At the calibration slopes (5 to 12.5 degrees) the same checks changed the slip by at most 0.03, so the calibrated fit itself is numerically stable (post hoc). No usable prediction of the steep slopes was obtained; the stage 24 outcome measures a numerically unreliable run, not the physics of the Soil Contact Model. The particle (DEM) check is prepared as a notebook for a free GPU (`notebooks/chrono_dem_gpu.ipynb`) and has not been run.

14.4 More dune fields against lidar (SIMULATED BY ME on PUBLISHED USGS 3DEP lidar; stage 20)

On 6 North American dune fields (Dumont, White Sands, Kelso, Great Sand Dunes, Algodones, Little Sahara), the 30 m map labelled TRAVERSABLE 36 % to 80 % of the cells that contain a steep face (7 % to 33 % at 10 m), and

16.3 % to 28.6 % of its TRAVERSABLE cells contained one (1.2 % to 3.6 % at 10 m). Detector AUC was 0.97 to 0.99 at 5 m and 0.64 to 0.86 at 30 m.

14.5 Arabian dunes from satellite laser profiles (PUBLISHED ICESat-2 data scored by SIMULATED BY ME code; stage 19)

Method check against lidar (registered bar). On passes near the lidar dates, the ICESat-2 along track slopes of 10 m segments correlated with the one metre lidar slope on the same line at r = 0.81 at Kelso (997 segments from 3 beam passes) but only 0.48 at White Sands (26269 segments over a mostly flat tile, where small slopes are dominated by noise); the median absolute difference was 3.3 degrees at Kelso (bar 2.0). On lidar slopes of 20 degrees or more, ICESat-2 recovered a median 0.91 of the lidar slope at Kelso (99 segments). The method therefore **failed its own bar** at both sites, and every Arabian number below is labelled "method not validated". The Kelso check rests on 3 beam passes only. Post hoc checks did not rescue the bar (averaging the lidar over the footprint or shifting it along the track left the Kelso difference at about 3 degrees). They also show that the steepest ICESat-2 segments overstate the slope: the steepest Kelso segment was 45.9 degrees where the steepest lidar slope on the same lines was 30.6, and for the 45 Kelso segments at 28 degrees or more the steepest lidar slope within 10 m along the track had a median of 27.9 degrees, close to but below the 28 degree threshold. With 20 m segments the same bar is met at both sites (sensitivity, `results/icesat2/validation_metrics_20m_sensitivity.csv`), although its steep segment part rests on 1 lidar segment at White Sands and 49 at Kelso.

Arabian dunes (method not validated). Over Liwa (A), the Rub' al Khali megadunes (B) and the linear dunes (C), 394, 429 and 433 beam passes from 2018 to 2026 gave 499965, 819332 and 601669 valid segments, of which 14217, 24231 and 1271 had an along track slope of 28 degrees or more. Of these face crossings, the original 30 m map placed 71 % (A), 47 % (B) and 67 % (C) in TRAVERSABLE cells; with the posterior parameter ranges, 26 %, 17 % and 26 %. These shares must be read against the base rate: 93 %, 92 % and 98 % of all valid segments fall in TRAVERSABLE cells, so a face crossing is placed there 0.77, 0.51 and 0.69 times as often as a random segment (post hoc): the map does flag faces more than chance, but leaves many in TRAVERSABLE cells. On the face crossings, the ICESat-2 slope exceeded the Copernicus 30 m slope along the same track by a median of 27.8, 18.5 and 24.5 degrees, and in Liwa the 95th percentile of slope was 22.9 degrees from ICESat-2 against 12.5 from Copernicus. The steepest slope of each run of steep segments had a median of 26.7 degrees raw and 30.1 corrected for the angle between track and face (Liwa; 28.0 to 42.7 when the assumed face direction is moved by 30 degrees either way). The face directions used for the correction come from the 30 m surface (stage 13) and are weak references (resultant lengths 0.58, 0.25 and 0.87).

Quality of the face set (post hoc). Some segments are steeper than loose sand can stand (the steepest raw slope was 87.4 degrees in A), which points to noise segments; 33 % (A), 22 % (B) and 8 % (C) of the face crossings were steeper than 40 degrees, and 53 %, 58 % and 80 % were isolated (no other steep segment of the same pass within 6 m, although segments overlap by half). Keeping only crossings between 28 and 40 degrees, the original map placed 65 %, 46 % and 67 % of them in TRAVERSABLE cells; keeping only crossings confirmed by a neighbouring steep segment and no steeper than 35 degrees (2580, 5245 and 188 segments), 52 %, 29 % and 49 % (0.56, 0.31 and 0.51 times the base rate). At Kelso, the gap is mostly not a resolution effect: on the 45 ICESat-2 segments at 28 degrees or more (2 passes), the median slope was 32.8 degrees from ICESat-2, 24.0 from the one metre lidar on the same line and 20.5 from the lidar averaged to 30 m (`results/icesat2/posthoc_kelso_gap.csv`): selecting steep ICESat-2 segments selects segments whose slope is overstated, and the 30 m surface lowers the slope on the same line by only 2.4 degrees there. This rests on one site and 2 passes and is not a comparison with Copernicus; with 20 m segments the Liwa gap is still 26.9 degrees, so the cause of the Arabian gap is not established. The segments use only high confidence photons and at least 10 photons per 10 m, which can thin the data on steep faces, and Copernicus (radar of about 2011 to 2015) and ICESat-2 (2018 to 2026) are years apart, so small dunes on the megadunes may have moved.

20 m segments (registered sensitivity). With 20 m segments, which do pass the lidar bar at both validation sites, 4859, 9027 and 225 segments are at 28 degrees or more, and the original map places 65 %, 27 % and 51 % of them in TRAVERSABLE cells, against 93 %, 92 % and 98 % of all segments. Longer segments average out noise but also smooth short faces, so the 20 m shares are a different view of the same ground, not a corrected one.

Reading. With a method that failed its own validation and a steep segment set that is about half isolated segments, these numbers are an indication, not a validation. What survives the post hoc screens is that the 30 m map leaves a large share (about 29 % to 52 %) of the steep crossings that pass the neighbour screen in cells it calls TRAVERSABLE, against more than 90 % of all segments; this does not contradict the North American lidar tests but cannot confirm them. The size of the slope gap between ICESat-2 and Copernicus is not established as a measure of what a 30 m surface misses.

14.6 Closed loop demonstration (SIMULATED BY ME on PUBLISHED lidar; stage 23 and a post hoc variant)

A simulated rover drove the stage 17 routes over the one metre lidar surface, with its true slip given by the exponential model at the stage 15 posterior medians. Its online slip estimate, from wheel odometry against position fixes with 0.3 m noise, had a median error of 0.029 (White Sands) and 0.032

(Kelso); that value follows almost by arithmetic from the assumed noise and exact odometry, and with 2 % odometry noise per metre it stayed at 0.030 and 0.032. In the registered loop, stopping on a high estimate and planning again cut entrapment on 30 m naive routes from 55.0 % to 12.5 % (White Sands) and from 55.0 % to 30.0 % (Kelso), but the midpoint review showed this came mostly from restarting with fresh random slip. In a post hoc variant with the slip field fixed in space, the rover kept at its real position and a control that retries without blocking, one slip field gave on 30 m naive routes 55.0 % open loop, 50.0 % retry and 42.5 % replan at White Sands, and 57.5 %, 55.0 % and 50.0 % at Kelso. After the final review the comparison was repeated over 5 independent slip fields with paired differences and bootstrap intervals over pairs and fields (`demo/closed_loop_multiseed.py`, `results/closed_loop/posthoc_multiseed/`):

Site, posting	Open loop	Retry	Replan	Replan minus retry, points (95 % interval)
White Sands, 30 m	54.0 %	49.0 %	43.0 %	minus 6.0 (minus 20.5 to plus 8.0)
Kelso, 30 m	55.0 %	53.0 %	55.0 %	plus 2.0 (minus 12.5 to plus 16.0)
White Sands, 10 m	25.5 %	43.0 %	33.0 %	minus 10.0 (minus 24.5 to plus 4.5)
Kelso, 10 m	15.0 %	25.5 %	30.5 %	plus 5.0 (minus 9.5 to plus 20.0)

Entrapped share of the 40 naive pairs per site, pooled over the slip fields. Every interval of replan minus retry includes zero. Against open loop, replanning raised entrapment on 10 m maps at Kelso by plus 15.5 points (interval plus 5.5 to plus 26.5), but retry alone raised it from 15.0 % to 25.5 %, so most of that rise is the restart geometry described next, not the blocking of cells. The retry control is not neutral: after a stop, the route restarts with a straight line from the rover's real position to the next cell centre, so path geometry changes outcomes independently of slip feedback, which is why retry can do worse than open loop on fine maps. The same loop runs as ROS 2 nodes (`demo/ros2/`); one smoke test reached the same outcome as the Python loop (arrived after 1 replan), on a pair that also arrives without replanning, with the random number order aligned to the Python loop; it shows that the nodes work together, not that the loops behave the same. These numbers describe the loop under its assumptions, not a vehicle: under them, slip feedback with replanning shows no benefit that can be told apart from stopping and retrying, and it is no substitute for a fine map.

14.7 Field kit (PLANNED)

`field_kit/` holds the data sheets, and `scripts/ingest_field_day.py` turns a field day into scored predictions in one command (tested on a synthetic fixture only). `docs/OSF_PREREGISTRATION_DRAFT.md`, `docs/DRONE_SURVEY_PROTOCOL.md` and `docs/SAND_LAB_PROTOCOL.md` are ready for the author; `research/second_rater_pack/` prepares the 20 row check for a second human rater.

15. The Dune Rover Planner (product, 2026-10-23)

> **SIMULATED BY ME.** A software tool built on the methods of sections 8 and 13 to 14. It is not field evidence, not a safety case, and not validated on any vehicle. User guide: `docs/PLANNER.md`.

15.1 What it does

The planner turns the study into a tool a robotics team can run on its own terrain. Given an elevation model as a GeoTIFF (a Copernicus GLO-30 tile, a USGS lidar tile or a drone survey) and a vehicle profile, it computes for every cell the slope, the share of slip parameter draws for which the cell is passable, the probability of a slip face hidden below the map's resolution, and the three class map (TRAVERSABLE, BLOCKED, UNKNOWN), and between two points it plans a route with each of the three policies of stage 17 (naive, conservative, risk weighted; the two cautious ones with a descent limit added for the product). It writes a self contained HTML report, a map figure, the class map as a GeoTIFF, the routes as GeoJSON and a summary file (`dunerover plan`, `src/dunerover/planner.py`, `src/dunerover/report.py`, `src/dunerover/cli.py`).

Three features carry the findings of the study into the tool:

- **Resolution gate.** Every assessment states what the map may claim. Coarser than 10 m, TRAVERSABLE is reported as not verified, with the measured miss rate: on 6 lidar dune fields a 30 m map called 36 % to 80 % of the face cells TRAVERSABLE (7 % to 33 % at 10 m). An input at 2 m or finer resolves faces directly, with the truth rule of the lidar tests. The numbers come from an evidence card generated from the result files (`scripts/build_evidence_card.py`, checked in CI), so the tool and the documents cannot disagree.
- **Vehicle profiles.** The tool never pretends to know the vehicle. Built in profiles are the assumption ranges of the study and the posterior from published curves of other vehicles; `dunerover vehicle calibrate` fits f_r and i0 to the vehicle's own slope runs up to 10 degrees from the field kit, with bootstrap pairs drawn jointly and the error on the steeper runs reported; that is the only profile with MEASURED parameters (its planning slip limit i_max stays an assumption).
- **Honest warnings.** Each route lists the UNKNOWN cells and face risk cells it crosses, flags grades close to the vehicle's climbing limit and descents of 25 degrees or more, and avoids cells without data; the two cautious policies never descend a grade of 28 degrees or more. In the

registered test on three lidar dune fields (stage 17), no policy was reliably safer than naive on 30 m maps, so the policies are choices for the user, not a guarantee.

15.2 Demonstration on lidar dune fields (not a registered test)

On 4 North American dune fields (Kelso, White Sands, Algodones, Great Sand Dunes), the same lidar ground was assessed as a 10 m and a 30 m map and one route per field was planned with each policy (`scripts/build_web_demo.py`, `results/planner_demo/metrics.json`, interactive page `demo/web/dune_rover_planner.html`). The maps called 43 % to 73 % of the true face cells TRAVERSABLE at 30 m and 7 % to 20 % at 10 m, in line with stages 12 and 20. At 10 m, the naive routes crossed 45 cells holding a true slip face, the conservative routes 16 and the risk weighted routes 12; at 30 m, 105, 51 (on the 3 fields where a conservative route existed) and 83. With one route per field this illustrates the tool; it does not measure a policy effect, and cell counts at 10 m and 30 m are not comparable because the cells differ in size.

15.3 How it becomes a field tool

The planned project (`docs/PROPOSAL_ROBOTICS.md`) closes the two gaps the tool cannot close itself: a drone survey of the test dune at about 10 cm lets the planner work in its RESOLVED mode on Arabian terrain, and the slope ladder runs give the rover its own MEASURED profile. The planner is then run on the surveyed dune with the calibrated profile, and its routes are driven with a safety driver and compared with what the rover meets.

16. Baselines and the conference benchmark (2026-10-27, stages 25 and 26)

> **SIMULATED BY ME (every detector, baseline, map and route) scored against PUBLISHED elevation data** (USGS 3DEP one metre lidar; Copernicus GLO-30). Registered in `docs/PREDICTIONS.md` before any metric was computed; the baseline code was written in the minutes that followed. Paper: `paper/paper.pdf`; scripts: `scripts/benchmark_detection.py`, `scripts/benchmark_planning.py`; code: `src/dunerover/benchmark.py`, `src/dunerover/benchmark_cnn.py`.

16.1 Why

Sections 13 and 14 compared the detector and the planner only with variants of themselves, on block means of the lidar, and scored routes with a rule that flagged nearly every route. This section adds the methods a practitioner

would use instead, a real 30 m product, a hazard outcome on face patches of 10 square metres or more, the shipped planner itself, and intervals.

16.2 Design

Six lidar windows (stages 12 and 20). Inputs: 10 m and 30 m block means of the lidar and Copernicus GLO-30 interpolated on the 30 m grid. Methods: the planner's synthetic trained detector (DET), DET recalibrated leave one site out, coarse slope, terrain ruggedness index, plane roughness, the steepest slope of a cubic interpolation, gradient boosted trees on 14 terrain features and a small dilated convolutional network, the last two trained leave one site out on lidar labels. Detection metrics with 300 m block bootstrap intervals. Planning: 600 origin and destination pairs drawn from the lidar only; the planner's own naive, conservative and risk weighted policies with each hazard layer swapped in; a route is unsafe when it touches a face patch (at least 10 connected one metre posts of 28 degrees or more).

16.3 Results

- **Detection.** Mean AUC at 10 m: 0.95 (DET), 0.97 (cubic interpolation), 0.97 (gradient boosting), 0.97 (network). At 30 m: 0.74, 0.82, 0.81 and 0.79, with plain slope at 0.70. On GLO-30: 0.65, 0.70, 0.72 and 0.73. The detector is not the best method at any input; at 30 m the interpolation, which needs no training, is best or tied best.

- **Planning.** Mission success (a route exists and touches no face patch) on 30 m maps: 6 % with the slip model alone, 7 % with the conservative policy and DET, at most 11 % with any computable layer, 59 % with a perfect cell label. On 10 m maps: 15 %, 24 %, 34 % (best computable layer) and 46 %. A 5 m face map: 100 %.

- **Predictions.** Stage 25: 7 hit, 2 missed; stage 26: 5 hit, 3 missed (`docs/PREDICTIONS.md`).

- **Post hoc, after a fresh review (`docs/REVIEW_2026-10-28.md`).** Marking every cell that holds a face patch post (the rule of the 5 m reference), the conservative policy finds a route for 87 % of pairs at 10 m and 83 % at 30 m (none can touch a face, by construction), so most of the gap between the oracles of stage 26 came from the label rule; the coarse grid still costs some routable pairs, and with the risk weighted policy the same face map gives only 55 % at 30 m. The length of route on a face patch at 30 m falls from 50 m per km with the slip model alone to 19 with the interpolation weighting. With face patches of at least 500 square metres, 30 m mission success is 34 % with the slip model alone and 47 % with the best computable layer; the order among the better layers changes, but all stay well below the face map.

16.4 Reading

At 30 m the face information is not in the elevation model: no model, physical or learned, recovers enough of it to plan face free routes, though the better layers halve the exposure, and the planner's resolution gate (section 15) is the right response. At 10 m all reasonable methods see most faces and the hazard layers help. Given a face map made from fine data, a hard gate on a coarse grid still finds face free routes for most pairs, so what a rover needs before driving on dunes is mainly face information from a drone survey or lidar, or onboard sensing. The fields are North American; the registered outcome counts any contact with a face patch as unsafe, which is strict; the vehicle model is an assumption range.

Sources and appendices

Appendix A lists every evidence row. Appendix B lists the UAE facts used. Appendix C lists the sources, one line each, with the rows that cite them. All are generated from `research/evidence.csv` and `research/raw/uae_facts.csv` by `scripts/build_report.py`.

Appendix A. Evidence table (all PUBLISHED; condensed, full fields in `research/evidence.csv`)

510 rows. Validation types (code in the CSV, then count): `field` 133, `lab` 166, `none` 113, `sim-vs-field` 23, `sim-vs-lab` 35, `simulation-only` 40. Values are as printed in the source; read the comparability warning before comparing any two rows. Verification status of each row is in the CSV (column `verification`).

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM-001	Wheel C: aluminium cylinder coated with simulant, radius 130 mm, aspe...	Quikrete 1962 Medium Sand (MS), silica, 0.3-0.8 mm, loosely packed be...	0 (level)	RFT-granular	Drawbar mean absolute error (RFT vs experiment)	5.55 N	lab	link	Table 5 (MS, 80 N, RFT drawbar row, Delta); text...	2026-10-11	Forced-sli wheel rig bed; slip s V/(omega error = me error over slip values about -1.. exper...
TM-002	Wheel C: aluminium cylinder coated with simulant, radius 130 mm, aspe...	Quikrete 1962 Medium Sand (MS), dry, loosely packed	0 (level)	RFT-granular	Drawbar mean absolute error (RFT vs experiment)	7.11 N	lab	link	Table 5 (MS, 190 N, RFT drawbar row, Delta)	2026-10-11	Forced-sli wheel rig bed; slip s V/(omega error = me error over slip values about -1.. exper...
TM-003	Wheel C: aluminium cylinder coated with simulant, radius 130 mm, aspe...	Mars Mojave Simulant (MMS), crushed basalt, fines present, dry	0 (level)	RFT-granular	Drawbar mean absolute error (RFT vs experiment)	15.11 N	lab	link	Table 5 (MMS, 190 N, RFT drawbar row, Delta)	2026-10-11	Forced-sli wheel rig bed; slip s V/(omega error = me error over slip values about -1.. exper...
TM-004	Wheel C: aluminium cylinder coated with simulant, radius 130 mm, aspe...	Mars Mojave Simulant (MMS), dry	0 (level)	RFT-granular	Sinkage mean absolute error (RFT vs experiment)	18.41 mm	lab	link	Table 5 (MMS, 190 N, RFT sinkage row, Delta); Sec...	2026-10-11	Forced-sli wheel rig bed; slip s V/(omega error = me error over slip values about -1.. exper...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM-005	Wheel C smooth cylinder, radius 130 mm, vertical load 120 N	Poppy seeds (about 1 mm, NOT sand), dense (phi about 0.605) and loose...	0 (level)	classical-terrame...	Drawbar mean absolute error (TM, compacted)	3.41 N	lab	link	Table 3 (compacted, TM drawbar, Delta); Sec. 9.1...	2026-10-11	Poppy see sand; TM came from (2.5 × 3.8 smaller tha wheel pat shear moc tuned; RFT error for...
TM-006	Wheel C smooth cylinder, radius 130 mm, vertical load 120 N	Poppy seeds (about 1 mm, NOT sand), dense state	0 (level)	classical-terrame...	Sinkage mean absolute error (TM, compacted)	26.07 mm	lab	link	Table 3 (compacted, TM sinkage, Delta)	2026-10-11	Poppy see sand; see parameter previous r
TM-007	Pairs of cylindrical and four-arm lug wheels of different sizes, mass...	Quikrete medium sand bed, dry; sandpaper-covered cylindrical wheels	0 (level)	RFT-granular	Deviation of best-fit slope of scaled big-vs-small wheel da...	3 % (max, over six datasets)	lab	link	Experiments section, text after Fig. 4 (lug-wheel...	2026-10-11	Tests scali invariance wheel pair absolute p error of slip/sinkag level terrai some devi at highest
TM-008	12-DOF hexapod, 4.04 kg, pointed dactyl feet (legged, free-walking)	Quikrete play sand in 15 cm deep sandbox, dry, settled for weeks	0 (level)	RFT-granular	5-step walking position error, no payload	6.84 %	sim-vs-lab	link	Sec. III-D (Speed and Sinkage of Walking Hexapod...	2026-10-11	Legged ro wheeled; l sandbox; s tuned/mat box; sim te underestir distance p 3.4-28%
TM-009	12-DOF hexapod, 4.04 kg plus payloads 0.5-2.277 kg	Quikrete play sand sandbox, dry	0 (level)	RFT-granular	Dactyl foot sinkage overestimate, mean across three payloads	2.6 %	sim-vs-lab	link	Sec. III-D, near Fig. 7E	2026-10-11	Experimer in Fig. 7E after a sing payload; f not wheel
TM-010	12-DOF hexapod, 4.04 kg	Sandbox (Quikrete), Beach 1 (near shore), Beach 2 (farther from shore...	0 (level; beaches n...	RFT-granular	Average percent error of speed and body height (Beach 1 / B...	10.5 (Beach 1); 11.5 (Beach 2) %	sim-vs-field	link	Sec. III-E (Comparison with other Contact Models)...	2026-10-11	Beach gra properties characteri zeta; the s sandbox z although b was about averaged speed+he legged...
TM-011	Bio-inspired L-leg and reversed L-leg rotated about hip (vertical-pla...	Poppy seeds (about 1 mm, NOT sand), loosely packed phi=0.58 via fluid...	0	RFT-granular	Vertical/horizontal force Fz,Fx vs leg angle, model vs meas...	within 15 %	lab	link	Sec. 3 text near Fig. 3c,d	2026-10-11	Poppy see sand; leg r loose state

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM-012	Small hexapedal robot (15 cm, 150 g) with L-legs	Poppy seeds (about 1 mm, NOT sand), loose	0	RFT-granular	Forward speed vs stride frequency	within 15 (L-legs); within 30 (reversed L-legs up to 3 Hz) %	sim-vs-lab	link	Sec. 4 text near Fig. 4b; leg-shape sweep within...	2026-10-11	Poppy see sand; tiny robot; spe
TM-013	Flat plate intruder on UR-10 arm (plate-level test)	0.8 mm glass beads (NOT sand), 18 cm deep bed	0	RFT-granular	Median relative force error /F1/ with beta-dependent alpha_...	1.2-6.5 %	lab	link	Sec. on plate validation text near Fig. 5 and Fig...	2026-10-11	Glass bea wheel; wit alpha_y th range is 1.
TM-014	Elliptical robotic foot driven by 2-link leg (human-gait replay)	Fine sand 0.2-0.5 mm (51×38×25 cm box with 20 cm depth was for plate...	0	RFT-granular	RMSE of vertical force Fz, slow gait (with correction vs wi...	21.4 +/- 19.5 (with); 25.8 +/- 18.9 (without) units not sta...	lab	link	Table I and text of Sec. IV-B (Robotic Foot-Terra...	2026-10-11	Foot intrus wheel; uni not printe Fz error SI equals the
TM-015	Smooth rigid wheel r=100 mm, b=75 mm, loads 14.9 / 47.7 / 63.9 N (tra...	Dry sand (donated by Shaw Brick) in 29 cm deep box, tests at slip 0.2...	0 (level)	classical-terrame...	Average error of mean of vertical force over all tests (as...	1.96 (SD 0.9) N	lab	link	Sec. 4 (Experimental Results), paragraph before F...	2026-10-11	Model cali the 14.9 N eta tuned eye; the pa focuses or amplitude, (ripples), r slip/drawb
TM-016	Grousered wheel (16 grousers, 23 mm), r=0.1 m, b=0.075 m, 15 N normal...	Dry sand (translated-terrain testbed)	0 (level)	classical-terrame...	RMS of mean-adjusted drawbar-pull oscillation: model 1.539...	0.536 units not stated in table	lab	link	Table 2 (Case Study #3, drawbar pull, slip 0.501...	2026-10-11	Metric is R oscillation mean (me adjusted), in mean dr units not p tuned on t case
TM-017	Grousered wheel (32 grousers, each 10 mm long), same rig, 15 N	Dry loose sandy soil (translated-terrain testbed)	0 (level)	classical-terrame...	RMS drawbar oscillation: model 1.708 vs experiment 0.285 at...	1.423 units not stated in table	lab	link	Table 4 (Case Study #5, drawbar pull, slip 0.75 r...	2026-10-11	Same cave previous r out-of-sa breakdow about 6x t measured
TM-018	Rigid toothed-grouser metallic wheel, 250 mm dia, 112 mm wide, 12 gro...	RMCS14 olivine sand (quartz substitute), dry; DLR TROLL single-wheel...	0 (level)	other	Loss of drawbar pull coefficient at 20% slip, speed 0.01 to...	8 %	lab	link	Sec. 3.1 text near Fig. 5a	2026-10-11	Empirical measurem model-vs experimer included b quasi-stat terramech assume sp independe method_fa 'other...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM-019	Rigid toothed-grouser metallic wheel, 250 mm dia, 112 mm wide, 12 gro...	RMCS14 olivine sand, dry	0 (level)	other	Loss of drawbar pull coefficient at 20% slip, speed 0.2 to...	63.7 %	lab	link	Sec. 3.1 text near Fig. 5a	2026-10-11	As previou gravity is chosen to roughly 3 wheel rove lunar grav
TM-020	Rigid grouser wheel D=150 mm, B=80 mm, 24 grousers of 8.5 mm, mass 10...	DEM model of FJS-1 lunar simulant (moderate cohesion), dry; coarse-gr...	-10 to +10 (slip ab...	DEM	RMSE of predicted vs direct lunar-g DEM slip ratio at theta...	0.015 slip ratio (dimensionless)	simulation-only	link	Sec. 4 text near Fig. 4a	2026-10-11	SIMULATI (DEM prec no physic target is th law, not DI to a real sc
TM-021	Two grousered wheels, 10 kg wheel load, single-wheel testbed (wheel d...	Toyoura sand (source does not call it dry); bed 3500 × 600 x 1200 mm	0 in tests (slope h...	learning	Slip-ratio estimate 'accuracy', average (undefined in the p...	85 % (definition of accuracy not given in text read)	lab	link	Abstract and Sec. 4 text before Table 2; Table 1...	2026-10-11	In-sample of fit, NOT prediction (verifier ar not stated authors); t slip-rate e
TM-022	Moonraker PFM2 rover about 7 kg, wheel radius 9 cm, width 8 cm, 15 gr...	Toyoura sand (spelled Toyura), 2.0 x 1.0 x 0.08 m tiltable sandbox; d...	10	classical-terrame...	Simulation error of the base model at a 10 degree slope (th...	14.76 %	sim-vs-lab	link	Sec. 4.1.2, Table 4 (thesis p. 62)	2026-10-11	Thesis, no reviewed; parameter from anotl measured sandbox; ratio of sic difference distan...
TM-023	Moonraker PFM2 rover about 7 kg, wheel radius 9 cm, width 8 cm, 15 gr...	Toyoura sand, tiltable sandbox; dry not stated	20	classical-terrame...	Simulation error in side slip over travelled distance, base...	25.91 %	sim-vs-lab	link	Sec. 4.1.2, Table 4 (thesis p. 62)	2026-10-11	Thesis, no reviewed; parameter from anotl measured sandbox; ratio of sic difference distan...
TM-024	Moonraker PFM2 rover about 7 kg, wheel radius 9 cm, width 8 cm, 15 gr...	Toyoura sand, dry, tiltable sandbox	20	classical-terrame...	Simulation error in side slip over travelled distance, afte...	0.49 %	sim-vs-lab	link	Sec. 4.1.2, Table 4 (thesis p. 62)	2026-10-11	Thesis, no reviewed; parameter from anotl measured sandbox; ratio of sic difference distan...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM-029	Rectangular plates 25×75 to 70×140 mm (plate sinkage, not a wheel)	Loose dry Tottori sand, 350 mm deep bin, bulk density about 1.5 g/cm3	0	classical-terrame...	Goodness of fit (Wong's definition) of predicted pressure-s...	95 % goodness of fit	lab	link	Results section text on Fig. 4b; Table 2	2026-10-11	Plate sinka small plate bin with bo beyond 60 goodness not a relati
TM-IS07-1	Single rigid paddled wheel, single wheel test bed	Lunar regolith simulant equivalent to FJS-1, 12 cm deep in test bed v...	not reported	terramechanics	wheel diameter	0.18 m	lab	link	Sec 3.1 p239 to p240 (wheel 0.18 m diameter, 0.11...	2026-10-13	Rigid whee regolith sir equivalent only; singl simulant, f slope, Eart parameter for this be
TM-IS07-2	Four wheeled rover test bed with rocker suspension, same wheels as si...	Lunar regolith simulant, 10 cm deep in flat 1.5 m by 2.0 m rectangula...	not reported	terramechanics	rover dimensions (length x width x height)	0.68 × 0.44 × 0.32 m	lab	link	Sec 4.1 p241 (dimensions and approx 35 kg total);...	2026-10-13	Rigid whee regolith sir equivalent only; singl simulant, f slope, Eart parameter for this be
TM-IS07-3	Single rigid paddled wheel, single wheel test bed	Lunar regolith simulant equivalent to FJS-1, 12 cm deep; slip ratio 0...	not reported	terramechanics	model versus experiment agreement for drawbar pull and side...	Differences relatively small for slip angles 5 to 20 deg; r...	sim-vs-lab	link	Sec 3.3 p241 with Figures 11 (p242) and 12 (p243)...	2026-10-13	Rigid whee regolith sir equivalent only; singl simulant, f slope, Eart parameter for this be
TM-IS07-4	Four wheeled rover test bed, front wheels steered 15 deg (case A) or...	Lunar regolith simulant, 10 cm deep, flat vessel 1.5 m by 2.0 m	not reported	terramechanics	position rms error, proposed model (range over 4 runs, case...	0.015 to 0.039 m	sim-vs-lab	link	Table V p249; Sec 5.2 p246; conditions in Sec 5.1...	2026-10-13	Rigid whee regolith sir equivalent only; singl simulant, f slope, Eart parameter for this be
TM-IS07-5	Four wheeled rover test bed during steering experiments	Lunar regolith simulant, 10 cm deep, flat vessel; wheel angular veloc...	not reported	terramechanics	observed slip ratio range and slip angle range during steer...	slip ratio 0.1 to 0.3; slip angle -7 to 15 slip ratio dimen...	lab	link	Sec 5.2 p246 to p247 (last paragraph before Sec 6...	2026-10-13	Rigid whee regolith sir equivalent only; singl simulant, f slope, Eart parameter for this be

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM-IS09-1	Four wheeled 4WD/4WS rover test bed, wheelbase 0.44 m, tread 0.30 m,...	Tiltable vessel 2.0 x 1.0 m filled to 8.0 cm depth with Toyoura Sand,...	5;10	terramechanics	rover mass	13.5 kg (about)	lab	link	Section 4.1 Experimental Setup, p. 272 (cases A 5...	2026-10-13	Rig descri Rigid lugg small 13.5 a preparec desired sp m/s (Secti a field resu
TM-IS09-2	Front left wheel, wheel load 29.8 N, radius 0.055 m, width 0.064 m	Toyoura Sand assumed in simulation	5	terramechanics	slip ratio needed for positive thrust at required cornering...	0.6 dimensionless (slip ratio)	simulation-only	link	Section 3.2.2, p. 270 to 271, Figures 4 and 5 (p....	2026-10-13	Pure mode with ideali Toyoura S parameter Figure 5 is wheel exa Fc assume not a rove result.
TM-IS09-3	Four wheeled 4WD/4WS rover test bed, wheelbase 0.44 m, tread 0.30 m,...	Toyoura Sand, 8.0 cm deep tiltable bed	10	terramechanics	measured wheel slip angle, case B (derived steering angle 1...	23 deg (around)	sim-vs-lab	link	Section 4.2 Table III, p. 274; Section 4.3 text,...	2026-10-13	Values are approxima ('around') not tabulat measurem soil param admitted b to mismat (p. 275). Comparisc
TM-IS09-4	Four wheeled 4WD/4WS rover test bed, wheelbase 0.44 m, tread 0.30 m,...	Toyoura Sand, 8.0 cm deep tiltable bed, desired speed 0.02 m/s	5;10	terramechanics	final state lateral distance error, average of four runs (r...	0.079 m (7.50 percent of travel distance)	lab	link	Table IV, p. 275 (average row); setup Section 4.1...	2026-10-13	Only four r per slope) typical by paper say number of done. Rep measure c performan
TM-IS09-5	Four wheeled 4WD/4WS rover test bed, wheelbase 0.44 m, tread 0.30 m,...	Toyoura Sand, 8.0 cm deep tiltable bed, desired speed 0.02 m/s	5;10	terramechanics	wheel slip ratio managed under feedback control	0.4 dimensionless (stated as less than 0.4)	lab	link	Section 6.3, p. 281 (slip ratio less than 0.4, wh...	2026-10-13	Value is ar bound stat prose, not measurem Feedback soil indepe soil param not used b controller.
TM-TK1	Slick (non-grouser) wheel, 180 mm diameter, PLA, 50 N vertical load	Tohoku Silica Sand No. 5, prepared bed, tilled and levelled, sand hei...	not reported	terramechanics	traction coefficient (drawbar pull / vertical load) at slip...	approximately 0.02 dimensionless	lab	link	Section 7.3, p. 7 (text describing Fig. 7a); Eq....	2026-10-13	Lunar simu only; sand silica sand bed. Value approxima quoted in t a table ent

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM-TK2	Slick (non-grouser) wheel, 180 mm diameter, PLA, 50 N vertical load	Tohoku Silica Sand No. 5, prepared bed, tilled and levelled, sand hei...	not reported	terramechanics	traction efficiency (Fx*vx* / *T*omega) versus translational...	approximately 0.03 at 1.0 m/s (slip above 0.25); approximat...	lab	link	Section 7.3, p. 7 (0.03) and Section 7.4, p. 8 (0...	2026-10-13	Approxima read from exact slip 0.5 and 0. values is n the senten lunar rego
TM-TK3	High grouser wheel (diameter 180 mm, grouser height 15 mm, 17 grouser...	Testbed and wheel geometry; Tohoku Silica Sand No. 5 in 6.85 m x 0.5...	not reported	terramechanics	wheel diameter (grouser height and grouser count in notes)	180 mm	lab	link	Sections 3.3, 3.4 and 3.5, p. 4; Table 1, p. 2 (t...	2026-10-13	Small PLA N is one q 24 kg rove scaled to a rover whe
TM-TK4	DEM simulated slick and high grouser wheels, 50 N vertical load	EDEM DEM bed with 2 mm diameter particles (Toyoura sand and aluminum...	not reported	DEM	simulated slick wheel traction coefficient across slip rati...	negative for all slip ratios at both speeds dimensionless	sim-vs-lab	link	Section 6.1, p. 5 (text describing Fig. 5a; Fig....	2026-10-13	Simulation significant predicts ex experimer positive cc near 0.02 and 1.0 m simulation Calib...
TM-TK5	DEM calibration of Toyoura sand against aluminum, applied to PLA whee...	Angle of repose calibration; silica sand assumed to behave like Toyou...	not reported	DEM	angle of repose used to calibrate DEM parameters	34 degrees	lab	link	Section 5.2, pp. 4 to 5 (sentence beginning on p....	2026-10-13	Parameter transferre different s wheel mat authors st recalibrati needed. A repose ma not validat
TM-YU1	Smooth rigid rover wheel: inner rim radius 62.5 mm, width 60.0 mm, gr...	Soil Direct #90 sand (moisture state not reported), the same soil use...	not reported	RFT	Mean absolute error of x force, Delta_x (smooth wheel)	2.8772 Table 4 does not state a unit for the mean absolute...	sim-vs-lab	link	Table 4, p. 7 (Eq. 10, Sec. 4.2, p. 5)	2026-10-13	Single san small rigid a gantry a forward ve lab bed on dune slope value is a = 45 slip r
TM-YU2	Grousered rigid rover wheel: inner rim radius 57.5 mm, width 60.0 mm,...	Soil Direct #90 sand (moisture state not reported), the same soil use...	not reported	RFT	Mean absolute error of x force, Delta_x (grousered wheel)	6.2461 Table 4 does not state a unit for the mean absolute...	sim-vs-lab	link	Table 4, p. 7	2026-10-13	Single san small rigid a gantry a forward ve lab bed on dune slope value is a = 45 slip r

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM-YU3	Smooth rigid rover wheel: inner rim radius 62.5 mm, width 60.0 mm, gr...	Soil Direct #90 sand (moisture state not reported), the same soil use...	not reported	RFT	Mean absolute error of vertical force, Delta_z (smooth whee...	6.4720 Table 4 does not state a unit for the mean absolute...	sim-vs-lab	link	Table 4, p. 7; text on vertical force in Sec. 4.2...	2026-10-13	Single san small rigid a gantry a forward ve lab bed on dune slop value is a = 45 slip r
TM-YU4	Grousered rigid rover wheel: inner rim radius 57.5 mm, width 60.0 mm,...	Soil Direct #90 sand (moisture state not reported), the same soil use...	not reported	RFT	Mean absolute error of lateral force, Delta_y (grousered wh...	2.0362 Table 4 does not state a unit for the mean absolute...	sim-vs-lab	link	Table 4, p. 7	2026-10-13	Single san small rigid a gantry a forward ve lab bed on dune slop value is a = 45 slip r
TM-YU5	Rover wheel towed at fixed sinkage (wheel dimensions for these trench...	Soil Direct #90 sand prepared with an automated rake and smoothing me...	not reported	RFT	Average error of predicted 2D trench profile versus LIDAR s...	2.2 mm	sim-vs-lab	link	Table 3, p. 6; Sec. 4.1, p. 5	2026-10-13	Soil surfac metric, no force accu metric. Se sinkages a cm while T Fig. 6 list h and 25 mr
TM3-001	not applicable (soil property of test bed)	GRC-1 (sub-angular manufactured silica sand mixture) in NASA GRC SLOP...	not reported	other	bulk density of loose GRC-1 used for slope climbing tests	1.63 g/cc	none	link	Test Setup, Table 1 (report p. 3; PDF p. 7)	2026-10-20	Manufactu strength si natural du values are from pene correlation density me very loose
TM3-002	not applicable (soil property of test bed)	GRC-1 (sub-angular manufactured silica sand mixture) in NASA GRC SLOP...	not reported	other	friction angle of loose GRC-1	31.5 deg	none	link	Test Setup, Table 1 (report p. 3; PDF p. 7)	2026-10-20	Estimated measured tested bec a sub-ang sand mixtu sands are rounder ar sorted.
TM3-003	Scarab rover, 4×4, 400 kg, fixed suspension (active suspension not us...	GRC-1 (sub-angular manufactured silica sand mixture) in NASA GRC SLOP...	0 to 15	empirical	slope angle above which the vehicle made no uphill progress...	9 deg	lab	link	Results and Discussion, paragraph after Figure 9...	2026-10-20	GRC-1 is manufactu strength si made mair sand, prep loose (rela 11%); Ear large 71 c wheels...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM3-004	Scarab rover, 4×4, 400 kg, fixed suspension (active suspension not us...	GRC-1 (sub-angular manufactured silica sand mixture) in NASA GRC SLOP...	15	empirical	uphill progress on a 15 deg slope at angles of attack 10 an...	no positive uphill progress qualitative	lab	link	Results and Discussion, paragraph after Figure 9...	2026-10-20	GRC-1 is manufactu strength si made mair sand, prep loose (rela 11%); Ear large 71 c wheels...
TM3-005	Scarab rover, 4×4, 400 kg, fixed suspension (active suspension not us...	GRC-1 (sub-angular manufactured silica sand mixture) in NASA GRC SLOP...	0 to 15	empirical	difference between front and rear wheel slip	less than 0.5 percent	lab	link	Results and Discussion, paragraph on Figure 11 (r...	2026-10-20	GRC-1 is manufactu strength si made mair sand, prep loose (rela 11%); Ear large 71 c wheels...
TM3-006	Scarab rover, 4×4, 400 kg, fixed suspension (active suspension not us...	Non slip surface (calibration run)	0 (level)	empirical	effective rolling radius of the 71 cm rigid wheel	35.6 cm	none	link	Metrics Used for Evaluation, after Equation (4) (...	2026-10-20	Calibratior slip definit shows tha definitions source use measured radius, wh comparisc valu...
TM3-007	Scarab rover, 4×4, 400 kg, fixed suspension (active suspension not us...	GRC-1 (sub-angular manufactured silica sand mixture) in NASA GRC SLOP...	0 to 15	empirical	angle of attack giving most efficient and most likely succe...	90 (direct uphill) deg	lab	link	Conclusions 1 and 2 (report p. 11); Figure 12 dis...	2026-10-20	GRC-1 is manufactu strength si made mair sand, prep loose (rela 11%); Ear large 71 c wheels...
TM3-008	Scarab rover (4×4, wheelbase variable for inching); tyres 28 in diame...	GRC-1 lunar terrain strength simulant (primarily silica sand) in soil...	0 (level)	empirical	maximum drawbar pull coefficient DP/W, pneumatic tyres, inc...	approximately 0.37 (inching) versus 0.28 (rolling) DP/W	lab	link	Experimental comparison, Results, paragraph befor...	2026-10-20	GRC-1 sili simulant, l 4×4 vehic needs an a chassis; le drawbar p indirectly r slope clim
TM3-009	Scarab rover (4×4, wheelbase variable for inching); tyres 28 in diame...	GRC-1 lunar terrain strength simulant (primarily silica sand) in soil...	0 (level)	empirical	increase in maximum drawbar pull, rigid tyres, inching rela...	about 30 percent	lab	link	Results, paragraph after Figure 5 (PDF p. 6)	2026-10-20	Same as T rigid tyres sandpaper added for

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM3-010	Scarab rover (4×4, wheelbase variable for inching); tyres 28 in diame...	GRC-1 lunar terrain strength simulant (primarily silica sand) in soil...	0 (level)	empirical	drawbar pull coefficient below which rolling gave lower tra...	0.25 DP/W	lab	link	Results, paragraph before Figure 5 ('much lower T...	2026-10-20	Same as T Compliant on a heavy small rove cannot rea low groun in the sam
TM3-011	Scarab rover, 400 kg, tyres 71 cm diameter x 18 cm wide, rigid or rub...	GRC-1 lunar terrain strength simulant (primarily silica sand) in soil...	0 (level)	empirical	maximum drawbar pull as share of vehicle weight, rigid tyre...	approximately 33 percent (inching) versus 25 percent (rolli...	lab	link	Previous Research Using This Technique, Drawbar P...	2026-10-20	Restateme 2012 tests 008, TM3 the same GRC-1 sili simulant; h vehicle.
TM3-012	Towed cart with four rigid wheels on bearings, wheel size roughly as...	GRC-1 soil bin (preparation as for drawbar pull tests)	0 (level)	empirical	total rolling resistance of free rolling rigid wheels	about 5 percent of vehicle weight	lab	link	Drawbar Pull Testing paragraph (report p. 3; PDF...	2026-10-20	Light loads of Scarab) say heavie would rais percentag level grou GRC-1 no sand.
TM3-013	Scarab rover, 400 kg, tyres 71 cm diameter x 18 cm wide, rigid or rub...	NASA GRC Sink Tank 12 m x 3 m x 0.5 m filled with Fillite (hollow alu...	0 (level)	empirical	average wheel sinkage after 80 s of rolling	17.6 (front); 24.6 (rear) cm	lab	link	Description of Terrain, repeatability paragraph (...	2026-10-20	Fillite is hc microsphe specific gr chosen as high sinka it is NOT s far weaker sand; use
TM3-014	Scarab rover, 400 kg, tyres 71 cm diameter x 18 cm wide, rigid or rub...	NASA GRC Sink Tank 12 m x 3 m x 0.5 m filled with Fillite (hollow alu...	0 (level)	empirical	average forward velocity while inching out of the entrapped...	about 0.010 m/s	lab	link	Entrapped Starting Condition, paragraph before Fi...	2026-10-20	Fillite is nc TM3-013 requires a single den run shown
TM3-015	not applicable (terrain preparation standard)	GRC-1 in NASA GRC SLOPE lab bins; T1 loosen, level, tamp (4.5 kg tamp...	not reported	other	bulk density of the three standard GRC-1 terrain conditions...	~1.71 (T1); ~1.66 (T2); ~1.63 (T3) g/cm3	none	link	Section 2.5, Table IV (report p. 15; PDF p. 21)	2026-10-20	Lunar stre simulant b silica sand interpolate measured conditions match Apo penetrome not dune s

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM3-016	Scarab rover with 71 cm diameter treadless rubber pneumatic tyres inf...	GRC-1 prepared to seven densities, loosest by loosening and levelling...	0 (level)	empirical	rate of change of travel reduction with cone index gradient...	on the order of -5 percent per (kPa/mm)	lab	link	Section 3.6, discussion of Figures 31 and 32 (rep...	2026-10-20	30 psi tyre to make sl sinkage hi simulant; l ground; th insensitive covers the states (G 1 kPa/mm).
TM3-017	Scarab rover with 71 cm diameter treadless rubber pneumatic tyres inf...	GRC-1 T3 (loose) condition, five repeat runs	0 (level)	empirical	maximum standard deviation of travel reduction across repea...	1.94 percent (TR)	lab	link	Section 3.7, text with Figures 33 and 34 (report...	2026-10-20	Repeatabi vehicle an one prepa simulant; s on how pr model erro judged ag tests.
TM3-018	Scarab rover with 71 cm diameter treadless rubber pneumatic tyres inf...	GRC-1 T3 (loose) condition	0 (level)	empirical	drive distance per DP step sufficient for steady state	0.8 m	lab	link	Section 3.2.5 (report p. 24) and Section 3.8 reco...	2026-10-20	Depends o size and w (0.71 m ty wheelbase say resear should be conservati repeat the other sizes
TM3-019	not applicable (terrain preparation repeatability)	GRC-1 prepared to T2 condition, 59 cone penetrometer readings over se...	not reported	other	median cone index gradient and standard deviation of T2 pre...	3.5 (SD 0.2807) kPa/mm	none	link	Section 2.5 (report p. 16; PDF p. 22)	2026-10-20	Preparatio repeatabili simulant b dune surfa vary far m
TM3-020	MGRU3 VIPER scarecrow rover digital twin, 73 kg, Earth gravity	GRC-3 lunar simulant parameter set in CRM (SPH continuum) simulation,...	30	other	steady state slip climbing a 30 deg slope on Earth	approximately 42 percent	sim-vs-lab	link	Section 4 discussion of Figure 22 (preprint p. 20)	2026-10-20	Simulation parameter lunar simu dune sand (gravity of authors ar Earth resu optimistic Moon.
TM3-021	MGRU3 VIPER scarecrow rover digital twin, 73 kg, Earth gravity	GRC-3 lunar simulant parameter set in CRM simulation, slope mode	10	other	steady state slip climbing a 10 deg slope on Earth	18 percent	sim-vs-lab	link	Section 4 discussion of Figure 22 (preprint p. 20)	2026-10-20	As TM3-0 GRC-3 pa set.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM3-022	88 kg MGRU3 digital twin, wheel angular velocity 0.8 rad/s (Earth) an...	GRC-1 parameter set in CRM simulation, slope mode	15	other	average rover velocity on a 15 deg slope, Earth versus Moon...	approximately 0.125 (Earth); 0.05 (Moon) m/s	simulation-only	link	Section 3.2 text after Figure 9 (preprint p. 13)	2026-10-20	Simulation scaling lav a measure gravity ca: relevant tc dunes exc check of g scaling.
TM3-023	LROP four wheeled test platform, dry mass 20.4 kg plus 11.3 kg payloa...	GRC-1 in NASA GRC SLOPE adjustable tilt bed, prepared to the loose co...	20	empirical	slope ascent outcome at 20 deg	Fail at all three centre of gravity positions qualitative	lab	link	Section 3.2 Slope Ascent, Table 3 (PDF p. 9)	2026-10-20	Smooth w without gr loose GRC sand simu categorica no slip nur number of
TM3-024	LROP four wheeled test platform, dry mass 20.4 kg plus 11.3 kg payloa...	GRC-1 in NASA GRC SLOPE adjustable tilt bed, prepared to the loose co...	10	empirical	slope ascent outcome at 10 deg by longitudinal centre of gr...	Half Pass (CG -10% and centred); Full Pass (CG +10% forward...	lab	link	Section 3.2, Table 3 and Conclusion (PDF p. 9)	2026-10-20	As TM3-0
TM3-025	ROMA-Sloper reconfigurable four wheel rover, 9.5 kg, L600 x W540 x H2...	Tiltable sandbox 1.2 m x 1.0 m x 0.15 m deep, loose sand levelled man...	0 to 30	empirical	maximum slope traversed (Mode II) versus mode switching thr...	30 (Mode II); about 16 (Mode I limit) deg	lab	link	Section 5.1, text with Figure 6 ('can traverse a...	2026-10-20	Cross slop not uphill sand unsp small sanc long and 0 deep; recc chassis nc representa fixed sm...
TM3-026	ROMA-Sloper reconfigurable four wheel rover, 9.5 kg, L600 x W540 x H2...	Tiltable sandbox 1.2 m x 1.0 m x 0.15 m deep, loose sand levelled man...	20	empirical	downhill slip angle in Mode I on a 20 deg slope	10.5 deg	lab	link	Section 5.2.1 Coarse Adjustment, text after Figur...	2026-10-20	As TM3-0
TM3-027	ROMA-Sloper reconfigurable four wheel rover, 9.5 kg, L600 x W540 x H2...	Tiltable sandbox 1.2 m x 1.0 m x 0.15 m deep, loose sand levelled man...	20	empirical	reduction of uphill to downhill wheel normal force differen...	77.5 percent	lab	link	Section 5.2.1, text after Figure 7 ('reduced to 1...	2026-10-20	As TM3-0
TM3-028	Serpentine edge rigid PLA wheel, 100 mm diameter, 5 mm thickness, 44...	Beach sand from Port Said (Egypt) in bin 1 m x 0.25 m x 0.12 m with t...	0 (level)	empirical	slip ratio at sinkage heights 10, 20, 40 percent of wheel d...	0.15 (10%); 0.25 (20%); 0.6 (40%) ratio	lab	link	Results, Table 2	2026-10-20	Very small wheel (10 unusual cc edge and sinkage im rather thar beach san sand; few

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM3-029	Serpentine edge rigid PLA wheel, 100 mm diameter, 5 mm thickness, 44...	Beach sand from Port Said (Egypt) in bin 1 m x 0.25 m x 0.12 m with t...	0 (level)	empirical	net traction force at sinkage heights 10, 20, 40 percent of...	3 (10%); 7 (20%); 9 (40%) N	lab	link	Results, Table 2 (columns Push force Fp and Net t...	2026-10-20	As TM3-0 traction m with statio carrier, so to a stall fo drawbar p slip.
TM3-030	Conventional rigid disc wheel of similar diameter (100 mm) and thickn...	Beach sand from Port Said (Egypt) in bin 1 m x 0.25 m x 0.12 m with t...	0 (level)	empirical	slip ratio of an ordinary thin rigid wheel when sunk in sand	100 (no forward motion) percent	lab	link	Results, paragraph after Table 2	2026-10-20	A 5 mm th an extrem wheel, not wheel; illus entrapmer narrow wh loose dry s
TM3-031	Locked (braked) rigid wheel with tactile sensor array, 0.17 m diamete...	Single wheel tester soil bin with loose Silica Sand No. 5 (dry), whee...	0 (level)	empirical	smallest contact (rear) angle of the locked wheel as sinkag...	-20 to 0 deg	lab	link	Section 6.2, Table 4	2026-10-20	Locked (n wheel pus through sa push pull l not a drive small light level bed.
TM3-032	Locked (braked) rigid wheel with tactile sensor array, 0.17 m diamete...	Single wheel tester soil bin with loose Silica Sand No. 5 (dry), whee...	0 (level)	other	bulk density of the loose Silica Sand No. 5 bed	1430 kg/m3	none	link	Section 7.3.3, Table 6	2026-10-20	Prepared l sand; cohe internal fri in the sam from a cite reference, measured
TM3-033	Locked (braked) rigid wheel with tactile sensor array, 0.17 m diamete...	Single wheel tester soil bin with loose Silica Sand No. 5 (dry), whee...	0 (level)	classical-terrame...	agreement of calculated with measured towing resistance	calculated larger than measured at small sinkage and displa...	lab	link	Section 7.3.4, Figure 20 caption and text	2026-10-20	Model use geometry same tests blind pred neglects w shear; num in Figure 2
TM3-034	Tractor mounted bevameter, circular sinkage plates 60 and 100 mm diam...	Sample soil classified as sand (sieve analysis) placed in a test box...	not reported	classical-terrame...	Bekker k_phi for sand (plates D60 and D100)	1979.17 kN/m^(n+2)	none	link	Section 3.1, Table 3 (Case #1) and Table 4	2026-10-20	Sand of ur origin, moi density 1.4 parameter on plate si TM3-035 mm deep r constrain sinkage.
TM3-035	Tractor mounted bevameter, plate pairs D40/D60, D60/D80, D80/D100	Sample soil classified as sand (sieve analysis) placed in a test box...	not reported	classical-terrame...	range of Bekker k_c across plate pairs	-66.63 to 82.99 kN/m^(n+1)	none	link	Section 3.1, Table 3	2026-10-20	Shows pla dependen Bekker pa the same s are not tra without sta size.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM3-036	Bevameter shear ring, outer/inner diameter 340/270 mm, 40 mm grousers...	Sample soil classified as sand (sieve analysis) placed in a test box...	not reported	classical-terrame...	angle of internal friction of the sand	23.12 deg	none	link	Section 3.2, Table 7	2026-10-20	Ring shear grousers a normal pre friction an a clean dr possibly d and grous geometry; unknown.
TM3-037	Bevameter shear ring (as TM3-036)	Sample soil classified as sand (sieve analysis) placed in a test box...	not reported	classical-terrame...	shear deformation coefficient K_w at normal pressures 14.04...	0.015; 0.007; 0.009 m	none	link	Section 3.2, Table 6	2026-10-20	Wong soft model coe from a gro not the Ja wheel; sm of tests.
TM3-038	Six wheeled Planetary Rover Prototype, lugged rigid wheel radius 155...	Indoor sandy terrain (ST) area; sand type, grain size and density not...	0 (level)	classical-terrame...	identified internal friction angle of the sand	29.67 deg	lab	link	Section 6.3, Table 6 (row T3 (ST))	2026-10-20	Back fitted parameter lugged wh unspecifie units for K printed loc inconsiste KPa is imp dry sand,...
TM3-039	Six wheeled Planetary Rover Prototype, lugged rigid wheel radius 155...	Indoor sandy terrain (ST) area; sand type, grain size and density not...	0 (level)	classical-terrame...	identified shear deformation modulus K (Janosi type) for sa...	6.6 mm	lab	link	Section 6.3, Table 6	2026-10-20	As TM3-0 value spec lug wheel.
TM3-040	Six wheeled Planetary Rover Prototype, lugged rigid wheel radius 155...	Indoor sandy terrain (ST) area; sand type, grain size and density not...	0 (level)	classical-terrame...	maximum relative error of estimated drawbar pull on sand	8.11 percent	lab	link	Section 6.3.6, Table 7 (column Figure 27)	2026-10-20	In sample (paramete from the s so this is a of fit, not a error.
TM3-041	Single wheel test platform with a lugged wheel of radius 140 mm, widt...	Hard slate, gravel, sand, gravel on slate, sand with gravel	0 (level)	other	recognition accuracy of wheel terrain interaction class in...	292 of 300 (about 97) samples (percent)	lab	link	Section 1 contributions paragraph; Section 4, Tab...	2026-10-20	Three inte classes or surface, s lugs enter a sand typ trafficabili
TM3-042	not applicable (sand property)	Tottori Sand Dunes (Japan), sample T-3 at the planned demonstration f...	not reported	other	friction angle of Tottori dune sand T-3 at relative density...	32.2 (Dr 0.3); 37.0 (Dr 0.8) deg	none	link	Section 2.5, Table 6	2026-10-20	Natural co sand (77.5 rounded b Arabian in sand; Eart laboratory shear on c samples.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM3-043	not applicable (sand property)	Tottori Sand Dunes, sample TS-1 (planned lunar field site outside the...	not reported	other	friction angle of Tottori dune sand TS-1 at relative densit...	32.7 (Dr 0.3); 38.4 (Dr 0.8) deg	none	link	Section 2.5, Table 6	2026-10-20	As TM3-0
TM3-044	not applicable (sand property)	JAXA Exploration Laboratory test field sand J-1 (Silica Sand No. 5)	not reported	other	friction angle of laboratory Silica Sand No. 5 (J-1) at rel...	34.9 (Dr 0.3); 42.2 (Dr 0.8) deg	none	link	Section 2.5, Table 6	2026-10-20	Processec sand, coar more unifc dune sand as the refe against wh sand differ
TM3-045	not applicable (sand property)	Tottori Sand Dunes surface samples TS-1 (two points) and T-3 field si...	not reported	other	in situ relative density of dune surface sand	0.268 and 0.442 (TS-1); 0.261 (T-3) Dr (fraction)	none	link	Section 2.5, Table 5 and paragraph on in situ den...	2026-10-20	Few points been comp heavy mac during cor and contai per the au surface sa only.
TM3-046	not applicable (sand property)	Tottori dune sand T-3, TS-1 and Silica Sand No. 5 J-1, dry, direct sh...	not reported	other	shear displacement at peak shear stress (relative density 0...	1 to 2 mm	none	link	Section 2.5, text discussing Figure 7	2026-10-20	Direct she displacem not a whee displacem only as an magnitude Janosi typ
TM3-047	Four wheeled small rover 1.5 kg (ballasted to equal mass), wheelbase...	Silica sand test field 1820 × 450 x 308 mm on lift table; Silica Sand...	0 to 30	empirical	steepest slope climbed on silica sand	30 (all seven wheel widths succeeded) deg	lab	link	Section 3.1, first paragraph and Figure 9	2026-10-20	Very light (one sixth lunar analo mm lugs; s bin about deep; Japa 30 deg wa maximum not...
TM3-048	Four wheeled small rover 1.5 kg (ballasted to equal mass), wheelbase...	Silica sand test field 1820 × 450 x 308 mm on lift table; Silica Sand...	0 to 30	empirical	effect of wheel width on running performance on silica sand	non monotonic: b30 worst; below 30 mm narrower was better;...	lab	link	Section 3.1 (Figure 9(d), 9(e)) and Section 4.2	2026-10-20	As TM3-0
TM3-049	Four wheeled small rover 1.5 kg (ballasted to equal mass), wheelbase...	FJS-1 lunar regolith simulant field 2000 x 1000 × 400 mm, tilting up...	0 to 30	empirical	steepest slope climbed on FJS-1 by wheel width	20 (b30, b45, b60; stuck at 25 and above); 25 (b10, b15; st...	lab	link	Section 3.1, second paragraph and Figure 10	2026-10-20	FJS-1 is a basaltic si sand; inclu because th rover and were teste sand, show the optima width re...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM3-050	Four wheeled small rover 1.5 kg (ballasted to equal mass), wheelbase...	FJS-1 lunar regolith simulant field 2000 x 1000 × 400 mm, tilting up...	20	empirical	increase in rear wheel sinkage over a 700 mm run on a 20 de...	9.8 (b3); 98 (b60) percent	lab	link	Section 3.2, Figure 16	2026-10-20	As TM3-0 not sand.
TM3-051	Copper beryllium flexible (spring) wheel 200 mm diameter x 100 mm wid...	Keisa No. 5 silica sand (JIS standard soil) in single wheel test bed,...	0 to 20	empirical	slip ratio at 20 deg, flexible wheel without grousers	around 0.7 to 0.8 ratio	lab	link	Section 5.2, text with Figure 16 (p. 1596)	2026-10-20	Flexible sp light load (slow (10 r values rea authors fr ('around'), sand.
TM3-052	Copper beryllium flexible (spring) wheel 200 mm diameter x 100 mm wid...	Keisa No. 5 silica sand (JIS standard soil) in single wheel test bed,...	20	empirical	slip ratio at 20 deg, flexible wheel with grousers (Figure...	around 0.2 to 0.3 ratio	lab	link	Section 5.2, text with Figure 17 (p. 1596)	2026-10-20	As TM3-0
TM3-053	Copper beryllium flexible (spring) wheel 200 mm diameter x 100 mm wid...	Keisa No. 5 silica sand (JIS standard soil) in single wheel test bed,...	20	empirical	slip ratio at 20 deg, grousers under limbs	around 0.5 to 0.7 ratio	lab	link	Section 5.2, text with Figure 18 (p. 1596)	2026-10-20	As TM3-0
TM3-054	Copper beryllium flexible (spring) wheel 200 mm diameter x 100 mm wid...	Keisa No. 5 silica sand (JIS standard soil) in single wheel test bed,...	20	empirical	slip ratio at 20 deg, grousers between limbs	0.3 to 0.5 ratio	lab	link	Section 5.2, text with Figure 19 (p. 1596 to 1597)	2026-10-20	As TM3-0
TM3-055	Boeing GM XIII wire mesh LRV wheel, 82.2 cm (32.4 in) diameter (nomin...	Yuma (Arizona) desert dune sand, fine sand, screeded and vibrated to...	0 (level)	empirical	wheel speed above which wheel efficiency on dune sand incre...	1.4 m/s	lab	link	Part III, paragraphs 33 to 35 (report p. 23 and 2...	2026-10-20	Few sand LRV wire r at 253 N; s above sma speeds; ol report with layer.
TM3-056	29 × 11.0-10 8 ply type III aircraft tyre, 74 cm diameter, 28 cm wide...	Langley landing loads track sand test bed: locally obtained poorly gr...	0 (level)	empirical	free rolling axle drag coefficient versus speed in sand at...	drag versus speed similar in shape to clay but smaller in m...	lab	link	Results, 'Comparison of clay and sand data' parag...	2026-10-20	MOIST sar percent wa dry dune s heavy, fas tyre (tens 95 knots); pressure e shown onl
TM3-057	Six wheel robot; five test wheels (Table 1): Wh1 r 157.4 mm, b 165 mm...	Sloped loose soil test bed (sand and lunar soil simulant, 'different...	0 to 25	empirical	wheel slip ratio range at which the climbing performance in...	0.1 to 0.4 ratio	lab	link	Section 3.1, text after Table 2 and Figure 3	2026-10-20	Soil for ea stated in te parameter textbook s not measu pooled acr wheels, sc slopes; on a...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM3-058	Automotive tyres 165/60R13 (535 mm diameter, 170 mm wide) with smooth...	Indoor soil bin 3015 × 480 x 605 mm filled with air-dried filter sand...	0 (level)	empirical	slip at and above which lug pattern gave significantly high...	17.8 percent slip	lab	link	Abstract; Section IV.3 and Table 1 (Welch test fo...	2026-10-20	Car tyre at dry filter s gravity; lev three repe point.
TM3-059	Automotive tyres 165/60R13 (535 mm diameter, 170 mm wide) with smooth...	Indoor soil bin 3015 × 480 x 605 mm filled with air-dried filter sand...	0 (level)	empirical	slip at self propulsion (zero drawbar pull, minimum running...	around 5 percent slip	lab	link	Section IV.2, paragraph on Figure 8 (running resi...	2026-10-20	As TM3-0
TM3-060	Automotive tyres 165/60R13 (535 mm diameter, 170 mm wide) with smooth...	Indoor soil bin 3015 × 480 x 605 mm filled with air-dried filter sand...	0 (level)	empirical	slip above which tyre sinkage increased almost linearly wit...	10 percent slip	lab	link	Section IV.2, paragraph on Figure 9 (sinkage and...	2026-10-20	As TM3-0 values onl 9.
TM3-061	not applicable (sand property)	Indoor soil bin 3015 × 480 x 605 mm filled with air-dried filter sand...	not reported	other	internal friction angle of the air-dried filter sand	21 deg	none	link	Section II.1, last paragraph (soil bin sand prope...	2026-10-20	Ring shear values at l stress; mu than direct friction an of dry qua (about 32 for examp 042...
TM3-062	not applicable (sand property); tests used 165/60R13 tyre, 980 N, 140...	Air-dried Toyoura sand in the same indoor soil bin, slip levels -6.5...	not reported	other	internal friction angle of air-dried Toyoura sand measured...	21.5 deg	none	link	Section III.1 Experimental condition (p. 16)	2026-10-20	Toyoura sa angle from is far belo 38 deg us Bekker-W models of sand (Carl in evidenc
TM3-063	Rigid smooth wheel model, nominal r 0.1 m, b 0.1 m (radius varied in...	Dry sand and lunar soil parameter sets in a Bekker-Wong-Janosi steady...	0 (level)	classical-terrame...	predicted sinkage ratio h/r at the tractive limit (entrapme...	h/r = 0.9 tan(phi) ratio	simulation-only	link	Section 4.2, Equation (10) and Figure 11; Conclus...	2026-10-20	Model only measurem nominal pa are secon textbook v kc and kph apparently in Table 1; wheel with grouse...
TM3-064	In-wheel sensor wheel 200 mm diameter x 122 mm wide with eight force...	Single wheel test bed (1440 mm long) filled with Silica Sand No. 5	0 (level)	learning	overall accuracy of three class slip classification (non st...	0.963; 0.965 fraction	lab	link	Section 4.2, Tables 2 and 3	2026-10-20	Imbalance (222 non s samples o single san classificat classes, n prediction ahead of t

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM3-065	Four wheel drive test rover 82 × 63 x 95 cm, 37.5 kg (9.4 kg per whee...	Indoor test field at ISAS/JAXA, about 5 m long and 3 m wide, filled w...	0 (level)	learning	overall accuracy of three class slip classification (non st...	0.921 fraction	lab	link	Section 5.2, Table 5	2026-10-20	Few quasi samples (t correctly c so the mac low; slip in braking re not by slop
TM3-066	Rigid wheel with rubber surface, dynamic load 217 N, peripheral speed...	Soil bin with air-dried Toyoura sand, layer 170 mm thick, homogeneous...	0 (level)	empirical	empirical slip sinkage relation on air-dried Toyoura sand	h = 0.11s + 0.47 (R2 = 0.979) h in mm (s as used in the pap...	lab	link	Section III.2(1), Equation (2); Table 1 (conditio...	2026-10-20	Only three slip unit in stated exp (percent li size not fo text; very s level bin.
TM3-067	Four track platform rover with robotic arm, 120 kg, L153 x W154 x H11...	JAXA Space Exploration Field (indoor, about 22.6 m x 17.7 m): level g...	0 (level) and 18	empirical	steady state track slip on level ground and on an 18 deg sa...	approximately 0.05 (level); approximately 0.3 (18 deg slope...	lab	link	Section 4.2.1, text with Figure 12	2026-10-20	TRACKED included f comparisc wheels; cc indoor hill, natural du difference arm postu
TM3-068	Four track platform rover with robotic arm, 120 kg, L153 x W154 x H11...	JAXA Space Exploration Field (indoor, about 22.6 m x 17.7 m): level g...	0 (level)	classical-terrame...	sum of normal forces on the four legs on level ground, expe...	709 (experiment) versus 650 (simulation), approximately 8%...	sim-vs-lab	link	Section 4.2.1, paragraph on Figure 11	2026-10-20	Tracked ve difference attributed author to e and harne missing in simulation mass book error rathe
TM3-069	Single degree of freedom crawler (one drive and one passive wheel) wi...	Toyoura sand driving area with guide rails, levelled before runs	0 (level)	empirical	reduction of forward slip ratio with 45 deg lug alignment r...	about 13 percent	lab	link	Section 3.2, text with Figures 5 and 6; Table 1	2026-10-20	Crawler (tr unit, not a ground on effect may wheel grou percent m relative rat percentag
TM3-070	Rigid circular wheel versus proposed flexible (spring) wheel, both 20...	Dried regolith like simulant soil, layer 0.07 m thick, slopes 0, 5, 1...	0 to 25	empirical	steepest slope traversed by the rigid circular wheel (flexi...	15 (failed at 20 and 25) deg	lab	link	Section 4.1 (soil), Table 1, Section 4.3 and Tabl...	2026-10-20	Regolith si 5 kPa adh cohesionle sand; thin layer may wheel feel light load; gravity.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
TM3-071	Model wheeled vehicle with independently driven front and back wheels...	Dry silica sand test lane, same preparation for both configurations	0 (level)	empirical	slip ratio of the wheeled vehicle without and with the inte...	1 (wheels only, did not move); 0.065 (with crawler) ratio	lab	link	Development report, text with Figures 2 and 3 (si...	2026-10-20	One page developm vehicle ma size and s properties hybrid wh track vehi lane only.
SIM-001	Single wheel on NASA Glenn TREC rig, 17.5 kg total load (half wheel,...	GRC-3 lunar simulant bed, cohesionless, flat, v=0.2 m/s, slip 0-0.8,...	0 (flat)	other	Drawbar pull (DBP) and traction slope vs slip, simulation v...	qualitative: described as good agreement; no numeric error...	sim-vs-lab	link	Sec 2.1 VV-mode single wheel experiments, pp.5-6...	2026-10-11	Simulant (natural sa agreemen visually fr each point run; lab da several re shown in f
SIM-002	MGRU3 (1/6-mass VIPER replica; 73 kg on GRC-3, 88 kg on GRC-1) full r...	GRC-1 and GRC-3 lunar simulants on tilted bed, wheel angular speed 0....	0 to 30 (steps of 5)	other	Slope-vs-slip (slope-mode) relationship, simulation vs SLOP...	qualitative: single-wheel and full-rover simulations match...	sim-vs-lab	link	Sec 2.2 slope-mode, pp.7-8 (Fig 7); count of 252...	2026-10-11	Slip define steady-st at fixed an velocity (s mode), no slip; simul dune sand 1/6-mass tested...
SIM-003	VIPER (440 kg) simulated on Moon gravity	GRC-3 simulant properties, ramp 30 deg, lunar gravity (simulation onl...	30	other	Predicted wheel slip, VIPER on Moon	approx. 85 % slip	simulation-only	link	Sec 3 text discussing Fig 13, p.12	2026-10-11	Authors st results co validated lunar test their discu p.13); sam ~42% slip kg MGRU3
SIM-004	VIPER (440 kg) simulated on Moon gravity	GRC-3 simulant properties, ramp 10 deg, lunar gravity (simulation onl...	10	other	Predicted wheel slip, VIPER on Moon	approx. 75 % slip	simulation-only	link	Sec 3 text discussing Fig 13, pp.12-13	2026-10-11	As SIM-0 same text ~18% slip (73 kg) on deg
SIM-005	Virtual bevameter plate-sinkage test: circular plates of 0.2 m and 0....	DEM soil bed generated by settling, DEM particle radius 0.005 m	0	multibody-SCM	Pressure-sinkage force: relative error of calibrated SCM vs...	<2 % of max force	simulation-only	link	Sec 3.1 (relative-error definition and result), p...	2026-10-11	SCM is co DEM, not t bevameter authors sa fidelity ver bevameter been repo (Conclusic soil is not.
SIM-006	Annulus shear test rig, loads 25 to 200 kg	Same DEM bed as SIM-005	0	multibody-SCM	Steady-state shear torque: relative error of calibrated SCM...	<5 %	simulation-only	link	Sec 3.2 text near Figs 20-22, p.14-15	2026-10-11	Sim-vs-si

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
SIM-007	Single wheels: cylindrical (R 0.47 m, W 0.3 m), HMMWV wheel (R 0.47 m...	DEM bin 16 × 1 x 0.25 m with ~4.6 million particles; slip 0 to 0.8, v...	0	DEM	Wall-clock for 15 s single-wheel run: DEM vs SCM	DEM 4800 s; SCM 30 s (real-time factor about 320 vs 2, deri...	simulation-only	link	Sec 4.1, pp.16-17	2026-10-11	GPU mode hardware: GPU card stated in tl passage); traction-s for SCM v: described comparinc with...
SIM-008	Yutu-type six-wheel grouser rover, ~120 kg, wheelbase 1.136 m	DEM bin 14 x 2.5 × 0.375 m with ~15 million particles; v=1 m/s, slip...	0	DEM	Wall-clock for 10 s full-rover run: DEM vs SCM	DEM 10000 s (about 3 h); SCM 20 s (RTF about 1000 vs 2, der...	simulation-only	link	Sec 4.2, p.20; Conclusions	2026-10-11	SCM vs DI agreemen slip and sl stated qua DEM is itse unvalidate physical rc
SIM-009	MGRU3 wheel (R 0.25 m, W 0.2 m, 24 grousers 25 mm high x 10 mm wide)...	GRC-1 simulant on ramp (gravity vector tilted), 5 x 0.8 × 0.25 m soil...	0 to 30	other	Wheel slip vs ramp angle: CRM vs experiment vs DEM	qualitative: close agreement with both lab data and publish...	sim-vs-lab	link	Sec 3.3, pp.9-10 (Fig 5a)	2026-10-11	Coarse SP resolution because o domain; si dune sand defined wi inclusive e radius
SIM-010	Cone penetration (not a wheel): 60 deg cone in glass beads and 30 deg...	Laboratory granular beds (glass beads; Ottawa sand), 4-inch Proctor-m...	0	other	Mean relative error of penetration depth vs experiment, ave...	glass beads 3.5; Ottawa sand 7.4 % mean relative error	sim-vs-lab	link	Sec 3.2 text and Fig 3 caption	2026-10-11	Not a whe test; error penetratio experimer cited sour retrieved), each
SIM-011	VIPER rover (1.24 M particles total, 0.28 M active) and M113 tracked...	Chrono CRM granular patches 4 × 2 × 0.1 m (VIPER) and 30 m S-path (tr...	0	other	Real-time factor (wall/sim; lower is faster)	VIPER: CRM 10.9 (optimised), SCM 50.7 default and 17.6 with...	simulation-only	link	Sec 4.3, Tables 3-4, p.21-22	2026-10-11	SCM grid CRM parti authors ca vs-SCM c a rough es values are specific
SIM-012	EX1 four-wheeled grouser rover, 21.63 kg, 0.82 × 0.52 x 0.67 m, passi...	JAXA Sagamihara 20 x 20 m sand field, ~0.3 m deep dry loose Tohoku Si...	0	empirical-field	Steady-state slip ratio vs wheel speed (regression)	s = 0.0265*vw + 0.0256 (vw in m/s) slip ratio (fraction)	lab	link	Sec III-A and III-B, Eq. 2	2026-10-11	Indoor pre field, not r dunes; slip grouser-ir effective r (excluding negative s 16 runs (a for...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
SIM-013	EX1 rover (21.63 kg) on slopes	Same Silica No.5 sand field, slopes 0 to 18 deg at speeds up to 0.47...	0 to 18 (fit claime...	empirical-field	Slip vs slope angle (regression term)	s(vw,alpha) = (0.00522*vw + 0.00105)*alpha^2 + s(vw) slip r...	lab	link	Sec III-B, Eq. 3	2026-10-11	Unit of alp printed an the paper read in ph Degrees is with radia term is ab at 18 deg,
SIM-014	EX1 rover in OmniLRS / NVIDIA IsaacSim rigid-body model (Coulomb fric...	Lunaryard virtual terrain; regression slip and compliant-contact sink...	0 to 20	empirical-field	Simulation error relative to the authors' own regression: s...	slip <0.02 (flat); 0.22 mean absolute (slopes up to 20 deg)...	simulation-only	link	Sec IV (Results and Analysis, first paragraphs)	2026-10-11	The quote against th model, not independe measurem field comp 7, one run m/s) is de only...
SIM-015	Rashid-1 micro-rover wheel (R 100 mm, W 80 mm, 14 grousers 20 mm, 1.1...	Single-wheel testbed sandbox, Toyoura sand and FJS-1 simulant at ~25%...	0	DEM	Traction coefficient and dynamic sinkage vs slip, DEM vs si...	qualitative: good agreement (Figs 5-6); no scalar error rep...	sim-vs-lab	link	Abstract; Sec 3 first paragraph; Table 3	2026-10-11	DEM parti 10-13x la real grains wheel is th sided test the flight c fixed-slip free-runni
SIM-016	Same Rashid-1 wheel DEM (1.382 M particles)	Toyoura/FJS-1 virtual soil bin 0.65 × 0.14 x 0.14 m, 30 s run, dynami...	0	DEM	Wall-clock for a 30 s single-wheel run	about 7 h (RTF about 840, derived by me) h wall per 30 s sim	simulation-only	link	Sec 2.5 (Simulation Procedure) last paragraph	2026-10-11	Cost depe frozen-pa dynamic d single GPU
SIM-017	Same Rashid-1 wheel DEM, same mass under Earth (24.5 N) and lunar (4....	As SIM-015, simulation-only extrapolation to lunar gravity	0	DEM	Simulated differences: lunar vs Earth gravity same mass; To...	Traction coefficient and tractive efficiency same Earth vs...	simulation-only	link	Abstract; Sec 3 (text before Table 4)	2026-10-11	No lunar-g validation simulation extrapolat
SIM-018	Grouser wheel R 125 mm, W 100 mm, 12 straight grousers 25 mm, 80 N lo...	Toyoura standard sand in lab sandbox with observation window; Earth g...	not reported (flat...	DEM	Steady-state wheel slip ratio: simulation vs measurement	sim 12.1 vs lab 14.7 +/- 3.0 (relative error about 18%); rut...	sim-vs-lab	link	Sec 4.1, Table 3 and following paragraphs; Table...	2026-10-11	2D single- free-slip r controlled speed; fric used as tu parameter window lir error metr er...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
SIM-019	Rigid DEM wheel D 71.27 cm, W 17.78 cm (wheel weight 1922 N in Table...	Dense sand, soil depth 20 cm in a 122 × 53 × 22 cm DEM box; lab cone...	0 (not stated)	DEM	Pull (drawbar) as % of wheel weight at 20% slip; gross trac...	Lab vs DEM: pull 17 vs 15; gross traction 31 vs 20; towed -...	sim-vs-lab	link	Sec 10 Table 2 and following text, p.11	2026-10-11	Authors' te towed diff percent' w table show lab load in 8230 N ar radius 41. versus 19
SIM-020	Same DEM rigid wheel	Same DEM setup (36 M particles)	0	DEM	Wall-clock to simulate 10 ms	26 h (RTF about 9 million, derived by me) h wall per 10 ms...	simulation-only	link	Sec 2 (DEM methodology), p.2-3	2026-10-11	2021 CPU cluster, fin not compa GPU DEM
SIM-021	Underactuated tendon-driven soft gripper foot (D 250 mm), 5 to 66 N l...	Dry silica sand (close to Toyoura), 446 x 332 × 218 mm sandbox filled...	0 (comparison shown...	DEM	Pressure-sinkage curve: mean difference of simulated vs mea...	13.49 (SD 9.15) overall; 9.25 (SD 5.61) below 40 N % differ...	sim-vs-lab	link	Sec 3.2 and Fig 13	2026-10-11	Foot (not v quasi-stat ramp over particle cli through gr above ~5(not UAE d
SIM-022	Same gripper foot, 66 N	Silica sand, sandbox tilted 0 deg	0	other	Maximum sinkage at 66 N	12.98 (5.2% of gripper diameter) mm	lab	link	Sec 2.3 (testing environment), Table 2; entry-spe...	2026-10-11	Slow manu ramp; aver tests; entr changes r ~20% on a (Sec 3.1)
SIM-023	Same gripper foot, 66 N	Silica sand, sandbox tilted 35 deg (about the angle of repose)	35	other	Maximum sinkage at 66 N	36.19 (14.5% of gripper diameter) mm	lab	link	Sec 3.1, Table 2	2026-10-11	Near angle the materi collapses; nearly trip this lab rov matching s
SIM-024	Same gripper EDEM+MotionSolve model	10 s, time step 0.01 s, 1.249 M particles	0	DEM	Wall-clock for a 10 s simulation	9 h (RTF about 3240, derived by me) h wall per 10 s sim	simulation-only	link	Sec 3.2, Table 3	2026-10-11	Hardware in the pass
SIM-025	Rashid-1 micro-rover wheel (R 100 mm, 14 grousers, 24.5 N) - lab data...	FJS-1 simulant, single-wheel lab data	0	classical-terrame...	Median error vs measured drawbar pull and sinkage	DBP 24; sinkage 28 % median error	sim-vs-lab	link	Sec 5.1, p.17-18	2026-10-11	Compared published (SIM-015 not own ex slip is an ir load fixed; metric is n point
SIM-026	Rigid smooth and grouser wheels, r=157 mm, 80 N (Ding et al. data)	Lunar simulant (full Bekker set reported by source)	0	classical-terrame...	Median error vs measured drawbar pull and sinkage	DBP 27; sinkage 52 (under-predicted) % median error	sim-vs-lab	link	Sec 5.1, p.18	2026-10-11	As SIM-02 error large rigid-whe under-pre sinkage

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
SIM-027	Rigid wheel r=85 mm, 59 N (Wang and Han data)	KLS-1 simulant compacted to ~60% relative density; wheel sinks 1-14 mm	0	classical-terrame...	Median error vs measured drawbar pull and sinkage (stress c...	DBP 135 over-predicted; sinkage 385 over-predicted % median...	sim-vs-lab	link	Sec 5.1, p.18	2026-10-11	Firm, dens loaded rec the kernel' author lab regime bo
SIM-028	Whole mission evaluation of a lunar micro-rover (<50 kg) including wh...	n/a (timings on one core of Apple M2 Pro, Python 3.12)	n/a	classical-terrame...	Computation cost	0.50 ms per wheel solve (median); 10.3 ms per mission evalu...	simulation-only	link	Abstract; Sec 3.2 'Soil parameters and runtime';...	2026-10-11	Single-ma timing, ker wheel and error banc 025 to SIN
SIM-029	RHex legged robot (27.4 kg) with 80 kg or 40 kg payload in Chrono	Digital twin of NASA Ames LHS-1 testbed (18 × 12 m grid, tamped/raked...	0 (straight path; t...	multibody-SCM	Qualitative outcome of traverse: immobilisation vs success	80 kg payload: about 100% slip and stop in medium-strength...	simulation-only	link	Sec 2.2.3 (simulation validation), p.9-10; Append...	2026-10-11	Authors ca 'validation checks the model aga with no ph measurem LHS-1 is l highland s dune sa...
SIM-030	Archimede four-wheel-steer rover (~0.17 m diameter low-grouser wheels...	DLR PEL sandbox, granular lava 1-5 mm (Test Setup A, low-grouser whee...	0	other	End-point relative position error between odometric estimat...	sim <5; experiment <10 % of path	sim-vs-lab	link	Sec 4.1 and Conclusions (OCR of PDF; reported as...	2026-10-11	Metric is o error, not deformabl simulation slope/slip 20 deg) ar experimer because n simulation fo...
LRN-001	KUSR (simulated)	Five heterogeneous terrain image samples (sample 4 = two lunar soils...	not reported	learning	MAE of slip-hazard estimate, samples 1-5	DeepLab+SlipNet 0.621/0.617/0.814/0.257/0.315; TerrainNet 0...	simulation-only	link	Table II (Sec. III Results)	2026-10-11	MAE is on scale, not slip-ratio l absolute v 0.6-0.8 or 1-3 are la definition i ambiguous comparab
LRN-002	KUSR (simulated)	Desert-sand-like terrain on path (a) of a heterogeneous test environm...	not reported	learning	predicted vs measured mean wheel slip ratio on desert-sand-...	predicted ~0.6; measured fluctuates (no summary number give...	simulation-only	link	Sec. III Results, discussion of Fig. 6 path (a)	2026-10-11	Qualitative plot; simul no error st reported fo large devia attributed speed cha modelled

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
LRN-003	LAGR research robot (2 front driven wheels, 2 rear castors) with ster...	Field sequences on soil, sand (flat), gravel (transverse slope), asph...	roll/pitch slopes f...	learning	slip-in-X prediction RMS, terrain type known: soil train/te...	soil 7.23%/11.8%; gravel 7.16%/27.5%; all datasets test app...	field	link	Sec. IV-B Results, Fig. 8 caption values and text...	2026-10-11	Test = late same one- sequence split), not l terrain; gra error high roll/pitch combinati absent f...
LRN-004	LAGR research robot	Composite field test set (~2000 frames) spanning soil, sand, gravel,...	not reported	learning	end-to-end slip prediction error (whole composite set); wit...	21.8% overall; 11.2% when terrain classified correctly % sl...	field	link	Sec. V-B Results, Fig. 10 and following text	2026-10-11	Test seque stated not training, b types (incl were seen error inflat misclassif soil/gravel with ~8...
LRN-005	LAGR research robot	Six-class patch classification (sand, soil, grass, gravel, asphalt, w...	not applicable	learning	terrain classification rate (pixels, 12×12 m map); per-clas...	76.4% overall; sand 0.85, soil 0.84, gravel 0.74, grass 0.3...	field	link	Sec. III, Fig. 7 (text labels extracted with pdft...	2026-10-11	Per-class are extrac confusion figure, trea approxima ~30 frame sequence, out locatio
LRN-006	Pluto research rover (JPL) with panoramic stereo imagery	JPL Mars Yard (55 m x 45 m): bedrock simulant, rocks, mixed sand/soil...	up to 30 (yard); pa...	learning	terrain classification accuracy, cell metric / pixel metric...	84.4% / 88.1% %	lab	link	Sec. 5 results, text after Fig. 12 (classificatio...	2026-10-11	Sand is me 'soil'; Mars managed images fro yard as tra (different perspectiv numeric sl reported ir
LRN-007	MSL flight-spare wheel on MIT single-wheel testbed (IMU on wheel)	Mars regolith simulant with small ripples in the wheel path, ~20 m to...	not reported (level...	learning	slip-class accuracy, SVM (filtered) vs CNN (raw)	83% vs 88% % accuracy (3-class)	lab	link	Sec. 4.1 (15,000 samples, three slip classes) and...	2026-10-11	Classificat bins, not c slip error; hold-out (60/40, 50 experimer test data a soil and ru
LRN-008	Curiosity hazcam/navcam imagery and Fitorobot tracked robot imagery	11 image classes: 3 Mars classes (clean, rocks, boulders) plus gravel...	not applicable	learning	image terrain classification accuracy, CNN raw vs best case	87% vs 92% % accuracy (11-class)	field	link	Sec. 4.2-4.3 (300 + 800 images) and Sec. 5.2 text...	2026-10-11	Only 1100 random ho mixed Mar imagery ar tracked-ro imagery; s class amo

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
LRN-009	MSL flight-spare wheel, MIT single-wheel testbed (bin 3.14 m x 1.2 m...	Mars regolith simulant with ripples in wheel path; slip 0 to ~100%	not reported (level)	learning	training/test sample counts and class accuracy from regress...	9235 train / 3958 test; low and high slip classes ~85%, mod...	lab	link	Sec. 4.1 dataset split (9235/3958) and Sec. 5.2 t...	2026-10-11	Random 7 out on one dataset (ir distributio MAE only not retriev
LRN-010	Mars 2020 Perseverance rover (flight data); Earth-analog rover test b...	~500 sols, 45 km, ~43,000 images; rocky hills, flat bedrock, sand rip...	tilt logged per sam...	learning	hazard AUROC / AP / precision / recall / F1 on 35 held-out...	AUROC 0.874; AP 0.878; precision 0.835; recall 0.695; F1 0....	field	link	Table II and Sec. V Results	2026-10-11	Mars flight label = /wh odometry visual-odo displacem WO distan threshold percentile binary cla.
LRN-011	Perseverance rover (flight data)	Held-out test sols including sand ripples, mixed rock and sand, a san...	not reported	learning	ablation hazard AUROC / quantile Spearman rho	full 0.874 / 0.721; no alignment 0.723 / 0.316; no Rank-N-C...	field	link	Table I and Sec. V-A	2026-10-11	As the mai Persevera Mars sand dune sand
LRN-012	Curiosity, Opportunity, Spirit NAVCAM/MASTCAM imagery	Four classes: soil, bedrock, sand, big rock; ~35K images, ~326K crowd...	not applicable	learning	overall pixel accuracy on MSL NAVCAM expert test set; sand-...	94.97% (random-label training) and 96.67% (merged-label tra...	field	link	Tables 3 and 4 (3-label-agreement test set); abst...	2026-10-11	Terrain-ty segmentat slip; test s (~1% of tr size) and across orb classes; M imagery
LRN-013	Simulated vehicle in Verti-Bench (Chrono SCM-type deformable terrain)	Randomised deformable terrain regions with per-region soil parameter...	not reported	learning	32-step (3.2 s) rollout MAE x/y/z (m): NeSAM vs plain Trans...	0.80/0.78/0.06 m vs 0.89/0.90/0.09 m; yaw 4.18 vs 6.76 deg...	simulation-only	link	Table I and text below it (Sec. V-A)	2026-10-11	Trajectory kinodynan not slip rat out 20% o trajectorie same simu distributio
LRN-014	Verti-4-Wheeler (scaled 4-wheel vehicle), Verti-Arena	Grass, dirt and deformable sand regions with uneven geometry; referen...	not reported	learning	rollout MAE x/y/z; roll/pitch/yaw; closed-loop completion	0.068/0.059/0.012 m vs 0.082/0.070/0.019 m; yaw 5.79 vs 7.0...	lab	link	Table III, Table IV and Sec. V-B	2026-10-11	physical d trajectorie 30 s (abou 15,000 tra 80/20 spli trajectory
LRN-015	Boston Dynamics Spot quadruped (joint position, velocity, effort)	Lava-tube floors (steep, sandy sloping floors in Mammoth Cave), limes...	described as steep/...	learning	slip prediction precision / recall / F-score (current time)...	95% / 80% / 0.84; 95% / 77% / 0.82 at 720 ms ahead %, F-sco...	field	link	Sec. IV Time-delay embedding and Future predictio...	2026-10-11	Binary slip detection, magnitude CV with 8(out within from same terrain hel sandy floo

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
LRN-016	Simulated 10 kg rocker-bogie rover, wheel radius 0.4 m, width 0.1 m	Loose sand and compact sand per Wong equations	not reported (no sl...	learning	rover velocity prediction error (largest, linear model on l...	<5 % velocity error	simulation-only	link	Sec. 4 Experiments (text around Fig. 3-4)	2026-10-11	Held-out band 0.7- friction an deg were testing bu from the s analytic W used for tr
LRN-017	Three-wheeled rover prototype (two lugged driving wheels plus a follo...	Sand, stony sand, gravel, bedrock, rock; CTEM dataset 2375 images spl...	not reported	learning	segmentation accuracy (mIoU) and full-scale error of predic...	93.0% mIoU; 12.5% and 10.8% full-scale error; ground truth...	lab	link	Sec. V-A segmentation result and Sec. V-C text be...	2026-10-11	Predicts s mechanic not slip; sp (better tha split) but s authors sa accurate e control
LRN-018	Rover wheel with 3 piezoelectric sensors on hook-shaped spokes, drive...	Six terrains in total: flat, fine sand (<1 mm), small stone, small pe...	not reported (flat)	learning	terrain classification accuracy (1.5 s window, 120 random t...	90 % accuracy	lab	link	Sec. IV text around Fig. 11 (confusion matrix)	2026-10-11	Single spe platform; r trials not t held-out; confused stones (at slip on sto
LRN-019	ERNEST 4-wheel rover with Active Gimbal Suspension (JPL)	M90 Mars soil simulant sand bed, 20 deg slope, dry and wet	20	learning	travel reduction (loss of expected no-slip distance) and co...	dry 81.9% vs 43.0%; wet 100% (entrapped) vs 30.0%; CoT dry...	lab	link	Table 6 and Sec. V text before it	2026-10-11	Control po prediction reduction but define integrated number of stated; lab
LRN-020	Simulated cargo rover (20 kg + 0 to 80 kg payload), Project Chrono SCM	Single loose-soil SCM condition; grid of slope, heading, speed, mass	-15 to 10 (Table 1)	multibody-SCM	fit RMSE against SCM simulation: slip ratio and slip angle	0.0293 slip ratio; 0.0279 rad slip angle slip ratio; rad	simulation-only	link	Sec. 2 around Eq. 9-10 and Table 1/2	2026-10-11	RMSE is fi training si held-out s parametric not a neur slopes onl uphill
LRN-021	Single grousered wheel on carriage test bed (10 kg wheel load), bin 3...	Toyoura sand; constant slip-rate runs (slip rates +/-1, 2, 5, 10, 20...	not applicable for...	learning	adjusted R-square of per-slip-rate regression; mean slip-ra...	adj R2 0.695 to 0.973 (lowest at rate -20 and 1; highest at...	lab	link	Table 1 and Sec. 4 text; abstract	2026-10-11	In-sample no held-o regressior model; RM unit not st recorded
LRN-022	Quasi-direct-drive quadruped, 2 gaits (Crawl N' Sense vs Trot-Walk),...	Damp construction sand at low/medium/high compaction; 300 micron beac...	0 (level)	other	crust rupture detection: Crawl gait specificity/sensitivity...	96%/63%; 12.5%/100% %	lab	link	Sec. V-B (Effect of gait design on detecting subs...	2026-10-11	Not learnir small samp crawl, 3 tr level artifi shows gai changes proprioce reliability c

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
LRN-024	WVU Pathfinder, 15 kg four-wheel skid-steer	Mars-analog field site at Point Marion, PA; ~140-150 m runs (one ~650...	not reported	other	slip detection accuracy at ~150 m (abstract)	>92 % accuracy	field	link	Abstract; Sec. 5 Table 2 (confusion matrices not...	2026-10-11	Filter-bas learned, a not a sand medium a bins often as low slip degrades ~500 m
FLD-001	MER Opportunity (6 wheels, ~26 cm wheel diameter, ~180 kg per Li et a...	Purgatory ripple, Meridiani Planum, Mars (sol 446); nondescript pile...	not reported	empirical-field	track length vs wheel-rotation distance during blind drive	2 m of track vs ~50 m of wheel travel m	field	link	Sec 5.1 text and Figure 10 caption (sol 446 Purga...	2026-10-11	Mars data data from g), NOT Ea Slip not cc authors fo ratio is imp two length percen...
FLD-002	MER Opportunity	Purgatory ripple escape drives, sols 463-483	not reported	empirical-field	VO-measured slip rate	98.9 to 99.5 %	field	link	Sec 5.1 paragraph beginning 'successful estimatio...	2026-10-11	Mars data data from g), NOT Ea Slip here is level slip f odometry command odometry not sing...
FLD-003	MER Opportunity	Sandy slope near obstacle Wopmay, Endurance Crater, Mars (around sol...	17 to 23	empirical-field	slope range of sandy terrain where rover spent weeks failin...	17-23 deg	field	link	Figure 6 caption	2026-10-11	Mars data data from g), NOT Ea No numeri for this ep is as state authors (lo regional d
FLD-004	MER Opportunity	Eagle Crater, loose soil and steep crater wall, Mars (crater diameter...	not reported	empirical-field	accumulated slippage over traverse	18.71 m slip over 91.42 m (20.5 %) %	field	link	Introduction paragraph [3]	2026-10-11	Mars data data from g), NOT Ea Cumulativ over >60 s slopes and not a per- Secondary
FLD-005	MER Opportunity, first crater-slope attempt, Meridiani	Sandy crater slope Meridiani Planum, Mars (Feb 2004)	not reported	empirical-field	slip (wheel revolutions that did not produce progress)	50 %	field	link	Paragraphs on Lindemann's description of 4 wheel...	2026-10-11	Mars data data from g), NOT Ea Press-rele descriptio not a peer measurem angle not t...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
FLD-006	SSTB-lite (engineering model of MER, 'same weight and size, identical...	Variable-tilt platform with 6 inches of dry loose sand ('similar to c...	20 (uphill)	other	uphill slip at 20 degrees	90 cm of 100 cm commanded lost (10 cm progress) %	lab	link	'How Far Will The Rovers Slip and Slide' section	2026-10-11	Earth grav bed; relea state whet mass was Mars weig defined by distance p command Press r...
FLD-007	SSTB-lite	Same tilt platform, loose dry sand	25 (uphill limit)	other	uphill climbing limit (dead end) for Opportunity's sand	25 deg	lab	link	Baumgartner quote: dead-end point where rover can...	2026-10-11	Limit quot driver for ' sand Oppo in'; derive tilt-platfor which wer retrieved. gravity lab
FLD-008	SSTB-lite	Same tilt platform, loose dry sand (downhill)	15 and 20 (downhill)	other	extra downslope slide while driving downhill	25 cm per m at 15 deg; 55 cm per m at 20 deg cm/m	lab	link	Same section, 'at a 15-degree angle facing down'	2026-10-11	Earth grav bed; negat (skid) regi comparab slip percer
FLD-009	MER Spirit	Gusev Crater, Mars; slope >25 degrees (sol 206)	>25	empirical-field	maximum VO-measured slip	125 %	field	link	Sec 5 (introductory text before Sec 5.1), text ab...	2026-10-11	Mars data data from g), NOT Ea Slip here is level slip f odometry command odometry not sing...
FLD-010	MER Spirit	Loose sand near 'Pot of Gold', Husband Hill base, Gusev, Mars	10 to 13	empirical-field	slip (reduction of distance traveled vs commanded)	up to 50 %	field	link	Sol ~200 section on repositioning at Pot of Gold	2026-10-11	Mars data data from g), NOT Ea Authors de 1 - achieved/ distance (text).
FLD-011	MER Spirit	Columbia Hills, Gusev, Mars; appearance-identical terrain	19	empirical-field	slip	15 to 20 %	field	link	Paragraph 'extremely difficult to predict slip ba...	2026-10-11	Mars data data from g), NOT Ea Paired with row: PanC showed no difference the two sit

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
FLD-012	MER Spirit	Columbia Hills, Gusev, Mars; ~30 m beyond the previous site, no visib...	16	empirical-field	slip	95 %	field	link	Same paragraph	2026-10-11	Mars data data from g), NOT Ea Shows slip predictabl slope or in alone: 95 16 deg vs percent at
FLD-013	MER Spirit	Clovis outcrop approach, Husband Hill, Gusev, Mars	up to 30 (tilt)	empirical-field	slip in one case	over 100 %	field	link	Paragraph on Clovis (sol ~203 onward)	2026-10-11	Mars data data from g), NOT Ea Value is a reported c up to 30 d mix of roc not pure s
FLD-014	MER Spirit	Husband Hill traverse, Gusev, Mars; 'fine grain' class (short VO segm...	10 to 15 (onset)	empirical-field	longitudinal slippage on fine-grain terrain	approaches 100 at slopes as early as 10-15 deg %	field	link	Sec 5.2 paragraph [40], Figure 11a	2026-10-11	Mars data data from g), NOT Ea Terrain cla assigned f images; sl local DEM, fit; slippag per segme
FLD-015	MER Spirit	Husband Hill traverse, long segments, sand-sized class	0 to 5	empirical-field	relative longitudinal slippage	4.0 (49 segments, 528.56 m) %	field	link	Table 6 (sand-sized rows)	2026-10-11	Mars data data from g), NOT Ea Segment- over mixed conditions note highe sometimes smaller sli
FLD-016	MER Spirit	Husband Hill traverse, long segments, sand-sized class	5 to 10	empirical-field	relative longitudinal slippage	9.9 (36 segments, 239.49 m) %	field	link	Table 6 (sand-sized rows)	2026-10-11	Mars data data from g), NOT Ea Same cave previous r
FLD-017	MER Spirit	Husband Hill traverse, long segments, fine-grain class	5 to 10	empirical-field	relative longitudinal slippage	16.6 (9 segments, 61.78 m) %	field	link	Table 6 (fine grain rows)	2026-10-11	Mars data data from g), NOT Ea Few segm fine-grain above 15 (authors).

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
FLD-018	MSL Curiosity (6 wheels; ~0.5 m wheel diameter per Li and Lewis 2022...	Megaripple field, Gale Crater, Mars (sol 1181 controlled crossing)	slightly uphill (va...	empirical-field	rover 3D slip over ~8 m of shallow megaripple sand	exceeded 70 in places %	field	link	Left column bullets, sol 1181-1185 crossing exper...	2026-10-11	Mars data data from g), NOT Ea Slip here is level slip fi odometry command odometry not sing...
FLD-019	MSL Curiosity	Uphill sandy terrain, Gale Crater, Mars (sol 1703 drive)	9.5 (rover pitch)	empirical-field	slip at fault	61.2 (after 2.88 m) %	field	link	Flight data-set comparison, Figure 25 (after the...	2026-10-11	Mars data data from g), NOT Ea Pitch is the body pitch terrain-slo measurem method pr VO-correc
FLD-020	MSL Curiosity	Gale Crater traverses (flight software default)	n/a	other	default autonomous-drive slip limit	60 %	field	link	Same passage: 'default 60% slip limit'	2026-10-11	Mars data data from g), NOT Ea Operationa a measure
FLD-021	MSL Curiosity	Gale Crater traverses (as cited by Kilic et al., secondary)	n/a	other	stop thresholds: single slip measurement; consecutive slip...	single 70-90 percent; consecutive 40-70 percent %	field	link	Slip ratio discussion in Sec 3.1	2026-10-11	Mars data data from g), NOT Ea Secondary Arvidson e (full text n retrieved); in the sour
FLD-022	MSL Curiosity with Traction Control enabled vs disabled	Sandy/rocky drives sols 1664-1707, Gale Crater, Mars	not reported (terra...	empirical-field	change in average max rover slip with Traction Control	2.1 % lower with enabled (absolute vs relative not specifie...	field	link	Discussion of Figure 25	2026-10-11	Mars data data from g), NOT Ea Small sam randomise the sol 17 excessive drive was from the...
FLD-023	Perseverance ENav tilt-limit setting, derived from Scarecrow (Curiosi...	Jezero planning limit referencing Curiosity-class Scarecrow test in p...	16 (max allowed til...	other	max terrain tilt allowed for Perseverance autonomous nav (a...	16 deg	field	link	'Flight Software Update' section (text partly gar...	2026-10-11	Design lim by Earth d with the S rover (cite Heverly 2(retrieved) applied to measured sink...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
FLD-024	Perseverance (Mars 2020)	Summerland Trail, Jezero crater rim hill, Mars (sols 1283-1305)	15 to 20	empirical-field	autonav slip limit needed (Slip Tables)	above 50 %	field	link	Mobility Challenges / Slip Issues section	2026-10-11	Mars data data from g), NOT Ea Slip here is level slip fi odometry command odometry not sing...
FLD-025	Perseverance (Mars 2020)	Midtoya, west Jezero crater rim, very sandy terrain with small loose...	about 20	empirical-field	slip	in excess of 94 %	field	link	Mobility Mechanism Issues section	2026-10-11	Mars data data from g), NOT Ea Slip here is level slip fi odometry command odometry not sing...
FLD-026	Perseverance (Mars 2020)	Small sand ripples near preferred path, Jezero crater floor, Mars (Ra...	not reported	empirical-field	wheel slip over small ripples	not quantified; described as insignificant n/a	field	link	Section 6/7 text around 'sand ripples near the pr...	2026-10-11	Mars data data from g), NOT Ea No numbe small/low firm subst equivalent megarippl f...
FLD-027	Scarab rover (400 kg, 4 rigid wheels 71 cm x 18 cm, sandpaper-covered...	NASA Glenn SLOPE lab tilt bed, 6 × 4.5 m, 0.23 m deep GRC-1 lunar reg...	20	other	longitudinal slip at 90 and 75 deg angle of attack	exceeded 90 %	lab	link	Sec 3.4.1 (Figures 3.6, text after Fig 3.5)	2026-10-11	Lunar simu dune sand gravity wit rover; rigic without gr definition i based at c wheel spe
FLD-028	Scarab rover (same)	SLOPE lab tilt bed, GRC-1 simulant	25	other	longitudinal slip at 90 deg angle of attack (direct ascent)	100 (no angle of attack made uphill progress) %	lab	link	Sec 3.4.1, Figure 3.7 discussion	2026-10-11	Same as p row.
FLD-029	Andy 2 (small 4-wheel skid-steer lunar rover)	loose soil (not tied to GRC-1 for this claim)	15	other	slip when ascending	20 to 30 %	none	link	Sec 2.2.2 (subsection Experiment setup and proced...	2026-10-11	Unreferen capability in Sec 2.2 measurem tied to GR use as a c
FLD-030	El-Dorado-II (35 kg, 4 wheels each 178 mm dia x 100 mm wide, no grous...	Tiltable sandbox 2 × 1 m filled with dry Toyoura standard sand, Earth...	4	other	slip ratio, direct uphill ascent	close to 0.9 (text); read from figure description ratio	lab	link	Appendix B.2.2	2026-10-11	Light rover wheels; To is uniform not natura Slip ratio v based.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
FLD-031	El-Dorado-II with 12 grousers (15 mm)	Same Toyoura sand slope bed	4	other	slip ratio, direct uphill ascent	around 0.2 ratio	lab	link	Appendix B.2.2	2026-10-11	Same as p row. More lowered sl but less sc deg.
FLD-032	Four-wheel rover, 2 driven front wheels, 23 kg, wheel 296 mm dia x 90...	Indoor sandbox, 0.725 m3 dry Toyoura sand 250 mm deep, incline 0-30 d...	10	other	maximum sinkage angle, 20 mm grouser	127 deg	lab	link	Section 4.1 (Figure 12b)	2026-10-11	Only front driven; sin reported a subtendec wheel cen slip ratio p not stated
FLD-033	Same rover, 80 mm fixed grousers	Same sandbox	10	other	maximum sinkage angle	66 deg	lab	link	Section 4.1 (Figure 12b)	2026-10-11	Same as p
FLD-034	Same rover, 80 mm fixed grousers	Same sandbox	20	other	maximum sinkage angle (smallest across grouser lengths)	118 deg	lab	link	Section 4.1 (Figure 12c)	2026-10-11	Same as p Authors ju deg as stu current lim runs near
FLD-035	Same rover, all grouser lengths 20-80 mm	Same sandbox	30 (at angle of rep...	other	forward progress per wheel revolution at slope = angle of r...	less than 10 mm (wheels spun in place) mm/rev	lab	link	Section 4.1 (Figure 12d discussion)	2026-10-11	Same as p
FLD-036	SpaceBok quadruped (legged), planar vs point feet	RUAG Space ExoMars locomotion testbed 6 × 6 m, 20 t ES-3 Mars soil si...	0 to 25	other	max foot sinkage with point feet	up to 120 mm	lab	link	Section 4.2.1	2026-10-11	Legged ro simulant, r compensa qualitative table.
FLD-037	Hexapod robot (350 g, 22 × 10 cm, C-legs)	Tiltable fluidised bed of 300 micron glass beads, 127 × 65 cm, Earth...	0	other	step-averaged speed on level bed	15.7 +/- 0.97 cm/s	lab	link	Section III-A	2026-10-11	Glass bea small legg speed not
FLD-038	Hexapod robot (same)	Same tilt bed	10 (11.0 +/- 0.14 c...	other	step-averaged speed at 20 deg	4.98 +/- 0.32 cm/s	lab	link	Section III-A	2026-10-11	Same as p
FLD-039	SSTB-Lite (MER model) stripped to Mars-weight; Scarecrow (Curiosity-c...	Dumont Dunes OHV Area near Baker, California (Earth dunes), May 2012	not reported (dune...	other	uphill vs downhill ground speed at equal wheel rate	uphill much slower than downhill (qualitative) n/a	field	link	Article body	2026-10-11	Qualitative statement, slower-tha observatic SSTB-Lite slope or sl Mars-weig at Earth gr
FLD-040	Opportunity (MER)	Purgatory ripple, Mars	not reported	other	ripple height and entrapment duration	30 cm high; several weeks embedded cm	field	link	Figure 1 caption	2026-10-11	Mars data data from g), NOT Ea Review/pe source, de

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
FLD-041	ESA/DFKI rover field campaign (Mana, Minnie, SherpaTT)	Sahara dunes near Erfoud, Morocco (Dec 2018 field test)	not reported	other	vision-feature tracking on smooth dunes	difficult; algorithms behaved unexpectedly (qualitative) n/a	field	link	Quote paragraph by G. Visentin	2026-10-11	ESA news slip, sinka numbers. only to the VO on feat dune face
FLD-042	Husky UGV (Graz University of Technology) in AMADEE-18	Dhofar desert base camp area, Oman (Feb 2018); loose soft sand	not reported	other	stuck events and dead-reckoning error	rover got stuck from time to time; dead-reckoning error lar...	field	link	Section 10.8 Summary	2026-10-11	Only Arabi wheeled-r evidence f slip, sinka numbers. proof-of- mapping r mobility st
FLD-HV13-1	Scarecrow (Mars weight MSL test rover, 340 kg, 50 cm diameter rigid g...	Aeolian cohesionless sand dunes, Dumont Dunes, Mojave Desert CA; eigh...	4 to 21 (fit validi...	empirical	Cubic fit coefficients c0; c1; c2; c3 (slip vs slope)	26.197; -8.849; 1.281; -0.034 dimensionless polynomial coef...	field	link	Section 6.3, p. 843, text accompanying Figure 12	2026-10-13	Fit valid or to 21 deg upslope dr this vehicl c_i not sta transferab wheels or
FLD-HV13-2	Scarecrow (Mars weight MSL test rover, 340 kg, 50 cm diameter rigid g...	Aeolian cohesionless sand dunes, Dumont Dunes, Mojave Desert CA; upsl...	16	empirical	Slip at 16 deg slope (approximate, as stated in text)	75 percent (approximate, text says approximately 75%)	field	link	Section 6.3, p. 843; repeated in Section 10, p. 8...	2026-10-13	Approxima values; sin and wheel
FLD-HV13-3	Scarecrow and DTM (Dynamic Test Model, 437 kg, same wheels, 128% of M...	Artificially built cohesionless sand slopes in the JPL Marsyard, 5 m...	about 10 (marginal...	empirical	Slope at which both vehicles embed in sand (transition from...	approximately 17 deg	lab	link	Section 6.3, p. 842 text accompanying Figure 11;...	2026-10-13	Prepared t (counted a moisture c packing be grousers a sinkage th cohesionle would not sparse poi tra...
FLD-HV13-4	Scarecrow (50 cm wheel diameter, EGP 5.75 kPa) vs SSTB-Lite (MER test...	Cohesionless sand, Dumont Dunes, Mojave Desert CA; side by side on sa...	12 and 15	empirical	Maximum slope climbed vs embedding (wheel diameter effect a...	Scarecrow climbed 15 deg slope and ascended 12 deg slope; S...	field	link	Section 7, p. 843 to 844, Figure 15 text and Tabl...	2026-10-13	Different s measurem methods f vehicles (I accurate f Lite); singl comparisc outcome i slip value

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
FLD-HV13-5a	Scarecrow (Mars weight MSL test rover, 340 kg, 50 cm diameter rigid g...	Constructed test beds in the JPL Marsyard: smooth flagstone simulated...	22 (bedrock, traver...	empirical	Maximum traversable slope on smooth bedrock; slip on Marsya...	Bedrock traversable up to 22 deg (Abstract); Marsyard cohes...	lab	link	Abstract p. 835; Section 6.1 p. 841 to 842 with F...	2026-10-13	Constructe Marsyard smooth fla described bounding bedrock; c soil is high
FLD-HV13-5b	Scarecrow (Mars weight MSL test rover, 340 kg, 50 cm diameter rigid g...	Cohesive soil of the desiccated lake bed at Tecopa, California (natur...	28	empirical	Slip at the steepest slope climbed	less than 50% slip at 28 deg (Sec 6.2, Tecopa); Section 10...	field	link	Section 6.2 p. 842 with Figure 10; Section 10 p....	2026-10-13	Single site vehicle; th gives two for 28 deg 50% in Se 30% in Se does not ti second to explicitly;.
FLD-HV13-6	Curiosity (MSL flight rover on Mars, 899 kg)	Hummocky well consolidated soil at Gale Crater, Mars, first 250 sols,...	not stated	empirical	Agreement of Curiosity slip vs slope data on Mars with Eart...	good agreement (qualitative; no numeric error reported) qua...	field	link	Abstract p. 835; Section 9 p. 844 to 845 and Figu...	2026-10-13	No quantit prediction in text; agr judged fro 17 overlay terrain and range on M
PLN-001	Zhurong (Chinese Mars rover, 6 drum shaped lugged wheels, 240 kg, whe...	Southern Utopia Planitia, Mars; flat plain with drifting soils, small...	~0.55 (average clim...	empirical-field	average wheel slip ratio during steady climb	0.056 slip ratio (dimensionless)	field	link	Main text paragraph on wheel slip (preprint p. 7...	2026-10-20	Mars grav field data, dunes. Lug wheel, 24 Slip definit lugged wh effective r lug coeffic
PLN-002	Zhurong (Chinese Mars rover, 6 drum shaped lugged wheels, 240 kg, whe...	Southern Utopia Planitia, Mars; sol 55 climb	~1.14	empirical-field	largest average slip ratio in first 60 sols	0.22 slip ratio (dimensionless)	field	link	Main text paragraph on wheel slip (preprint p. 8)	2026-10-20	Mars grav field data, dunes. Ver slope; slip at about 1 indicates s drifting so rather than
PLN-003	Zhurong (Chinese Mars rover, 6 drum shaped lugged wheels, 240 kg, whe...	Utopia Planitia, Mars; legible wheel tracks between waypoints X and A	not reported	empirical-field	average wheel slip ratio from wheel track imprint	0.05 slip ratio (dimensionless)	field	link	Main text (preprint p. 8) and Methods subsection...	2026-10-20	Mars grav field data, dunes. Imp method re intact lug i only rear w usable.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
PLN-004	Zhurong (Chinese Mars rover, 6 drum shaped lugged wheels, 240 kg, whe...	Utopia Planitia, Mars; Martian topsoil along first 60 sols	not reported	empirical-field	equivalent wheel sinkage	about 5 (range about 2 to approximately 15) mm	field	link	Main text paragraph on soil mechanical properties...	2026-10-20	Mars grav field data, dunes. Fir topsoil, no dune sand dominate t value.
PLN-005	Zhurong (Chinese Mars rover, 6 drum shaped lugged wheels, 240 kg, whe...	Utopia Planitia, Mars; topsoil	not reported	classical-terrame...	internal friction angle of topsoil (inferred)	~21 to 34 deg	sim-vs-field	link	Main text paragraph on shearing curves (preprint...	2026-10-20	Mars grav field data, dunes. Inf parameter on assume bounds an wheel mod topsoil is c
PLN-006	Zhurong (Chinese Mars rover, 6 drum shaped lugged wheels, 240 kg, whe...	Utopia Planitia, Mars; topsoil	not reported	classical-terrame...	equivalent stiffness Ks of topsoil (inferred)	1390 to 5872 kPa m^-n	sim-vs-field	link	Main text (preprint p. 10); Fig. 4a; abstract	2026-10-20	Mars grav field data, dunes. Inf bearing pa firm Martia dune sand
PLN-007	not applicable (dune observed by Zhurong NaTeCam)	First aeolian dune encountered by Zhurong on sol 50, Utopia Planitia;...	not reported	empirical-field	dune length, width and height	40 m length, 8 m width, 0.6 m height m	none	link	Main text paragraph on panorama and dunes (prepri...	2026-10-20	Mars grav field data, dunes. Sm transverse bedform, f than Arabi geometry mobility da
PLN-008	Zhurong (Chinese Mars rover, 6 drum shaped lugged wheels, 240 kg, whe...	Design specification for soft ground	>=20 (soft ground);...	other	maximum climbable slope on soft ground (design value)	>=20 deg	none	link	Extended Data Table 1	2026-10-20	Mars grav field data, dunes. De capability, measured value; soft defined.
PLN-009	MSL Curiosity (6 aluminium wheels, 50 cm diameter, 40 cm width, rocke...	Gale crater, Mars; all terrain driven sols 1 to about 2488 (bedrock,...	not reported	empirical-field	average wheel slip over the mission (VO steps)	8.44 %	field	link	Sec. 3.1 Measuring Slip, Table 7 (Overall Mission...	2026-10-20	Mars grav field data, dunes. Ave mixed terr used most hazardous terrain, so is biased t
PLN-010	MSL Curiosity (6 aluminium wheels, 50 cm diameter, 40 cm width, rocke...	Gale crater, Mars; all terrain driven sols 1 to about 2488	not reported	empirical-field	average of maximum rover slip per sol	16.69 %	field	link	Sec. 3.1, Table 7	2026-10-20	Mars grav field data, dunes. Mi per sol ma values.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
PLN-011	MSL Curiosity (6 aluminium wheels, 50 cm diameter, 40 cm width, rocke...	Hidden Valley, Gale crater, Mars; polygonal ripple sands (sol 709)	not reported	empirical-field	wheel slip during drive through ripple sand	45 growing to 77.5 over seven steps %	field	link	Sec. 7.3 (drives that failed due to mobility faul...	2026-10-20	Mars grav field data, dunes. Ma megarippl sand; Curi 50 cm dia
PLN-012	MSL Curiosity (6 aluminium wheels, 50 cm diameter, 40 cm width, rocke...	Hidden Valley, Gale crater, Mars; polygonal ripple sands (sol 710)	not reported	empirical-field	actual motion vs commanded distance before slow slip fault	7.78 m actual of over 17.35 m commanded m	field	link	Sec. 7.3, Limit Cycle subsection, paragraph begin...	2026-10-20	Mars grav field data, dunes. Ma sand; drive a 70% Slo exceeded consecuti
PLN-013	MSL Curiosity (6 aluminium wheels, 50 cm diameter, 40 cm width, rocke...	Hidden Valley, Gale crater, Mars; polygonal ripple sands after sol 71...	not reported	empirical-field	wheel sinkage after high slip in ripple sand	approximately 9 (about 7 grousers embedded per wheel) cm	field	link	Sec. 7.3, Limit Cycle subsection, paragraph begin...	2026-10-20	Mars grav field data, dunes. Sin estimated images, nc sinkage se cm wheel about 18 p diamete...
PLN-014	MSL Curiosity (6 aluminium wheels, 50 cm diameter, 40 cm width, rocke...	Hidden Valley, Gale crater, Mars; ripple sand (sol 711)	not reported	empirical-field	turn in place slip (turn timed out)	40.4 %	field	link	Fig. 43 caption (Sec. 7.3)	2026-10-20	Mars grav field data, dunes. Tur rotational, longitudina
PLN-015	MSL Curiosity (6 aluminium wheels, 50 cm diameter, 40 cm width, rocke...	Gale crater, Mars; right wheels in sand, left wheels on firmer terrai...	not reported	empirical-field	wheel slip with split sand and firm terrain under left and...	78.3 %	field	link	Sec. 2 heading error discussion (text under Fig....	2026-10-20	Mars grav field data, dunes. Asy terrain und sides of th
PLN-016	MSL Curiosity (6 aluminium wheels, 50 cm diameter, 40 cm width, rocke...	Gale crater, Mars; operations rule for sandy terrain	not reported	other	distance between slip checks in Slip Check mode	20 m	none	link	Sec. 4.1.1, VO modes list (Slip Check)	2026-10-20	Mars grav field data, dunes. Op parameter a 50 cm w scale with and toleral
PLN-017	MSL Curiosity (6 aluminium wheels, 50 cm diameter, 40 cm width, rocke...	Gale crater, Mars; partial embedding during backward drive (sol 672)	not reported	empirical-field	outcome of autonomous embedding detection	drive halted; single sol recovery qualitative	field	link	Sec. 10 Risks to Continued Mobility Success, Whee...	2026-10-20	Mars grav field data, dunes. Det depends c Curiosity's thresholds qualitative

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
PLN-018	MER Spirit (6 wheels, rocker bogie; right front wheel inoperative, dr...	Troy, near Home Plate, Gusev crater, Mars; thin sulfate rich soil cru...	not reported (rover...	empirical-field	wheel slippage during embedding drives	almost 100 %	field	link	Sec. 5 Embedding at Troy, paragraphs [16] to [17]...	2026-10-20	Mars grav field data, dunes. Fiv drive with right front crusted su unlike dry
PLN-019	MER Spirit (6 wheels, rocker bogie; right front wheel inoperative, dr...	Troy, Gusev crater, Mars; Ulysses disturbed soils	not reported	other	resisting force needed relative to available thrust of five...	30% higher than the thrust available % of available thrust	sim-vs-field	link	Sec. 5, paragraph [18]	2026-10-20	Mars grav field data, dunes. Mo to field out wheel driv
PLN-020	MER Spirit (6 wheels, rocker bogie; right front wheel inoperative, dr...	Troy, Gusev crater, Mars; extrication attempts after sand box experim...	not reported	empirical-field	progress from 21 forward drives (sols 2078 to 2142)	only a few centimeters of lateral motion cm	field	link	Sec. 7 Troy Extrication Attempts, paragraph [30]	2026-10-20	Mars grav field data, dunes. Em vehicle wit actuators actuator fa 2104).
PLN-021	MER Spirit (6 wheels, rocker bogie; right front wheel inoperative, dr...	Troy, Gusev crater, Mars; extrication attempts	not reported	empirical-field	progress during last ten backward drives (sols 2145 to 2165)	34 cm	field	link	Sec. 7, paragraph [31]; Fig. 11 caption	2026-10-20	Mars grav field data, dunes. Fou drive by th embedded free drivin
PLN-022	MER Spirit (6 wheels, rocker bogie; right front wheel inoperative, dr...	Gusev crater, Mars; flat soil, five wheel drive with dragged right fr...	0 (level) for 500 N...	other	resisting force on the inoperative right front wheel (pushe...	up to 500 (forward); 75 (backward) N	simulation-only	link	Sec. 2 Five Wheel Drive Dynamics, paragraphs [5]...	2026-10-20	Mars grav field data, dunes. Mo calibrated terrestrial tests, not independe validated i
PLN-023	MER Opportunity (6 wheels, 179 kg as stated in the paper)	Meridiani Planum, Mars; western side of an aeolian ripple, sol 2220,...	~8 (rover tilt magn...	empirical-field	slip when climbing ripple flank backward at ~25 deg to cres...	~58 %	field	link	Sec. 5.2 Mobility Issues, paragraph [27]; Figs. 1...	2026-10-20	Mars grav field data, dunes. Ma aeolian rip basaltic sa hematite c low angle the crest p
PLN-024	MER Opportunity (6 wheels, 179 kg as stated in the paper)	Meridiani Planum, Mars; ripple flank of sol 2220 event	~8 (rover tilt)	empirical-field	wheel sinkage at high slip location	~5 cm	field	link	Sec. 5.2, paragraph [27]	2026-10-20	Mars grav field data, dunes. Wh diameter r this paper, ratio not g authors.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
PLN-025	MER Opportunity (6 wheels, 179 kg as stated in the paper)	Meridiani Planum, Mars; ripple soil, model of sol 2220 event on an 8...	8	classical-terrame...	terramechanics parameters that reproduce observed sinkage a...	n = 1.1; k_phi = 75,000; kx = 65 mm mixed (dimensionless; u...	sim-vs-field	link	Sec. 5.2, paragraph [28]; Eqs. 1 and 2	2026-10-20	Mars grav field data, dunes. Fitt parameter first order kc; k_phi u printed in passage.
PLN-026	MER Opportunity (6 wheels, 179 kg as stated in the paper)	Meridiani Planum, Mars; Jammerbugt ripple	not reported	empirical-field	duration of extrication from ripple embedding	six sols (backed out on sol 841) sols	field	link	Sec. 3.2 Overview of Mission Activities, paragrap...	2026-10-20	Mars grav field data, dunes. Du reflects op cadence c mission, n time.
PLN-027	MER Opportunity (6 wheels, rocker bogie, straight line speed 3.75 cm/...	Eagle crater, Meridiani Planum, Mars (about 20 m across, 2 m deep); f...	not reported (egres...	empirical-field	slip during a 12.5 m segment driving straight uphill out of...	100 %	field	link	Sec. 2 Sols 1 to 60: Eagle Crater, final paragrap...	2026-10-20	Mars grav field data, dunes. Lo sand in a s slope for t not stated
PLN-028	MER Opportunity	Eagle crater floor, Mars; loose fine sand during wheel trenching	<5 (tilt)	empirical-field	rover sliding while trenching with one wheel	25 cm	field	link	Sec. 2, paragraph beginning The last observations...	2026-10-20	Mars grav field data, dunes. Tre maneuver, driving.
PLN-029	MER Opportunity	Eagle crater and Endurance crater, Mars; loose sandy slopes (Enduranc...	~20 (Endurance soft...	empirical-field	most effective heading on loose sandy slopes	roughly 45 deg to the upslope vector deg	field	link	Sec. 5 Sols 132 to 315: Inside Endurance, paragra...	2026-10-20	Mars grav field data, dunes. Qu operationa no slip nur for the dia strategy.
PLN-030	Rocky8 research rover (rocker bogie)	Sandy slopes in the Mojave desert (Earth analog field site)	-5 to 10 pitch; up...	learning	slip prediction error on sandy slopes	within 5%-7% of the traversed distance per step for a slip...	field	link	Sec. VI Experimental Results, paragraph beginning...	2026-10-20	Earth field on natural slopes; sm (about 10 examples) heavier an than a sma rover.
PLN-031	LAGR research robot (differential drive)	Flat sandy terrain in a park (Earth)	0 (level)	empirical-field	consistent slip on flat deep sand	about 80 %	field	link	Sec. VI Experimental Results, paragraph on LAGR d...	2026-10-20	Earth, 1 g; vehicle wh mass not g paper; dee park, not a
PLN-032	LAGR research robot	Soil, gravel transverse slope, flat sand, asphalt, woodchip (Earth)	various (from stere...	learning	average slip prediction error across datasets	about 3%-15% (gravel: 27% MSE, 16% absolute error) %	field	link	Sec. VI Experimental Results, end of paragraph on...	2026-10-20	Earth, 1 g; measured whose 1 s is 12% pe

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
PLN-033	not named in source (stepwise run, likely Rocky8 but not stated)	not reported (7.5 m test run, 80 frames, 0.1 m steps)	not reported	empirical-field	1 sigma VO error per step used as slip ground truth	12 % per step	none	link	Sec. II.C Visual Odometry validation (p. 3326) an...	2026-10-20	Measurem bound for Earth test.
PLN-034	MSL Curiosity (6 wheels, one wheel diameter 0.50 m)	Gale crater, Mars; route planning to Mount Sharp (Rapid Transit Route)	>12.5 (sandy) and >...	other	maximum sandy slope allowed in route hazard maps	12.5 deg	none	link	Sec. 2 First 360 Sols: Mission and Mobility Overv...	2026-10-20	Mars grav field data, dunes. Pla threshold, measured
PLN-035	MSL Curiosity (6 wheels, one wheel diameter 0.50 m)	Gale crater, Mars; drives during first 360 sols including approach to...	not reported	empirical-field	wheel sinkage during first 360 sols	~1 cm or less cm	field	link	Sec. 7 Modeling Curiosity Mobility, paragraph aft...	2026-10-20	Mars grav field data, dunes. Ha soils and b dominatec driving.
PLN-036	Yutu-2 (Chang'e-4 lunar rover, grouser wheels)	Longji site (45.44 S, 177.56 E), lunar farside, 41st lunar day traver...	up to ~8.86 along r...	empirical-field	wheel sinkage along traverse	about 8 on average; no larger than 15 mm	field	link	Results, paragraph on bearing characteristics; Fi...	2026-10-20	Lunar grav and lunar r Earth dune
PLN-037	Yutu-2 (Chang'e-4 lunar rover, grouser wheels)	Longji site, lunar farside	up to ~8.86	classical-terrame...	internal friction angle and cohesion of regolith (inferred)	21.5 to 42.0 deg; 520 to 3154 Pa deg; Pa	sim-vs-field	link	Abstract; Results paragraph on shear property; Fi...	2026-10-20	Lunar grav and lunar r Earth dune Inferred, m dependen
PLN-038	Yutu-2 (Chang'e-4 lunar rover, grouser wheels)	Longji site, lunar farside; descending route toward crater wall platf...	up to ~8.86 (5.38 a...	empirical-field	wheel slip ratio range	mostly skid (s < 0); s almost no less than -0.1 slip ratio...	field	link	Results paragraph on shear property (rover mostly...	2026-10-20	Lunar grav and lunar r Earth dune Downhill tr negative s dominates
PLN-039	Yutu-2 (Chang'e-4 lunar rover, grouser wheels)	Longji site, lunar farside; confined route between craters of 6.1 m a...	up to ~8.86	classical-terrame...	arrival position error vs planned destination	0.367 (actual); predicted endpoints within 0.762 m	sim-vs-field	link	Results, paragraphs on outbound traverse predicti...	2026-10-20	Lunar grav and lunar r Earth dune Short trave 11 m strai distance).
PLN-040	Yutu-2 (Chang'e-4 lunar rover, grouser wheels)	Longji site, lunar farside	up to ~8.86	classical-terrame...	external friction angle (lateral property of regolith, infe...	8.3 to 16.5 deg	sim-vs-field	link	Abstract; Results paragraph on lateral properties	2026-10-20	Lunar grav and lunar r Earth dune Inferred, m dependen
PLN-041	not applicable (Curiosity sampling at Namib dune, Gobabeb site)	Bagnold dune field, Gale crater, Mars; active barchan dunes Namib and...	not reported	empirical-field	grain size range of active dune sand	~45 to 500 um	none	link	Abstract; Sec. 3.1 Grain Scale Properties and Tab...	2026-10-20	Mars grav field data, dunes. Ba finer and c many Eart sands; pro measurem

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
PLN-042	not applicable (Curiosity sampling at Namib dune)	Namib dune, Gobabeb site, Bagnold dune field, Mars	not reported	empirical-field	maximum grain size at sampling site	typically 400 to 600 (grains larger than 1 mm typically abs...	none	link	Sec. 3.1 Grain Scale Properties; Table 2; Sec. 2....	2026-10-20	Mars grav field data, dunes. Pro measurem
PLN-043	MSL Curiosity wheel track (Weissrand target near High dune)	Wheel disturbed sand in rover tracks near High dune vs undisturbed ri...	not reported	empirical-field	grain size range inside rover wheel tracks	40 to 200 (all grains fine sand and smaller; some below res...	none	link	Sec. 3.1; Table 2 row 1182_Weissrand	2026-10-20	Mars grav field data, dunes. Gra from imag limited res property measurem
PLN-044	MSL Curiosity (wheel width approximately 40 cm)	Margins of Namib and High barchan dunes, Mars	not reported	other	driving onto the dunes with all wheels	precluded by vehicle safety considerations qualitative	none	link	Sec. 2.1 Measurement Locations, Strategy, and Sam...	2026-10-20	Mars grav field data, dunes. Op policy, not result.
PLN-045	DLR Lightweight Rover Unit (LRU), four wheeled planetary rover protot...	Mt. Etna (Cisternazza crater area), relatively new loose volcanic soi...	calibration area ~5...	empirical-field	calibrated six parameter slip model: p1 (slope dependent lo...	p1 0.6367; p2 0.1089; p3 0.2465 dimensionless	field	link	Sec. III.A Slip Model Calibration; Fig. 6	2026-10-20	Earth 1 g v soil, not du four whee calibration loosened l traffic.
PLN-046	DLR Lightweight Rover Unit (LRU), four wheeled planetary rover protot...	Mt. Etna volcanic soil; long range teleoperated drive of approx. 900...	up to 18 (site)	empirical-field	relative odometry error after slip calibration	median approx. -8% (calibrated); median below 1% with 90% o...	field	link	Sec. III.B Validation During a Long Range Drive;...	2026-10-20	Earth 1 g v soil; error a hand tun parameter DGPS kno
PLN-047	Apollo 15 Lunar Roving Vehicle (wire mesh wheels with chevron tread)	Hadley-Apennine region, Moon (Apollo 15 traverses, 3 EVAs)	0 to 12 (operated)	empirical-field	wheel slip bias used for navigation	2.3 %	field	link	LRV Mobility Performance at Hadley-Apennine secti...	2026-10-20	Lunar grav and lunar r Earth dune mesh whe vehicle; m regolith.
PLN-048	Apollo 15 Lunar Roving Vehicle (wire mesh wheels with chevron tread)	Hadley-Apennine region, Moon	0 to 12	empirical-field	wheel sinkage	average of the order of 1.25 at 2.3% slip; ranged from impe...	field	link	printed p. 54	2026-10-20	Lunar grav and lunar r Earth dune
PLN-049	Apollo 15 Lunar Roving Vehicle (wire mesh wheels with chevron tread)	Station 8 near ALSEP site, Hadley-Apennine, Moon; locally loose soil	not reported	empirical-field	wheel dig in depth during wheel spinout at high slip	approximately 13 (down to lower part of wheel rim) cm	field	link	printed p. 54	2026-10-20	Lunar grav and lunar r Earth dune Loosening due to astr activity at
PLN-050	Apollo 15 Lunar Roving Vehicle (wire mesh wheels with chevron tread)	Hadley-Apennine region, Moon; Apennine Front near Station 6	0 to 12 operated; 1...	empirical-field	fraction of estimated maximum slope climbing capability used	at most 62 %	field	link	printed p. 55	2026-10-20	Lunar grav and lunar r Earth dune

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
PLN-051	LRV wire mesh wheel (WES single wheel tests)	Lunar soil simulants LSS1 to LSS5, laboratory	0 (level) and slopi...	empirical	wheel slip at self propelled point on level soft soil; maxi...	2 to 5 % slip; 18 to 23 deg % ; deg	lab	link	Conclusions of WES wheel soil interaction studies...	2026-10-20	Earth 1 g t lunar soil s interprete loads.
PLN-052	MSL Curiosity (6 wheels, 0.5 m diameter)	Hidden Valley, Gale crater, Mars; sand (first sol 709)	max 7.4	empirical-field	maximum slip in sand at site	80.7 %	field	link	Sec. V; Table I	2026-10-20	Mars grav field data, dunes. Sa Hidden Va appeared thermal in
PLN-053	MSL Curiosity (6 wheels, 0.5 m diameter)	Moosilauke Valley, Gale crater, Mars; small ripple field (first sol 6...	max 3.5	empirical-field	maximum slip in sand at low slope	79.7 %	field	link	Sec. V; Table I	2026-10-20	Mars grav field data, dunes. On samples a
PLN-054	MSL Curiosity (6 wheels, 0.5 m diameter)	Dingo Gap, Gale crater, Mars; sand (first sol 528)	max 13.9	empirical-field	maximum slip in sand at site	46.4 %	field	link	Sec. V; Table I	2026-10-20	Mars grav field data, dunes. 13
PLN-055	MSL Curiosity (6 wheels, 0.5 m diameter)	Nine sand sites, Gale crater, Mars (135 samples, sols 1 to 986)	max slope per site...	learning	slip prediction RMSE in leave one site out cross validation...	30.3 (slope only); 22.7 (two expert THEMIS); 20.4 (two expe...	field	link	Sec. VI; Table II	2026-10-20	Mars grav field data, dunes. On sites; mod (best fit RI 15.9, 16.8
PLN-056	SSTB-Lite (MER surface system test bed, 3/8 mass: 65.5 kg vs 176.5 kg...	Dumont Dunes, Mojave Desert, California (Earth analog, May 2012); int...	2 to 12	empirical-field	wheel slip vs dune face slope (uphill)	approximately 0.3 at 2 to 4 deg rising to approximately 0.9...	field	link	Sec. 4.2 Deformable Soil Rover Experiments at the...	2026-10-20	Earth 1 g v mass rove Mars whe dune sand moisture a model); va figure rea large scatt
PLN-057	SSTB-Lite (MER surface system test bed, 3/8 mass: 65.5 kg vs 176.5 kg...	Dumont Dunes, Mojave Desert, California (Earth analog, May 2012); int...	2 to 12	empirical-field	slip measurement error of cleat imprint method	~0.6 %	field	link	Sec. 4.2, paragraph on slip determination; Eq. 24...	2026-10-20	Method ac a wheel w cleat mark transferab tests with wheels.
PLN-058	SSTB-Lite (MER surface system test bed, 3/8 mass: 65.5 kg vs 176.5 kg...	Dumont Dunes, Mojave Desert, California (Earth analog, May 2012); int...	2 to 12	classical-terrame...	shear deformation modulus kx of dune sand (tuned)	29 mm	sim-vs-field	link	Sec. 4.2; Table II (Soil parameters for modeling...	2026-10-20	Earth dune and phi nc but taken literature f dune sand moisture; drawbar d constrain

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
PLN-059	MER Opportunity	Meridiani Planum, Mars; sol 2143 crossing of a 5 m wide, 0.4 m high w...	not reported (pitch...	classical-terrame...	shear deformation modulus kx of Martian ripple sand (tuned)	25 mm	sim-vs-field	link	Sec. 4.3 Simulation of Opportunity's Ripple Cross...	2026-10-20	Mars grav field data, dunes. Tu bedrock fl used kx 1! mm.
PLN-060	Mars 2020 rover (MSL heritage mobility system)	Candidate Mars 2020 landing sites; large scale sand dunes vs bedrock...	~10 (sand dunes); ~...	other	maximum slope the rover can traverse up on large scale sand...	approximately 10 deg	none	link	Sec. 2 Problem Description, M2020 Rover Capabilit...	2026-10-20	Mars grav planning a for a 50 cr rover; unc sentence, derived fr programm et al. cited
PLN-061	not applicable (orbital terrain classifier)	HiRISE images of candidate Mars 2020 landing sites; 11 terrain classe...	not applicable	learning	per pixel terrain classification performance	94.9 %	none	link	Sec. 4 Automated Terrain Classification (p. 4)	2026-10-20	Orbital ima Mars at Hi resolution; measured labels only note true performan unlabeled to be char
PLN-062	Yutu (Chang'E-3 lunar rover, 6 wheels, 135 kg, 22.50 kg per wheel, wh...	Chang'E-3 landing site, Mare Imbrium (Sinus Iridum), Moon; track segm...	not reported	empirical-field	wheel sinkage from track images	0.56 to 1.15 cm	field	link	Sec. 2.2 (modern method), Table 2 (manuscript p....	2026-10-20	Lunar grav and lunar r Earth dune Single sho segment; photoclinc depends c illuminatio assumed r
PLN-063	Lunokhod-1 (Luna 17, 94.50 kg per wheel, wheel diameter 51 cm)	Luna 17 landing site, Moon; track segment of 2 to 3 m	not reported	empirical-field	wheel sinkage from track images	1.07 to 2.54 cm	field	link	Sec. 2.2, Table 2 (manuscript p. 10)	2026-10-20	Lunar grav and lunar r Earth dune Archive im analysis d after the tr
PLN-064	Yutu, Lunokhod-1 and Apollo 15 LRV (comparison)	Chang'E-3, Luna 17 and Apollo 15 landing sites, Moon	not reported	classical-terrame...	median soil stiffness kD^0.39	0.55 (CE-3); 0.5 (Luna 17); 2.45 (Apollo 15) N/cm^2.8	field	link	Sec. 2.2 and 2.3, Table 2 and Fig. 3	2026-10-20	Lunar grav and lunar r Earth dune Depends c assumed r penetrome n was 0.5 Luna 17 a 15.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
PLN-065	Mars 2020 Perseverance rover	Jezero crater, Mars; crossing westward over a north to south oriented...	not reported	empirical-field	slip that stopped a planned 200 m drive after 20 m	65.5 %	field	link	Sec. 3 Rover Mobility, The Next Year of Driving s...	2026-10-20	Mars grav field data, dunes. Ma ripple; san been cros 428 with r slip.
PLN-066	Mars 2020 Perseverance rover	Same sandy ripple, Jezero crater, Mars (sol 551 second attempt)	not reported	empirical-field	average slow slip that faulted the drive	78% over a 1.5 m distance %	field	link	Sec. 3 Rover Mobility, The Next Year of Driving s...	2026-10-20	Mars grav field data, dunes.
PLN-067	Mars 2020 Perseverance rover	Same sandy ripple, Jezero crater, Mars (sol 555 third attempt)	not reported	empirical-field	peak slip while successfully crossing the ripple	78 %	field	link	Sec. 3 Rover Mobility, The Next Year of Driving s...	2026-10-20	Mars grav field data, dunes.
PLN-068	Mars 2020 Perseverance rover	All drives covered by the paper (first year and a half of Mars 2020 o...	not reported	empirical-field	number of excessive VO slip faults, all in sand	3 count	field	link	Sec. 3, Mobility Faults, Unexpected Drive Faults...	2026-10-20	Mars grav field data, dunes. Sm fault thres set per dri operators 60%).
PLN-069	Mars 2020 Perseverance rover	Route choice around Seitah sand ripples, Jezero crater, Mars	not reported	other	expected drive rate through sand without Autonav relative t...	four times slower ratio (qualitative estimate)	none	link	Sec. 3, paragraph on route options after sol 362	2026-10-20	Mars grav field data, dunes. Pla expectatio measured
PLN-070	Mars 2020 Perseverance rover	Crater rim climb, Jezero crater, Mars (sols 1251 to 1264); high tilt...	high tilt (Autonav...	empirical-field	drives faulted early due to excessive slip	4 out of 5 count	field	link	Sec. 3 Mobility Challenges, High Slopes subsection	2026-10-20	Mars grav field data, dunes. Slo per drive r combinati pitch up d loose sanc
PLN-071	Mars 2020 Perseverance rover	Complex terrain including tall sand ripples in a river valley, Jezero...	not reported	empirical-field	mean drive distance per sol in ripple laden complex terrain...	42.89 (sd 22.63) for 35 human planned drives vs 129.59 (sd...	field	link	Sec. 3 Mobility Challenges, paragraph on Autonav...	2026-10-20	Mars grav field data, dunes. Slo due to Aut to find saf complex te only to sar

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
LEG-001	SandBot RHex-class hexapod (total mass 2.3 kg; c-legs R=3.55 cm, hip...	Lab fluidized-bed trackway of ~1 mm poppy seeds, volume fraction 0.60...	0 (level)	RFT-granular	mean forward speed	2 cm/s	lab	link	Sec. 3 (Methods and Results), Fig. 2c caption (HG...	2026-10-11	Poppy see fluidized b tethered-p hexapod; s size (1 mn packing c not repres dune sand normal...
LEG-002	SandBot RHex-class hexapod (2.3 kg), soft-ground clock (SGK)	Lab fluidized-bed trackway of ~1 mm poppy seeds, volume fraction 0.605	0 (level)	RFT-granular	mean forward speed	8 cm/s	lab	link	Fig. 2c caption (SGK ~8 cm/s); Sec. 3 states opti...	2026-10-11	As LEG-0 parameter hand-tune substrate packing.
LEG-003	SandBot hexapod (2.3 kg), tuned rotary-walking gait, leg frequency 16...	Lab fluidized-bed trackway of ~1 mm poppy seeds, volume fraction 0.633	0 (level)	other	mean forward speed	20 cm/s	lab	link	Results, text around Fig. 1d,e (vx ~20 cm/s at ph...	2026-10-11	Model gra medium; p fraction c available i sand; hexa 2.3 kg.
LEG-004	SandBot hexapod (2.3 kg), same gait, leg frequency 16 rad/s, loosely...	Lab fluidized-bed trackway of ~1 mm poppy seeds, volume fraction 0.600	0 (level)	other	mean forward speed	2 cm/s	lab	link	Results, text around Fig. 1d,e (vx ~2 cm/s at phi...	2026-10-11	As LEG-0 change in fraction ch speed ~5- packing-s
LEG-005	DASH hexapod (22.1 g, ~10 cm), sprawled posture	Lab fluidized-bed trackway of poppy seeds, loosely packed, volume fra...	0 (level)	other	mean forward speed (normalized)	5 body lengths/s	lab	link	Sec. 4 Results, paragraph after Fig. 7 (about 5 b...	2026-10-11	~22 g, 10 thrust-to- higher tha vehicle; p not dune s not compa m/s of larg
LEG-006	RoACH hexapod (24.9 g, ~10 cm), upright compliant legs	Lab fluidized-bed trackway of poppy seeds, loosely packed, volume fra...	0 (level)	other	mean forward speed	5 cm/s	lab	link	Sec. 4 Results (RoACH loses >90% of performance,...	2026-10-11	~25 g robo seeds not
LEG-007	Hexapod with C-shaped legs (350 g, 22 cm x 10 cm; leg radius 4 cm, wi...	Lab tiltable fluidized bed of 300 um glass beads, loosely packed, dep...	0	RFT-granular	step-averaged speed parallel to surface	15.7 cm/s	lab	link	Sec. III-A (v = 15.7 +/- 0.97 cm/s at 0 deg)	2026-10-11	350 g robo beads in 1 bed; slope of uniform bed, not n with variab compactio max 24 de an...
LEG-008	Same 350 g hexapod	Same tiltable glass-bead bed (300 um)	10	RFT-granular	step-averaged speed parallel to surface	11.0 cm/s	lab	link	Sec. III-A (v = 11.0 +/- 0.14 cm/s at 10 deg; 10....	2026-10-11	As LEG-0

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
LEG-009	Same 350 g hexapod	Same tiltable glass-bead bed (300 um), near angle of repose	24	RFT-granular	step-averaged speed parallel to surface	5.6 cm/s	lab	link	Sec. III-A (v = 5.6 +/- 1.6 cm/s at 24 deg; 4.98...	2026-10-11	As LEG-0 is at the be of repose, extrapolab
LEG-010	SpaceBok quadruped (22 kg, hip height 0.5 m), point feet	Lab Mars-analog soil testbed, 6×6 m, 20 t of ES-3 (coarse, poorly-sor...	0-25 (ascent/descen...	other	maximum foot sinkage observed	120 mm	lab	link	Sec. 4.2.1 (point feet sank up to 120 mm into the...	2026-10-11	Simulant E Mars-ana lab bin, no dune sand legged rob hand-tune sinkage re qualitative maximum.
LEG-011	SpaceBok planar adaptive foot, 110 cm2, with 12 mm grousers (single-f...	Lab single-foot traction test on ES-3 simulant	not reported for tr...	other	traction coefficient increase with grousers vs same foot wi...	22 percent	lab	link	Sec. 3.3.2 (traction coefficient raised by 27% on...	2026-10-11	Foot-pad a legged f wheel grou single-foo full vehicle
LEG-012	SpaceBok quadruped (22 kg), dynamic (trot) vs static walk, planar and...	Lab Mars-analog ES-3 slope testbed	0-25	other	energy per distance, trot relative to static walk	>2 (trot more efficient by more than 2x) ratio	lab	link	Sec. 4.2.3, text before Fig. 15 (dynamic gait mor...	2026-10-11	Ratio only; cost of tra extracted. tuned gait
LEG-013	SpaceBok quadruped (22 kg), planar and point feet, static and trot	Lab Mars-analog ES-3 slope testbed (max tilt 25 deg)	25	other	maximum slope traversed (testbed limit)	25 deg	lab	link	Sec. 4.2.1 and Table 2 (up to 20 deg no significa...	2026-10-11	25 deg is t maximum, failure limi Qualitative scoring (p numerical rate not re Slope was prepa...
LEG-014	Single-leg hopper (2 kg robot, direct-drive) with active damping cont...	Lab sandbox of 3.4 mm glass beads, 0.16 m deep, controlled compaction	0 (vertical jump)	other	reduction in energy per jump vs compression-extension contr...	15 percent (10-15 range across foot size and stiffness)	lab	link	Sec. 3.2 / Table 4 (savings 14, 15, 10, 11% acros...	2026-10-11	Single hop rail in glas jump ener horizontal transport; glass beac sand.
LEG-015	Unitree G1 humanoid (hardware), RFT-trained RL policy vs rigid-contac...	Beach volleyball court dry sand and basalt, outdoor hardware tests	0 (level; slope not...	learning	highest commanded speed tested (2.5 m/s) at which RFT-train...	2.5 m/s	field	link	Sec. V-B (Results, Hardware Experiments), Fig. 7	2026-10-11	Humanoid command measured success d staying up without hu sand untes slope; num success ra extract...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
LEG-016	Four-module wheel-paddle (wheel-leg hybrid) robot, mass 24 kg, hybrid...	Outdoor sand field at Ritsumeikan University BKC, Shiga, Japan (Sand...	0 (level; slope not...	other	traversing speed approaching 1 body length/s when wheel joi...	1 body lengths/s (approx.)	field	link	Sec. 4.4.2 text around Fig. 4.15; sand-least-effi...	2026-10-11	Single test inhomoge body leng extracted, robot; whe hybrid wit paddles (r only).
LEG-017	Four inline rigid lugged wheels, D=116 mm (lug 5 mm, 24 lugs, width 1...	Lab sand box of Toyoura sand	12 (equivalent slop...	other	slip ratio	0.741 -	lab	link	Slide 38 ('Wheels with different dimensions' and...	2026-10-11	Slope is er horizontal actual incl vertical loa stated on slides; slid peer-revie
LEG-018	Four inline rigid lugged wheels, D=327 mm (lug 15 mm, 24 lugs, width...	Lab sand box of Toyoura sand	12 (equivalent slop...	other	slip ratio	0.056 -	lab	link	Slide 38	2026-10-11	As LEG-0 Diameter a height var together, s are not sep
LEG-019	Four inline rigid lugged wheels, D=202 mm, width 50 mm	Lab sand box of Toyoura sand	14 (equivalent slop...	other	slip ratio	0.471 -	lab	link	Slide 40 (width series at D=200 mm)	2026-10-11	As LEG-0 emulated.
LEG-020	Four inline rigid lugged wheels, D=202 mm, width 150 mm	Lab sand box of Toyoura sand	14 (equivalent slop...	other	slip ratio	0.247 -	lab	link	Slide 40	2026-10-11	As LEG-0 emulated.
LEG-021	Rigid grousered wheel, radius 114 mm, width 57 mm, 13 mm grousers, 20...	Lab glass-walled soil bin with GRC-1 lunar simulant (not dune sand)	0 (level)	classical-terrame...	number of grousers beyond which drawbar pull stops improving	24 count	lab	link	Sec. IV and Fig. 11, Table II (13 mm: 24 predicte...	2026-10-11	Lunar simu wheel at fi slip, whee scale.
LEG-022	Flexible metallic grousered wheel vs rigid wheel, both D=250 mm, W=11...	Lab soil bin with olivine sand RMCS14	0 (level)	other	drawbar pull and tractive efficiency of flexible vs rigid w...	>2 (over two times higher) ratio	lab	link	Abstract and Sec. 4 discussion (flexible wheel at...	2026-10-11	Single-wh at 20% sli load; flexit a metallic not a non- tire for a h vehicle.
LEG-023	OriWheelBot origami-wheeled robot (1348.5 g), wheel width 22 mm vs 72...	Lab soft sand bed with subsidence pit (starts 30 mm sunk)	0 (level)	other	distance travelled in 8 s after starting sunk in sand, narr...	60 mm	lab	link	Sec. on soft sand walking (dpt about 60, 130, 330...	2026-10-11	Small 1.3 k with defor origami wh uncharact start-from not steady
LEG-024	OriWheelBot (1348.5 g), wheel width 72 mm vs 22/38/52 mm	Lab sand slope, about 17 deg	17	other	climbs 17 deg sand slope (success only at widest wheel)	success at 72 mm; failure at 22, 38, 52 mm pass/fail	lab	link	Sec. on slope climbing, Fig. 5b caption	2026-10-11	One slope trial class, result; san uncharact

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MOB-001	not applicable (soil characterisation)	Fine sand from the desert near Yuma, Arizona, used in WES soil bins	not applicable	other	median grain size d50 of Yuma sand	0.12 mm	none	link	Fig. 1 (printed p. 4, PDF p. 15), Soil Properties...	2026-10-20	Single US sand; grac uniform (C fine; Arabi sands hav grading ar measured
MOB-002	not applicable (soil characterisation)	Yuma sand, soil condition S1 (air-dry, loose), before traffic	not applicable	other	average friction angle, soil condition S1, by test method (...	37.1; 30.0; 17.1; 27.4 deg	none	link	Table 1, Page 1 of 5 (PDF p. 65), rows Average Fr...	2026-10-20	Lab prepa loose sanc spread sh friction an any model strongly o method; v not interch
MOB-003	not applicable (soil characterisation)	Yuma sand, soil condition S1 (air-dry, loose)	not applicable	classical-terrame...	Bekker soil values, average (kc; kphi; n)	-0.01; 11.41; 0.91 kc (kN/m)(cm^-n); kphi (kN/m2)(cm^-n); n...	none	link	Table 1, Page 1 of 5 (PDF p. 65), rows Bekker Soi...	2026-10-20	Plate sizes as printed and cm wi in bracket zero and r average re cohesionl parameter
MOB-004	Single lunar prototype wheels (pneumatic, Bendix, Boeing-GM, Grumman,...	Yuma sand soil bins, five strength levels	0 (level)	empirical	sand mobility number definitions as printed (N1; N2)	N1 = (G b d^2 / W)(1 - 2 delta/d)^-8 ; N2 = G/W * Ac^3/2 di...	lab	link	Notation, printed p. ix (PDF p. 7)	2026-10-20	Laboratory Yuma (Ari desert dur g. Wheel l 670 N anc prototype (about 81 diameter) larger thar
MOB-005	Pneumatic wheel, single wheel tests (load range 67 to 670 N)	Yuma sand, soil condition S1: phi 37.1 deg, cohesion 0.0 kN/m2, Dr 32...	0 (level)	empirical	pull coefficient at 20 percent slip, P20/W (average over lo...	0.448 dimensionless	lab	link	Summary, printed p. xii, tabulation of P20/W, PNs...	2026-10-20	Laboratory Yuma (Ari desert dur g. Wheel l 670 N anc prototype (about 81 diameter) larger thar
MOB-006	Boeing-GM I wire mesh wheel, single wheel tests	Yuma sand, soil condition S1 (air-dry, loose)	0 (level)	empirical	pull coefficient at 20 percent slip, P20/W (average over lo...	0.274 dimensionless	lab	link	Summary, printed p. xii, tabulation	2026-10-20	Laboratory Yuma (Ari desert dur g. Wheel l 670 N anc prototype (about 81 diameter) larger thar

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MOB-007	Bendix, Boeing-GM and Grumman wheels, single wheel at assumed 220 N l...	Loose air-dry Yuma sand, level surface	0 (level)	empirical	power consumption rate (Bendix; Boeing-GM; Grumman)	4; 6; 10 whr/km	lab	link	Summary printed p. xii; Conclusions para. 44d (pr...	2026-10-20	Laboratory Yuma (Ari: desert dur g. Wheel l 670 N anc prototype (about 81 diameter) larger thar
MOB-008	Original Bendix, Boeing-GM and Grumman lunar wheels	Yuma sand, level single wheel tests in soil bins (estimate, not slope...	35	empirical	maximum slope that could be relied on (none reached 35 deg)...	Bendix 28 to 30; Boeing-GM and Grumman 15 to 20 deg	lab	link	Summary printed p. xii; Conclusions para. 44d (pr...	2026-10-20	Laboratory Yuma (Ari: desert dur g. Wheel l 670 N anc prototype (about 81 diameter) larger thar
MOB-009	4×4 vehicle (310 N wheel load) and 6×6 SLRV vehicle, compared with pn...	Yuma sand slopes, wet sand (C2) shown	0 to 35	empirical	bias of slope climbing prediction from single wheel level p...	about 1 to 2 deg	lab	link	Summary printed p. xiii; paras. 36 and 37 (PDF p....	2026-10-20	Laboratory Yuma (Ari: desert dur g. Wheel l 670 N anc prototype (about 81 diameter) larger thar
MOB-010	6×6 SLRV vehicle	Wet Yuma sand slopes in soil bins	0, 15 and 25 tested...	empirical	maximum slope actually climbed relative to prediction from...	approximately 3 deg (less steep than predicted)	lab	link	Para. 40 (printed p. 43, PDF p. 54)	2026-10-20	Laboratory Yuma (Ari: desert dur g. Wheel l 670 N anc prototype (about 81 diameter) larger thar
MOB-011	4×4 and 6×6 vehicles	Yuma sand slopes in soil bins	not reported	empirical	reduction in ultimate slope climbing ability when steering...	1 to 2 deg	lab	link	Para. 43 (PDF pp. 54 and 57); Conclusions para. 4...	2026-10-20	Laboratory Yuma (Ari: desert dur g. Wheel l 670 N anc prototype (about 81 diameter) larger thar

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MOB-012	4×4 vehicle, pneumatic wheels at 3.03 kN/m2 inflation pressure	Yuma sand, level and 25 deg slope	25	empirical	load transferred from front to rear axle on a 25 deg slope	approximately 20 percent	lab	link	Para. 39 (PDF pp. 51 and 54)	2026-10-20	Laboratory Yuma (Ari: desert dur g. Wheel l 670 N anc prototype (about 81 diameter) larger thar
MOB-013	Single lunar wheels, loads 67 to 670 N	Yuma sand, all five soil conditions	0 (level)	empirical	wheel load below which pull coefficient did not vary with l...	about 220 N	lab	link	Summary printed p. xi; Conclusions para. 44a and...	2026-10-20	Laboratory Yuma (Ari: desert dur g. Wheel l 670 N anc prototype (about 81 diameter) larger thar
MOB-014	Pneumatic tires (Freitag 1965; Green 1967) used as point of departure...	Yuma sand soil bins	0 (level)	empirical	functional relation for pull coefficient in sand as printed	P/W = f(G(bd)^3/2 / W * delta/h) dimensionless	lab	link	Para. 137 (printed p. 113, PDF p. 129)	2026-10-20	Laboratory Yuma (Ari: desert dur g. Wheel l 670 N anc prototype (about 81 diameter) larger thar
MOB-015	Pneumatic, Bendix, Boeing-GM, Grumman, SLRV wheels	Air-dry and damp Yuma sand	0 (level)	empirical	goodness of fit statistic reported for the sand mobility nu...	not reported (visual line of best fit only) not applicable	lab	link	Para. 137 to 138 (printed pp. 113 to 116); Figs....	2026-10-20	Laboratory Yuma (Ari: desert dur g. Wheel l 670 N anc prototype (about 81 diameter) larger thar
MOB-016	Modified Bendix and Grumman wheels with grousers; Boeing-GM wheel wit...	Yuma sand	0 (level)	empirical	position of grousered wheel data relative to the sand mobil...	above the trend qualitative	lab	link	Para. 139 (printed p. 116, PDF p. 132); Fig. 66b	2026-10-20	Laboratory Yuma (Ari: desert dur g. Wheel l 670 N anc prototype (about 81 diameter) larger thar
MOB-017	Boeing-GM 50 percent chevron covered wire mesh LRV wheel (81 cm diame...	Lunar soil simulant: crushed basalt (not sand), soil bin	not reported	empirical	estimated maximum slope climbing capability	of the order of 20 deg	lab	link	Conclusions para. 42e; rule stated in para. 29	2026-10-20	Crushed b simulant, r sand; 1 g s lunar whe only for th from 20 p rule.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MOB-018	Three vehicle types (AMC-71 verification)	Five test sites (not described as sand specifically)	not reported	empirical-field	share of AMC-71 predictions considered correct	about 70 percent	sim-vs-field	link	Section 3.1 (printed p. 23 area; text line beginn...	2026-10-20	Secondary citing Schi Willoughb terrain mix to sand; fu military ve
MOB-019	NRMM (wheeled and tracked military vehicles)	General cross country terrain	not applicable	empirical	stated limitations of NRMM	qualitative not applicable	none	link	Executive summary (printed p. iii) and Section 1....	2026-10-20	Review sta full size ve numbers.
MOB-020	Several military wheeled and tracked vehicles	Traverse test courses (mixed terrain)	not reported	empirical	overall relative deviation (prediction error) of traverse s...	30.1 percent	sim-vs-field	link	Abstract	2026-10-20	Full size m vehicles (M M35A2, M M60 amor US test co g; coarse validation mostly silt (SM) while
MOB-021	Military wheeled and tracked vehicles (M151, M35A2, M113A1, M60; all...	Coarse grained soils, mostly silty sands (SM)	0 (level)	empirical	weighted average relative deviation of drawbar pull, coarse...	17.9 percent	sim-vs-field	link	Para. 45	2026-10-20	Full size m vehicles (M M35A2, M M60 amor US test co g; coarse validation mostly silt (SM) while
MOB-022	Military vehicles (coarse grained soil tests)	Coarse grained soils, mostly silty sands (SM)	0 (level)	empirical	weighted average relative deviation of motion resistance, c...	32 percent	sim-vs-field	link	Para. 47	2026-10-20	Full size m vehicles (M M35A2, M M60 amor US test co g; coarse validation mostly silt (SM) while
MOB-023	M113A1 tracked vehicle (and M60 tank)	Sand slopes	40 percent (M113A1...	empirical	deviation of predicted from measured maximum negotiable slo...	73; 109 percent	sim-vs-field	link	Paras. 52 and 53	2026-10-20	Full size m vehicles (M M35A2, M M60 amor US test co g; coarse validation mostly silt (SM) while

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MOB-024	M35A2 wheeled truck	Soil slopes	not reported	empirical	cone index at which predictions change from conservative to...	100 cone index	sim-vs-field	link	Para. 50	2026-10-20	Full size m vehicles (M M35A2, M M60 amor US test co g; coarse validation mostly silt (SM) while
MOB-025	M151 1/4 ton wheeled vehicle	Slope tests	not reported	empirical	measured go tests predicted as no-go	four tests	sim-vs-field	link	Para. 49	2026-10-20	Full size m vehicles (M M35A2, M M60 amor US test co g; coarse validation mostly silt (SM) while
MOB-026	Single pneumatic tires and field military wheeled vehicles	Air-dry Yuma sand (from an active dune near Yuma, Arizona) and field...	0 (level)	empirical	constant vertical offset between single wheel and field veh...	2.5 percent of wheel load or vehicle weight	sim-vs-field	link	Para. 53 (p. 27)	2026-10-20	Full size m vehicles (a 50 t) on U beach and sands at 1 index (WE 6 in layer) strength d a...
MOB-027	2-1/2 ton 6×6 truck, 11.00-20 tires (worked example)	Sand with G = 5.4 N/cm2/cm	21 percent grade (p...	empirical	predicted pull coefficient P/W (sand mobility number 23.5)	0.216 dimensionless	none	link	Para. 55, Example 1 (pp. 29 to 30)	2026-10-20	Full size m vehicles (a 50 t) on U beach and sands at 1 index (WE 6 in layer) strength d a...
MOB-028	3/4 ton M37 4×4 truck, run as 4×4 and as 4×2	Sand with G = 5.4 N/cm2/cm	7 percent (4×2 limi...	empirical	predicted pull coefficient P/W (4×4; 4×2)	0.230; 0.070 dimensionless	none	link	Example 4, para. 58 (pp. 34 to 35)	2026-10-20	Full size m vehicles (a 50 t) on U beach and sands at 1 index (WE 6 in layer) strength d a...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MOB-029	Pneumatic tires	Air-dry Yuma sand	not applicable	empirical	range of deflection number for which the sand mobility numb...	roughly 0.1 to 0.4 delta/h	lab	link	Para. 43 (p. 23)	2026-10-20	Full size m vehicles (a 50 t) on U beach and sands at 1 index (WE 6 in layer) strength d a...
MOB-030	Single pneumatic tires 4.00-7, 4.00-20, 6.00-16, 9.00-14 (basic) plus...	Air-dry Yuma sand, limited tests in mortar sand	0 (level)	empirical	range of conditions behind the basic sand prediction term G...	G 2.3 to 27.7; load 100 to 1350; delta/h 0.15 to 0.35 psi/i...	lab	link	Para. 17 and 18; Summary; Notation	2026-10-20	Laboratory Yuma (Ari desert dur g. Tire loa 1350 lb, ti larger thar rover whe coefficient p...
MOB-031	Full size wheeled vehicles versus single tire laboratory relations	Field sands	not reported	empirical	field pull relative to laboratory prediction	significantly worse qualitative	sim-vs-field	link	Summary	2026-10-20	Full size m vehicles (a 50 t) on U beach and sands at 1 index (WE 6 in layer) strength d a...
MOB-032	Single pneumatic tires at 20 percent slip	Yuma sand versus mortar sand	0 (level)	empirical	pull coefficient in mortar sand relative to Yuma sand at eq...	consistently smaller qualitative	lab	link	Paras. 31 and 32	2026-10-20	Laboratory Yuma (Ari desert dur g. Full size coefficient percent sli sand is fin mm) and u
MOB-033	13 commercial tires from a 1.75-26 bicycle tire to a 16×15-6R Terra-T...	Air-dried Yuma desert sand in large soil bins	0 (level)	empirical	number of first pass single wheel tests analysed	709 tests	lab	link	Summary; para. 2; Conclusions para. 42e	2026-10-20	Laboratory Yuma (Ari desert dur g. Tires an mostly lar small rove smallest a pull coeffi
MOB-034	not applicable (soil characterisation)	Yuma desert sand	not applicable	other	direct shear friction angle range, air-dry Yuma sand	35.1 to 38.2 deg	none	link	Para. 5	2026-10-20	Single US sand; inde only.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MOB-035	4.50-18 tire single (455 lb) versus dual (910 lb)	Yuma sand, cone index 40	0 (level)	empirical	pull coefficient (single 4.50-18; dual)	about 0.33; about 0.46 dimensionless	lab	link	Para. 17	2026-10-20	Laboratory Yuma (Ari: desert dur g. Full size coefficient percent sli sand is fin mm) and u
MOB-036	Pneumatic tires, diameters about 14 to about 40 in., deflections 5 to...	WES field and soil bin data on dry sands (Yuma sand, mortar sand)	0 (level)	empirical	agreement of semiempirical sand coefficient with Bernstein...	agreement in form; quantitative rather than qualitative dif...	none	link	Paras. 54 to 55	2026-10-20	Full size m vehicles (a 50 t) on U beach and sands at 1 index (WE 6 in layer) strength d a...
MOB-037	Pneumatic tires, field vehicles and soil bin tests	Dry sands	0 (level)	empirical	agreement between laboratory and field data on the sand num...	high degree of agreement qualitative	sim-vs-field	link	Para. 64(b); limits in para. 57	2026-10-20	Full size m vehicles (a 50 t) on U beach and sands at 1 index (WE 6 in layer) strength d a...
MOB-038	2-1/2 ton 6×6 trucks with 11.00-20 tires versus 9.00-20 tires, 10 psi	Sand with cone index 60	21 percent grade ve...	empirical-field	maximum climbable slope (11.00-20 tires; 9.00-20 tires)	about 21; about 13 percent	field	link	Para. 15	2026-10-20	Field tests trucks (1/4 ton) on US dunes, coa beaches a and Pacifi sands at 1 index aver the...
MOB-039	Military wheeled trucks, tire pressures 10 to 45 psi	Desert dunes and coastal beach and dune sands (quartz, coral, volcani...	not reported	empirical-field	effect of lowering inflation pressure on slope climbing; pr...	markedly improved; about 10 qualitative; psi	field	link	Para. 13 (pp. 6 to 7); vehicles table para. 5	2026-10-20	Field tests trucks (1/4 ton) on US dunes, coa beaches a and Pacifi sands at 1 index aver the...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MOB-040	Five military wheeled trucks (M38A1, M37, M211, M34/M135, M41)	Relatively soft desert dune and beach sands	35 percent grade	empirical-field	steepest slope climbed by any tested vehicle on the soft sa...	35 percent	field	link	Para. 23	2026-10-20	Field tests trucks (1/ ton) on US dunes, co beaches a and Pacifi sands at 1 index aver the...
MOB-041	2-1/2 ton truck with 11.00-20 tires	Desert dune and beach sands	not reported	empirical-field	residual spread in slope at the same dimensionless vehicle...	as great as a 3 percent slope	field	link	Paras. 13 and 20	2026-10-20	Field tests trucks (1/ ton) on US dunes, co beaches a and Pacifi sands at 1 index aver the...
MOB-042	Single self-propelled military wheeled vehicles (885 tests in program...	Yuma dune area (natural active dunes, in California about 20 miles we...	not reported for th...	empirical-field	cone index range of Yuma dune test area (0 to 6 in.)	21 to l41 (as OCR printed) cone index	none	link	Part II paras. 44 and 45; test count Part III par...	2026-10-20	Full size m vehicles (a 50 t) on U beach and sands at 1 index (WE 6 in layer) strength d a...
MOB-043	Military wheeled vehicles (and tracked for comparison)	natural and harrowed sand, level (pilot towing tests reported in the...	0 (level)	empirical-field	maximum drawbar pull on level sand as fraction of gross veh...	20 and 40; 50 and 60 percent (ranges between)	field	link	Para. 9a (p. 4)	2026-10-20	Field tests Island, Brit Mississipp Cape Cod Dunes (be coastal du sands, no in this sup 1...
MOB-044	Self-propelled military wheeled vehicles	Sand, same sand and vehicle conditions on level and slope	maximum slope negot...	empirical-field	maximum towing force on level sand relative to maximum slop...	about 2 percent greater	field	link	Summary (p. vii); para. 78 (pp. 34 to 35)	2026-10-20	Field tests Island, Brit Mississipp Cape Cod Dunes (be coastal du sands, no in this sup 1...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MOB-045	Military wheeled vehicles	quartz sands on inland areas and beaches of the United States (13th S...	not reported	empirical-field	most significant single vehicle characteristic for wheeled...	tire-inflation pressure qualitative	field	link	Para. 6e	2026-10-20	Field tests Island, Brit Mississipp Cape Cod Dunes (be coastal du sands, no in this sup 1...
MOB-046	not applicable (site characterisation)	Warren Dunes State Park, Lake Michigan, beach and unstabilized dunes	level to 75 percent...	other	cone index range, Warren Dunes site (uniform fine dune sand...	16 to 110 cone index	none	link	Paras. 27 and 25	2026-10-20	Field tests Island, Brit Mississipp Cape Cod Dunes (be coastal du sands, no in this sup 1...
MOB-047	Notional tracked armoured personnel carrier and two unit articulated...	LETE Sand (described as a firm terrain)	not reported	classical-terrame...	agreement of NTVPM predicted with measured drawbar pull on...	0.922; 0.850; 3.55; 0.120 dimensionless; dimensionless; kN;...	sim-vs-field	link	Section 5.1(A)(b)(i) to (ii); Table 1	2026-10-20	Tracked m vehicles (a personnel class) on l at 1 g; phy Bekker typ not wheel as the refe compa...
MOB-048	Notional tracked armoured personnel carrier	LETE Sand	not reported	empirical	average measured MMP / calculated MMP by Rowland's empirica...	3.91 ratio	sim-vs-field	link	Section 7; Table 6 row LETE Sand	2026-10-20	Tracked m vehicles (a personnel class) on l at 1 g; phy Bekker typ not wheel as the refe compa...
MOB-049	Standard military tracked and wheeled vehicles	Coastal sabkha (salt flat), Jubail region, Saudi Arabia	0 (level)	empirical-field	penetrometer reading below which sabkha is not trafficable...	271 Pascals or 200 pounds per square inch as printed (inter...	field	link	p. 49 (page inferred from OCR folios)	2026-10-20	Coastal sa flats near Saudi Arat dune sand for standa wheeled a vehicles; i because s
MOB-050	Standard military tracked and wheeled vehicles	Sabkha algal mats, Jubail region	0 (level)	empirical-field	trafficability of algal mat surfaces	untrafficable unless shallowly underlain by cap rock qualit...	field	link	p. 49	2026-10-20	Coastal sa flats near Saudi Arat dune sand for standa wheeled a vehicles; i because s

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MOB-051	Wheeled military vehicles	Low strength soils (fine grained; sands with fines, poorly drained)	0 (level)	empirical	wheeled vehicle mobility index formula as printed (Equation...	MIw = (CPF*WF)/(TEF*GF) + WLF - CF(TF) dimensionless	none	link	Section 4.9, Equation 4 and Table 2 nomenclature	2026-10-20	Test proce size militar the MI anc relations w from fine g tests (criti table assu grained s..
MOB-052	Wheeled military vehicles (single pass)	Low strength fine grained soils	0 (level)	empirical	single pass vehicle cone index VCI1 equations as printed (E...	MIw < 115: VCI1 = [11.48 + 0.2MI - 39.2/(MI +3.74)] * DCF;...	none	link	Section 4.9(d), Equation 6 (p. 28); DCF Equation 5	2026-10-20	Test proce size militar the MI anc relations w from fine g tests (criti table assu grained s..
MOB-053	Wheeled and tracked military vehicles	Natural, unprepared test lanes	not applicable	empirical-field	test lane requirement for VCI testing	natural, unprepared lanes only qualitative	none	link	Sec. 1.3 b, p. 3 (related: Sec. 4.8 b, p. 26)	2026-10-20	Test proce size militar the MI anc relations w from fine g tests (criti table assu grained s..
MOB-055	Modified jeep station wagon (4×4), 40 tests, compared with 4.50-18 an...	Yuma and mortar sands, soil bins	0 (level)	empirical	agreement of 4×4 vehicle performance with single wheel pred...	reasonable accuracy qualitative	lab	link	Paras. 41, 44, 45; Plates 25 to 27	2026-10-20	Soil bin sir tests and a jeep statio (4×4) in Y sand and r sand at 1 g tires (4.50 14); deflec
MOB-056	Single pneumatic tires	Yuma desert sand versus mortar sand at equal cone index	0 (level)	empirical	pull at equal cone index, Yuma relative to mortar sand	higher (Yuma) qualitative	lab	link	Para. 61g; paras. 50, 55, 60	2026-10-20	Soil bin sir tests and a jeep statio (4×4) in Y sand and r sand at 1 g tires (4.50 14); deflec
MOB-057	Towed rigid model wheels, 0.047 m wide (0.25 m and 0.3 m diameter) an...	Dry Calne sand, laboratory soil bin	0 (level)	empirical	predicted versus observed benefit of dual wheels in rolling...	predicted benefit considerably more than that actually obse...	lab	link	p. 85, Discussion; p. 82 wheel sizes (page placed...	2026-10-20	Rover scal wheels in (1 g, but to (unpowere Calne sand not given i paper; OC numbers i discussior

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MOB-058	Pneumatic tires 9.00-14, 6.00-16, 4.00-7, 4.00-20 and others; deflect...	Yuma sand, WES soil bins (27 tests, Table 1)	0 (level)	classical-terrame...	agreement of calculated with measured pull coefficient in s...	excellent agreement (visual comparison, no error statistic)...	sim-vs-lab	link	Section VIII, p. 29 continuing on p. 36; Figs. 3...	2026-10-20	Semi empi (Bekker st model con WES soil b full size tir to 9.00-1. Yuma sand agreement
MOB-059	Pneumatic tires (all sizes in the study)	soil not stated for Eq. 6 (relation given for all tire sizes; Section...	0 (level)	classical-terrame...	limiting pressure as a function of inflation pressure (Eq....	P1(psi) = 0.64 x pi + 4 psi	sim-vs-lab	link	Section VIII, p. 37, Eq. 6; List of Symbols	2026-10-20	Semi empi (Bekker st model con WES soil b full size tir to 9.00-1. Yuma sand agreement
MOB-060	WES mobility cone (not a vehicle)	Granular soil, friction angle 45 deg case	not applicable	other	increase in cone index for a 25 fold increase in shear modu...	approximately a factor of 6 ratio	none	link	Part III, Part B, para. 10 (pp. 11 to 12); Figs....	2026-10-20	Analytical penetratio compared cone index Yuma and sand only; vehicle tes
MOB-061	WES mobility cone	Yuma sand loose (relative density 21.5 percent) and very dense (87.5...	not applicable	other	agreement of predicted with experimental cone index curves...	compare very favorably qualitative	sim-vs-lab	link	Part IV, para. 17 (p. 24); Figs. 9 and 10	2026-10-20	Analytical penetratio compared cone index Yuma and sand only; vehicle tes
MOB-062	M1025 HMMWV, M923A1 5 ton truck, MTVR, BV206 SUSV (UNIMOG run without...	Test Course 1, level tilled sand, Nevada Automotive Test Center	0 (level)	empirical	cone index of sand test course on demonstration day	80 cone index	none	link	p. 7; p. 11; Table 6; Appendix B p. B1	2026-10-20	Full size m vehicles (M HMMWV, MTVR, BV Nevada te at 1 g; the course firm cone index did not tes
MOB-063	M1025, M923A1, MTVR, BV206	Nevada test courses including slopes (not sand specific)	not reported	empirical	direction of NRMM slope prediction error	conservative (underpredicts slope performance) qualitative	sim-vs-field	link	p. 18; p. 17; p. 11	2026-10-20	Full size m vehicles (M HMMWV, MTVR, BV Nevada te at 1 g; the course firm cone index did not tes

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MOB-064	Single pneumatic tyres	Dry Cresswell sand, laboratory	0 (level)	empirical	agreement of measured tyre performance with the WES system	complete disagreement qualitative	lab	link	Abstract (p. I)	2026-10-20	Laboratory tests with and rigid v dry UK sar (Cresswel Buzzard) p reanalysis Yuma and sand data,
MOB-065	Single tyres and rigid wheels	Cresswell and Leighton Buzzard sands; WES Yuma and mortar sand data	0 (level)	empirical	transferability of a cone gradient (G) based traction syste...	not likely to give the same correlation for different sands...	lab	link	Chapter 8 Conclusions 19 and 20 (printed p. 40)	2026-10-20	Laboratory tests with and rigid v dry UK sar (Cresswel Buzzard) p reanalysis Yuma and sand data,
MOB-066	Tyres (WES revised system)	Different sands at equal relative density	0 (level)	empirical	validity of normalising sands to Yuma sand at equal relativ...	generally incorrect qualitative	lab	link	Chapter 8 Conclusions 18 and 21 (printed p. 40)	2026-10-20	Laboratory tests with and rigid v dry UK sar (Cresswel Buzzard) p reanalysis Yuma and sand data,
MOB-067	6.00-16 2PR tyre at 14 psi, load W = 2002 N, 20 percent slip, two pas...	Leighton Buzzard sand: very dense, medium dense, very loose	0 (level)	empirical	coefficient of traction first pass to second pass, very den...	0.187 to 0.089 dimensionless	lab	link	Chapter 8 Conclusion 22 (printed pp. 40 to 41)	2026-10-20	Laboratory tests with and rigid v dry UK sar (Cresswel Buzzard) p reanalysis Yuma and sand data,
MOB-068	Rigid wheels and tyres (WES data reanalysed)	Sands in WES data set	0 (level)	empirical	numeric that reduces WES rigid wheel data to a narrower band	G b d^2 / W (better than G (bd)^3/2 / W) dimensionless	lab	link	Chapter 8 Conclusions 24 and 25 (printed p. 41)	2026-10-20	Laboratory tests with and rigid v dry UK sar (Cresswel Buzzard) p reanalysis Yuma and sand data,

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MOB-069	Tyres and rigid wheels at 20 percent slip	Dry sands	0 (level)	empirical	upper limit of coefficient of traction at high sand number	DP/W tends to tan(phi_cv) dimensionless	lab	link	Chapter 8 Conclusions 21, 26 and 28 (printed pp....	2026-10-20	Laboratory tests with and rigid v dry UK sa (Cresswel Buzzard) p reanalysis Yuma and sand data,
MOB-070	not applicable (soil property review)	Granular media from glass beads to crushed quartz and crushed glass (...	not applicable	other	range of critical state friction angle phi_cv (glass beads...	24 to 30; 38 to 43 deg	none	link	Chapter 8 Conclusions 6, 7 (printed p. 38) and 10...	2026-10-20	Laboratory tests with and rigid v dry UK sa (Cresswel Buzzard) p reanalysis Yuma and sand data,
MOB-071	Single tyres on two sands	Dry sands, laboratory	0 (level)	empirical	effects of inflation pressure and load on tyre performance...	separate independent effects qualitative	lab	link	Chapter 8 Conclusion 23 (printed p. 41); Abstract...	2026-10-20	Laboratory tests with and rigid v dry UK sa (Cresswel Buzzard) p reanalysis Yuma and sand data,
SCL-001	2D DEM rigid rotating bar "wheels" (fully rough bar about its centre)...	LAMMPS planar packing of polydisperse spheres (radii 0.254 to 0.381 m...	0 (level)	DEM	agreement of dimensionless velocity V/sqrt(Lg) and power P/...	strong overlap qualitative	simulation-only	link	p. 6, section Potential extraplanetary applicatio...	2026-10-21	2D DEM w millimetre and bar sh "wheels" c centimetre value or sl agreemen only graph gravity inc d...
SCL-002	Generic rigid wheel of shape f, width D, radius scale L, mass M	Any granular bed obeying RFT or Coulomb plasticity	not reported	RFT-granular	validation status of the drawbar force extension (Eq. 9), w...	It would be interesting future work to experimentally valid...	none	link	p. 8, Discussion, Eq. (9) and the sentence after...	2026-10-21	No data; th and incline proposal c source
SCL-003	Grouser wheel D 150 mm, B 80 mm, 24 grousers 8.5 mm high, 2.0 mm thic...	DEM (ANSYS Rocky) bed 2400 × 200 x 300 mm tilted as a whole, FJS-1 lu...	10 (climbing)	DEM	slip ratio when climbing a 10 deg slope at 2 pi rad/s (luna...	approximately 0.9 slip ratio (dimensionless)	simulation-only	link	p. 4, Section 4, paragraph after Fig. 3 ("The whe...	2026-10-21	Simulation against DE physical te gravity; m cohesive l simulant m dry dune s wheel des with alp...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
SCL-004	Grouser wheel D 150 mm, B 80 mm, 24 grousers 8.5 mm high, 2.0 mm thic...	DEM (ANSYS Rocky) bed 2400 × 200 x 300 mm tilted as a whole, FJS-1 lu...	-10 (descending)	DEM	slip ratio when descending a 10 deg slope at pi rad/s (luna...	approximately -0.2 slip ratio (dimensionless)	simulation-only	link	p. 4, Section 4, paragraph after Fig. 3	2026-10-21	Simulation against DE physical te gravity; m cohesive l simulant m dry dune s wheel des with alp...
SCL-005	Grouser wheel D 150 mm, B 80 mm, 24 grousers 8.5 mm high, 2.0 mm thic...	DEM (ANSYS Rocky) bed 2400 × 200 x 300 mm tilted as a whole, FJS-1 lu...	-10 to 10	DEM	RMSE of predicted against simulated wheel sinkage over 10 c...	0.8 to 8.9 mm	simulation-only	link	Fig. 4b, p. 5; text p. 5 ("The error was larger w...	2026-10-21	Simulation against DE physical te gravity; m cohesive l simulant m dry dune s wheel des with alp...
SCL-006	Grouser wheel D 150 mm, B 80 mm, 24 grousers 8.5 mm high, 2.0 mm thic...	DEM (ANSYS Rocky) bed 2400 × 200 x 300 mm tilted as a whole, FJS-1 lu...	-10 to 10	DEM	RMSE of predicted against simulated wheel driving power ove...	0.17 to 0.63 W	simulation-only	link	Fig. 4c, p. 5	2026-10-21	Simulation against DE physical te gravity; m cohesive l simulant m dry dune s wheel des with alp...
SCL-007	MGRU3 VIPER digital twin (88 kg) and 22 kg single wheel, CRM simulati...	GRC-1 lunar simulant parameter sets in CRM (continuum SPH) simulation...	0 to 30 (5 deg step...	other	effect of wheel angular velocity (0.33 versus 0.8 rad/s) on...	almost identical qualitative	simulation-only	link	p. 14, text before Fig. 12 ("The results are almo...	2026-10-21	Moon grav simulation lunar test simulant, r sand; cont model; aut the insens only up to ang...
SCL-008	MGRU3 VIPER digital twin and single wheel (about one quarter of rover...	GRC-1 and GRC-3 lunar simulant parameter sets, Earth gravity, slope m...	0 to 30	other	agreement of single wheel and full rover simulations with p...	match well; the single wheel simulation closely predicted t...	sim-vs-lab	link	p. 13 (points i and ii after Fig. 8 text) and p....	2026-10-21	Lunar simu indoor slo rover of 7 with large wheels; ac stated qua authors re single whe
SCL-009	Double helix Archimedes screw propelled vehicle (ABS screws of three...	Soda lime glass bead bed (laboratory) and its EDEM model; DEM only fo...	0 (level)	DEM	lunar gravity translational velocity as a percentage of the...	102.6%, 105.9% and 102.7% percent of predicted	simulation-only	link	Section 3.3, paragraph after Eq. (11) and Fig. 7	2026-10-21	Screw pro vehicle, nc glass beac sand; DEM DEM; level

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
SCL-010	Double helix Archimedes screw propelled vehicle (ABS screws of three...	Soda lime glass bead bed (laboratory) and its EDEM model; DEM only fo...	0 (level)	DEM	Ceres gravity velocity as a percentage of the scaling law p...	67.5% (5 rpm); 79.5% (20 rpm) percent of predicted	simulation-only	link	Section 3.3, paragraph after Fig. 7; Section 4 (c...	2026-10-21	As SCL-0(gravity (ab percent of
SCL-011	Double helix Archimedes screw propelled vehicle (ABS screws of three...	Soda lime glass bead bed (laboratory) and its EDEM model	0 (level)	DEM	slope of actual craft velocity against no slip velocity: ex...	0.657; 0.761 ratio of actual to no slip velocity (called sl...	sim-vs-lab	link	Section 3.2, paragraphs with Figs. 4 and 5	2026-10-21	Screw veh beads; slip against sc advance, r circumfere
SCL-012	Grousered rigid wheel, outer diameter 212 mm, free locomotion with we...	Poppy seed bed (model granular medium), laboratory	0 (level)	RFT-granular	upper limit of the quasi static domain where translation ve...	omega < 30 rotations per minute (RPM) (corresponding to ome...	lab	link	Results, Wheel locomotion experiments, second par...	2026-10-21	Poppy see sand; one ground; th is for this v material or
SCL-013	Grousered rigid wheel, outer diameter 212 mm, MPM simulations	Poppy seed continuum model (MPM, plane strain)	0 (level)	RFT-granular	form of psi in delta h = r psi(r omega^2/g)	psi is well approximated by the identity function quoted te...	simulation-only	link	Understanding the dynamic structural correction,...	2026-10-21	MPM cont one wheel seeds; lev
SCL-014	Lunar vehicle wheel candidates (pneumatic, Bendix, Boeing-GM, Grumman...	Air dry and damp Yuma (Arizona desert) sand in soil bins, Earth gravi...	0 (level)	empirical	slip at which pull and torque had reached their plateau in...	20 percent slip	lab	link	Para. 44, printed p. 28 (PDF p. 44)	2026-10-21	Lunar whe candidate three 100 diameter v Yuma sand gravity; pl varied bet (about 15 typical)
SCL-015	Lunar vehicle wheel candidates (pneumatic, Bendix, Boeing-GM, Grumman...	Air dry and damp Yuma (Arizona desert) sand in soil bins, Earth gravi...	0 (level)	empirical	range of the sand mobility number (G b d^2/W)(1 - 2 delta/d...	The abscissa (numeric) extends to 1800 dimensionless numeric	lab	link	Para. 138, printed p. 113 (PDF p. 129)	2026-10-21	Fit by eye, goodness of G as in (MN/m3 o gravity Yu
SCL-016	Lunar vehicle wheel candidates (pneumatic, Bendix, Boeing-GM, Grumman...	Air dry and damp Yuma (Arizona desert) sand in soil bins, Earth gravi...	0 (level)	empirical	contact pressure form of the sand relation	P/W = f((G/W) . A_c^3/2) functional relation	lab	link	Para. 140, printed p. 116 (PDF p. 132); Figs. 67a...	2026-10-21	A_c not ad soil operat printed); fi Earth grav
DUN-001	not applicable (terrain property)	Wahiba Sands, Oman, interior high dunes (linear megaridges)	not reported	field observation	spacing of linear ridges; width of interdune swales (wales)	1 to 3 (ridge spacing); 1 to 3 (swale width) km	none	link	Overview and Quaternary History sections	2026-10-20	Newslette of sympos on the 198 Wahiba Sa Project, nc reviewed p methods a counts are

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
DUN-002	not applicable (terrain property)	Rub al Khali, four study areas	not reported	remote sensing	maximum dune wavelength (bedform extent perpendicular to cr...	0.05; 3.34; 1.22 km	none	link	Table 1; text pages 11 to 12	2026-10-20	Remote se measurem sharpenec and SRTM SRTM vert accuracy the author absolute, relative), n
DUN-003	not applicable (terrain property)	Rub al Khali interdune corridors and flats, four study areas	not reported	remote sensing	interdune width: minimum; maximum; mean; interdune length m...	0.06; 6.22; 1.49; 6.97 km	none	link	Table 2 Interdune data summary	2026-10-20	Remote se measurem sharpenec and SRTM SRTM vert accuracy the author absolute, relative), n
DUN-004	not applicable (terrain property)	Rub al Khali, central dune field versus margin	not reported	remote sensing	mean amplitude of along crest sinuosity, centre versus oute...	2.5 to less than 0.5 km	none	link	Text page 12 (Fig. 9d)	2026-10-20	Remote se measurem sharpenec and SRTM SRTM vert accuracy the author absolute, relative), n
DUN-005	not applicable (terrain property)	Central and eastern Rub al Khali, five meteorological stations	not applicable	meteorological da...	prevailing resultant drift direction	to the southwest direction	none	link	Methods, accepted manuscript page 9 (Figure 5)	2026-10-20	Direction potential n printed, ar stations ar Arabia (Al Riyadh, Wa Dawaser, Sharurah compa...
DUN-006	not applicable (terrain property)	Sharurah station, southern margin of the Rub al Khali, Saudi Arabia	not applicable	meteorological da...	wind direction frequency at Sharurah: NE; E; S; N	30; 30; 12; 10 percent	none	link	Figure 2.5 and its table, thesis page 35 (PDF pag...	2026-10-20	Direction f of all wind of observa source WeatherO sand movi above thre not a Fryb potential;

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
DUN-007	not applicable (terrain property)	Al-Hasa station, north eastern Saudi Arabia (upwind of the Rub al Kha...	not applicable	meteorological da...	wind direction frequency at Al-Hasa: N; NW; S	32; 26; 11 percent	none	link	Figure 2.5 table, thesis page 35 (PDF page 56)	2026-10-20	Direction f of all wind of observa source WeatherO sand movi above thre not a Fryb potential;
DUN-008	not applicable (terrain property)	Wadi Ad-Dawaser and Najran stations, western margin of the Rub al Kha...	not applicable	meteorological da...	dominant direction frequency: Wadi Ad-Dawaser E; Najran NE	43; 31 percent	none	link	Figure 2.5 table, thesis page 35 (PDF page 56)	2026-10-20	Direction f of all wind of observa source WeatherO sand movi above thre not a Fryb potential;
DUN-009	not applicable (terrain property)	Barchan dune fields, south east Qatar (Umm Said area)	not reported	field observation	barchan height: maximum; typical	up to 40; mostly less than 10 m	none	link	Section 2 Study area, page 6 (paragraph before Se...	2026-10-20	Barchans carbonate and sabkh Qatar; mig measured on 0.8 to 1 at 5 refere per dune. method nc
DUN-010	not applicable (terrain property)	16 barchans (B1 to B16), south east Qatar, time window 1965 to 1995	not reported	remote sensing	migration rate range across the 16 dunes, 1965 to 1995	0.4 (B1) to 2.1 (B12) m/month	none	link	Section 4.1 Barchan migration, page 9, Fig. 8; me...	2026-10-20	Barchans carbonate and sabkh Qatar; mig measured on 0.8 to 1 at 5 refere per dune.
DUN-011	not applicable (terrain property)	Same 16 barchans, time window 1995 to 2006	not reported	remote sensing	migration rate range across the dunes, 1995 to 2006	0.3 to 1.7 m/month	none	link	Section 4.1, page 9, Fig. 8	2026-10-20	Barchans carbonate and sabkh Qatar; mig measured on 0.8 to 1 at 5 refere per dune.
DUN-012	not applicable (terrain property)	Barchan B8, south east Qatar, 1965 to 2015	not reported	remote sensing	total displacement of barchan B8 and direction	ca. 587 towards SSE m	none	link	Section 3, page 7 (Fig. 5 example)	2026-10-20	Barchans carbonate and sabkh Qatar; mig measured on 0.8 to 1 at 5 refere per dune.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
DUN-013	not applicable (terrain property)	Compound linear dunes and linear sand sheets near Ash Shariqah (Sharj...	not applicable	meteorological da...	annual drift potential DP; RDP/DP; wind regime class (Sharj...	280; .60; acute bimodal vector units; ratio; class	none	link	Chapter F, Table 18 (localities used to evaluate...	2026-10-20	Fryberger drift poten vector uni 1970s sta summarie 12 knots a height, cha page 146 single stat
DUN-014	not applicable (terrain property)	Ash Shariqah (Sharjah), UAE coast	not applicable	meteorological da...	resultant drift direction by season at Sharjah	toward the east December through May; toward the southeast...	none	link	Chapter J, Arabian sand seas, text page 373 and F...	2026-10-20	Fryberger drift poten vector uni 1970s sta summarie 12 knots a height, cha page 146 single stat
DUN-015	not applicable (terrain property)	Al Ubaylah station, northern Rub al Khali, Saudi Arabia	not applicable	meteorological da...	annual drift potential at Al Ubaylah; resultant drift direc...	approximately 200; toward the south southwest vector units;...	none	link	Chapter J, page 373 (resultant direction) and pag...	2026-10-20	Fryberger drift poten vector uni 1970s sta summarie 12 knots a height, cha page 146 single stat
DUN-016	not applicable (terrain property)	Al Masirah island station, Oman coast (south east of the Wahiba Sands)	not applicable	meteorological da...	peak monthly drift potential (July, southwest monsoon); res...	299; generally toward the north vector units; direction	none	link	Chapter J, page 373 (direction) and page 375 (Jul...	2026-10-20	Fryberger drift poten vector uni 1970s sta summarie 12 knots a height, cha page 146 single stat
DUN-017	not applicable (terrain property)	Northern Arabian sand seas: Saudi Arabia and Kuwait (An Nafud, north)...	not applicable	meteorological da...	average annual drift potential (10 stations); energy class...	489; low < 200, intermediate 200 to 399, high at least 400...	none	link	Chapter F, Table 16 and text, report page 150	2026-10-20	Fryberger drift poten vector uni 1970s sta summarie 12 knots a height, cha page 146 single stat

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
DUN-018	not applicable (terrain property)	Bahrain station, Arabian Gulf (example of a wide unimodal regime)	not applicable	meteorological da...	DP; RDP; RDP/DP (Bahrain)	540; 435; 0.81 vector units; vector units; ratio	none	link	Chapter F, Fig. 96 caption and values, report pag...	2026-10-20	Fryberger drift poten vector uni 1970s sta summarie 12 knots a height, cha page 146 single stat
DUN-019	not applicable (terrain property)	Compound linear dunes, southern Wahiba Sands, Oman (Landsat sample ar...	not reported	remote sensing	mean dune width; mean spacing; mean ridge length; maximum r...	1.1; 1.6; 31.3; as much as 100 km	none	link	Chapter J, The Wahiba Sands, Interpretation of La...	2026-10-20	Landsat er measurem sample are heights. Li described Pleistocen not aligne present wi Glennie 19
DUN-020	not applicable (terrain property)	Shifting compound longitudinal dunes, Taklimakan Desert hinterland (T...	not reported	remote sensing	Copernicus DEM vertical RMSE; SD; height difference range a...	2.72; 2.42; -8.79 to 8.96 m	field	link	Section 3.1, pages 9 to 10, Fig. 4a	2026-10-20	Taklimaka compounc longitudina with 3 to 2 relief and m spacing 2.1), much than Liwa Khali mega
DUN-021	not applicable (terrain property)	Taklimakan Desert hinterland dune field, ICESat-2 ATL08 control points	not reported (all s...	remote sensing	RMSE range across the six DEMs; best RMSE (AW3D30); mean er...	3.63 to 6.07; 3.63; -2.15 to 0.61 m	field	link	Section 3.1, page 8 to 9, Fig. 3	2026-10-20	Taklimaka compounc longitudina with 3 to 2 relief and m spacing 2.1), much than Liwa Khali mega
DUN-022	not applicable (terrain property)	Taklimakan Desert hinterland dune field	0 to 15 versus abov...	remote sensing	slope above which DEM errors rise and fluctuate markedly	15 deg	field	link	Abstract; Section 3.2, pages 12 to 13, Fig. 6	2026-10-20	Taklimaka compounc longitudina with 3 to 2 relief and m spacing 2.1), much than Liwa Khali mega
DUN-023	not applicable (terrain property)	Eastern fringe of the Rub al Khali dune field, UAE (dune ridges from...	not reported	field observation	maximum height of large scale dune ridges; ridge orientatio...	up to 70; SW-NE; NNW-SSE m; direction; direction	none	link	Section 2.1 Unconsolidated dune forms, manuscript...	2026-10-20	Descriptiv from regio geological the UAE; n height det not given; distributio

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
DUN-024	not applicable (terrain property)	Central Abu Dhabi emirate, downwind of the coast and north of Liwa	not reported	field observation	height of large coalesced transverse dunes	15 to 60 m	none	link	Section 2.1, manuscript page 4	2026-10-20	Descriptiv from regio geological the UAE; n height det not given; distributio
DUN-025	not applicable (terrain property)	Liwa oasis megabarchanoid dune field, UAE	not reported	field observation	width of interdune sabkhas between Liwa megadunes	2 to 3 km	none	link	Section 2.1, manuscript page 4	2026-10-20	Descriptiv from regio geological the UAE; n height det not given; distributio Interdune can h...
DUN-026	not applicable (terrain property)	Isolated barchan dunes beside the Jeddah to Makkah highway, Saudi Ara...	not reported	field observation	barchan height above base; length parallel to wind; width a...	1.5 to 3; 3 to 8; 6 to 10 m	none	link	Geomorphological features, A Barchan dunes, journ...	2026-10-20	Small dune Red Sea c near Jedd the UAE; t the PDF is scan, so s are uncert Number of
DUN-027	not applicable (terrain property)	Longitudinal (seif) dunes, Jeddah to Makkah highway, southern Cornich...	not reported	field observation	longitudinal dune height; length	1 to 6; 3 to 180 m	none	link	Geomorphological features, C Longitudinal or seif...	2026-10-20	Small dune Red Sea c near Jedd the UAE; t the PDF is scan, so s are uncert
SND-001	not applicable (terrain property)	Active dunes, palaeodunes, fluvial deposits and Miocene sandstones ac...	not reported	lab property meas...	average heavy mineral assemblage: ferromagnesian minerals t...	73.5; 35.7; 25.8; 12.6 percent of heavy mineral fraction	none	link	Results, Sample mineralogy, manuscript lines 324...	2026-10-20	Average o analysed s (Miocene, and mode active dun heavy min few perce sand, so th litt...
SND-002	not applicable (terrain property)	Modern dune sands, United Arab Emirates (coastal to inland transects...	not reported	lab property meas...	dominant particle size class of modern dune sand	predominantly coarse grained (> 125 µm) micrometre (size cl...	none	link	Section 4.8 Modern dunes, manuscript lines 432 to...	2026-10-20	Only three classes ar mean size statistic is it cannot b d50. Samp per class r this pass..

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
SND-003	not applicable (terrain property)	Gulf coast dunes versus inland dunes including the Liwa megabarchans,...	not reported	field observation	coastal dune sand composition and sorting; inland Liwa sand...	coastal: fine to medium grained, poorly sorted carbonate sa...	none	link	Section 4.8 Modern dunes, manuscript lines 419 to...	2026-10-20	Qualitative descriptio carbonate coastal sa more angu bioclasts) rich Liwa s differ in fri crushabilit
SND-004	not applicable (terrain property)	Rub al Khali desert sands of the UAE (northern margin of the sand sea)	not reported	lab property meas...	estimated share of UAE Rub al Khali dune sand derived ultim...	50 to 60; 15 to 20 each percent	none	link	Discussion, manuscript lines 721 to 724 (PDF page...	2026-10-20	Interpretiv not a meas mechanic relevant o evidence t UAE dune dominatec sand.
SND-005	not applicable (terrain property)	Five sand dunes (long and short dunes in areas A and B, one coastal d...	not reported	lab property meas...	graphic mean grain size Mz, area A (16 samples) range and a...	A: 1.17 to 2.12, average 1.78; B: 1.61 to 2.20, average 1.9...	none	link	Section 3.1.1 Average size, page 5; Table 2 (page...	2026-10-20	Surface gr (depth not laser diffra mg can dif sieving. Q are low ba carbonate not the lar
SND-006	not applicable (terrain property)	Southern Qatar dunes, areas A and B (as above)	not reported	lab property meas...	sorting (inclusive graphic standard deviation): area A rang...	A: 0.37 to 0.79, average 0.53; B: 0.52 to 0.73, average 0.6...	none	link	Section 3.1.1 Standard deviation, page 5 (right c...	2026-10-20	Area B tex appears to the three dune samp 2 gives 0.8 phi for san 35, moder sorted).
SND-007	not applicable (terrain property)	Southern Qatar dunes, areas A and B	not reported	lab property meas...	quartz content: area A range and average; area B range and...	A: 49.00 to 90.00, average 70.63; B: 52.00 to 88.00, averag...	none	link	Section 3.1.3 Mineralogical composition by XRD, p...	2026-10-20	Semi quan XRD; numl samples a XRD per a restated in passage (I 5).
SND-008	not applicable (terrain property)	Southern Qatar dunes, areas A and B	not reported	lab property meas...	calcite content: area A range and average; area B range and...	A: 7.00 to 26.00, average 17.50; B: 7.00 to 18.00, average...	none	link	Section 3.1.3, page 6	2026-10-20	Carbonate softer thar carbonate this order Gulf coast not of inla sand.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
SND-009	not applicable (terrain property)	Southern Qatar dunes, areas A and B	not reported	lab property meas...	SiO2 range (area A; area B) and CaO average (A; B)	SiO2 58.1 to 74.5 (A), 63.7 to 74.9 (B); CaO average 18.3 (...	none	link	Section 3.1.2 Chemical composition by XRF, page 6	2026-10-20	Doubt: the the SiO2 a 74.50 and percent, e stated ma verifier sh Fig. 3.
SND-010	not applicable (terrain property)	Southern Qatar dunes (SEM subset of eleven representative samples)	not reported	lab property meas...	share of grains subrounded to rounded; shape range	approximately 35 % subrounded to rounded; shapes from angul...	none	link	Section 3.1.4 Morphology and chemical composition...	2026-10-20	Qualitative quartz gra friction an to rounded no shape i reported.
SND-011	not applicable (terrain property)	Liwa dune field, UAE (reddish barchanoid megadunes); crest of largest...	not reported	lab property meas...	quartz Q; feldspar F; carbonate lithic grains Lc; chert Lh...	73; 15; 6; 3 percent of framework grains	none	link	Table 1, row Liwa (DUNEFIELDS block, PDF page 53)...	2026-10-20	Bulk petro (quartz, fe lithic share framework not a mecl property; samples o dune at ea Small sam
SND-012	not applicable (terrain property)	Coastal dunes, UAE (Trucial coast, creamy yellow carbonate dunes)	not reported	lab property meas...	quartz Q; feldspar F; carbonate lithic grains Lc (coastal U...	24; 9; 58 percent of framework grains	none	link	Table 1, row Coastal UAE (DUNEFIELDS block, PDF p...	2026-10-20	Bulk petro (quartz, fe lithic share framework not a mecl property; samples o dune at ea Small sam
SND-013	not applicable (terrain property)	Northeastern Rub al Khali megadunes, Shaybah area (Saudi Arabia, near...	not reported	lab property meas...	quartz Q; feldspar F (Shaybah dunes, n = 7)	81; 14 percent of framework grains	none	link	Table 1, row Shaybah (PDF page 53); text lines 30...	2026-10-20	Bulk petro (quartz, fe lithic share framework not a mecl property; samples o dune at ea Small sam
SND-014	not applicable (terrain property)	Southwestern Rub al Khali dunes (Saudi Arabia); 12 dune samples, sand...	not reported	lab property meas...	quartz Q; feldspar F (SW Rub al Khali, n = 12)	86; 9 percent of framework grains	none	link	Table 1, row SW Rub al Khali (PDF page 53)	2026-10-20	Bulk petro (quartz, fe lithic share framework not a mecl property; samples o dune at ea Small sam

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
SND-015	not applicable (terrain property)	Northeastern Rub al Khali and western Jafurah dune fields	not reported	lab property meas...	quartz share of bulk sediment: northeastern Rub al Khali du...	78±12; 93.2±0.4 percent	none	link	Discussion, manuscript lines 541 to 545 (PDF page...	2026-10-20	Bulk petro (quartz, fe lithic share framework not a mecl property; samples o dune at ea Small sam
SND-016	not applicable (terrain property)	Barchan sand dune, coastal Eastern Province, Saudi Arabia; two 10 m b...	not reported	lab property meas...	mean grain size range over both boreholes	0.22 to 0.40 mm	none	link	Section 3.5 Sand texture, page 8, Fig. 7; samplin...	2026-10-20	One coast in the East Province c Arabia; tw only; samp from the d which the argue diffe surfa...
SND-017	not applicable (terrain property)	Same barchan, crest and limb boreholes	not reported	lab property meas...	sorting coefficient range	0.54 (moderately well sorted) to 1.07 (poorly sorted) phi	none	link	Section 3.5 Sand texture, page 8, Fig. 9	2026-10-20	One coast in the East Province c Arabia; tw only; samp from the d which the argue diffe surfa...
SND-018	not applicable (terrain property)	Same barchan, crest and limb	not reported	lab property meas...	quartz; feldspars; calcite; heavy minerals	86.8; 9.4; 2.9; 0.9 percent	none	link	Section 3.2 Mineral composition of sands, page 5,...	2026-10-20	One coast in the East Province c Arabia; tw only; samp from the d which the argue diffe surfa...
SND-019	not applicable (terrain property)	Same barchan; vadose zone profile from surface to 10 m (crest) and 6....	not reported	lab property meas...	maximum gravimetric water content: dune limb; dune crest	3.72; 3.31 percent (mass of water per mass of dry sand)	none	link	Section 3.3 Water content, page 6, Fig. 5	2026-10-20	One coast in the East Province c Arabia; tw only; samp from the d which the argue diffe surfa...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
SND-020	not applicable (terrain property)	Same barchan, limb borehole	not reported	field observation	thickness of dune sand in limb; depth to water table	7; 7.6 m	none	link	Section 3.1 Lithology, page 5, Fig. 3	2026-10-20	One coast in the East Province c Arabia; tw only; samp from the d which the argue diffe surfa...
SND-021	not applicable (terrain property)	Natural dune sand, Dhahran area, Eastern Province, Saudi Arabia (ligh...	not reported	lab property meas...	D10; D30; D60; coefficient of uniformity Cu; specific gravi...	0.146; 0.2; 0.29; 1.98; 2.68 mm; mm; mm; dimensionless; dim...	none	link	Table 5-1 Properties of natural dune sand, thesis...	2026-10-20	One dune the Dhahra Eastern Pr Saudi Arat reconstitu laboratory situ dune measurem
SND-022	not applicable (terrain property)	Same dune sand (natural, about 1 percent silt)	not reported	lab property meas...	maximum dry density; minimum dry density; maximum void rati...	1.84; 1.63; 0.68; 0.48 g/cm3; g/cm3; dimensionless; dimensi...	none	link	Table 5-1 and Table 5-3 (1 percent silt, natural...	2026-10-20	One dune the Dhahra Eastern Pr Saudi Arat reconstitu laboratory situ dune measurem surface du
SND-023	not applicable (terrain property)	Same dune sand reconstituted at relative density 40, 60 and 90 percen...	0 (level)	lab property meas...	peak friction angle at Dr 40, 60, 90 percent (natural sand,...	45; 46; 48 deg	none	link	Table 5-4 Summary of direct shear test results of...	2026-10-20	One dune the Dhahra Eastern Pr Saudi Arat reconstitu laboratory situ dune measurem angles fro
SND-024	not applicable (terrain property)	Compacted dune sand backfill, large housing project area in Al Jubail...	0 (level)	field property me...	average moisture content; average dry density (point near s...	4.64; 1.62 percent; g/cm3 (printed as gram/cm2)	none	link	Table 5-8, thesis page 123 (Section 5.2.1 Al-Juba...	2026-10-20	Engineere compacte dune sand dune surfa moisture r conditions reproduce data from rep...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
SND-025	not applicable (terrain property)	Eastern Province (Dhahran area) dune sand with 1, 4 and 8 percent sil...	0 (level)	lab in situ test	DCP index as relative density rises from 40 to 90 percent	63 to 8 mm/blow	none	link	Section 3 Results and Discussions, page CON33-5;...	2026-10-20	Same sand programm 2013 thes independe Chamber p vibration, index is a p resistance wheel...
SND-026	not applicable (terrain property)	Same chamber tests	0 (level)	empirical correla...	friction angle correlation for natural sand (1 percent silt)	phi = 40 - 0.126(DCPI) (R2 0.99); phi = 0.14(Dr) + 27 (R2 0...	none	link	Table 4, page CON33-6 to CON33-7, Fig. 7	2026-10-20	Doubt: the correlatior to 39.6 de Dr 40 to 9 (derived b reviewer), inconsiste 45 to 48 d angles in..
SND-027	not applicable (terrain property)	Dune sand from the northern Wahiba Sands (Sharqiya Sands), Oman, abou...	not reported	lab property meas...	specific gravity Gs; fines content	2.62; 3.5 dimensionless; percent	none	link	Section 2.1 Dune sand, page 3 to 4; Table 2 row DS	2026-10-20	Single bull depth not Classified graded fin grains rou semi angu smooth su
SND-028	not applicable (terrain property)	Same Wahiba dune sand	not reported	lab property meas...	maximum dry density; optimum water content	1.69; 17 g/cm3; percent	none	link	Section 2.1, page 3 to 4; Table 3 gives MDD 16.54...	2026-10-20	Proctor de moist com sand is an bound for fill, not the density of dune surfa
SND-029	not applicable (terrain property)	Wahiba Sands, Oman, surface sand near the project field base	not reported	field property me...	surface sand moisture content: first millimetre; third mill...	1.8; 0.6 percent wet weight	none	link	Section Moisture in the Air and in the Ground, pa...	2026-10-20	Newsletter of sympos on the 198 Wahiba Sa Project, nc reviewed p methods a counts are Winter and
SND-030	not applicable (terrain property)	Wahiba Sands, Oman, dune top and interdune depressions after infiltra...	not reported	field property me...	wetting front depth 24 hours after saturating a dune top to...	90; 120 cm	none	link	Section Moisture in the Air and in the Ground, in...	2026-10-20	Newsletter of sympos on the 198 Wahiba Sa Project, nc reviewed p methods a counts are

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
SND-031	not applicable (terrain property)	Inland (desert) dunes along the Jeddah to Makkah highway, Saudi Arabi...	not reported	lab property meas...	graphic mean size Mz range of desert dune samples; sorting...	2.32 to 2.9; 0.4 to 0.55 phi	none	link	Table 2 (statistical grain size parameters), samp...	2026-10-20	Small dun Red Sea c near Jedd the UAE; t the PDF is scan, so s are uncert OCR show
MAP-001	Simulated planetary rover (point model, reference speed 0.1 m/s)	Ambiguous Appearance (AA) synthetic dataset: heterogeneous deformable...	not reported (fract...	learning	path success rate (no simulated slip of 1 or minus 1 on the...	95 versus 11 % of test instances	simulation-only	link	Table I, Ambiguous Appearance (AA) Dataset block,...	2026-10-20	Synthetic grid maps grid with f geometry invented s functions; sand, no p rover; suc simulated
MAP-002	Simulated planetary rover (point model, reference speed 0.1 m/s)	Standard (Std) synthetic dataset: four of ten terrain classes per map...	not reported (fract...	learning	path success rate, proposed versus single GP with expected...	96 versus 71 % of test instances	simulation-only	link	Table I, Standard (Std) Dataset block, column Suc...	2026-10-20	Synthetic grid maps grid with f geometry invented s functions; sand, no p rover; suc simulated
MAP-003	Simulated planetary rover (point model, reference speed 0.1 m/s)	Erroneous Slip (ES) synthetic dataset: slip measurement noise grows w...	not reported (fract...	learning	path success rate, proposed versus single GP with expected...	77 versus 60 % of test instances	simulation-only	link	Table I, Erroneous Slip (ES) Dataset block, colum...	2026-10-20	Synthetic grid maps grid with f geometry invented s functions; sand, no p rover; suc simulated
MAP-004	Simulated planetary rover (point model, reference speed 0.1 m/s)	Ambiguous Appearance (AA) synthetic dataset	not reported (fract...	learning	maximum slip experienced along executed path, 100 x max(se)...	63.7 ± 21.6 versus 92.9 ± 27.4 % slip	simulation-only	link	Table I, AA Dataset block, column smax	2026-10-20	Synthetic grid maps grid with f geometry invented s functions; sand, no p rover; suc simulated

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MAP-005	Simulated planetary rover (reference speed 0.1 m/s)	Unfamiliar geometry and appearance (UGA) synthetic test subset: crate...	17.5 to 30 (out of...	learning	path success rate, proposed with adaptation versus segmenta...	90 versus 66 % of test instances	simulation-only	link	Table I, UGA columns, rows Ours w/ DA and Seg+EV	2026-10-20	Synthetic grid maps grid with f geometry invented s functions; sand, no p rover; succ simulated
MAP-006	Simulated planetary rover	Four synthetic test subsets: in domain, unfamiliar geometry, unfamili...	in domain mostly wi...	learning	slip prediction mean absolute error scaled by 100, UGA subs...	16.3 ± 2.3 before; 14.3 ± 2.1 after slip x 100 (mean absolu...	simulation-only	link	Table II, UGA column	2026-10-20	Synthetic grid maps grid with f geometry invented s functions; sand, no p rover; succ simulated
MAP-007	Mars 2020 rover (MSL heritage), orbital analysis	HiRISE orbital images at nominal 0.3 m per pixel, candidate landing s...	not applicable	learning	per pixel classification performance	94.9 %	field	link	Sec. 4 Automated Terrain Classification, paragrap...	2026-10-20	Mars orbit at 0.3 m p ground rov ground tru are sparse authors sa performan unlabelled wa...
MAP-008	MSL Curiosity (odometry) versus orbital route planner	Gale crater traverse compared with distance optimal route planned on...	not reported	empirical-field	difference between planned shortest distance and actual dri...	~30 % longer actual distance	field	link	Sec. 8 Analysis Results, paragraph beginning 'Fir...	2026-10-20	Mars, Curi rover, incl detours fo well as ob avoidance map resol 0.3 m per specific to
MAP-009	Mars 2020 rover, orbital rock mapping	HiRISE images at 0.3 m per pixel after blind deconvolution	not applicable	other	smallest resolvable shadow size	~5 pixels (1.5 meters at 0.3m resolution) pixels (m)	none	link	Sec. 5 Automated Rock Detection, sentence on blin...	2026-10-20	Orbital det on Mars; a can be sto much sma ripple cres orbital ma resolution hazards at
MAP-010	Stanley, modified Volkswagen Touareg with five roof lidars	reference dataset of labelled terrain (location and course not specif...	not reported	other	fraction of known drivable area classified as obstacle	12.6 %	field	link	Sec. 5 Laser Terrain Mapping, paragraph after Fig...	2026-10-20	Full size ca (Volkswag Touareg) a speed on l desert trac lakebeds, rover on lo sand; drive obstacle la

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MAP-011	Stanley, modified Volkswagen Touareg with five roof lidars	Mojave desert roads, independent test data set	not reported	learning	false positive rate (as defined in the paper, area labelled...	drops from 12.6% to 0.002% %	field	link	Sec. 5, end of paragraph 'The probabilistic analy...	2026-10-20	Full size ca (Volkswag Touareg) a speed on l desert trac lakebeds, rover on lo sand; drive obstacle la
MAP-012	Stanley, colour camera vision module	Flat desert roads, 2004 race course independent test data	not reported	learning	drivable terrain detection rate at 10 to 20 m and at more t...	10 to 20 m: 93.25 / 90.46; 50+ m: 87.13 / 69.42 %	field	link	Table II (Sec. 6 Computer Vision Terrain Analysis)	2026-10-20	Full size ca (Volkswag Touareg) a speed on l desert trac lakebeds, rover on lo sand; drive obstacle la
MAP-013	Stanley lidar mapper versus vision mapper	2005 Grand Challenge desert course	not reported	other	effective maximum obstacle detection range of laser mapper;...	approximately 22 m (laser); beyond 70 m (vision) m	field	link	Sec. 6 first paragraph and Figure 14 text	2026-10-20	Full size ca (Volkswag Touareg) a speed on l desert trac lakebeds, rover on lo sand; drive obstacle la
MAP-014	Stanley, modified Volkswagen Touareg	2005 Grand Challenge course, Mojave desert, including the mountainous...	not reported	empirical-field	share of course where the vehicle slowed down because of ru...	18.1 % of course	field	link	Sec. 10 Development Process and Race, subsection...	2026-10-20	Full size ca (Volkswag Touareg) a speed on l desert trac lakebeds, rover on lo sand; drive obstacle la
MAP-015	not applicable (terrain map derivation)	Steep test area (100 m by 100 m), Stordalen catchment, lidar DEM resa...	mean about 22 to 30...	other	mean estimated slope at 0.5 m versus 5 m DEM resolution (re...	30.327 to 24.341 (19.74%) degrees (assumed, see warning); %...	none	link	Sec. 3.2, paragraph beginning 'Generally a reduce...	2026-10-20	Airborne li (about 13 square me subarctic p two 100 m test areas; dunes. Co between a
MAP-016	not applicable (terrain map derivation)	Steep test area, lidar DEM at 0.5 m and 5 m	mean about 23	other	mean and standard deviation of estimated slope, 0.5 m versu...	mean 23.783 to 23.284; STD 11.424 to 8.170 degrees (assumed)	none	link	Table 1 (continued), Steep, 3FD row, columns R05m...	2026-10-20	Airborne li (about 13 square me subarctic p two 100 m test areas; dunes. Co between a

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MAP-017	not applicable (terrain map derivation)	Flat test area (elevation change max 0.7 m), lidar DEM at 0.5 m and 5...	mean about 0.3 to 6...	other	reduction of mean estimated slope from 0.5 m to 5 m resolut...	84% (6.7 to 1.039) Max; 83.6% (5.056 to 0.826) SimpleD % re...	none	link	Sec. 3.2, paragraph beginning 'Generally a reduce...	2026-10-20	Airborne li (about 13 square me subarctic p two 100 m test areas; dunes. Co between a
MAP-018	not applicable (terrain map derivation)	Steep test area, lidar DEM at four resolutions	mean about 22 to 30	other	difference between highest (Max) and lowest (Quadsurface) m...	2.360 (5 m resolution) to 7.991 (0.5 m resolution) degrees...	none	link	Sec. 3.2, same paragraph, sentence beginning 'The...	2026-10-20	Airborne li (about 13 square me subarctic p two 100 m test areas; dunes. Co between a
MAP-019	not applicable (orbital terrain maps)	Mars test sites with HiRISE DEM (1 to 2 m spacing) and CTX DEM (20 m)	not reported (site...	other	average and maximum slope difference, HiRISE versus CTX, be...	5.32 (average); 51 (max) degrees	none	link	Table 1 (Sec. 3), row HiRISE and CTX, columns Dif...	2026-10-20	Mars orbit (HiRISE 1 20 m, HRS MOLA 463 pixel) over valleys an area; base coarser th rove...
MAP-020	not applicable (orbital terrain maps)	Mars test sites, HiRISE DEM versus MOLA DEM (463 m per pixel)	not reported	other	average slope difference HiRISE versus MOLA, before and aft...	6.37 before; 1.96 after degrees	none	link	Table 1, row HiRISE and MOLA; Sec. 4 conclusion i...	2026-10-20	Mars orbit (HiRISE 1 20 m, HRS MOLA 463 pixel) over valleys an area; base coarser th rove...
MAP-021	not applicable (orbital terrain maps)	Seven HiRISE DEM sets (flat, crater, dune, valley floor areas)	not reported	other	root mean square deviation of amplified low resolution slop...	2.8 degrees	none	link	End of Sec. 2.3, sentence 'The Root-Mean-Square D...	2026-10-20	Mars orbit (HiRISE 1 20 m, HRS MOLA 463 pixel) over valleys an area; base coarser th rove...
MAP-022	not applicable (terrain map derivation)	South Fork of the Iowa River agricultural watershed; DEMs at 3 m (lid...	average 0.0115 to 0...	other	average slope at 3 m versus 90 m DEM	0.0301 (3m) versus 0.0115 (90m) m/m	none	link	Poster Table 1 (Values assigned by the GIS interf...	2026-10-20	Gentle agr watershed context; p than peer paper. Sho direction a resolution

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MAP-023	MSL Curiosity rover	Martian sand at 9 sites in Gale crater (Hidden Valley, Ripple Crossin...	up to 16.4 (site ma...	learning	slip prediction RMSE, leave one site out cross validation,...	30.3 versus 20.4 % slip (RMSE)	field	link	Table II and Sec. VI Results and Discussion	2026-10-20	Curiosity (kg, six 0.5 on Martiar 0.38 g; on slip sampl sites (sols visual odo over drive
MAP-024	MSL Curiosity rover	Hidden Valley sand ripples, Gale crater (first sol 709), lowest GTS t...	7.4 (maximum slope...	empirical-field	maximum slip at site, with maximum slope of the samples	80.7 % slip	field	link	Table I, row Hidden Valley (50 samples)	2026-10-20	Curiosity (kg, six 0.5 on Martiar 0.38 g; on slip sampl sites (sols visual odo over drive
MAP-025	MSL Curiosity rover, orbital versus rover borne thermal sensing	all 9 Curiosity sand sites in Gale crater, sols 1 to 986 (leave one s...	up to 16.4 (site ma...	learning	slip prediction RMSE, cross validation with resampling, THE...	24.4 versus 18.6 % slip (RMSE)	field	link	Table II (Resampling, CV column) and Sec. VI para...	2026-10-20	Curiosity (kg, six 0.5 on Martiar 0.38 g; on slip sampl sites (sols visual odo over drive
MAP-026	Camera based perception for off road ground robots (no vehicle tested)	4,176 annotated off road images (512 × 512) from arid desert and semi...	not applicable	learning	mean IoU and pixel accuracy on validation set, DesertFormer...	mIoU 64.4% versus 41.0%; PA 86.1% versus 76.5% %	none	link	Abstract; Table I (model comparison on the valida...	2026-10-20	Image seg of vegetat ground cla sand stren dataset or location ar specified; on the vali not...
MAP-027	Crewed surface vehicle acting as remotely guided robot (Haughton Mars...	Haughton Crater region, Devon Island (polar desert analog); planned w...	not reported	empirical-field	impassable locations found during traverse and how many wer...	Four impassable locations found during data collection phas...	field	link	PDF page 8 (slide numbered 7), 'Investigation Res...	2026-10-20	Rocky pol not sand; presentati not a peer paper; veh not given i
MAP-028	Angled spoke wheel robot (DODO), small scale	River sand track in a tilting test bench, robot guided along a carbon...	minus 25 to 10 (san...	empirical	traversable difficulty at 10 degree uphill on sand, 400 rpm...	4.17 (400 rpm) to 19.48 (1000 rpm) dimensionless (1/(1-S))	lab	link	Table 1 (Traversable difficulty in granular media...	2026-10-20	Small angl (non circu robot on ri a tilting tes traversabl is the ratio not a stan slip mea...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MAP-029	Angled spoke wheel robot (DODO), simulated	Four simulated maps mixing granular media and rigid support, costs fr...	planner admits gran...	other	ratio of d-hat (varying wheel speed) planner result to dist...	maximum 135.94 % longer total actual distance; minimum 44.7...	simulation-only	link	Abstract; Results, Fig. 6 and Table 3	2026-10-20	Small angl (non circu robot on ri a tilting te traversabl is the ratio not a stan slip mea...
MAP-030	Zoe rover (198 kg, four 0.75 m wheels, solar powered)	Atacama Desert, Chile, 2004 field season; satellite maps at 30 m reso...	not reported	empirical-field	share of traverses ending in success or non fault; share en...	45%; 5% (14/272) % of traverses	field	link	Sec. 4.1 Long-distance Traverse, paragraph 'Funct...	2026-10-20	Zoe is a 1 powered f rover with wheels an articulatec Atacama t mostly roc gravelly de per...
MAP-031	Zoe rover (198 kg, four 0.75 m wheels, solar powered)	Atacama Desert, 2005 field season, harder terrain than 2004 with scie...	not reported	empirical-field	traverse outcome shares: success; likely success; faults re...	47% success; 37% likely-success; 15% required intervention;...	field	link	Sec. 4.1 Long-distance Traverse, paragraph 'Opera...	2026-10-20	Zoe is a 1 powered f rover with wheels an articulatec Atacama t mostly roc gravelly de per...
MAP-032	Zoe rover (Atacama survey concept of operations)	Atacama Desert daily survey traverse planned on a 30 m satellite elev...	not reported	empirical-field	planned traverse length on 30 m DEM and expected excess of...	plan 1.2km; actual odometric distance 10-20% longer km; %	none	link	Fig. 9 caption (Sec. 5.2 to 5.3)	2026-10-20	Zoe is a 1 powered f rover with wheels an articulatec Atacama t mostly roc gravelly de per...
MAP-033	Small drone (remotely piloted aircraft) photogrammetry versus terrest...	Coastal dunefield, southern Brazil, February 2019; 810 photos from 10...	not reported	other	vertical accuracy of drone DEM against terrestrial lidar DE...	RMSE of 0.08 m and MAE of 0.06 m m	field	link	Sec. 3.1 TLS and SfM-MVS, paragraph beginning 'Th...	2026-10-20	Coastal du southern E surveyed day after li the sand w and textur footprints, tracks and
MAP-034	not applicable (map currency for route planning)	Same coastal dunefield, airborne lidar DEM of October 2010 compared w...	not reported	empirical-field	mean crest displacement and implied migration rate	mean length of ≈44.5 m in 9 years; ≈5 m/year m; m per year	field	link	Sec. 3.2 ALS and SfM-MVS, paragraph on dune crest...	2026-10-20	Coastal du southern E surveyed day after li the sand w and textur footprints, tracks and

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
MAP-035	Mawson six wheel rocker bogie research rover	Mars analogue terrain facility (indoor), solid and loose soil, slopes...	not reported (slope...	learning	failed runs (stuck on rocks or in the crater, or wheels bog...	10 (100%); 1 (9.1%); 0 (0%) failed runs (% of runs)	lab	link	Sec. 7 Experimental Results, Planning; Table 4	2026-10-20	Indoor Ma yard in a n with solid soil, rocks crater; six rocker boc rover (Ma making 0.: crabbing r
MAP-036	Mawson six wheel rocker bogie research rover	Same Mars analogue yard, successful runs only (10 per method)	not reported	learning	mean executed path length; mean accumulated terrain cost re...	16.04 m versus 15.87 m; 39.34% cost reduction; 46.50 versus...	lab	link	Sec. 7, Table 5 and the following paragraphs	2026-10-20	Indoor Ma yard in a n with solid soil, rocks crater; six rocker boc rover (Ma making 0.: crabbing r
ENV-001	Ouster OS-1 lidar (static), the sensor whose data appear in Figure 1c...	Indoor garage test, artificial dust cloud (flour, cement, calcium car...	not applicable	empirical	target detectability behind dust cloud, low versus high dus...	low density: car and truck remain detectable; high density:...	lab	link	Sec. 2 Experimental results, paragraph on Figures...	2026-10-20	Indoor arti not wind b sand or du a rover's c densities r quantified
ENV-002	Velodyne VLP-16 lidar (static), off road vehicle context	Outdoor scene with human, trees and background objects; dust blown by...	not applicable	empirical	share of dust points with intensity 1; share of non dust po...	about 71% (dust, intensity 1); almost 88% (non dust, intens...	field	link	Sec. 3.3, discussion of Figure 3 histograms	2026-10-20	Dust raise blower on distances from a VLF was 4 per point clou evaluated natural de
ENV-003	Velodyne VLP-16 lidar (static), off road vehicle context	Experiments 1 and 3 of Table 1 (lidar to dust 4 m and 8 m; lidar to t...	not applicable	other	F1 score of dust removal, LIDROR versus SOR	97.55 versus 0.26 %	field	link	Table 5 (Evaluation results) and Sec. 4 text	2026-10-20	Dust raise blower on distances from a VLF was 4 per point clou evaluated natural de
ENV-004	Unitree A1 quadruped (electric joint motors) with 3 kg payload on its...	Outdoor flat terrain, commanded forward speed 1 m/s, continuous walki...	0 (level)	learning	time until motor overheat protection stopped the robot (bas...	approximately 7 minutes versus over 27 minutes min	field	link	Abstract; Sec. IV B Real-world Locomotion Results...	2026-10-20	Legged ro small joint a wheeled ambient te and motor value not r initial mot temperatu betw...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
ENV-005	Clearpath Husky with Velodyne HDL-32E lidar (static recordings)	Outdoor scenes with dust and fog particles, scan matching test on 10...	not applicable	other	translational ICP error after convergence, with particles v...	17cm versus 2.5cm cm	field	link	Sec. 5.5.1 Pose Estimation Task and Fig. 8	2026-10-20	Static Hus with Velod 32E; artifi and fog in scenes; nc rover in na dust.
ENV-006	Clearpath Husky with Velodyne HDL-32E lidar	Outdoor scenes with dust and fog; 2D occupancy grid 30 m by 30 m at 0...	not applicable	learning	false obstacle cells prevented; true obstacle cells removed	96% prevented, 0.4% removed (point); 89% prevented, 0.1% re...	field	link	Table 5 and Sec. 5.5.2 Obstacle Detection Task	2026-10-20	Static Hus with Velod 32E; artifi and fog in scenes; nc rover in na dust.
ENV-007	Clearpath Husky with Velodyne HDL-32E lidar	16,247 labelled scans, 19 outdoor scenes with dust and fog	not applicable	learning	F1 score for airborne particle classification; dust class F...	94% (particle versus non particle); 92% (dust class) %	field	link	Sec. 5.2 Point Classification Evaluation (Table 3...	2026-10-20	Static Hus with Velod 32E; artifi and fog in scenes; nc rover in na dust.
ENV-008	Stationary PV modules (proxy for a solar powered rover panel)	Inland UAE desert climate, Al Ain, spring to summer 2018	not applicable	empirical-field	electric power loss after 10, 20, 30 and 90 days without cl...	3%, 5%, 7% and 13% % power loss	field	link	Sec. 4 Conclusions, second paragraph; abstract	2026-10-20	Fixed PV r glass coup Ain, UAE, August 20 major sanc rover pane ground an moving, vi
ENV-009	Glass coupons mimicking PV cover glass	Inland UAE desert climate, Al Ain, spring to summer 2018	not applicable	empirical-field	average radiation penetration loss after 10, 20, 30 and 90...	10%, 15%, 19% and 32% % radiation loss	field	link	Sec. 3, paragraph on Figure 7; Sec. 4 Conclusions...	2026-10-20	Fixed PV r glass coup Ain, UAE, August 20 major sanc rover pane ground an moving, vi
ENV-010	Weather record at the PV test site	Al Ain, UAE, typical days during April to August 2018	not applicable	empirical-field	ambient temperature on a typical day after the 3rd month of...	39 °C and 45 °C °C	field	link	Sec. 3, paragraph on Figure 4 (ambient temperatur...	2026-10-20	Air temper one inland surface ar componer temperatu rover are r higher; on day per pe

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
ENV-011	MER Opportunity rover solar arrays	Meridiani Planum, Mars; 4.95 Martian years of daily energy, dust fact...	not applicable	empirical-field	correlation between energy generated and dust factor; lag e...	0.84; 21 sol lag explains approximately 60% of variance cor...	field	link	Sec. III A (computational analysis), paragraphs a...	2026-10-20	Mars surfa atmospher dust, 0.38 space gra junction ce desert soil dew, heav loads and sand...
ENV-012	MER Opportunity rover solar arrays	Meridiani Planum, Mars, 4.95 Martian years	not applicable	empirical-field	permanent array degradation rate	0.002±0.116 percent per Martian year (95% confidence below...	field	link	Sec. III A, paragraph beginning 'The year-over-ye...	2026-10-20	Mars surfa atmospher dust, 0.38 space gra junction ce desert soil dew, heav loads and sand...
ENV-013	Triple junction solar cell with cover glass (CIC), lab abrasion	JSC 1A Mars simulant blown at about 40 m/s (4 PSI nozzle) for 5 minut...	not applicable (cel...	empirical	power conversion efficiency reduction after 5 min, cell at...	10% (60) and 13.54% (45) % PCE reduction	lab	link	Sec. III B Experimental Results, paragraph on Fig...	2026-10-20	Mars surfa atmospher dust, 0.38 space gra junction ce desert soil dew, heav loads and sand...
ENV-014	Panther mobile robot carrying an Ouster OS0 lidar (moving sensor subs...	Outdoor moving robot recordings with blower raised dust (the most rea...	not reported	learning	dust F1 score, best learned variant (SITdv, with odometry c...	0.82 versus 0.30 F1 (dust class, 0 to 1)	field	link	Sec. 4.4 Results in Real Environments with Moving...	2026-10-20	Mining and scenes wi introduced blower (m subset) in is under 1 points in a Ouster OS
ENV-015	Panther mobile robot carrying an Ouster OS0 lidar	Generalisation test: trained on static mining centre subsets (interio...	not reported	learning	dust F1 score on unseen moving robot subset, best variant (...	0.76 versus 0.30 F1 (dust class, 0 to 1)	field	link	Table 11 (Generalization results, without odometr...	2026-10-20	Mining and scenes wi introduced blower (m subset) in is under 1 points in a Ouster OS

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparab warning
ENV-016	Six parked passenger vehicles (silver mid size sedans, silver economy...	Hot dry desert city (Tempe, Arizona), summer days between 25 June and...	not applicable	empirical-field	average final dashboard surface temperature, sun versus sha...	68.9°C versus 47.8°C °C	field	link	Sec. 3.1 In-Vehicle Temperatures and Heating Rate...	2026-10-20	Closed ca not a rove electronic ambient ai temperatu test days i the paper' not used h Arizona su co...
ENV-017	Camera based object detection for road vehicles (RT-DETR baseline det...	DAWN dataset of real traffic scene images in fog, rain, snow and sand...	not applicable	learning	mAP50 of baseline RT-DETR in the sand (sandstorm) subset ve...	Sand 65.0%; Fog 86.1%; Rain 83.6%; Snow 77.8% % mAP50	field	link	Sec. 4.6, Table 6 and the paragraph after it	2026-10-20	Road traffi cars and p terrain cla sandstorm come from web collec with few ir condition;

Appendix B. UAE facts (from `research/raw/uae_facts.csv`)

ID	Fact	Source	Status
UAE01	Rashid 1 rover mass was about 10 kg.	link	verified-from-source
UAE02	Rashid 1 had four wheels; wheels designed to climb slopes up to 20 degrees and rocks up to 10 cm high.	link	verified-from-source
UAE03	Rashid 1 instruments: two wide-field navigation cameras, a microscopic imager and a thermal imager (Langmuir probe also listed by a secondary source).	link	partially-verified
UAE04	A UAE wheel-terrain paper gives the Rashid wheel as 0.1 m diameter, 7.5 mm wide, 12 fixed grousers 10 mm high, wheel weight 1 kg.	link	verified-from-source
UAE05	The same paper modelled soil with a modified Reece pressure-sinkage model plus Wong shear model, using an MBRSC regolith sample representing Lacus Somniorum; simulator was Vortex Studio.	link	verified-from-source
UAE06	Single-wheel rig results at slip 0, 0.25, 0.5, 0.75: experimental sinkage 0.010, 0.0158, 0.0169, 0.0192 m; drawbar pull 5.2, 10.1, 16.77, 24.1 N.	link	verified-from-source
UAE07	Khalifa University master's thesis (2022, Mariam Alhammadi) analysed Rashid-1 wheels with terramechanics and FEA; reports peak tractive efficiency at 10-15% slip and that larger grousers raise drawbar pull while general...	link	verified-from-source
UAE08	A 2022 IAC paper (Alhammadi et al.) used finite element analysis on Rashid wheel and grouser geometry on lunar regolith; beyond a point larger diameter, more or taller grousers do not further reduce required drawbar pul...	link	verified-from-source
UAE09	Samples from ESA, CREST (Universite libre de Bruxelles), Cambridge Graphene Centre, York University and others were bonded to Rashid 1's wheels for a Material Adhesion and Abrasion Detection experiment.	link	verified-from-source
UAE10	Rashid 1 was carried by ispace's Hakuto-R Mission 1 lander, which failed to land on 2023-04-26 (JST); the rover never reached the surface.	link	verified-from-source
UAE11	ispace attributed the failure to software: the altitude sensor reading over a crater rim was discarded, so the lander believed it was at zero altitude when about 5 km up and ran out of propellant.	link	verified-from-source
UAE12	Rashid 1 landing target was Atlas crater (47.5N, 44.4E) on the southeastern edge of Mare Frigoris; UAE planned to change the site for the second rover.	link	verified-from-source

ID	Fact	Source	Status
UAE13	MBRSC's Director General said in May 2023 the team had not yet decided any technical improvements for the next rover, because Rashid 1 never operated on the Moon.	link	verified-from-source
UAE14	Rashid 2 will fly on Firefly Aerospace's Blue Ghost Mission 2 to the Moon's far side; Firefly said the rover will test various wheel materials for durability in lunar dust.	link	verified-from-source
UAE15	MBRSC completed Rashid 2 and shipped it to the US for preparation with Firefly; funded by the TDRA ICT Fund.	link	verified-from-source
UAE16	Rashid 2 testing included drive-off and deployment trials with Firefly in the US, antenna/arm deployment under simulated lunar gravity and vibration tests at EARTH (an EDGE subsidiary) in the UAE, and thermal-vacuum tes...	link	verified-from-source
UAE17	Rashid 2 launch timing sources conflict: Gulf News/Khaleej Times (Nov-Dec 2025) say 2026; Wikipedia Blue Ghost Mission 2 cites March 2026 reporting of NET Dec 2026 and infobox 2027; SpaceQ (2026-10-07) says Firefly list...	link	partially-verified
UAE18	Blue Ghost Mission 2 also carries Lunar Pathfinder, LuSEE-Night, Fleet Space SPIDER and other payloads; destination is the lunar far side.	link	partially-verified
UAE19	Hope (Emirates Mars Mission) is an orbiter, not a rover; launched from Tanegashima on an H-IIA on 19 July 2020 (UTC date varies by source) with Mars orbit insertion 9 Feb 2021.	link	verified-from-source
UAE20	Hope carries three instruments: a colour imager, infrared spectrometer and ultraviolet spectrometer, studying atmosphere and escape; no surface mobility role.	link	verified-from-source
UAE21	Mars Science City (announced Sept 2017): planned 177,000 sq m simulated Mars settlement, AED 500 million, domes of 3D-printed desert sand; the article does not mention rover or terrain testing.	link	verified-from-source
UAE22	TII (Abu Dhabi) presented an unmanned ground vehicle described as built to operate in desert without human intervention; named lead was Enrico Natalizio of TII's Autonomous Robotics Research Centre.	link	partially-verified
UAE23	xRift, an autonomous off-road vehicle by SteerAI (ATRC ecosystem, supported by SASC; TII engineers developed the base technology), debuted 20 Jan 2026 at UMEX for Gulf deserts and rocky terrain.	link	verified-from-source
UAE24	xRift specs reported: 4200×1880×1400 mm, 1,350 kg, 500 kg payload, 225 hp, max unmanned speed 50 km/h; production units expected by end of 2026.	link	verified-from-source
UAE25	Space42 (Abu Dhabi) showed a UGV at UMEX 2026 with 114 mm ground clearance, 25 degree slope capability and 10-80 km/h speed range; whether wheeled or tracked is not stated.	link	verified-from-source
UAE26	Emirates Robotics Competition (4th edition, RIT Dubai) task is a TIAGo Pro library book-retrieval challenge in the Mohammed Bin Rashid Library; Phase 1 in Gazebo simulation (deadline 2026-09-15), competition day Nov 202...	link	verified-from-source
UAE27	A recent arXiv scout-rover study on dune and lunar-simulant terrain (USC, UPenn, NASA) was tested at White Sands, New Mexico, not in the UAE.	link	verified-from-source

Appendix C. Sources (176 distinct URLs; retrieval dates 2026-10-11, 2026-10-13, 2026-10-20, 2026-10-21, per row in `research/evidence.csv`)

#	Title	URL	Evidence rows
1	Modeling of the interaction of rigid wheels with dry granular media (Agarwal et al., J. Terramechanics 2019 / arXiv 2019)	arxiv.org	TM-001, TM-002, TM-003, TM-004, TM-005, TM-006
2	General scaling relations for locomotion in granular media (Slonaker et al., arXiv 2016/PRE 2017)	arxiv.org	TM-007, SCL-001, SCL-002
3	Simulating Robotic Locomotion in Sand: Resistive Force Theory in an Open-Source Physics Engine (Brown, Treers, Daltorio; arXiv 2606.19504, under revi...	arxiv.org	TM-008, TM-009, TM-010
4	A resistive force model for legged locomotion on granular media (Li, Zhang, Goldman; arXiv 1911.00079)	arxiv.org	TM-011, TM-012
5	Granular Resistive Force Theory Implementation for Three-Dimensional Trajectories (Treers, Cao, Stuart; IEEE RA-L 2021)	edg.berkeley.edu	TM-013

#	Title	URL	Evidence rows
6	Foot Shape-Dependent Resistive Force Model for Bipedal Walkers on Granular Terrains (arXiv 2403.03460)	arxiv.org	TM-014
7	Dynamic Wheel-Soil Model for Lightweight Mobile Robots with Smooth Wheels (Irani, Bauer, Warkentin; J. Intell. Robot. Syst. 2012)	carleton.ca	TM-015
8	Application of a Dynamic Pressure-Sinkage Relationship for Lightweight Mobile Robots (Irani, Bauer, Warkentin; Int. J. Veh. Auton. Syst., preprint)	carleton.ca	TM-016, TM-017
9	The effects of increasing velocity on the tractive performance of planetary rovers (Rodriguez-Martinez, Buse, Van Winnendael, Yoshida; arXiv 2306.021...	arxiv.org	TM-018, TM-019, LEG-022
10	Granular Scaling Laws for Accurate Prediction of Wheel Mobility on Slopes in Low-Gravity Environments (Omura, Ishigami; EasyChair preprint 14770 / IS...	easychair.org	TM-020, SCL-003, SCL-004, SCL-005, SCL-006
11	Modeling of slip rate-dependent traversability for path planning of wheeled mobile robot in sandy terrain (Sakayori, Ishigami; Frontiers in Robotics...	www.frontiersin.org	TM-021, LRN-021
12	Planetary rover mobility on loose soil: terramechanics theory for side slip prediction and compensation (Carletti, MSc thesis, Politecnico di Milano...	www.politesi.polimi.it	TM-022, TM-023, TM-024
13	Experimental Device for Measuring Sandy Soil Sinkage Parameters (Nguyen Van, Matsuo, Koumoto, Inaba; Bull. Fac. Agr. Saga Univ. 2007)	saga-u.repo.nii.ac.jp	TM-029
14	Terramechanics-based model for steering maneuver of planetary exploration rovers on loose soil	doi.org	TM-IS07-1, TM-IS07-2, TM-IS07-3, TM-IS07-4, TM-IS07-5
15	Slope traversal controls for planetary exploration rover on sandy terrain (Ishigami, Nagatani, Yoshida, J. Field Robotics 26(3) 264-286, 2009)	doi.org	TM-IS09-1, TM-IS09-2, TM-IS09-3, TM-IS09-4, TM-IS09-5
16	Grouser wheel high-speed traction performance: DEM simulation and experimental result (Takehana et al., J. Terramechanics 120, 101084, 2025)	doi.org	TM-TK1, TM-TK2, TM-TK3, TM-TK4, TM-TK5
17	Modeling wheeled locomotion in granular media using 3D-RFT and sand deformation	doi.org	TM-YU1, TM-YU2, TM-YU3, TM-YU4, TM-YU5
18	Effect of Angle of Attack on Slope Climbing Performance (Creager, Jones, Smith; NASA/TM-2017-219549, prepared for ASCE Earth and Space 2016)	ntrs.nasa.gov	TM3-001, TM3-002, TM3-003, TM3-004, TM3-005, TM3-006 (+1)
19	Benefit of "Push-pull" Locomotion for Planetary Rover Mobility (Creager, Moreland, Skonieczny, Johnson, Asnani, Gilligan; ASCE Earth and Space 2012)	ntrs.nasa.gov	TM3-008, TM3-009, TM3-010
20	Push-Pull Locomotion for Vehicle Extrication (Creager, Johnson, Plant, Moreland, Skonieczny; NASA/TM-2014-218431)	ntrs.nasa.gov	TM3-011, TM3-012, TM3-013, TM3-014
21	Drawbar Pull (DP) Procedures for Off-Road Vehicle Testing (Creager, Asnani, Oravec, Woodward; NASA/TP-2017-219384)	ntrs.nasa.gov	TM3-015, TM3-016, TM3-017, TM3-018, TM3-019
22	A Study Demonstrating that Using Gravitational Offset to Prepare Extraterrestrial Mobility Missions is Misleading (Hu, Li, Rogg, Schepelmann, Chandle...	ntrs.nasa.gov	TM3-020, TM3-021, TM3-022, SCL-007, SCL-008
23	Lunar Rover Optimization Platform for Wheel Traction Studies (Gerdts, Breckenridge, Johnson; ASCE Earth and Space 2020)	ntrs.nasa.gov	TM3-023, TM3-024
24	Modeling and Analysis of a Reconfigurable Rover for Improved Traversing over Soft Sloped Terrains (Lyu, Zhang, Yao, Zhu, Jia; Biomimetics 2023, 8(1):...	doi.org	TM3-025, TM3-026, TM3-027
25	Design of a special rigid wheel for traversing loose soil (Elsheikh; Scientific Reports 2023, 13:171)	doi.org	TM3-028, TM3-029, TM3-030
26	Measuring the Normal Stress Distribution Acting on a Locked-Wheel of Push-Pull Locomotion Rovers via a Wheel Sensor System (Fujiwara, Oshima, Iizuka;...	doi.org	TM3-031, TM3-032, TM3-033
27	Development and Performance Evaluation of a Bevameter for Measuring Soil Strength (Kim, Im, Choi, Oh, Park, Biswas; Sensors 2021, 21(4):1541)	doi.org	TM3-034, TM3-035, TM3-036, TM3-037
28	Vibration-Based Recognition of Wheel-Terrain Interaction for Terramechanics Model Selection and Terrain Parameter Identification for Lugged-Wheel Pla...	doi.org	TM3-038, TM3-039, TM3-040, TM3-041
29	Evaluation and Comparison of Sand Mechanics in Tottori Sand Dunes and Moon (Takehana, Suzuki, Tanaka, Nakamura; Journal of Evolving Space Activities...	doi.org	TM3-042, TM3-043, TM3-044, TM3-045, TM3-046

#	Title	URL	Evidence rows
30	Effects of wheel width on running performance of four-wheeled small lunar rover (Watanabe, Maru, Kawaguchi, Tanaka; Transactions of the JSME 2026, 92...	doi.org	TM3-047, TM3-048, TM3-049, TM3-050
31	Study on Effect of Grousers Mounted Flexible Wheel for Mobile Rovers (Iizuka, Kubota; Journal of Asian Electric Vehicles 2012, 10(1):1591 to 1597)	doi.org	TM3-051, TM3-052, TM3-053, TM3-054
32	Performance of the Boeing LRV Wheels in a Lunar Soil Simulant, Report 2: Effects of Speed, Wheel Load, and Soil (Melzer; U.S. Army Engineer Waterways...	ntrs.nasa.gov	TM3-055
33	Aircraft Tire Behavior During High-Speed Operations in Soil (Leland, Smith; NASA TN D-6813, August 1972)	ntrs.nasa.gov	TM3-056
34	Coordinated Control of Slip Ratio for Wheeled Mobile Robots Climbing Loose Sloped Terrain (Li, Wang; The Scientific World Journal 2014, 396382)	doi.org	TM3-057
35	Experimental Analysis of Tread Pattern Effects on Tire Tractive Performance on Sand using an Indoor Traction Measurement System with Forced-slip Mech...	doi.org	TM3-058, TM3-059, TM3-060, TM3-061
36	Tri-axial Contact Reaction at the Tire-Soil Interface (Kasetani, Nakashima, Shinone, Shimizu, Miyasaka, Ohdoi; Engineering in Agriculture, Environmen...	doi.org	TM3-062
37	Prediction of Tractive Limitations of a Rigid Wheel on Loose Soil (Nagaoka, Mizukami, Kubota; Journal of Asian Electric Vehicles 2012, 10(1):1583 to...	doi.org	TM3-063
38	Wheel Slip Classification Method for Mobile Robot in Sandy Terrain Using In-Wheel Sensor (Omura, Ishigami; Journal of Robotics and Mechatronics 2017,...	doi.org	TM3-064, TM3-065
39	Prediction of Wheel Traveling Performance Using Ground Contact Stress Models (Saengprachatanarug, Fukami, Ueno, Taira; Engineering in Agriculture, En...	doi.org	TM3-066
40	Development and Evaluation of Mobility and Excavation Rover Toward Lunar Base Construction (Sutoh; Journal of Robotics and Mechatronics 2024, 36(2):3...	doi.org	TM3-067, TM3-068
41	A study on the alignment angle of the lugs for the lunar rover's crawler mechanism and its mobility performance (Maeda, Tsujita; Mechanical Engineeri...	doi.org	TM3-069
42	Development of a Flexible Wheel That Allows Rovers to Traverse Lunar Soil (Iizuka, Kubota; Journal of Asian Electric Vehicles 2009, 7(1):1231 to 1237)	doi.org	TM3-070
43	Performance of Wheel-Typed Vehicle with Crawlers in Between Front and Rear Wheels on Soft Ground (Eto, Satomi, Takahashi; Journal of Robotics and Mec...	doi.org	TM3-071
44	Using physics-based simulation towards eliminating empiricism in extraterrestrial terramechanics applications (Hu et al. 2024)	arxiv.org	SIM-001, SIM-002, SIM-003, SIM-004
45	Using high-fidelity discrete element simulation to calibrate an expeditious terramechanics model in a multibody dynamics framework (Zhang et al. 2024)	arxiv.org	SIM-005, SIM-006, SIM-007, SIM-008
46	A Physics-Based Continuum Model for Versatile, Scalable, and Fast Terramechanics Simulation (Unjhawala et al. 2025)	arxiv.org	SIM-009, SIM-010, SIM-011
47	Data-Driven Terramechanics Approach Towards a Realistic Real-Time Simulator for Lunar Rovers (Kern et al. 2026)	arxiv.org	SIM-012, SIM-013, SIM-014
48	Traction Performance Evaluation for a Rashid-1 Rover Wheel (Hurrell et al. 2025, Space Science Reviews 221:37)	doi.org	SIM-015, SIM-016, SIM-017
49	Soil Flow Analysis for Planetary Rovers Based on Particle Image Velocimetry and Discrete Element Method (Ono, Lichtenheldt, Yoshida; ISTVS 2020)	elib.dlr.de	SIM-018
50	DEM Analysis of Contact Forces and Tractive Performance of Rigid Wheel in Granular Media (Jelinek et al.; ISTVS 2021 preprint)	www.cavs.msstate.edu	SIM-019, SIM-020
51	Sinkage Study in Granular Material for Space Exploration Legged Robot Gripper (Candalot, Hurrell et al.; ISTVS 2024 paper 6422)	arxiv.org	SIM-021, SIM-022, SIM-023, SIM-024
52	RoverDevKit: An open, physics-grounded tradespace toolkit for conceptual design of lunar micro-rovers (Reifschneider 2026)	arxiv.org	SIM-025, SIM-026, SIM-027, SIM-028

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53	Scout-Rover cooperation: online terrain strength mapping and traversal risk estimation for planetary-analog explorations (Liu et al. 2026)	arxiv.org	SIM-029
54	The Archimede Rover: A Comparison between Simulations and Experiments (Caruso et al., Robotics 2023, 12(5), 125)	doi.org	SIM-030
55	SlipNet: Enhancing Slip Cost Mapping for Autonomous Navigation on Heterogeneous and Deformable Terrains	arxiv.org	LRN-001, LRN-002
56	Slip Prediction Using Visual Information	m.roboticsproceedings.org	LRN-003, LRN-004, LRN-005
57	Terrain Adaptive Navigation for Planetary Rovers	www-robotics.jpl.nasa.gov	LRN-006
58	DeepTerramechanics: Terrain Classification and Slip Estimation for Ground Robots via Deep Learning	arxiv.org	LRN-007, LRN-008
59	Slippage prediction for off-road mobile robots via machine learning regression and proprioceptive sensing	hal.univ-grenoble-alpes.fr	LRN-009
60	Learning to Drive on Mars: Visual Multimodal Traversability Estimation for Off-World Navigation	arxiv.org	LRN-010, LRN-011
61	AI4MARS: A Dataset for Terrain-Aware Autonomous Driving on Mars	openaccess.thecvf.com	LRN-012
62	NeSAM: Neuro-Symbolic Kinodynamics with Soil Adaptation for Off-Road Mobility	arxiv.org	LRN-013, LRN-014
63	PrePARE: Predictive Proprioception for Agile Failure Event Detection in Robotic Exploration of Extreme Terrains	arxiv.org	LRN-015
64	Adaptive Meta-Learning for Identification of Rover-Terrain Dynamics	arxiv.org	LRN-016
65	Predicting Terrain Mechanical Properties in Sight for Planetary Rovers with Semantic Clues	arxiv.org	LRN-017
66	Physical Reservoir Computing in Hook-Shaped Rover Wheel Spokes for Real-Time Terrain Identification	arxiv.org	LRN-018
67	Learning All-Terrain Locomotion for a Planetary Rover with Actively Articulated Suspension	arxiv.org	LRN-019
68	Slip estimation model for traversability-based motion planning of cargo rover on extraterrestrial surface	www.frontiersin.org	LRN-020
69	Effect of Gait Design on Proprioceptive Sensing of Terrain Properties in a Quadrupedal Robot	arxiv.org	LRN-022
70	Proprioceptive Slip Detection for Planetary Rovers in Perceptually Degraded Extraterrestrial Environments	arxiv.org	LRN-024
71	Two Years of Visual Odometry on the Mars Exploration Rovers (Maimone, Cheng, Matthies; J. Field Robotics 2007)	robotics.jpl.nasa.gov	FLD-001, FLD-002, FLD-003, FLD-009
72	Characterization of traverse slippage experienced by Spirit rover on Husband Hill at Gusev crater (Li et al., JGR 2008)	authors.library.caltech.edu	FLD-004, FLD-014, FLD-015, FLD-016, FLD-017
73	Slip Sliding Away On The Red Planet (JPL news release reposted by Mars Daily, 2004-03)	www.marsdaily.com	FLD-005, FLD-006, FLD-007, FLD-008
74	Mars Exploration Rover Surface Operations: Driving Spirit at Gusev Crater (Leger et al., 2005)	www-robotics.jpl.nasa.gov	FLD-010, FLD-011, FLD-012, FLD-013
75	Curiosity Rover Mobility Issues: Crossing Martian Megaripple Fields (Arvidson, Maimone, Gildner, Hines; LPSC 2016 poster abstract 1137)	www-robotics.jpl.nasa.gov	FLD-018
76	Traction Control Design and Integration Onboard the Mars Science Laboratory Curiosity Rover (Toupet et al., IEEE Aerospace 2018)	www-robotics.jpl.nasa.gov	FLD-019, FLD-020, FLD-022
77	Proprioceptive Slip Detection for Planetary Rovers in Perceptually Degraded Extraterrestrial Environments (Kilic, Gu, Gross; arXiv 2207.13629)	arxiv.org	FLD-021
78	Roving on the Edge: Robotic Operations Power Perseverance's Ascent of Jezero Crater Rim (Maimone et al., AAS 08-138)	robotics.jpl.nasa.gov	FLD-023, FLD-024, FLD-025
79	Driving Strategies Used During Perseverance Rapid Traverse Campaign (Rankin et al., 2023)	robotics.jpl.nasa.gov	FLD-026
80	Analysis of Angle of Attack for Efficient Slope Ascent by Rovers (Inotsume, CMU-RI-TR-15-22 MS thesis)	publications.ri.cmu.edu	FLD-027, FLD-028, FLD-029, FLD-030, FLD-031
81	The Effect of Assistive Anchor-Like Grousers on Wheeled Rover Performance over Unconsolidated Sandy Dune Inclines (Sensors 2016, 16(9):1507)	www.ncbi.nlm.nih.gov	FLD-032, FLD-033, FLD-034, FLD-035
82	Traversing Steep and Granular Martian Analog Slopes With a Dynamic Quadrupedal Robot (Kolvenbach et al., arXiv 2106.01974)	arxiv.org	FLD-036

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83	Failure Mechanisms and Risk Estimation for Legged Robot Locomotion on Granular Slopes (Liao and Qian, arXiv 2603.06928)	arxiv.org	FLD-037, FLD-038
84	Rovers in the desert (Emily Lakdawalla, The Planetary Society, 2012-05-14)	www.planetary.org	FLD-039
85	The Need for and Feasibility of Alternative Ground Robots to Traverse Sandy and Rocky Extraterrestrial Terrain (Li and Lewis, Adv. Intell. Syst. 2022...	arxiv.org	FLD-040
86	Self-driving rovers tested in Mars-like Morocco (ESA, 2018-12-19)	esa.int	FLD-041
87	AMADEE-18 Mission Report V1.2 public version (Austrian Space Forum, Oman Dhofar Mars analog, Feb 2018)	oewf.org	FLD-042
88	Traverse Performance Characterization for the Mars Science Laboratory Rover (Heverly et al. 2013, J. Field Robotics 30(6), 835-846)	doi.org	FLD-HV13-1, FLD-HV13-2, FLD-HV13-3, FLD-HV13-4, FLD-HV13-5a, FLD-HV13-5b (+1)
89	Localization and Surface Characterization by Zhurong Mars Rover at Utopia Planitia (Ding et al., Research Square preprint 2021; published as Surface...	www.researchsquare.com	PLN-001, PLN-002, PLN-003, PLN-004, PLN-005, PLN-006 (+2)
90	Mars Curiosity Rover Mobility Trends During the First Seven Years (Rankin, Maimone, Biesiadecki, Patel, Levine, Toupet; J. Field Robotics 2021, accep...	www-robotics.jpl.nasa.gov	PLN-009, PLN-010, PLN-011, PLN-012, PLN-013, PLN-014 (+3)
91	Spirit Mars Rover Mission: Overview and selected results from the northern Home Plate Winter Haven to the side of Scamander crater (Arvidson et al.,...	repository.si.edu	PLN-018, PLN-019, PLN-020, PLN-021, PLN-022
92	Opportunity Mars Rover mission: Overview and selected results from Purgatory ripple to traverses to Endeavour crater (Arvidson et al., JGR 2011)	authors.library.caltech.edu	PLN-023, PLN-024, PLN-025, PLN-026
93	Mars Exploration Rover Surface Operations: Driving Opportunity at Meridiani Planum (Biesiadecki et al., IEEE SMC 2005)	www-robotics.jpl.nasa.gov	PLN-027, PLN-028, PLN-029
94	Learning to Predict Slip for Ground Robots (Angelova, Matthies, Helmick, Sibley; IEEE ICRA 2006)	authors.library.caltech.edu	PLN-030, PLN-031, PLN-032, PLN-033
95	Terrain physical properties derived from orbital data and the first 360 sols of Mars Science Laboratory Curiosity rover observations in Gale Crater (...	authors.library.caltech.edu	PLN-034, PLN-035
96	Lunar rock investigation and tri-aspect characterization of lunar farside regolith by a digital twin (Ding et al., Nature Communications 2024)	www.nature.com	PLN-036, PLN-037, PLN-038, PLN-039, PLN-040
97	Chemistry, mineralogy, and grain properties at Namib and High dunes, Bagnold dune field, Gale crater, Mars: A synthesis of Curiosity rover observatio...	pmc.ncbi.nlm.nih.gov	PLN-041, PLN-042, PLN-043, PLN-044
98	Slip Modeling and Estimation for a Planetary Exploration Rover: Experimental Results from Mt. Etna (Bussmann, Meyer, Steidle, Wedler; IEEE/RSJ IROS 2...	elib.dlr.de	PLN-045, PLN-046
99	Mobility Performance of the Lunar Roving Vehicle: Terrestrial Studies, Apollo 15 Results (Costes, Farmer, George; NASA TR R-401, 1972)	ntrs.nasa.gov	PLN-047, PLN-048, PLN-049, PLN-050, PLN-051
100	Improving Slip Prediction on Mars Using Thermal Inertia Measurements (Cunningham, Nesnas, Whittaker; Robotics: Science and Systems XIII, 2017)	roboticsproceedings.org	PLN-052, PLN-053, PLN-054, PLN-055, MAP-023, MAP-024 (+1)
101	Simulations of Mars Rover Traverses (Zhou, Arvidson, Bennett, Trease, Lindemann, Bellutta, Iagnemma, Senatore; J. Field Robotics 2014)	www.kiss.caltech.edu	PLN-056, PLN-057, PLN-058, PLN-059
102	Data-Driven Surface Traversability Analysis for Mars 2020 Landing Site Selection (Ono, Rothrock, Almeida, Ansar, Otero, Huertas, Heverly; IEEE Aerosp...	www-robotics.jpl.nasa.gov	PLN-060, PLN-061, MAP-007, MAP-008, MAP-009
103	Lunar soil strength estimation based on Chang'E-3 images (Gao, Spiteri, Li, Zheng; Advances in Space Research 2016, accepted manuscript)	openresearch.surrey.ac.uk	PLN-062, PLN-063, PLN-064
104	Results from the First Year and a Half of Mars 2020 Robotic Operations (Verma, Maimone, Graser, Rankin, et al.; IEEE Aerospace Conference 2023)	www-robotics.jpl.nasa.gov	PLN-065, PLN-066, PLN-067, PLN-068, PLN-069
105	Robotic Operations During Perseverance's First Extended Mission (Verma, Maimone, Kaplan, Thiel, Rothenberger, et al.; IEEE Aerospace Conference 2025)	www-robotics.jpl.nasa.gov	PLN-070, PLN-071
106	The effect of limb kinematics on the speed of a legged robot on granular media	arxiv.org	LEG-001, LEG-002

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107	Sensitive dependence of the motion of a legged robot on granular media	arxiv.org	LEG-003, LEG-004
108	Systematic study of the performance of small robots on controlled laboratory substrates	goldmanlab.gatech.edu	LEG-005, LEG-006
109	Failure Mechanisms and Risk Estimation for Legged Robot Locomotion on Granular Slopes	arxiv.org	LEG-007, LEG-008, LEG-009
110	Traversing Steep and Granular Martian Analog Slopes With a Dynamic Quadrupedal Robot	arxiv.org	LEG-010, LEG-011, LEG-012, LEG-013
111	Virtual Energy Management for Physical Energy Savings in a Legged Robot Hopping on Granular Media	www.frontiersin.org	LEG-014
112	Learning Terrain-Adaptive Humanoid Locomotion on Granular Terrain	arxiv.org	LEG-015
113	Evaluation of Locomotion Performance for a Wheel-Paddle Robot (doctoral thesis, Ritsumeikan Univ.)	ritsumei.repo.nii.ac.jp	LEG-016
114	Traction Performance of Wheel and Track for Soft-Soil Traversal (Yoshida, ICRA 2010 planetary rovers workshop slides)	ewh.ieee.org	LEG-017, LEG-018, LEG-019, LEG-020
115	A Grouser Spacing Equation for Determining Appropriate Geometry of Planetary Rover Wheels	publications.ri.cmu.edu	LEG-021
116	OriWheelBot: An origami-wheeled robot	arxiv.org	LEG-023, LEG-024
117	Performance Evaluation of Wheels for Lunar Vehicles (Summary Report), Miscellaneous Paper M-70-4 (Freitag, Green, Melzer; Waterways Experiment Statio...	ntrs.nasa.gov	MOB-001, MOB-002, MOB-003, MOB-004, MOB-005, MOB-006 (+7)
118	Performance Evaluation of Wheels for Lunar Vehicles, Technical Report M-70-2 (Freitag, Green, Melzer; Waterways Experiment Station, March 1970)	ntrs.nasa.gov	MOB-014, MOB-015, MOB-016, SCL-014, SCL-015, SCL-016
119	Performance of Boeing LRV Wheels in a Lunar Soil Simulant, Report 1: Effect of Wheel Design and Soil, Technical Report M-71-10 (Green, Melzer; Waterw...	ntrs.nasa.gov	MOB-017
120	ET-148 Next-Generation NATO Reference Mobility Model (NRMM), STO Technical Report (Dasch, Jayakumar eds.; NATO AVT Exploratory Team ET-148)	www.mtu.edu	MOB-018, MOB-019
121	Validation of the AMC-71 Mobility Model, Technical Report M-76-5 (Schreiner, Willoughby; Waterways Experiment Station, March 1976)	apps.dtic.mil	MOB-020, MOB-021, MOB-022, MOB-023, MOB-024, MOB-025
122	Performance of Soils Under Tire Loads, Report 5: Development and Evaluation of Mobility Numbers for Coarse-Grained Soils, Technical Report 3-666 (Gre...	apps.dtic.mil	MOB-026, MOB-027, MOB-028, MOB-029
123	Performance of Soils Under Tire Loads, Report 8: Application of Test Results to Tire Selection for Off-Road Vehicles, Technical Report 3-666 (Turnage...	apps.dtic.mil	MOB-030, MOB-031, MOB-032
124	Performance of Soils Under Tire Loads, Report 2: Analysis of Tests in Yuma Sand Through August 1962, Technical Report 3-666 (Powell, Green; Waterways...	apps.dtic.mil	MOB-033, MOB-034, MOB-035
125	A Dimensionless Consolidation of WES Data on the Performance of Sand Under Tire Loads, Contract Report 3-130 (Nuttall; December 1965)	apps.dtic.mil	MOB-036, MOB-037
126	A Technique for Estimating the Slope-Climbing Ability of Wheeled Vehicles in Sand, Miscellaneous Paper 4-535 (Waterways Experiment Station, October 1...	apps.dtic.mil	MOB-038, MOB-039, MOB-040, MOB-041
127	Trafficability of Soils: Tests on Coarse-Grained Soils with Self-Propelled and Towed Vehicles, 1956 and 1957, Technical Memorandum 3-240, Fifteenth S...	apps.dtic.mil	MOB-042
128	Trafficability of Soils: Tests on Coarse-Grained Soils with Self-Propelled and Towed Vehicles, 1958-1961, Technical Memorandum 3-240, Seventeenth Sup...	apps.dtic.mil	MOB-043, MOB-044, MOB-045, MOB-046
129	Evaluation of the computer simulation model NTVPM for assessing military tracked vehicle cross-country mobility (Wong, Jayakumar, Preston-Thomas; Pro...	apps.dtic.mil	MOB-047, MOB-048
130	Sabkha Trafficability (Valentine, Cromwell; Tetra Tech for the Office of Naval Research; field work May to July 1980)	apps.dtic.mil	MOB-049, MOB-050
131	Test Operations Procedure (TOP) 02-2-619A, Soft Soil Mobility (U.S. Army Test and Evaluation Command; Automotive Directorate, Aberdeen Test Center; 1...	apps.dtic.mil	MOB-051, MOB-052, MOB-053
132	Performance of Soils Under Tire Loads, Report 4: Analysis of Tests in Sand from September 1962 Through November 1963, Technical Report 3-666 (Turnage...	apps.dtic.mil	MOB-055, MOB-056

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133	The Rolling Resistance and Sinkage of Towed Dual Wheel Combinations in Sand (Hetherington, Littleton; ISTVS International Conference on the Performan...	apps.dtic.mil	MOB-057
134	Development of Mathematical Model for Pneumatic Tire-Soil Interaction, TACOM Technical Report 11900 (LL147), Grumman RE-479 (Karafiath; July 1974)	apps.dtic.mil	MOB-058, MOB-059
135	Correlation of Mobility Cone Index with Fundamental Engineering Properties of Soil, Miscellaneous Paper SL-81-4 (Rohani, Baladi; Waterways Experiment...	apps.dtic.mil	MOB-060, MOB-061
136	A Limited NRMM Validation Study for ISTVS, Miscellaneous Paper GL-94-20 (Murphy, Randolph; Waterways Experiment Station, May 1994)	apps.dtic.mil	MOB-062, MOB-063
137	Traction on Sand (Jose Manuel Nobre de Oliveira Peca; PhD thesis, October 1982; institutional repository copy)	dspace.ncl.ac.uk	MOB-064, MOB-065, MOB-066, MOB-067, MOB-068, MOB-069 (+2)
138	Helically-driven granular mobility and gravity-variant scaling relations (Thoesen, McBryan, Marvi; RSC Advances 9, 12572 to 12579, 2019)	doi.org	SCL-009, SCL-010, SCL-011
139	Surprising simplicity in the modeling of dynamic granular intrusion (Agarwal, Karsai, Goldman, Kamrin; Science Advances 7(17) eabe0631, 2021)	doi.org	SCL-012, SCL-013
140	The Wahiba Sand Sea, Oman: report of the Royal Geographical Society symposium of 7 October 1987 (R. Western; Emirates Natural History Group Bulletin...	enhg.org	DUN-001, SND-029, SND-030
141	Remote sensing of spatial variability in aeolian dune and interdune morphology in the Rub al Khali, Saudi Arabia (Al-Masrahy and Mountney; Aeolian Re...	eprints.whiterose.ac.uk	DUN-002, DUN-003, DUN-004, DUN-005
142	Fluvial and aeolian interactions in modern and ancient dryland continental sedimentary systems: implications for reservoir heterogeneity (M. A. J. Al...	etheses.whiterose.ac.uk	DUN-006, DUN-007, DUN-008
143	Migration of barchan dunes in Qatar: controls of the Shamal, teleconnections, sea level changes and human impact (Engel, Boesl, Brueckner; Geoscience...	mdpi-res.com	DUN-009, DUN-010, DUN-011, DUN-012
144	A study of global sand seas (McKee, ed.; USGS Professional Paper 1052, 1979): chapter F, Dune forms and wind regime (Fryberger with Dean) and chapter...	pubs.usgs.gov	DUN-013, DUN-014, DUN-015, DUN-016, DUN-017, DUN-018 (+1)
145	Multi-source DEM vertical accuracy evaluation of Taklimakan Desert hinterland based on ICESat-2 ATL08 and UAV data (Wang, Li, Liu, Li; Remote Sensing...	mdpi-res.com	DUN-020, DUN-021, DUN-022
146	Developing a framework of Quaternary dune accumulation in the northern Rub al Khali, Arabia (Farrant, Duller, Parker, Roberts, Parton, Knox, Bide; Qu...	nora.nerc.ac.uk	DUN-023, DUN-024, DUN-025
147	Geomorphological features and sedimentological aspects of some coastal and inland sand dunes, Jeddah region, Saudi Arabia (Gheith and Abou Ouf; Arab...	agjsr.agu.edu.bh	DUN-026, DUN-027, SND-031
148	Gone with the wind: dune provenance and sediment recycling in the northern Rub al Khali, United Arab Emirates (Farrant, Mounteney, Burton, Thomas, Ro...	nora.nerc.ac.uk	SND-001, SND-002, SND-003, SND-004
149	Physicochemical characterization and origin of aeolian sand dunes in southeastern Qatar: a comparative study with Mediterranean sand dunes (Al-Awah a...	adgeo.copernicus.org	SND-005, SND-006, SND-007, SND-008, SND-009, SND-010
150	Tracing transcontinental sand transport: from Anatolia Zagros to the Rub al Khali sand sea (Garzanti, Vermeesch, Al-Ramadan, Ando, Limonta, Rittner,...	discovery.ucl.ac.uk	SND-011, SND-012, SND-013, SND-014, SND-015
151	Integrated geological, hydrogeological, and geophysical investigations of a barchan sand dune in the Eastern Region of Saudi Arabia (Benaafi, Hanafy,...	mdpi-res.com	SND-016, SND-017, SND-018, SND-019, SND-020
152	Assessment of density and shear strength of Eastern Saudi sands using dynamic cone penetration test (DCPT) (A. M. Hamid, MSc thesis, November 2013)	eprints.kfupm.edu.sa	SND-021, SND-022, SND-023, SND-024
153	Effect of fine material on penetration resistance (A. M. Hamid; CSCE Annual Conference 2017, Vancouver, paper CON33)	legacy.csce.ca	SND-025, SND-026
154	Use of reservoir sediments to improve engineering properties of dune sand in Oman (Qureshi, Alsaidi, Aziz, Chang, Rasool, Kazmi; Applied Sciences 11(...	mdpi-res.com	SND-027, SND-028

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155	Risk-aware Path Planning via Probabilistic Fusion of Traversability Prediction for Planetary Rovers on Heterogeneous Terrains (Endo, Taniai, Yonetani...	arxiv.org	MAP-001, MAP-002, MAP-003, MAP-004
156	Deep Probabilistic Traversability with Test-time Adaptation for Uncertainty-aware Planetary Rover Navigation (Endo, Taniai, Ishigami; arXiv 2409.0064...	arxiv.org	MAP-005, MAP-006
157	Stanley: The Robot that Won the DARPA Grand Challenge (Thrun et al., Journal of Field Robotics 23(9), 2006)	pages.cs.wisc.edu	MAP-010, MAP-011, MAP-012, MAP-013, MAP-014
158	Estimating slope from raster data: a test of eight algorithms at different resolutions in flat and steep terrain (Tang, Pilesjo, Persson; Geodesy and...	doi.org	MAP-015, MAP-016, MAP-017, MAP-018
159	Improved large-scale slope analysis on Mars based on correlation of slopes derived with different baselines (Wang, Wu; ISPRS Archives XLII-3/W1, 2017)	doi.org	MAP-019, MAP-020, MAP-021
160	The Effect of DEM Resolution on Slope Estimation and Sediment Predictions (Sadeghi, Beeson, Lang, Tomer; conference poster, SWAT conference)	swat.tamu.edu	MAP-022
161	DesertFormer: Transformer-Based Semantic Segmentation for Off-Road Desert Terrain Classification in Autonomous Navigation Systems (Chebolu et al.; ar...	arxiv.org	MAP-026
162	Traverse Planning Experiments for Future Planetary Surface Exploration (Hoffman, Voels, Mueller, Lee; ASCE Earth and Space 2012 presentation, NASA NT...	ntrs.nasa.gov	MAP-027
163	Slip considered path planning of a novel angled-spoke based robot in a terrain mixture of granular media and rigid support (Lee, Kim, Ryu, Seo; Scien...	doi.org	MAP-028, MAP-029
164	Long-Distance Autonomous Survey and Mapping in the Robotic Investigation of Life in the Atacama Desert (Wettergreen, Wagner, Jonak, Baskaran, Deans,...	longhorizon.org	MAP-030, MAP-031
165	Second Experiments in the Robotic Investigation of Life in the Atacama Desert of Chile (Wettergreen et al.; Proc. iSAIRAS 2005, ESA SP-603)	longhorizon.org	MAP-032
166	Aeolian dune modelling from airborne LiDAR, terrestrial LiDAR and Structure from Motion-Multi View Stereo (Grohmann, Garcia, Affonso, Albuquerque; ar...	arxiv.org	MAP-033, MAP-034
167	Enhanced Stochastic Mobility Prediction on Unstructured Terrain using Multi-Output Gaussian Processes (Lui, Peynot, Fitch, Sukkarieh; Advances in Int...	opus.lib.uts.edu.au	MAP-035, MAP-036
168	Modelling of LIDAR sensor disturbances by solid airborne particles (Hadj-Bachir, de Souza, Nordqvist, Roy; arXiv 2105.04193, 2021)	arxiv.org	ENV-001
169	Design of Dust-Filtering Algorithms for LiDAR Sensors Using Intensity and Range Information in Off-Road Vehicles (Afzalaghaeinaeini, Seo, Lee, Lee; S...	doi.org	ENV-002, ENV-003
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